

OpenAI's 719 AI-generated math papers, explained from zero

On 2026-10-06 OpenAI published **722 math manuscripts (372 result families)** generated by an unreleased internal model. Among the claims: the quasi-Riemann hypothesis, the Unique Games Conjecture, matrix-multiplication exponent $\omega \leq 9/4$, non-amenability of Thompson's group F, and the plane not being 5-colorable — a run of problems open for decades. The next day three manuscripts were withdrawn over a sign error. Mathematicians dubbed it the "Mathocalypse." The goal of this deck: using **only linear algebra, calculus and high-school math**, take 10 representative results far enough that you can judge how big each is, how hard it was, why it matters, and how far it has been verified.

First edition 2026-10-08 (Part II outlook §17–§22 and Part III explainers §23–§32 added the same day) · repo snapshot `openai/math@fd4aeeb` (includes the Oct 7 withdrawals and fixes) · every fact is sourced; [P] = primary material read (paper / repo / Lean source), [R] = secondary reporting, Δ = unverified or conflicting · English edition of `notes.html` (Chinese)

ORIENTATION

0. What this deck will and won't give you

YOU WILL GET

A "ladder" understanding of each of 10 headline results: start from a toy example you can compute by hand and climb level by level to the real statement; learn how long the problem was stuck and where; see exactly what OpenAI claims, how much the Lean formalization covers, and what human experts have said so far; and what it means for mathematics, for AI, and for AI safety.

YOU WON'T GET

The ability to judge whether any proof is correct. As of 2026-10-08, **none** of these 10 results has a human domain expert publicly saying "I have read and checked it."

"Verification" in this deck means two things only: (1) a static review of the Lean source and statements, and (2) a summary of public reactions. Both are labeled with sources and confidence.

0.1 Reading path

§1 fills in three basics (what mathematicians do, what an "open problem" is, how to judge how big a result is). §2–§3 cover the event and what Lean actually guarantees — **these two sections determine how you should read every "claim" that follows**. §4–§14 are the 10 results, each with the same structure: *ladder* → *problem* → *why it was hard* → *what OpenAI*

claims → *verification status* → *if true*; read them in any order. §15–§16 give cross-cutting patterns and implications for AI. **Part II (§17–§22) looks ahead**: three trend curves (task length, open-problem solve rate, formal-proof cost), why bottlenecks move from proofs to statements and specs, what AI math and coding imply for debate, singular learning theory, Guaranteed Safe AI and neurosymbolic systems, and a T0–T4 timeline with falsifiers. **Part III (§23–§32) has the from-zero explainers**: one chapter for each of the 10 results in §4–§13, built up from the most basic concepts, with hand-checkable examples and [Check] exercises, about ten times the length of the matching Part I section. If a Part I section loses you, go to its explainer. §33 collects reactions, §34 lists open questions and a learning path, §35 is the glossary and sources.

0.2 Cheat sheet

§	RESULT (FAMILY NO.)	OPEN FOR	LEAN	ONE LINE
4	Quasi-Riemann hypothesis (003)	~125 yrs	yes	ζ has no zeros at all in $\text{Re } s > 7/8$
5	Irrationality exponent of $\pi = 2$ (017)	~70 yrs	yes Δ	π cannot be approximated "unusually well" by fractions
6	Matrix multiplication $\omega \leq 9/4$ (107)	57 yrs	yes	a drop of 0.12 at once; the previous 12 years gave 0.0017
7	Unique Games Conjecture (102)	24 yrs	yes	optimality of many approximation algorithms becomes theorem
8	$L = RL = BPL$ (103)	~30 yrs	yes	randomness is useless for small-memory computation
9	The plane is not 5-colorable (158)	76 yrs (last lower-bound step 2018)	yes	chromatic number of the plane $\in \{6, 7\}$
10	Thompson's group F non-amenable (248)	47 yrs	yes	famous for "proofs" in both directions later withdrawn
11	Navier–Stokes blowup (separate Sept release)	Millennium Prize problem	yes	only the "with forcing" alternative; credit disputed
12	Relativistic Vlasov–Maxwell (362)	40 yrs	yes	large-data global smoothness for a plasma model (one species)
13	Mézard–Parisi formula (221)	25 yrs	yes	free-energy formula for sparse spin glasses (even-arity case)
14	Hodge-related withdrawals (032 etc.)	—	no	one sign error, three papers withdrawn: the risk of no formalization

1. What do mathematicians actually do?

In one sentence: **they study the structure of abstract objects and establish truths about them by proof.** What separates mathematics from science and engineering is *how you know something is right*: mathematics by logical proof (once proven, true forever), science by experiment (can be overturned), engineering by "it works and measures well enough."

1.1 Four words

WORD	MEANING	EXAMPLE FROM THIS EVENT
conjecture	a statement widely believed but unproven	the Unique Games Conjecture (Khot 2002)
theorem	a statement that has been proven	the Goemans–Williamson 0.878 algorithm
counterexample	a concrete object showing that a "for all ..." conjecture is false	the 9-dimensional Borsuk counterexample; about 20% of this batch are counterexamples/disproofs
formalization	writing a proof as code a computer can check step by step (here Lean 4)	about 300/719 top-line results have Lean proofs

1.2 Two cultures of mathematics

In *The Two Cultures of Mathematics* (2000), Fields medalist Tim Gowers splits mathematicians into **problem-solvers** (who crack specific hard problems, like Erdős problems) and **theory-builders** (who invent new concepts and frameworks, like "group" or "manifold"). Thurston's *On Proof and Progress in Mathematics* (1994) goes further: the point of mathematics is not to produce theorems but to **advance human understanding**. These two essays are the key to this controversy — what AI has broken open at scale is the problem-solving culture, and that is what Tao means by "the end of Math 1.0."

1.3 Five questions for judging how big a result is

- 1. How long was it open, and who tried?** A problem open for 50 years with repeated failures by top experts weighs very differently from one posed two years ago.
- 2. Last step or giant leap?** Moving ω from 2.3713 to 2.3711 is the former; to 2.25 is the latter.
- 3. How much does it unlock?** If UGC holds, hundreds of "if UGC then ..." papers become theorems at once.

4. **Is the method new?** New methods often matter more than the result, because they can be reused.
5. **Who chose the problem, and how many attempts failed?** The most overlooked question in this event; see §2.3 and §15.

THE EVENT · TIMELINE AND NUMBERS

2. What happened

DATE	EVENT
2025-07	An OpenAI experimental model and Google DeepMind's Gemini Deep Think reach gold-medal scores at IMO 2025 (35/42) — competition problems with known answers fall. [R]
2026-09-08	OpenAI announces that ~10,000 agents over 88 hours proved finite-time blowup "with forcing" for 3D Navier–Stokes (166 pages + Lean), sparking a credit dispute (§11). [P] [R]
late 2026-09	The Advisory Group on Mathematics and AI hosted by the Institute for Advanced Study (members include Terence Tao) publishes recommendations for releasing AI-generated mathematics. [R]
2026-10-06	openai/math released: 722 manuscripts / 372 result families, Apache-2.0, with a Lean library and 10 abridged reasoning summaries. [P]
2026-10-07	3 manuscripts withdrawn (one sign error, cascading), 14 repaired, 13 re-cited, 6 new formalizations → 719 manuscripts , "300/719 ≈ 42% of top-line results formalized." A statement from the Association for Human Mathematics (AHM) is reposted on Tao's blog; Scott Aaronson posts "The Mathocalypse". [P] [R]
2026-10-08	Tao's blog runs a guest post by Lozano-Robledo, What should we tell our students? [P]

2.1 How the results were produced

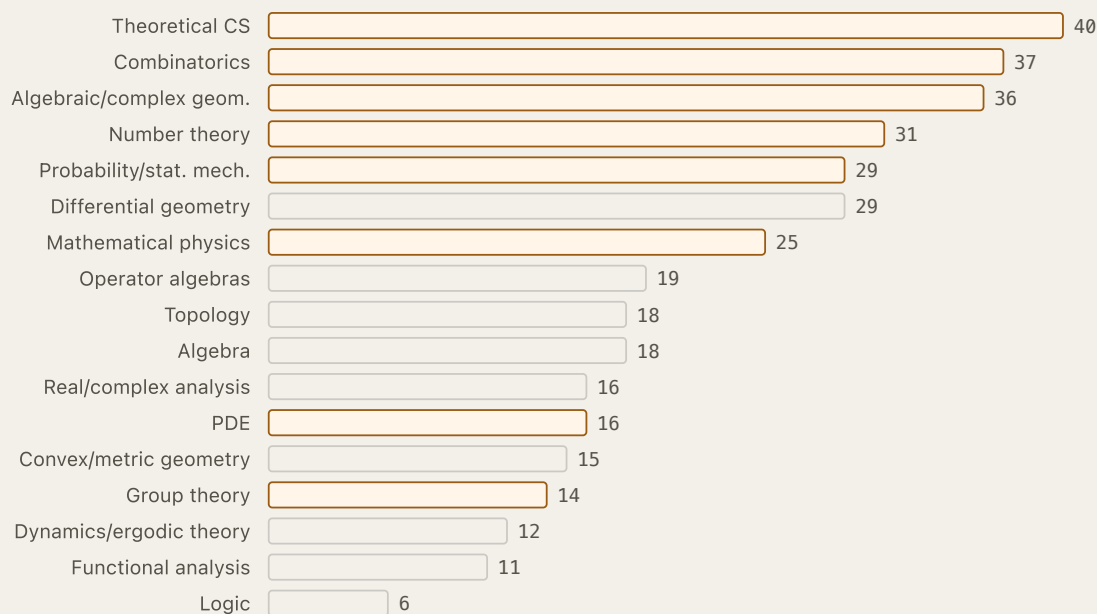
From the README [P]: "The vast majority of results were obtained with the same procedure using an unreleased internal OpenAI model. On average, each result used three hours of ChatGPT Pro thinking compute... the model was posed approximately 4,000 problems... requiring an appropriate level of significance led to the catalog." In short: **~4,000 open problems → filtered for significance → 372 families**. An OpenAI spokesperson told Scientific American that almost everything came from "a single prompt handed to a single AI agent" [R]; MIT's Andrew Sutherland responded that until the model can be replicated, "one-shotting" claims should be treated as unverified — "We should ask for receipts."

Exceptions: the README concedes that the zero-free region for ζ (§4) and the Hodge conjecture for CM abelian varieties (§14) were *not* produced by the fixed procedure, and that the $\text{Re } s > 11/12$ write-up was human-edited. What exactly differed (more compute? human guidance?) is not disclosed $\triangle$.

2.2 Reconciling the numbers

CLAIM	WHAT IT ACTUALLY MEANS
"722" vs "719"	722 at release; 719 after the three Oct 7 withdrawals. Early news used 722.
"372 families" vs "377"	The catalog is numbered 001–377 with 045/061/070/123/163 unused: 372 actual families.
"~42% formalized"	$300 \text{ top-line results} \div 719 \text{ manuscripts}$ — numerator and denominator in different units. By family, $242/372 \approx 65\%$ have a Lean scope document.
"162 fully + 73 partially formalized"	Reported by some outlets; source not found △ .

372 result families by field (orange = this deck covers at least one result in depth)



Source: the 17 subject sections of [openai/math overview.pdf](#), counted entry by entry (total 372). Catalog numbers 045/061/070/123/163 are unused; no reason given.

2.3 The one thing to keep in mind: selection effects

Epoch AI's FrontierMath Erdős (2026-09) took 68 Erdős problems **pre-registered** as open and required Lean proofs; GPT-6 Astra solved 2 ($\approx 3\%$), all other models 0. OpenAI's release is **post-hoc selection**: of $\sim 4,000$ problems it kept the "significant" ones and published no failure list. A rough hit rate is $\leq 9\%$ (by family) — but problem difficulty is not comparable.

Both numbers are right; they answer different questions: the first asks "given an arbitrary open problem, can it solve it?", the second "given 4,000, how many good results can it pick out?"

3. What "has a Lean proof" actually guarantees

Lean is a proof assistant: you write a theorem and its proof as code, and a small **kernel** checks every inference step. If the kernel has no bugs and only accepted axioms are used, passing the check means "this Lean statement is true." The difference from human refereeing is like the difference between a compiler with a type system and code review — the latter misses things; the former doesn't miss *what it checks*.

What "has a Lean proof" guarantees: the trust stack, bottom up

① **Lean kernel checks every proof step**

this repo: sorry = 0, added axioms = 0, native_decide = 0

strong

② **Only the 3 standard axioms**

all 416 Comparator configs allow only propext / Quot.sound / Classical.choice

strong

③ **Lean statement = what the paper claims?**

challenge files written by OpenAI (agents); a human must read the definitions

human

④ **Are definitions in deps and patches right?**

24 patches (largest ~2 MB) + 11 lana-agents/* dependencies of unknown origin

unaudited

⑤ **The ~58% with no formalization**

no machine check at all; the 3 Hodge-related papers withdrawn on 10-07 were here

none

①② are guaranteed by the machine; ③④⑤ still need humans. Audit data: source grep of `lean/OAI/` (122,458 .lean files, ~26M lines) plus Comparator config counts (2026-10-08, commit fd4aeeb); not compiled.

3.1 Our static audit of the Lean library [P]

CHECK	RESULT	MEANING
sorry (unfinished proof)	0	no "fill in later" steps
axiom declarations	0 (all hits are in comments)	no smuggled-in axioms
native_decide / implemented_by / extern	0	no "trust the compiler" backdoors
decide +kernel	≈1.31M lines, 5,848 files	large finite computations done by the kernel — still trusted, but the proofs are enormous
Comparator configs	416, all allowing only the three standard axioms; only 2 enable the second independent kernel nanoda	see below
lean/patches/	24 patches to third-party dependencies (not the Lean kernel, not Mathlib), largest ~2 MB	patches change definitions and proofs in dependencies △
dependencies	30, of which 11 are <code>lana-agents/*</code> repos of unknown origin	an open point in the trust chain △
formalization.yaml	<code>review: status: unchecked</code>	OpenAI's own label: not reviewed

Scope: lean/OAI/ has 122,458 .lean files, ~26M lines; we only grepped the source and counted configs, without compiling (a full build needs many GB of memory and a Linux sandbox). Independent re-checks: Dave Goldblatt ran the ζ 7/8 proof through Comparator and the nanoda kernel (~63 minutes) [repo]; a third party reviewed the statement of the 9/4 matrix-multiplication theorem [repo].

3.2 Comparator: what it guards against, and what it doesn't

The Lean FRO's Comparator takes a "challenge file" (only the theorem statement, with `sorry` as the proof, importing only trusted libraries such as Mathlib) and a "solution module": the theorem proven in the solution must be **the same type** as the challenge statement, using only allowed axioms. It guards against swapping the statement, changing definitions or adding axioms inside the proof. It does **not** guard against the challenge statement itself failing to say what the paper claims — and the challenge files were written by OpenAI (its agents). That is why the "verification status" of each section below focuses on **reading the statement line by line**.

A LIVE COUNTEREXAMPLE

A footnote in OpenAI's plane-coloring paper mentions a May 2026 manuscript that claimed the chromatic number of the plane is 7 with a "formalized theorem" — but that theorem **took a false density bound as a hypothesis**. The formalization was fine; the proven statement was the problem. [P, as reported in the OpenAI paper; original not read △] This is the

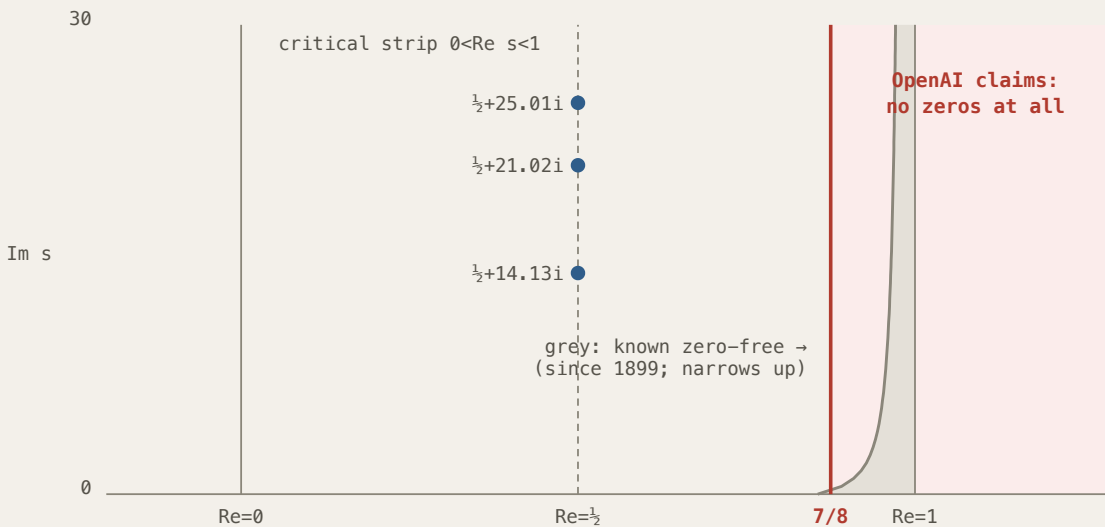
same kind of problem as specification gaming in AI: the system perfectly satisfies the written spec, but the spec is not what you wanted.

NUMBER THEORY · FAMILY 003

4. The quasi-Riemann hypothesis: a ceiling on the noise in the primes

Lost? → §23 Explainer 01: the same result explained from zero.

- Level 0** Count primes by hand: 25 below 100, 168 below 1000, 78,498 below 10^6 . Let $\pi(x)$ = "number of primes $\leq x$."
-
- Level 1** **Prime number theorem:** $\pi(x) \approx \text{Li}(x) = \int_2^x dt/\ln t$. $\text{Li}(10^6) \approx 78,626.5$ vs the true 78,498 — the ratio tends to 1 (proved 1896). The real question is **how big the error $\pi(x) - \text{Li}(x)$ is**.
-
- Level 2** **The ζ function encodes all the primes:** $\zeta(s) = \sum 1/n^s = \prod_p 1/(1-p^{-s})$. The identity holds because every integer factors uniquely into primes. By hand at $s = 2$: using only $p = 2, 3, 5$ gives 1.5625; all primes below 100 give 1.6419; the true value is $\pi^2/6 \approx 1.6449$.
-
- Level 3** Extend s to complex numbers $s = \sigma + it$. A **zero** is a point where $\zeta(s) = 0$. The non-trivial zeros all lie in the "critical strip" $0 < \text{Re } s < 1$; the first three are at $\frac{1}{2} + 14.13i$, $\frac{1}{2} + 21.02i$, $\frac{1}{2} + 25.01i$. **Riemann hypothesis (1859):** all of them lie on $\text{Re } s = \frac{1}{2}$.
-
- Level 4** **Why zeros control the error:** in the explicit formula each zero $\rho = \beta + i\gamma$ contributes an "oscillating noise" term of size about x^β . All zeros with $\text{Re } \rho \leq \theta \Rightarrow$ error about x^θ . The Riemann hypothesis $\Leftrightarrow$ error about $\sqrt{x}$. Think of the zeros as the "spectrum" of the primes, with β setting how fast each frequency's amplitude grows.
-
- Level 5** **Previously known zero-free regions:** de la Vallée Poussin (1899) $\sigma > 1 - c/\log t$; Vinogradov–Korobov (1958) somewhat wider. **Key point: their width shrinks to 0 as the height $t \rightarrow \infty$.** No one had ever proven that a *fixed* $\delta > 0$ exists with no zeros in the whole strip $\text{Re } s > 1 - \delta$.



Blue dots: the first three non-trivial zeros of ζ (all on $\text{Re } s = \frac{1}{2}$; the Riemann hypothesis says every one is). Grey band: schematic de la Vallée Poussin-type zero-free region (its width shrinks to 0 as the height t grows; curve shape illustrative, constant 0.1). Red region $\text{Re } s > \frac{7}{8}$: the fixed-width zero-free region claimed by OpenAI family 003. Nobody had previously proven any fixed-width zero-free strip.

4.1 The problem

Quasi-Riemann hypothesis: there is a fixed vertical line $\text{Re } s = \theta < 1$ with no zeros to its right. It is a weak form of the Riemann hypothesis ($\theta = \frac{1}{2}$), but before 2026 it was just as far out of reach.

4.2 Why it was hard

Classical methods start from positivity of the Euler product for $\sigma > 1$ (e.g. the trick $3 + 4\cos \theta + \cos 2\theta \geq 0$), and that positivity weakens with height, so the regions obtained necessarily get narrower. The other route, zero-density estimates (e.g. Guth–Maynard 2024, arXiv:2405.20552), can say "there are few zeros with $\text{Re } s > \sigma$," but not "there are none." A fixed width needs a genuinely new mechanism.

4.3 What OpenAI claims

CLAIM [P]

Main paper (2026-09-30, 199 pages): "We prove that all finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ and all Dirichlet L-functions are zero-free in the half-plane $\Re s > 7/8$... In particular, the Riemann zeta function is zero-free in this half-plane." The paper stresses: "The Riemann hypothesis remains open."

The family also includes a separate proof of 11/12 (49 pages, written with human assistance) and a 9-page "uniform exclusion of Landau–Siegel zeros."

Method sketch (introduction only): pass to $\mathbb{Q}(\sqrt{-3})$ and build a "completed sum" from Fourier coefficients of the cubic theta series, which can be evaluated two ways (a reflection formula and Poisson summation); a zero with $\beta > 7/8$ would make the two estimates contradict each other. Finally the factorization $L_F(s, \chi \circ N) = L(s, \chi) \cdot L(s, \chi \chi_{-3})$ transfers the result to all Dirichlet L-functions.

4.4 Verification status

Machine

- The Lean challenge statement uses only standard Mathlib definitions, e.g. `theorem riemannZeta_ne_zero_of_seven_eighths_lt_re {s : ℂ} (hs : 7/8 < s.re) : riemannZeta s ≠ 0`. [P]
- Solution ~2,926 files, 487k lines; `sorry/axiom/native_decide` all 0. [P]
- Independent re-check: Goldblatt ran it under two kernels; `#print axioms` lists only the three standard axioms. [P]
- ⚠ The dependency patch `PrimeNumberTheoremAnd-lean4341.patch` is ~40k lines and replaces two upstream sorries with proofs — part of the "dependency library" is OpenAI's own code.
- The paper's later applications (Section 6 corollaries) are outside the formalization.

Humans

- Alex Kontorovich (Rutgers; founder of the PrimeNumberTheoremAnd formalization project): "Quasi-RH?!?!?! Are you kidding me? If a human did this, it would be an instant Fields Medal..." [R]
- Levent Alpöge: "Big, big, big, big props for quasiriemann and no Siegel zeroes." [R]
- As of 10-08, **no human expert has publicly said they read all 199 pages.** ⚠

Assessment: as long as Mathlib's definition of `riemannZeta` is correct and the Lean kernel has no bug, the theorem is "proven in the machine sense" — an order of magnitude more

trustworthy than an unrefereed paper. The remaining risk sits in the build chain (patches and dependencies).

4.5 If true

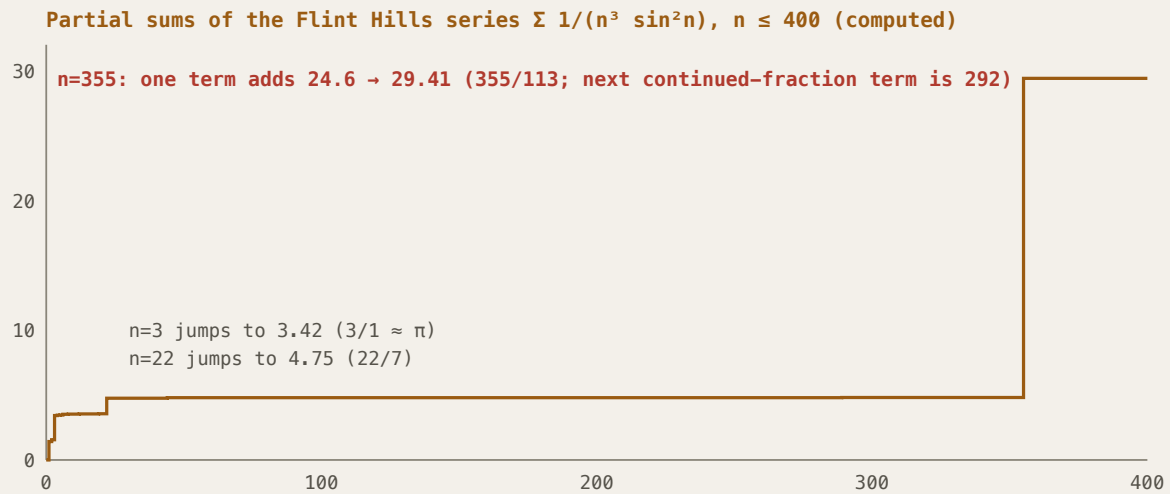
- **A power saving in the prime number theorem:** $\pi(x) - \text{Li}(x) = O(x^{7/8} \log^2 x)$. Going from "width shrinking to 0" to "fixed width" is a qualitative change; but between $7/8$ and $1/2$ everything is still unknown.
- **Siegel zeros ruled out:** an obstacle that since 1935 made many constants ineffective disappears (e.g. effective lower bounds for class numbers of imaginary quadratic fields).
- **Algorithmic corollaries** (paper Cor. 1.2): least quadratic non-residue $\leq C(\log p)^{32} \Rightarrow$ square roots mod p and the Miller primality test become *unconditional* deterministic polynomial-time algorithms. **No impact on practical cryptography** (randomized algorithms are used in practice, AKS 2002 already gave unconditional deterministic primality testing, and RSA and elliptic-curve security are unaffected).

NUMBER THEORY · FAMILY 017 (+ 005)

5. How well can fractions approximate π ?

Lost? → §24 Explainer 02: the same result explained from zero.

- | | |
|----------------|--|
| Level 0 | 22/7 differs from π by 1.26×10^{-3} ; 355/113 (Zu Chongzhi, 5th century) by only 2.67×10^{-7} . Compare the benchmark $1/q^2$: $1/113^2 \approx 7.8 \times 10^{-5}$ — 355/113 beats it by nearly 300x. |
| Level 1 | Continued fractions: $\pi = [3; 7, 15, 1, 292, 1, 1, \dots]$. The larger the next term, the "unusually better" the previous truncation. 355/113 is followed by 292, which is why it is so accurate. |
| Level 2 | Dirichlet (1842): every irrational number has infinitely many p/q with $ x - p/q < 1/q^2$ (pigeonhole principle). So "exponent 2" is always achievable. |
| Level 3 | Irrationality exponent $\mu(x)$ = the largest v such that $ x - p/q < q^{-(v)}$ has infinitely many solutions. Rationals: 1; Liouville numbers: ∞ ; algebraic irrationals (e.g. $\sqrt{2}$): exactly 2 — Roth's theorem (1955), Fields Medal 1958; e : 2; almost every real number: 2. The hard part is proving it for a <i>specific number</i> . |
| Level 4 | History of upper bounds for π: Mahler 1953: 42 → Mignotte 1974: 21 (20 for large enough denominators) → Hata 1993: 8.016 → Salikhov 2008: 7.606 → Zeilberger–Zudilin 2020: 7.103. Seventy years took it from 42 to 7.1 — still far from 2. |



Each jump happens when n is the numerator of an unusually good fraction approximating π (so $\sin n$ is tiny). Whether the series converges depends on how often and how large these jumps are, which is governed by the irrationality exponent of π . Values computed by the build script of this deck.

5.1 What OpenAI claims

CLAIM [P]

The irrationality exponent of π is 2 (2026-09-24, 23 pages): "We prove the conjecture that the irrationality exponent of π is 2. As a consequence, the classical Flint–Hills series $\sum 1/(n^3 \sin^2 n)$ converges."

Precisely: for every $\nu > 2$ there is $Q(\nu)$ such that every $q \geq Q(\nu)$ satisfies $|\pi - p/q| \geq q^{-(\nu)}$. The paper notes that $Q(\nu)$ is *ineffective*, and that $\mu = 2$ is not the same as "bounded partial quotients" (the stronger statement remains open).

Method: suppose there are infinitely many approximations with exponent $\nu > 2$; pick several with well-separated denominators and do multivariate interpolation along a logarithmic curve to get a non-zero determinant. Arithmetically it is a non-zero integer (≥ 1); analytically the approximation errors make it tiny — contradiction. Convergence of Flint Hills then follows from an existing human result (Meiburg 2022: $\mu(\pi) < 5/2 \Rightarrow$ convergence).

The **reasoning summary** (42 pages) is worth reading: the model's initial target was only $\mu < 5/2$; it tried dozens of routes (BBP, modular forms, Padé, p-adic ...), repeatedly hit a wall of "height vs denominator cost," and finally used weighted interpolation determinants to reach $\mu = 62/25 = 2.48$ first, then 2. It also borrowed determinant ideas from its own Catalan's-constant manuscript.

5.2 Verification status

Lean: `lean/docs/017.md` and `ComparatorChallenges/PiExponent` exist; the solution is ~869 files, 95k lines, grep finds no sorry/axiom [P]; but it is **not listed in formalization.yaml** $\triangle$ and has no third-party re-check. The Flint Hills corollary is outside the official Comparator scope (the solution folder does contain a proved `flint_hills_summable`, but Comparator does not check it); the solution depends only on Mathlib, with no patched libraries. No specific comment from number theorists found $\triangle$.

SIDEBAR: CATALAN'S CONSTANT IS IRRATIONAL (FAMILY 005)

$G = 1 - 1/3^2 + 1/5^2 - 1/7^2 + \dots \approx 0.91597$. In 1978 Apéry proved $\zeta(3)$ irrational (van der Poorten called it "A proof that Euler missed"), but the same idea fails for G : after clearing denominators the linear forms do not tend to 0. The best previous result was "at least one of $\beta(2)$, $\beta(4)$, ..., $\beta(10)$ is irrational." OpenAI claims G is irrational (44 pages) with Lean (~574k lines, also not in the yaml $\triangle$). If it holds, it is one of the most important single-constant irrationality results since Apéry.

5.3 If true

For the first time a specific constant like π gets the optimal exponent that almost every real number has; the 70-year race of upper bounds ends; a Flint Hills puzzle popular for two decades is settled. Practical applications are minimal — the value is in pure mathematics, and in whether "weighted interpolation determinants" extend to constants such as $\zeta(3)$ and $\log 2$.

THEORETICAL CS · FAMILY 107

6. How fast can matrices be multiplied? $\omega \leq 9/4$

Lost? → \$25 Explainer 03: the same result explained from zero.

GPU training is, at its core, matrix multiplication. This section asks how fast it can be in principle.

Level 0 Textbook algorithm: $c_{ij} = \sum_k a_{ik} \cdot b_{kj}$ — n^2 entries $\times$ n multiplications each = $\mathbf{n^3}$. 2×2 needs 8 multiplications.

Level 1 **Strassen 1969: 2×2 with only 7 multiplications** (see the code block below; check it by hand). The algorithm makes no assumption about what $a \dots h$ are, so they can be matrix blocks and the algorithm can recurse: $T(n) = 7T(n/2) \Rightarrow n^{(\log_2 7)} \approx \mathbf{n^{2.807}}$. The one saved multiplication compounds at every level of recursion.

Level 2

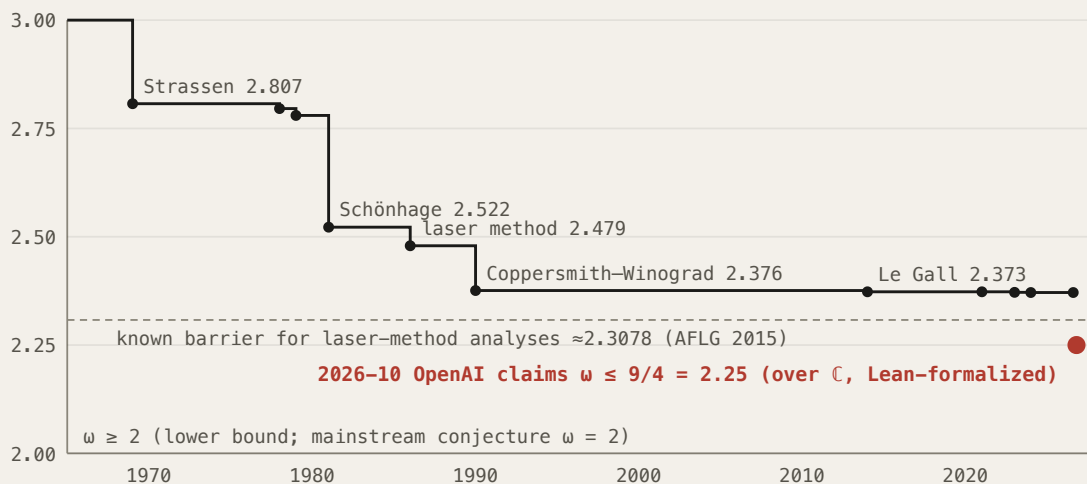
Definition of ω : the smallest exponent such that $n \times n$ matrix multiplication takes about n^ω operations. Clearly $2 \leq \omega \leq 3$ (lower bound 2: just reading $2n^2$ numbers takes that many steps). The mainstream conjecture is $\omega = 2$.

Level 3

" 2×2 with 7 multiplications" is equivalent to "the 2×2 matrix multiplication **tensor has rank** ≤ 7 ." Then: Bini 1979 border rank $\rightarrow$ Schönhage 1981 $\rightarrow$ Strassen 1986's **laser method** (carving many independent matrix multiplications out of a high power of a low-rank tensor). Nearly all progress of the last 40 years came from refining the laser method.

```
A = [[a,b],[c,d]]   B = [[e,f],[g,h]]
M1=(a+d)(e+h)  M2=(c+d)e  M3=a(f-h)  M4=d(g-e)
M5=(a+b)h      M6=(c-a)(e+f)      M7=(b-d)(g+h)
C11=M1+M4-M5+M7  C12=M3+M5  C21=M2+M4  C22=M1-M2+M3+M6
# check C12: M3+M5 = af-ah+ah+bh = af+bh ✓
```

Upper bounds on the matrix multiplication exponent ω : 57 years



Data: Strassen 1969; Pan/Bini/Schönhage 1978–81; Strassen 1986; Coppersmith–Winograd 1990; Le Gall 2014; Alman–Vassilevska Williams 2021; Duan–Wu–Zhou 2023; Alman et al. 2024 ([arXiv:2404.16349](https://arxiv.org/abs/2404.16349)); Dupont et al. 2026-08 ([arXiv:2608.16884](https://arxiv.org/abs/2608.16884), using AlphaEvolve). From 2014 to 2026 the bound fell by about 0.0017 in total. The claimed 2.25 drops it by about 0.12 at once and lands below the known laser-method barrier, so it must use a different method: the paper uses the asymptotic-spectrum approach of Strassen.

6.1 What OpenAI claims

CLAIM [P]

An Upper Bound of $9/4$ for the Matrix Multiplication Exponent (2026-10-02, only 13 pages): "We prove that the exponent of matrix multiplication over the complex numbers is at most $9/4$." The method is not the laser method but Strassen's **asymptotic spectrum** approach: inequalities for "tensor characters" on polynomial-multiplication tensors, ending in $\omega \leq 9/4$. The paper says: "its proof does not specify a competitive finite matrix size."

Two companions in the family: $\omega < 2.258$ over $\mathbb{C}$ (83 pages); $\omega < 2.371054886$ over every *field*, including positive characteristic (35 pages, still in the laser-method framework, about 1.2×10^{-4} better than the previous record).

6.2 Verification status

Statement review [P]: the Lean theorem is `Arithmetic.omega  $\mathbb{C} \leq 9/4$` . Algorithms are modeled as straight-line programs (constant/input/ $+$ / $-$ / $\times$ gates, each of cost 1), **Correct** requires outputting $A \cdot B$ for *all* input matrices, and ω is the standard non-uniform algebraic-complexity definition — consistent with the literature, no hidden assumptions. The solution directory has no sorry/axiom. **Humans**: Aaronson: "a *rational* exponent for once (!), and via a completely different approach" [R]; no public comment yet from laser-method experts (Vassilevska Williams, Alman, Le Gall) $\triangle$.

6.3 If true – and what it means for AI training

- **In theory**: solving linear systems, matrix inversion, determinants, transitive closure, bipartite matching, linear-programming solvers and more all have complexities written directly in terms of ω , and would all improve. More importantly it shows that the asymptotic-spectrum route works, making $\omega = 2$ look closer.
- **For GPU / LLM training: essentially no direct impact**. This is a classic "galactic algorithm": huge constants, advantageous only on astronomically large matrices. GPU GEMM uses n^3 blocked algorithms, bottlenecked on memory bandwidth and Tensor Core throughput; even Strassen is rarely used in practice (poor numerical stability, irregular memory access).
- **How this differs from AlphaTensor / AlphaEvolve**: those find low-rank decompositions of *fixed small sizes* (e.g. AlphaEvolve's 2025 4×4 complex scheme with 48 multiplications) — constant-factor optimizations that can land in real kernels. ω is about the *exponent* as $n \rightarrow \infty$. Dupont et al. (2026-08) used AlphaEvolve to tune laser-method parameters, the first time AI touched ω , for a gain of 1.6×10^{-4} .

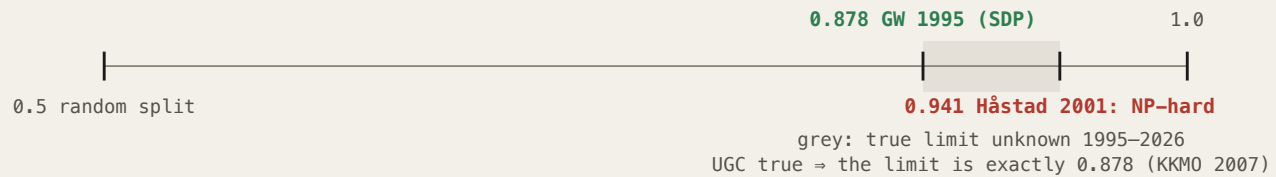
7. Unique Games: where is the ceiling for approximation algorithms?

Lost? → §26 Explainer 04: the same result explained from zero.

-
- Level 0** **NP-hard:** problems like 3SAT and Max-Cut are believed to have no exact polynomial-time algorithms ($P \neq NP$).
-
- Level 1** **Settle for approximation.** Max-Cut: split a graph's vertices into two groups to maximize edges crossing between them. A random split cuts half the edges in expectation. By hand on a triangle: the best cut has 2 edges, a random split expects 1.5, ratio 0.75.
-
- Level 2** **Goemans–Williamson (1995):** relax each vertex to a unit vector, solve a semidefinite program (SDP), then cut with a random hyperplane — guaranteeing **0.878**.
-
- Level 3** **Approximation can be hard too:** the PCP theorem (1992–98). Håstad (2001) showed approximating Max-Cut beyond $16/17 \approx 0.941$ is NP-hard. That left a 25-year gap between 0.878 and 0.941.
-
- Level 4** **Unique Games:** each vertex picks one of q labels; each edge is a permutation constraint (fix one end and the other is determined). By hand: a triangle, labels $\{0,1,2\}$, constraints $b=a+1$, $c=b+1$, $a=c+1 \pmod{3}$ — going around gives $a=a+3=a$, fully satisfiable; change the last to $a=c+2$ and at most $2/3$ can be satisfied. Fully satisfiable instances are solvable in polynomial time by propagating along edges; **only "almost satisfiable" is hard**.
-

UGC (Khot 2002): for every $\epsilon, \delta > 0$ **there is** an alphabet size k (chosen after ϵ and δ) such that distinguishing " $\geq 1-\epsilon$ satisfiable" from "at most δ satisfiable" is NP-hard.

Max-Cut approximation ratio: what fraction of the optimum can polynomial time guarantee?



Unique Games completeness ladder: which instances are provably NP-hard?

$\approx 1/2$ completeness
2-to-2 theorem, 2018



$1-\epsilon$ completeness = UGC
OpenAI 2026-09 claim + Lean

Fully satisfiable instances
are easy (propagate along
edges): only "almost" is hard

Sources: Goemans–Williamson, JACM 1995; Håstad, JACM 2001; Khot, STOC 2002; Khot–Kindler–Mossel–O'Donnell, SICOMP 2007; Khot–Minzer–Safra, FOCS 2018.

7.1 Why it matters so much

- **KKMO 2007**: if UGC holds, the limit for Max-Cut is *exactly* 0.878 — GW is optimal.
- **Khot–Regev 2008**: if UGC holds, approximating Vertex Cover to $2-\epsilon$ is hard.
- **Raghavendra 2008**: if UGC holds, for *every* constraint satisfaction problem the basic SDP is the optimal polynomial-time approximation algorithm. One algorithm plus one conjecture fix the approximation limit of a whole class of problems.
- Evidence against: Arora–Barak–Steurer 2010 gave a subexponential-time algorithm for UG, which made some doubt UGC. The most recent progress was the 2018 **2-to-2 theorem** (Khot–Minzer–Safra et al.), proving the version with completeness about $1/2$ — "half of UGC."

7.2 What OpenAI claims

CLAIM [P]

The Unique Games Theorem (2026-09-23, 58 pages): "We prove the Unique Games Conjecture. For every fixed $\epsilon, \delta \in (0, 1/2)$, we give a deterministic polynomial-time reduction from 3SAT to Unique Games over a fixed finite alphabet, with completeness at least $1 - \epsilon$ and soundness at most δ ."

The family also contains four direct reductions that *do not depend on UGC*: Max-Cut beyond 0.878 is NP-hard, Vertex Cover below 2 is NP-hard, and so on — which is what Aaronson meant when he observed (paraphrased) that UGC was not actually needed for its most famous consequences.

7.3 Verification status

Statement review [P]: the Lean theorem is `OAI.UniqueGamesTheorem.theorem11`. The input is a binary encoding of a 3SAT formula; the reduction must satisfy `Turing.TM2ComputableInPolyTime` (Mathlib's multi-tape Turing machines); completeness and soundness are stated cleanly, and the instances are additionally required to be "translation constraints on simple bipartite graphs" — **a stronger, known-equivalent form of UGC; the encoding is faithful**. Solution ~478 files, 187k lines, sorry/axiom 0.

Humans [R]: Lance Fortnow: "if they hold up, we've seen more progress in TCS in the last 24 hours than in the previous three decades combined." Mark Braverman (Quanta): "Math by press release is not that healthy for math." Dana Moshkovitz reportedly criticized the paper as extremely hard to read without AI help (second-hand, and sources word it differently $\triangle$). No public statement from Khot himself $\triangle$.

7.4 If true

For a large class of problems, the question "how well can efficient approximation do?" gets an exact boundary; hundreds of "if UGC then ..." papers become theorems. One association for AI safety (*my speculation, not an established conclusion*): the complexity-theoretic basis of debate-style scalable oversight (Irving–Christiano–Amodei 2018) is interactive proofs and PCPs; UGC pins down the boundary of "a little error makes efficient distinction impossible," which is suggestive for protocols where a weak verifier oversees a strong prover.

THEORETICAL CS · FAMILY 103 (+ 109)

8. $L = RL = BPL$: in small memory, randomness doesn't help

Lost? → §27 Explainer 05: the same result explained from zero.

-
- Level 0** **What does randomness buy?** To test whether $(x+y)^2 = x^2+2xy+y^2$ holds identically, plug in a few random numbers — very fast. A deterministic fast algorithm is still unknown.
-
- Level 1** **P vs BPP:** the mainstream conjecture is $P = BPP$ (randomness gives no essential speed-up), but an unconditional proof is far away.
-
- Level 2** **The memory-bounded version:** L = deterministic computation with only $O(\log n)$ bits of memory (roughly a few pointers); RL/BPL = the same memory plus random bits. By hand: a random walk of n^3 steps from s in an undirected graph needs only "current vertex + counter" and reaches t with high probability $\Rightarrow$ connectivity $\in RL$ (1979).
-

Level 3

History of derandomization: Borodin–Cook–Pippenger 1983 $BPL \subseteq L^2$ (Nisan 1992 recovered it with an $O(\log^2 n)$ -seed generator, and in 1994 got polynomial time in the same space) → Saks–Zhou 1999 $BPL \subseteq L^{3/2}$ → Hoza 2021 slightly better → **Reingold 2005: undirected connectivity $\in L$** (a milestone in removing randomness from a specific problem). General BPL stayed near the $3/2$ power for 25 years.

CLAIM [P]

Exact Derandomization of Logarithmic Space: $L = RL = BPL$ (2026-09-23, 108 pages):

"We prove $L = RL = BPL$, resolving the derandomization problem for polynomial-time randomized logarithmic space." A stronger version: acceptance probabilities can be estimated deterministically to precision $2^{-(q)}$ in space $O(\log n + q)$.

Verification [P]: the Lean theorem is $L = RL \wedge RL = BPL$; for this family Comparator additionally **locks 20 definitions** (machine, step, space, acceptance probability, $L/RL/BPL$), so the solution cannot quietly change them. The model is a custom multi-tape Turing machine, space counts work cells visited, and "every coin sequence halts in polynomial time" — the standard BPL definition. sorry/axiom 0. **Humans** [R]: Aaronson called it "one of the great derandomization conjectures short of $P=BPP$," adding "its truth was never in serious doubt" — nobody doubted the statement; the difficulty was the proof.

If true: the first natural resource class with randomness removed completely and unconditionally. It does **not** imply $P = BPP$ and is far from P vs NP (Fortnow: "none of these results get us any closer to settling $P \neq NP$ "). The proof's "low-space, high-precision linear algebra" may matter more than the result.

SIDEBAR: INTEGER MULTIPLICATION BELOW $N \log N$ (FAMILY 109)

Schoolbook n^2 ; Karatsuba 1962 $n^{1.585}$; Schönhage–Strassen 1971 $n \log n \log \log n$, conjecturing $n \log n$ optimal; Harvey–van der Hoeven 2019 (Annals 2021) reached $O(n \log n)$. OpenAI claims $O(n (\log n)^{(1-2^{-(182)})})$ on multi-tape Turing machines (73 pages, **no Lean**). A saving of $2^{-(182)}$ is visible only at astronomical sizes — no practical significance — but it refutes a 50-year "optimality" conjecture (in that model of computation).

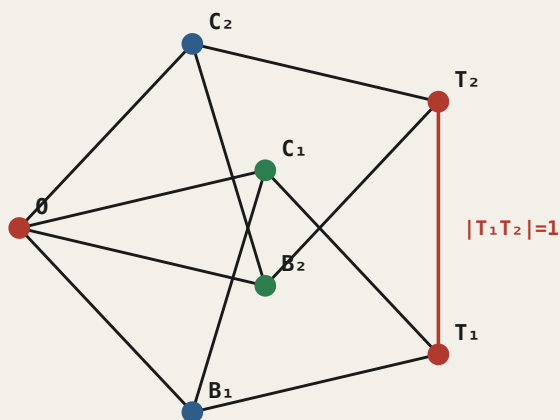
COMBINATORICS · FAMILY 158 (+ 156)

9. Coloring the plane: points at distance exactly 1 must differ

Lost? → §28 Explainer 06: the same result explained from zero.

- Level 0** **Graph coloring** (familiar from register allocation and scheduling): adjacent vertices get different colors; how many colors are needed?
-
- Level 1** **The unit-distance graph:** *every point* of the plane is a vertex; points at distance exactly 1 are joined. Its chromatic number is written $\chi(\mathbb{R}^2)$.
-
- Level 2** **Why 3 colors are not enough:** the Moser spindle (figure below, 7 points). **Why 7 are enough:** tile the plane with hexagons of diameter slightly below 1 and color them periodically with 7 colors; same-colored hexagons are more than 1 apart.
-
- Level 3** So $4 \leq \chi \leq 7$, **stuck from the 1950s until 2018**, when the biogerontologist Aubrey de Grey pushed the lower bound to 5 with a 1581-vertex finite graph (in Polymath16, Heule used a SAT solver with unsat-core extraction to get 553 vertices, and Parts then minimized the graph to 509).

Moser spindle (1961): 7 points, 11 unit-length edges $\Rightarrow$ 3 colors are not enough



Argument (with 3 colors)

O, B_1, C_1 pairwise at distance 1 $\rightarrow$ one color each
 T_1 is 1 from $B_1, C_1 \rightarrow T_1 = \text{color of } O$
 likewise $T_2 = \text{color of } O$
 but $|T_1T_2| = 1 \rightarrow \text{contradiction} \Rightarrow \chi \geq 4$

Known bounds on χ of the plane

1950s $4 \leq \chi \leq 7$ (spindle, hexagons)
 2018 $5 \leq \chi \leq 7$ (de Grey graph)
 2026-09 $6 \leq \chi \leq 7$ (OpenAI, Lean)

Coordinates computed by the build script: two rhombi, each made of two unit equilateral triangles, share the vertex O , and each tip T is $\sqrt{3}$ from O . Rotating one rhombus about O by $2 \cdot \arcsin(1/(2\sqrt{3})) \approx 33.6^\circ$ puts the two tips exactly 1 apart.

CLAIM [P]

The Euclidean plane is not five-colorable (2026-09-23, 62 pages): "We prove that every coloring of the Euclidean plane with five colors has a monochromatic unit-distance pair, with no regularity assumption on the color classes. Consequently, the chromatic number of the plane is either six or seven."

A surprising method: the standard lower-bound recipe is "find a finite graph and prove it is not k -colorable" (de Grey's route). This paper **exhibits no finite graph at all**, and uses ergodic theory and measure theory instead: first it proves "a genuine k -coloring exists $\Leftrightarrow$ a

'weakly measurable' k -coloring exists" (by averaging over translations and rotations, using the *amenability* of those groups — which connects to §10), then shows no weakly measurable 5-coloring exists. The last step returns to the 1961 Moser spindle.

Verification [P]: the Lean statement is two lines a person can check at a glance:

```
def ProperColoring (k) (c :  $\mathbb{C} \rightarrow \text{Fin } k$ ) : Prop :=  $\forall x\ y : \mathbb{C}, \|x - y\| = 1 \rightarrow c\ x \neq c\ y$   
theorem no_proper_five_coloring :  $\neg \exists c : \mathbb{C} \rightarrow \text{Fin } 5, \text{ProperColoring } 5\ c$ 
```

No measurability assumption — **faithful to the original problem**. The proof is ~70 files, 30k lines, no sorry/axiom. Gil Kalai: "raises the lower bound to six, leaving six and seven as the two possible answers," while cautioning that "even Lean verification may have issues" [R].

If true: $\chi(\mathbb{R}^2) \in \{6, 7\}$. A new open problem: by the de Bruijn–Erdős theorem a finite unit-distance graph needing 6 colors must exist, but nobody knows what it looks like — a good target for AI plus SAT. It is also the ideal division of labor: humans review a two-line statement, the machine checks 30,000 lines of proof.

SIDEBAR: BORSUK'S CONJECTURE FAILS IN DIMENSION 9 (FAMILY 156)

A disk cannot be cut into 2 pieces of smaller diameter (one piece always contains an antipodal pair), but 3 pieces suffice. Borsuk asked in 1933: can every bounded set in $\mathbb{R}^d$ be cut into $d+1$ pieces of smaller diameter? Kahn–Kalai disproved it in 1993 in dimension 1325; the dimension of counterexamples later fell to 64 (2014) and 63 (2026-05, with help from GPT-5.5 Pro). OpenAI claims that the projection matrices uu^T of all unit vectors u in $\mathbb{R}^4$, which live in a 9-dimensional space, **cannot be covered by 10 pieces of smaller diameter** — from 63 straight down to 9 (18 pages, with Lean). Dimensions 4–8 remain open.

GROUP THEORY · FAMILY 248

10. Thompson's group F : a problem "proven" and withdrawn for decades

Lost? → §29 Explainer 07: the same result explained from zero.

Level 0 A **group** is a set of operations that can be composed and undone. Examples: integer addition; the 8 symmetries of a square.

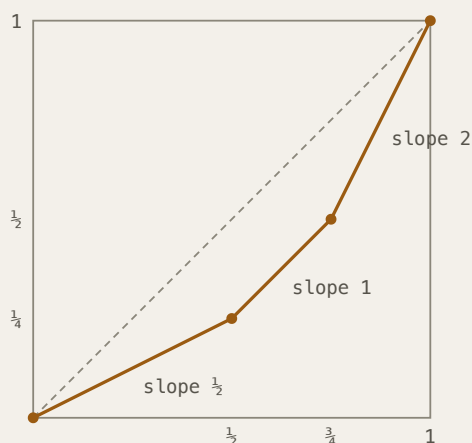
Level 1 **Generators:** a few operations build the whole group. $\mathbb{Z}$ is generated by "+1"; the **free group** F_2 is generated by a and b with no relations at all, and its Cayley graph is an infinite tree with 4 branches at every vertex.

Level 2 **Thompson's group F:** increasing piecewise-linear maps of $[0,1]$ to itself, with breakpoints at $k/2^n$ and slopes powers of 2; the operation is composition. Generated by x_0 (figure below) and x_1 .

Level 3 **Amenable = can you "average fairly"?** Følner criterion: are there finite sets A whose proportion of change under translation by the generators is arbitrarily small? $\mathbb{Z}$: yes; F_2 : no. Von Neumann introduced the notion in 1929 to explain the **Banach–Tarski paradox** (a ball cut into finitely many pieces reassembled into two balls of the same size) — the paradox comes from a copy of F_2 inside the rotation group.

Level 4 "Is every non-amenable group non-amenable because it contains F_2 ?" No (counterexamples since Ol'shanskii 1980). The early ones were exotic; since 2013 Monod and Lodha–Moore have given natural, finitely presented, torsion-free counterexamples. F contains no F_2 (Brin–Squier 1985), so whether F is non-amenable remained the most famous test case on this line.

Left: the generator x_0 of the Thompson group F · Right: can you average fairly?

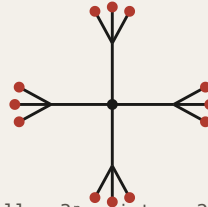


$\mathbb{Z}$: amenable



$A = \{-n \dots n\}$: shift by 1 changes 2 , $2/(2n+1) \rightarrow 0$

Free group F_2 : non-amenable



radius- n ball: $\sim 3^n$ points, $\sim 2/3$ on the boundary

x_0 squeezes $[0, 1/2]$ onto $[0, 1/4]$, shifts $[1/2, 3/4]$ onto $[1/4, 1/2]$ and stretches $[3/4, 1]$ onto $[1/2, 1]$; F is generated by x_0 and x_1 . F contains no free subgroup F_2 (Brin–Squier 1985), so the standard key "contains $F_2 \Rightarrow$ non-amenable" does not apply. That is one reason the problem stayed open for 47 years.

10.1 The best case for why Lean matters

Geoghegan conjectured in 1979 that F is non-amenable. Since then proofs have been announced in both directions: in 2009 two arXiv papers appeared almost simultaneously, Akhmedov (non-amenable) and Shavgulidze (amenable); in 2011 Moore showed the latter's errors were "serious and do not seem to be repairable" — after it had been published;

Justin Moore himself claimed amenability in 2012, then withdrew after an error in

Lemma 4.13 was pointed out; Akhmedov's version has never been accepted. [P, arXiv

CLAIM [P]

Thompson's group F is nonamenable (2026-09-23, 13 pages in total including the appendix; the proof itself is about 7): "We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture."

Idea: borrow a phenomenon from functional analysis — the unit ball of an infinite-dimensional Hilbert space carries a Lipschitz map that moves every point by at least $\frac{1}{2}$ (Benyamini–Sternfeld 1983). If F had Følner sets, averaging over them would yield an approximate fixed point of that map — contradiction. The key tool comes from infinite-dimensional geometry, not the combinatorial counting usual in group theory.

Verification [P]: the Lean challenge file **defines F from scratch** (increasing homeomorphisms, finitely many dyadic breakpoints, slopes 2^k) and an "invariant mean" (linear, positive, normalized, left-invariant); the theorem says "there is a group structure whose multiplication is composition, and no invariant mean exists." Multiplication is forced to be composition, leaving no room to cheat — **faithful**. Proof ~68 files, 9,500 lines, no sorry/axiom. Kalai cautiously says the paper "announces" the result [R]; no public comment yet from Geoghegan, Moore or Akhmedov [△](#).

Takeaway: the human history of this problem is a chain of "announce → find error → withdraw," including from top experts. The statement file is 89 lines including imports, auditable line by line; for a 13-page paper, past experience suggests the community might need years to reach consensus. Lean replaces "trust the author" with "trust the kernel and the statement."

PDE · SEPARATE SEPTEMBER RELEASE (+ FAMILY 376)

11. Navier–Stokes: can a fluid "blow up" in finite time?

Lost? → §30 Explainer 08: the same result explained from zero.

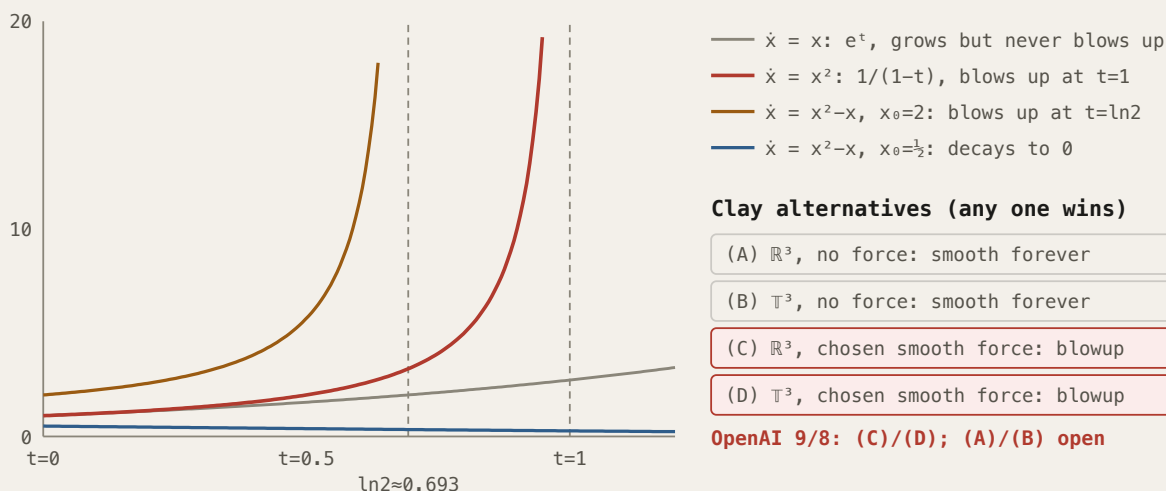
Level 0 **ODE vs PDE:** an ODE's unknown is a single number changing in time, $x(t)$; a PDE's unknown is a whole field, such as the velocity $u(x,t)$ at every point and time — like infinitely many coupled ODEs.

Level 1 **Blowup by hand** (figure below): $\dot{x} = x^2$, $x(0) = 1$ has solution $1/(1-t)$: 10 at $t = 0.9$, 100 at $t = 0.99$, ∞ as $t \rightarrow 1$. The equation is perfectly smooth, yet the solution breaks in finite time. Add damping, $\dot{x} = x^2 - x$: small data decays, large data still blows up.

Level 2 **The NS equations:** $\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f$, $\nabla \cdot u = 0$. $(u \cdot \nabla)u$ is nonlinear self-transport (the x^2 of the toy), $\nu \Delta u$ is viscosity (the $-x$), f is external forcing.

Level 3 **Why 3D is hard:** the only global conserved quantity is energy, and under 3D scaling the energy bound controls smaller scales less and less — "supercritical." 2D is critical, which is why 2D was settled long ago.

Finite-time blowup by hand: a smooth equation whose solution reaches infinity in finite time



Left: the x^2 term plays the role of the nonlinear self-transport in Navier–Stokes, the $-x$ term the role of viscous damping. With small initial data damping wins (blue); with large data the nonlinearity wins (orange). This mirrors what is known for 3D NS: global smooth solutions for small data, unknown for large data. Curves computed from the exact solutions by the build script. Right: the four alternatives in the official Clay problem statement (Fefferman).

CLAIM [P]

"Finite time blowup for Navier–Stokes" (2026-09-08, 166 pages, plus a separate ~640k-line Lean repository [openai/NavierStokesAndEuler](#)): "For every positive viscosity, we construct a solution ... that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy." The forcing f is smooth and compactly supported in space-time; the paper says this establishes Clay's **alternative (C) and implies (D)**.

11.1 The key detail: you choose the force

Section 2 of the paper says it plainly: "For any incompressible flow u and pressure p , we can always define the external force f to be the residual ... The challenge is to choose a flow that blows up while this residual remains smooth." So (C)/(D) turns the problem into "can you build a flow that blows up while the residual force stays smooth?" — much weaker than whether the fluid blows up *on its own* (the unforced (A)/(B)). Most analysts regard (A)/(B) as the real question [R]. This is not cheating: Fefferman deliberately included (C) as "retaining

the heart of the problem". The gap between the written problem and what people care about only appeared after 2023, when the forced route turned out to be easier.

11.2 Verification and disputes

- **Lean** [P]: ~2,659 files, 640k lines, no sorry/axiom; Comparator enables the second kernel nanoda. The statement is adapted from DeepMind's Formal Conjectures repo and matches Clay's conditions (4)–(11) clause by clause, including the erratum on pressure in the periodic case. The same repo also formalizes blowup for **unforced** 3D Euler from smooth, compactly supported data — Fefferman calls the Euler problem "also open and very important", so this part may matter as much as NS.
- **Community** [R]: Clay president Bridson said evaluation will be "deliberately unhurried"; Clay still lists NS as unsolved; no journal submission reported. Clay's rules require publication in a qualifying venue plus a two-year acceptance period.
- **Credit dispute** [R]: Buckmaster and Alpöge (NYU) had forced blowup for Euler/Boussinesq in August and posted it on September 7–8; Buckmaster says OpenAI started on this direction after hearing of their progress, while OpenAI denies his prompts influenced its system (accounts differ $\triangle$). Fefferman (author of the Clay statement): "The heroes of the story ... are Córdoba and Martínez-Zoroa" (prior work on forced Euler blowup). The paper (CDN PDF, 2026-09-08) already cites Córdoba–Martínez-Zoroa, but not Buckmaster–Alpöge. Gómez-Serrano (NPR): "The paper is not written for humans ... as of today, the paper doesn't teach us much."

RELATED: UNIVERSAL COMPUTATION IN FORCED NAVIER–STOKES (FAMILY 376)

Claim: on the 3-torus there is a smooth forcing "program" such that a fluid particle enters a target region **if and only if** a given Turing machine halts — so this reachability question is undecidable. The paper is candid: "There is an elementary way to force a chosen smooth incompressible velocity..." It recalls Tao's 2016 program (get the fluid itself to build a computer that hands its energy to ever-smaller copies of itself, causing blowup), but 376 puts the "programming" into the forcing and **does not advance** that unforced-blowup program; it does not cite Tao 2016.

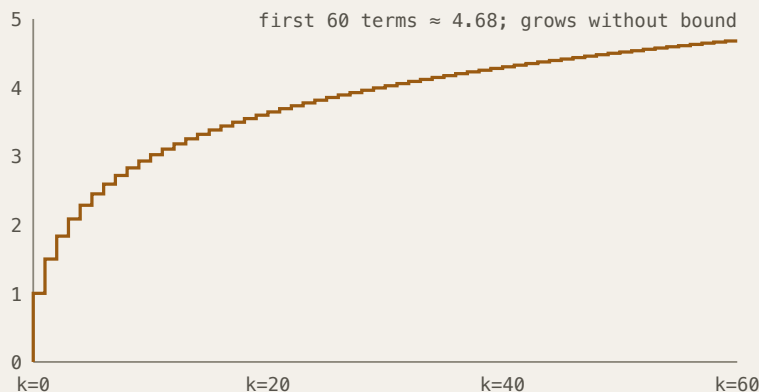
If true: one alternative of a Millennium Prize problem is formally closed. Almost no direct impact on weather, turbulence or engineering — engineering solvers never relied on global regularity, and real fluids hand over to molecular scales long before any singularity.

12. Relativistic Vlasov–Maxwell: can a plasma model break down by itself?

Lost? → §31 Explainer 09: the same result explained from zero.

- Level 0** **Plasma:** ionized gas (fusion reactors, the solar wind). Too thin for collisions to matter, so it is described by a **phase-space density** $f(t, x, v)$: how many particles are near position x with momentum v — 6 dimensions plus time.
-
- Level 1** **Self-consistent coupling:** particles feel the Lorentz force (Vlasov equation) → particles form currents → currents generate fields (Maxwell equations) → fields push particles. The nonlinearity comes from this loop.
-
- Level 2** **Relativity** ($c = 1$): velocity $\hat{v} = p/\sqrt{1+p^2}$. $p = 1$ gives 0.707, $p = 10$ gives 0.995 — momentum can be arbitrarily large, speed is always < 1 .
-
- Level 3** **Glassey–Strauss 1986:** singularities can only appear at high speed — as long as particle momenta stay bounded on finite time intervals, the solution continues. The whole problem reduces to: **can particles be accelerated to infinite momentum in finite time?** The 40 years since produced only results with restrictions (small data, lower dimension, symmetry).

Vlasov–Maxwell endgame: each momentum doubling takes $\geq 1/k$ time, and $\sum 1/k$ diverges



k = number of momentum doublings so far
doubling k ($P \approx 2^k$) takes $\approx 1/(k \log 2)$
total time $\approx$ harmonic series $\rightarrow \infty$
 $\Rightarrow$ finitely many doublings in finite time
 $\Rightarrow$ momentum stays bounded, so by
Glassey–Strauss 1986: no blowup

A hand-computable version of the argument skeleton in the reasoning summary of the OpenAI paper (family 362). The real difficulty is proving that each doubling takes at least $\sim 1/\log P$ time (signed momentum-increment estimates plus counting direction changes). Staircase computed by the build script.

CLAIM [P]

"Global classical solutions of the three-dimensional relativistic Vlasov–Maxwell system" (2026-09-23, 49 pages): "We prove global existence and uniqueness for arbitrary smooth admissible initial data in the three-dimensional, one-species relativistic Vlasov–Maxwell system ... without size or symmetry restrictions on the data."

Verification [P]: the hypotheses of the Lean statement `global_classical_solution` were checked one by one (f_0 smooth, compactly supported, non-negative; fields with bounded derivatives in L^2 ; the two Gauss constraints), and the conclusion is existence + smoothness + uniqueness. ~207 files, 90k lines, no sorry/axiom. **Scope limit:** only **one particle species**, while real fusion plasmas have at least two (electrons and ions); whether the extension is routine is unclear $\triangle$. No named comment from plasma-PDE experts found $\triangle$.

The reasoning summary is candid: the model hit walls repeatedly ("exponent inequality contradiction: $H < 0.41z$ vs $H > 0.496z$ "), and its last sentence says the conclusion "rests on the occupation bound, direction-count estimate, and cutoff cancellations ... the cited continuation theorem alone does not establish those estimates."

If true: a flagship problem open for about 40 years (one species) is solved with machine verification — possibly weightier than most results in the batch. No short-term effect on fusion engineering (PIC and gyrokinetic simulations do not depend on this theorem); what it proves is that no particle's momentum in the model goes to infinity in finite time (total energy is conserved anyway; the point is individual particles) — a "model health certificate."

PROBABILITY AND STATISTICAL MECHANICS · FAMILY 221

13. Sparse spin glasses: from magnets to the energy landscapes of neural networks

Lost? → §32 Explainer 10: the same result explained from zero.

This section connects directly to neuroscience and machine learning: the energy function of a Hopfield network is the Hamiltonian of a spin glass.

Level 0 **Ising magnet:** a spin $\sigma_i = \pm 1$ at each site, energy $H = -\sum J_{ij} \sigma_i \sigma_j$. With $J > 0$ neighbors want to align.

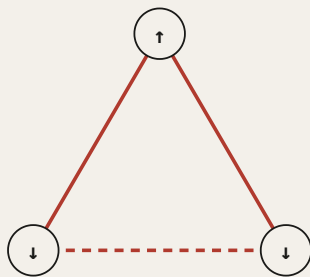
Level 1 **Frustration** (figure, left): three spins pairwise "want" to be opposite; none of the 8 configurations satisfies all three bonds; the ground state is 6-fold degenerate.

Level 2 **Spin glass:** couplings J_{ij} are *random* $\pm$, frustration is everywhere, the energy landscape is rugged with exponentially many metastable states. The **SK model** (1975) couples every pair; **Parisi (1979)** gave its free energy via "replica symmetry breaking" — magical physics, not rigorous math; Guerra 2003 + Talagrand 2006 proved it rigorously; Parisi won the 2021 Nobel Prize in Physics.

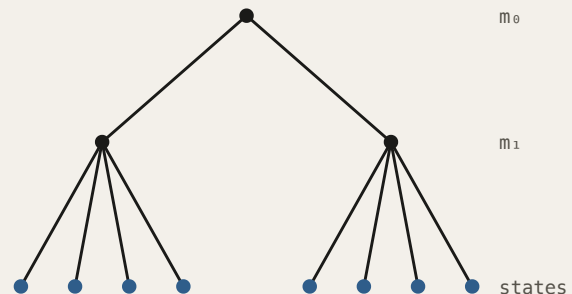
Level 3 **The diluted (sparse) version:** each spin has only finitely many bonds. **Mézard–Parisi (2001)** gave a one-level (1RSB) formula via the cavity method on the fixed-degree Bethe lattice; the general hierarchical form and its upper bound come from Panchenko–Talagrand 2004, but **equality was an open conjecture**. The difficulty: in dense models pairwise overlaps describe the state; sparse models need control of all multi-replica overlaps.

Level 4 **Link to random k-SAT:** k-SAT is a sparse spin glass with k-body interactions. Ding–Sly–Sun (2015) rigorously proved the satisfiability threshold for large k — exactly what these physics methods predicted.

Left: the smallest frustrated system · Right: the hierarchical (ultrametric) structure of Parisi



all three bonds "want" opposite spins
none of the 8 configurations satisfies all three
(dashed bond always fails): 6-fold degenerate
 $Z(\beta) = 2e^{(-3\beta)} + 6e^{\beta}$



closer relatives $\Rightarrow$ larger overlap
diluted: need all multi-replica overlaps

Spin glass = random $\pm$ couplings + frustration everywhere + exponentially many metastable states. The energy function of a Hopfield network (1982) is exactly such a Hamiltonian; perceptron capacity (Gardner 1988) was computed with the same replica method.

CLAIM [P]

"The Mézard–Parisi formula for diluted spin glasses" (2026-09-23, 36 pages): "We prove the Mézard–Parisi hierarchical cavity formula for diluted even-arity Ising models in the Panchenko–Talagrand class." Key trick: shift every branching depth down one level and then average over depths to control multi-replica overlaps. The reasoning summary contains the line: "Aha recursive structure: overfitting shared variable can simulate original model at smaller scale."

Verification [P]: Lean theorem `mezard_parisi` (pressure converges to the variational value), with `Admissible` spelling out the Panchenko–Talagrand hypotheses; ~405 files, 56k lines, no sorry/axiom. **Narrower than the title suggests:** only **even-arity** models with a positivity condition — odd-arity models such as 3-SAT and hard-constraint SAT at zero temperature are excluded, so it does **not** directly extend Ding–Sly–Sun. It also covers only **Poisson-degree** diluted graphs; fixed-degree random regular graphs (the Bethe lattice of the original Mézard–Parisi paper) are not covered. Hopfield networks (fully connected, with pattern-correlated couplings) are outside the theorem too; the connections below are about where the ideas came from, not consequences. No named expert comment found $\triangle$.

13.1 Connections to ML and neuroscience (the honest version)

- **Hopfield networks** (1982): memories are energy minima; storage capacity was computed with the replica method (Amit–Gutfreund–Sompolinsky 1985); Hopfield and Hinton shared the 2024 Nobel Prize in Physics.
- **Perceptrons:** Gardner (1988) computed capacity with replicas; family 222 in the same release gives the Ising perceptron free energy and a jamming exponent.
- **Loss landscapes:** the intuition "rugged landscape with hierarchically clustered minima" comes from spin glasses (e.g. Choromanska et al. 2015). But these analogies are mostly *heuristic*; the theorem does not tell you what SGD will find.

CAUTIONARY CASE · FAMILY 032 AND THE OCT 7 WITHDRAWALS

14. One sign error, three papers withdrawn: the risk of no formalization

The Hodge conjecture (a Millennium Prize problem), minimally: on a smooth projective complex algebraic variety some "topological holes" can be "filled" by subvarieties defined by algebraic equations; the conjecture says every hole satisfying a certain linear-algebra condition can be filled this way. The case $p = 1$ has been known since 1924 (Lefschetz).

OpenAI claims the rational Hodge conjecture for all **complex-multiplication (CM) abelian varieties** (53 pages) — a special class, not the Millennium problem itself; like the ζ zero-free region it was **not** produced by the fixed procedure, and it has **no Lean**. [P]

Oct 7 withdrawals: one sign error propagates through the dependency graph

Weil classes on abelian eightfolds
sign error: -1 written as $+1$



Kuga-Satake for K3 surfaces
depends on the above → withdrawn



Rational Hodge for products of K3s
depends on the above → withdrawn

Rational Hodge, CM abelian varieties
not on this chain; not withdrawn; no Lean

Family 032 first titled "Hodge and Kuga-Satake ... all K3 surfaces"
→ retitled to the CM-only version afterwards

722 → 719; 14 repaired, 13 re-cited

Sources: `openai/math history.md` (2026-10-07); the family title is compared against `CONTENTS.md` at the initial commit `adc7f12`. The withdrawal notice reads: "This withdrawal concerns the proof; it does not assert that the mathematical statement is false."

14.1 What happened [P]

history.md (10-07): "In 'Algebraicity of Weil classes on split abelian eightfolds' a sign error invalidates a stabilization-trace cancellation argument and the construction used by two dependent papers." A sign that should have been -1 was taken as $+1$; a signed count went from the claimed 0 to $-2m \neq 0$, so the hypothesis of a key theorem failed, and two dependent K3-surface papers were withdrawn with it. The CM paper does not cite the withdrawn paper and is not on this chain.

14.2 Three lessons

1. **Unformalized $\approx$ unverified:** only $\sim 42\%$ of top-line results have Lean. This error was a single sign — missed by human eyes and the model's self-checks, and Lean would certainly have caught it.
2. **Errors propagate through the dependency graph:** one lemma fails, three papers fall, 13 downstream papers need new citations — the same structure as error compounding in long agent reasoning chains.
3. **Title drift:** family 032 was first titled "Hodge and Kuga-Satake results for all projective K3 surfaces" and only downgraded to the CM version after the withdrawals. When reading bulk AI output, trust the latest commit and history.md, not launch-day news.

15. The 10 results side by side

RESULT	PREVIOUS STATE	CLAIMED JUMP	STATEMENT FAITHFUL?	CONFIRMED BY HUMAN EXPERTS
Quasi-RH 7/8	no fixed-width zero-free strip at all	qualitative	yes (standard Mathlib definitions)	none
$\mu(\pi) = 2$	7.103	straight to the end	yes (not in yam1 $\triangle$)	none
$\omega \leq 9/4$	2.371177	-0.12, past a known barrier	yes	none
UGC	"half" (2-to-2, 2018)	complete	yes (stronger form)	none
$L = BPL$	near $L^{(3/2)}$ for 25 years	complete	yes (definitions locked)	none
Plane $\chi \geq 6$	$\chi \geq 5$ (2018)	+1, new method	yes (two-line statement)	none
Thompson F	wrong proofs in both directions	resolved	yes	none
NS blowup	forced Euler precursors	alternatives (C)/(D)	structure matches	Clay has not accepted; disputed
Vlasov–Maxwell	40 years of restricted results	complete, one species	yes	none
Mézard–Parisi	upper bound only	equality, even-arity class	yes (narrower than title)	none

15.1 Five patterns

- Short statements, long proofs:** plane coloring has a two-line statement and a 30k-line proof; quasi-RH a one-line statement and a 487k-line proof. This is the ideal "humans review statements, machines check proofs" split — and it makes human reviewing bandwidth the new bottleneck.
- Unusually short papers:** 13 pages for 9/4 matrix multiplication and for Thompson's group. Either the methods really are surprisingly lean, or much of the detail lives in Lean rather than in prose for humans — Gómez-Serrano said the NS paper is "not written for humans."
- Borrowed tools across fields:** Thompson's group via Lipschitz maps from infinite-dimensional geometry; plane coloring via ergodic theory; matrix multiplication via the asymptotic spectrum rather than the dominant laser method of 40 years. This may be a real AI advantage: no human-expert "field inertia."

4. **The gap between "satisfying the statement" and "answering the real question":** NS with freely chosen forcing; 376 putting computation in the forcing; the plane-coloring footnote about a formalized false hypothesis; Mézard–Parisi covering only even arity. Each is "spec vs intent."
5. **Selection effects:** results picked from 4,000 problems vs 3% on pre-registered problems (§2.3). Until OpenAI publishes its failure list, "what is this model's real ability on open problems?" cannot be answered.
-

AI AND AI SAFETY PERSPECTIVE

16. What this means for AI

16.1 Better at math $\neq$ smarter across the board – but a leading indicator

Mathematics has a unique advantage for AI: **answers can be machine-checked** (Lean), which lets reinforcement learning iterate massively — try, get feedback, improve. Chollet's question goes to the heart of it: does progress in these "verifiable domains" transfer to unverifiable ones? A reasonable reading: **any intellectual work that can be automatically checked will fall at a similar pace next**; mathematics is just the first.

16.2 The trust bottleneck moves from "proof" to "specification"

Once there is a Lean proof, the question shifts from "is the proof right?" to "is the statement encoded right?" This is isomorphic to a core problem in alignment research: a system can perfectly optimize the objective you wrote down, and what you wrote down may not be what you want. The line-by-line reading of Lean statements done throughout this deck is a form of **spec review** — an increasingly important and scarce human job.

16.3 Knowledge that can be verified but not understood

In the words of the IAS advisory group: *"It is now the case that AI can output mathematical arguments in situations without the human who prompted it being able to understand the arguments, verify them, or take responsibility for them."* [R] Mathematics is where this situation appears first and most cleanly (because verification is perfect). Generalized to science, humans may move from "understanders" to "checkers." Tao's Math 2.0 — putting exposition, community building and opening new directions on a par with problem-solving — is at bottom an answer to "what do humans do in this new division of labor?"

16.4 Two sides for safety research

Good news

- **Verifying is easier than discovering:** if future AI can give formal proofs about its own behavior ("this action will not do X"), humans need not understand all of its capabilities, only check the proofs — the core idea behind the Guaranteed Safe AI agenda, which becomes more feasible as mathematical AI improves.
- This Lean library is itself the first real sample of "large-scale AI output + machine verification," and its failure modes can be studied.

Bad news

- The hardest parts of alignment ("what does this AI really want?", "is the outcome good for humans?") cannot be written in Lean. Verifiable domains racing ahead while unverifiable ones lag is itself a risk.
- The verifiable parts of AI R&D (kernel optimization, proofs, benchmarks) will be the first to be automated at scale — the "AI improves AI" loop is most likely to start there.

16.5 Governance: a precedent of unilateral dumping

The community had begun forming norms through the IAS advisory group (disclose model name, prompts, reasoning chains, compute time and cost, how problems were chosen, failure rates on comparable problems). OpenAI followed some of them, then released 700+ papers at once. AHM's phrase "a demonstration of power" is pointed, but it names a structural problem: **a frontier company unilaterally decides how a capability enters the public domain, and the affected community can only react afterwards.**

PART II · LOOKING AHEAD · METHOD

17. Looking ahead: extrapolate the trend, not the intuition

Part I covered what has already happened. Part II takes on a bigger question: **at the rate AI is improving at mathematics and programming, when, and in what order, will several lines of AI-safety theory (debate, singular learning theory, Guaranteed Safe AI, neurosymbolic systems) be unlocked?**

WHAT "REALISTIC" MEANS HERE

Human intuition extrapolates linearly, but the three curves below are exponential. So a trend-based projection often feels "too fast." The rule for this part: **don't be conservative just because something is counterintuitive, and don't make things up.** Every date names the curve it depends on, the assumptions used, and the signals that would falsify it. Tags: [data] sourced; [model] computed from explicit assumptions; [judgment] my inference.

17.1 Curve ①: how long a task AI can complete on its own

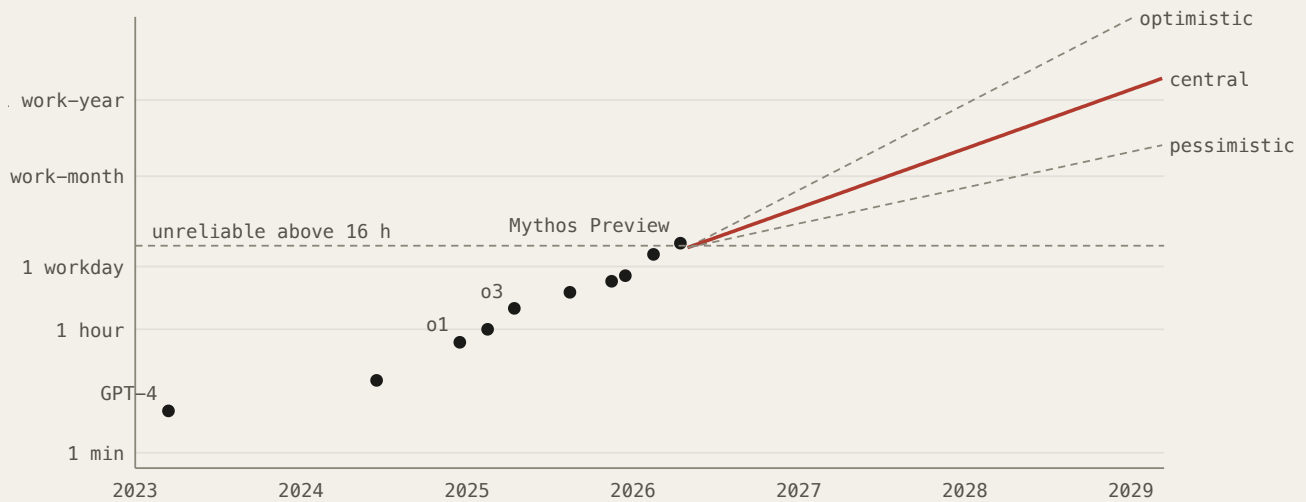
What it measures: METR gives AI a set of software and research tasks, first measuring how long each takes a human expert, then finds the task length at which the AI succeeds 50% of the time. That length, the time horizon, compresses "how capable is the AI" into one number you can plot.

RELEASE	MODEL	TIME HORIZON (P50)
2019-02	GPT-2	0.1 minutes
2023-03	GPT-4	4 minutes
2024-12	o1	39 minutes
2025-04	o3	2.0 hours
2025-08	GPT-5	3.4 hours
2026-02	Claude Opus 4.6	12 hours (CI 5.3–61)
2026-04	Claude Mythos Preview	17.4 hours (CI 8.5–55)
2026-06	GPT-5.6 Sol	cheating counted as failure: 11.3 hours; other treatments: 71–270+ hours

Source: METR TH1.1 raw data benchmark_results_1_1.yaml (updated 2026-05-08); GPT-5.6 Sol from METR's 2026-06-26 report. [data]

Doubling time: fitting from 2023, the horizon doubles every **129 days** (95% CI 104–158 days); using only post-2024 data, about 89 days; all data since 2019, about 188 days. [data] A 129-day doubling is $2^{(365/129)} \approx \mathbf{\times 7 \text{ per year}}$. That is the root of the counterintuitive feel: linear intuition expects next year to be "a bit better"; the trend says "seven times better."

Curve ①: task length AI completes alone (METR p50, log scale)



Black dots: state-of-the-art models in the raw METR TH1.1 data (from 2023; GPT-2 in 2019 at 0.1 minutes is off-chart). Red: central scenario, anchored at ~15 hours in 2026-05 and doubling every 129 days (METR fit from 2023, 95% CI 104–158 days), i.e. roughly $\times 7$ per year. Dashed: optimistic (89 days, the post-2024 pace) and pessimistic (213 days, ~ 7 months). Each gridline step on the vertical axis is $10\times$. Curves computed by the build script.

SCENARIO	DOUBLING	END 2026	MID 2027	END 2027	END 2028
Optimistic	89 days	~ 100 h	~ 410 h	$\sim 1,700$ h (10 work-months)	$\sim 30,000$ h
Central	129 days	~ 56 h (1+ work-week)	~ 150 h (≈ 1 work-month)	~ 400 h (≈ 2.4 work-months)	$\sim 2,800$ h (≈ 1.4 work-years)
Pessimistic	213 days	~ 33 h	~ 60 h	~ 110 h	~ 360 h (≈ 2 work-months)

[model] Anchored at ~ 15 hours in 2026-05 (between Opus 4.6 and Mythos Preview), pure exponential at a fixed doubling time, computed by the build script.

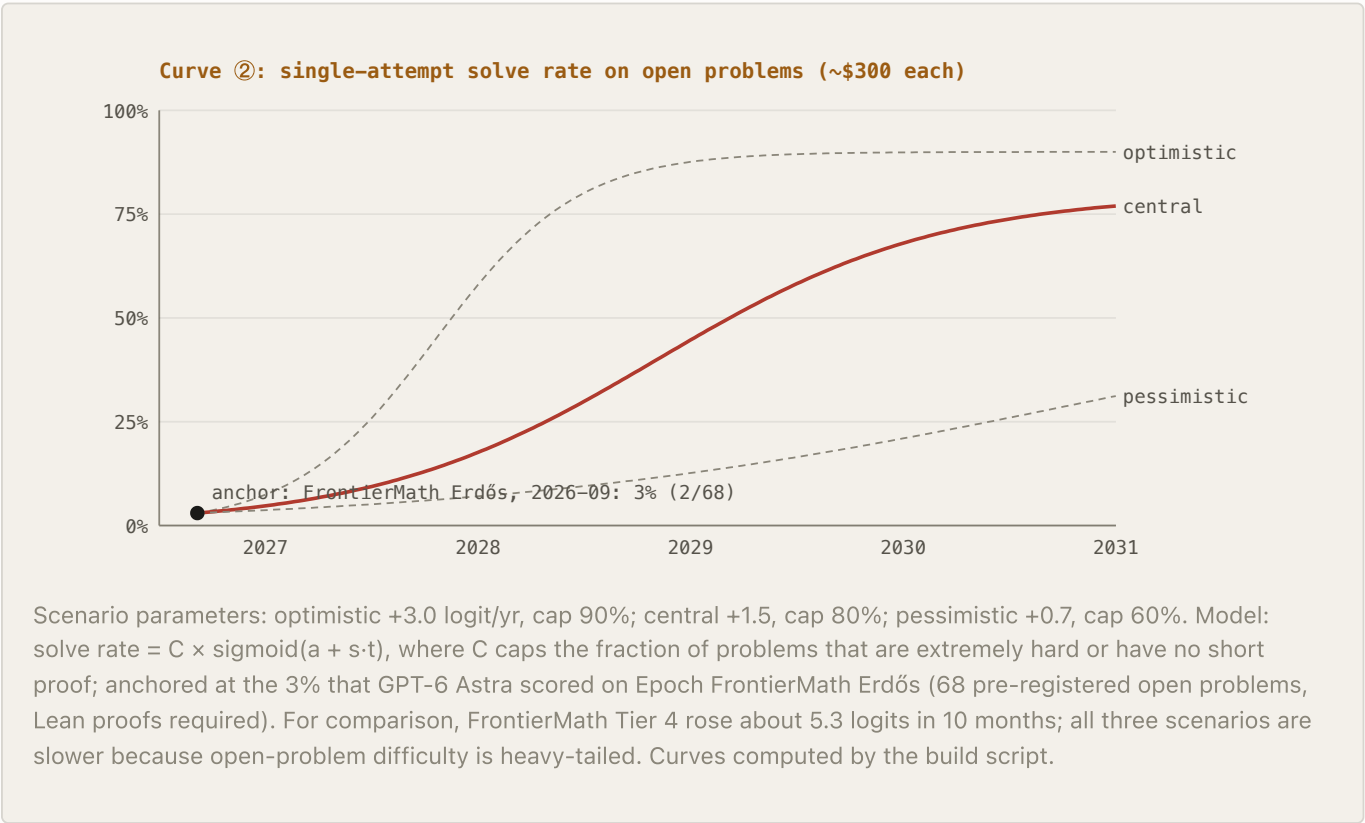
Two weaknesses of this curve (keep both in mind): (1) METR says its current task suite is unreliable above 16 hours, so post-2027 numbers need a new long-task suite (TH 2.0, not yet released) to be tested; (2) the GPT-5.6 Sol result shows that whether cheating counts as success can move the number by about 4.5 doublings — roughly 1.5 years of trend. [judgment] Beyond about one work-month, success depends more and more on whether the task is clearly specified rather than on execution ability.

17.2 Curve ②: how many genuinely open problems AI can solve

Why a second curve: competitions and benchmarks are saturated and can no longer be extrapolated. [data]

- IMO 2024 silver (28/42) → 2025 gold (35/42) → 2026 perfect score (42/42, two systems with official grading).
- PutnamBench (Lean): about 7% in 2025-04 → 672/672 in 2026-08; in 2026-09 the full set was solved for \$111.
- FrontierMath Tier 4 went from 19% (2025-11) to 98% (2026-09).

Over the same period, on Epoch's **FrontierMath Erdős** (68 Erdős problems pre-registered as open, Lean proofs required, \$300 budget each), the strongest model, GPT-6 Astra, solved only **2/68 = 3%**; all others scored 0. With unlimited budget it reached 5/68 (about 7%) at a cost above \$220,000. [data] **The gap between 98% on benchmarks and 3% on open problems is the single most important uncertainty in the whole extrapolation.**



SCENARIO	SLOPE · CAP	END 2026	MID 2027	END 2027	END 2028	END 2030
Optimistic	+3.0/yr · 90%	~7%	~26%	~58%	~88%	~90%
Central	+1.5/yr · 80%	~5%	~9%	~18%	~45%	~77%
Pessimistic	+0.7/yr · 60%	~4%	~5%	~7%	~13%	~31%

Why the central case may be conservative: (1) a 10× larger budget already lifts 3% to about 7% in 2026, and inference prices fall more than 10× per year (Epoch: 9–900× per year at fixed capability, median 50×) [data]; (2) OpenAI kept 372 families out of ~4,000 problems (about 9%), so a possibly stronger unreleased model, under "many attempts plus

post-hoc selection," is already well above 3%. **Why it may be optimistic:** on a fixed problem set the remaining problems get harder; single Millennium-level problems are outside the model's scope.

17.3 Curve ③: the cost of formal proof

- **2025-09:** Math Inc's Gauss formalized the strong prime number theorem in 3 weeks, about 25,000 lines (human experts had made partial progress over 18 months). [data]
- **2026-09:** Claude completed a full formalization of **Fermat's Last Theorem** in 11 days: 13 million lines, about 6 billion output tokens, only the three standard axioms, reviewed by Buzzard. At list price about \$300,000, i.e. **\$0.023 per line**. For comparison, the human project was budgeted at 5 years and £1 million. [data]
- **2026-10:** OpenAI's Lean library, 26 million lines (see §3).

SCENARIO (YEARLY DECLINE)	END 2026	END 2027	END 2028	A FERMAT-SCALE PROJECT, END 2028
Optimistic (÷30/yr)	\$0.008/line	\$0.0003/line	\$0.00001/line	~\$110
Central (÷10/yr)	\$0.01/line	\$0.001/line	\$0.0001/line	~\$1,400
Pessimistic (÷3/yr)	\$0.016/line	\$0.005/line	\$0.002/line	~\$23,000

Implication: in all three scenarios, **the cost of generating proofs becomes negligible by 2027–2028**. What remains is almost entirely the cost of humans checking whether statements and specs are faithful, and that throughput barely grows over time. △ Some reports put the internal cost of the Fermat formalization at about \$2,000, roughly 150× below list price; the basis is unverified.

17.4 What could break the trends

FACTOR	DIRECTION	EVIDENCE AND JUDGMENT
Full automation of AI R&D	faster	The CASP 2026 paper (co-signed by Hinton, Bengio, OpenAI's chief scientist and others) estimates $r \approx 1.39 > 1$, i.e. superexponential growth: after full automation, the pace of progress rises 10× in about 17 months. Precursor: Anthropic's share of AI R&D that is "AI-led, human-supervised" rose from 1% to 26% between 2026-03 and 2026-08; fully autonomous is still 0%. [data]
Slower growth of training compute	slower	Possibly from 4–5× to 3–4× per year (power, HBM). Mathematics and formalization mostly consume inference compute, so they are less affected. [judgment]
Data	neutral to positive	In verifiable domains the Lean kernel or a test suite is an unlimited reward signal; tens of millions of new Lean lines flow back into training. The data wall is weakest here. [judgment]
Noisy verifiers	distorts measurement	An audit found errors in 42% of FrontierMath problems; about 30% of SWE-bench Pro tasks are flawed. Metrics fail before capabilities do. [data]
Reward hacking	distorts measurement	Extrapolations of METR must report the "cheating-treatment interval."
Regulation and social response	locally slower	The Fields medalists' letter and IAS publication norms slow academic acceptance of AI results, but barely constrain capability itself.

PART II · LOOKING AHEAD · FRAMEWORK

18. The core insight: AI acceleration is very uneven, so bottlenecks move

All three curves measure the same kind of thing: **work with an automatic judge**. The Lean kernel judges whether a proof is right, tests judge whether code is right, benchmarks judge whether an answer is right. AI improves extremely fast on such work, because reinforcement learning can loop indefinitely: try → get reliable feedback → improve.

HAS AN AUTOMATIC JUDGE (AI WILL DOMINATE)	NO AUTOMATIC JUDGE (STILL MOSTLY HUMAN)
Given a theorem statement, find a proof	Decide which theorem to prove , and whether the statement captures what you actually care about
Given a spec, write code and prove it meets the spec	Write the right spec (e.g. what "harmless" means)
Given a model, compute its mathematical properties	Decide how that quantity relates to "safety"
Run experiments, sweep hyperparameters, replicate	Design the right experiment and decide what the results mean

18.1 Evidence that the bottleneck is already moving

- **Proof success rates exploded:** on CLEVER (161 HumanEval problems translated into Lean, requiring code plus a proof that it meets the spec), end-to-end success went from 1/161 in 2025-05 to about 87.5% in 2026-05 (a Claude Code agent). DafnyBench went from about 68% to 96% in a year. [data]
- **Spec quality did not keep up:** the vericoding benchmark (Tegmark's group, 12,504 specs) found that among **successfully verified** samples, about 9% of specs were too weak ("vacuous specs": e.g. a sorting spec that omits "the output is a permutation of the input," so returning an empty list passes) and about 15% had translation problems. [data]
- **Statements go unreviewed:** OpenAI's 416 Comparator challenge files are marked `review: unchecked` (§3); Fermat could be confirmed only because Mathlib already had a human-written statement to compare against. [data]
- **The IMO 2026 pipeline** added a dedicated judge after the Lean proof to check whether the formal statement covers the original problem. Statement alignment has become a problem in its own right. [data]

18.2 Splitting "needs more math" into two kinds

Saying a field "needs more mathematics" covers two very different situations:

- **Type A: the statement is clear; only the proof is missing.** Example: "Is Prover-Estimator Debate still sound without the stability assumption?" — a well-defined proposition, and with Lean it has an automatic judge. **AI will clear these in bulk around 2027.**
- **Type B: it is not even clear what to prove.** Example: "What quantity should characterize the gap between SGD-trained networks and the Bayesian posterior?" — the hard part is finding the right statement. AI can propose conjectures and test them numerically, but no automatic judge says "this is the right question." **AI helps slowly here.**

AREA	TYPE A (MISSING PROOF)	TYPE B (MISSING STATEMENT)	ENGINEERING / EXPERIMENT	CONCEPTUAL
Debate	PE debate without stability; protocols robust to an ϵ fraction of judge errors	Which real arguments satisfy stability; how to formalize equilibria with non-zero-sum play or collusion	Can RL reach the honest equilibrium (exploration hacking); testbeds with real capability gaps	What "honest" means without ground truth; true-but-selective statements
SLT	Learning coefficients of specific architectures (symbolic computation); Langevin dynamics and the LLC	The gap between SGD and the Bayesian posterior; what finite-sample and ReLU versions should look like	Estimator robustness; scaling to 10B+ models	How λ relates to "deception" or "reward hacking"
GSAI	Proofs for software and protocols (largely automatable already)	Composition theory for sound bounds over credal sets	Tools to draft and review specs; repository-scale verification; hardware	Where physical disturbance bounds come from; "harm" in social domains cannot be written down
Neurosymbolic	(mostly engineering already)	Is there a sound check for the translation layer?	"LLM proposes, solver judges" as the default architecture	Which tasks can be decomposed this way

PART II · LOOKING AHEAD · AREA 1

19. AI safety via debate

Intuition To decide whether the number of primes between 1 and 2^{60} is odd, a human cannot count. Rules: Alice states a total, splits the range in half and states the count for each half; Bob challenges one half; they keep halving... After about 60 rounds a single number remains, and the judge checks whether it is prime. If Alice lies, **at least one half is wrong**, and Bob can always follow the error down. So a single cheap check by the judge indirectly verifies an exponential computation.

Theorem Irving–Christiano–Amodei 2018: with optimal play, debate extends the problems a judge can decide from NP ("give me a certificate and I'll check it") to **PSPACE**. Debate is zero-sum, and for a true claim honesty is a winning strategy, so the equilibrium is honest — that is what "the Nash equilibrium of debate is honesty" means. The authors call it an analogy and do not expect practical ML to reach all of PSPACE. [data]

Hidden assumptions

(a) strong enough play; (b) a judge who is reliable on each small step; (c) problems that decompose recursively with checkable steps; (d) zero-sum, no collusion; (e) training that actually converges to the equilibrium.

2023 refinement

Doubly-efficient debate (Brown-Cohen, Irving, Piliouras): the honest side needs only polynomially many steps to beat a liar allowed exponentially many. "The honest side is no stronger than its opponent" is enough.

19.1 Where it is stuck

1. **Obfuscated arguments (2020)**: a liar decomposes a simple problem into many subproblems, *one of which contains an error*, but **neither side can find which** (finding the error is computationally hard). The 2018 theorem assumed the honest side can locate errors. **Prover-Estimator Debate** (2025, 2506.13609) partly fixes this: the estimator no longer picks what to attack but assigns probabilities to each subclaim; it needs a **stability assumption** — the conclusion does not flip under small changes in those probabilities. **Whether real arguments are stable, nobody knows.**
2. **Learnability of the equilibrium**: a 2026-04 paper (2604.28182, with GDM researchers) shows models can be trained to resist RL exploration, and frontier models that know they are in training explicitly reason about "suppressing exploration." [data]
3. **Systematic judge errors**: humans err consistently on some sub-distributions. Irving proposed (2025-05) making arguments robust to an ϵ fraction of errors; whether a version exists that liars cannot exploit remains open.
4. **True but selective statements** (2609.29189): the verdict is right, but selection and framing serve a hidden goal. Debate's correctness guarantee does not reach this layer.
5. **Collusion**: when both debaters are copies of one model, what keeps the game zero-sum? 2603.05293 finds that with identical training corpora debate collapses to RLAIIF.

19.2 Evidence

WORK	SETTING	RESULT
Michael et al. 2023 (2311.08702)	human debaters, judge without the source text	judge accuracy 84% with debate vs 74% with a single consultant
Khan et al. 2024 (ICML best paper)	LLM debaters	non-expert LLM judges 48% → 76%; humans 60% → 88%
Kenton et al. 2024 (GDM)	math, code, logic, multimodal	debate > consultancy everywhere; versus the judge answering directly, better only on information-asymmetry tasks
Engels...Tegmark 2025 (2504.18530)	oversight success vs capability gap	nested-oversight success at a 400-Elo gap: debate 51.7% (Mafia 13.5%, backdoored code 10.0%)
Kenton et al. 2026-08 (GDM, 2608.17776)	math RL with debate as the reward signal	the RLAIF baseline quickly learns to exploit the judge; debate training keeps the judge effective, recovering 45% of the performance gap

[judgment] The evidence supports debate as **a reward signal that is harder to hack**; it does not yet support "debate lets humans reliably oversee superhuman systems." Every experiment's capability gap is far smaller than the theory imagines.

19.3 What AI mathematics and coding can do

- **Type A, now:** formalize and prove the theorems about debate protocols. Irving said in a 2026 80,000 Hours interview: "theory is easier to automate than empirics, because proofs give a verifiable reward." This is the core bet of Resolution, where he is chief scientist. [data]
- **A new opportunity: debate experiments with real capability gaps, built on formal mathematics.** AI's formal-math ability already far exceeds most humans', so "a human judge overseeing an AI's mathematical argument" is a natural superhuman-oversight experiment, with Lean providing perfect ground truth — much closer to the real problem than QuALITY-style hidden information. [judgment]
- **A counterintuitive point:** where things can be formalized, Lean is already a perfect judge and debate is unnecessary. **Debate's value lies precisely outside the formalizable boundary**, which is exactly where AI helps least.
- **What it cannot help with:** defining "honesty" for value judgments; selective statements.

20. Singular learning theory (SLT)

Classical theory	Classical statistics assumes the best parameter is a single point surrounded by a bowl. Then the "effective number of parameters" is the true count d , and generalization error is about $d/(2n)$.
Toy example	Model $y = a \cdot b \cdot x$ with data from $y = 0$. The loss is about $(ab)^2$, which is optimal whenever $a = 0$ or $b = 0$. So the optimum is a cross , not a point; at the crossing the Hessian is all zeros and classical approximations fail. This is called singular .
Counting effective parameters	Look at how the volume of "nearly optimal" parameters $V(\epsilon)$ shrinks as $\epsilon \rightarrow 0$: $V \approx \epsilon^\lambda$. One-parameter bowl a^2 : $\lambda = 1/2$; two-parameter bowl $a^2 + b^2$: $\lambda = 1$; singular a^2b^2 : $\lambda = 1/2$. Two nominal parameters behave statistically like one. λ is the learning coefficient (RLCT).
Theorem	Watanabe (2009, 2018) proved that for Bayesian inference the free energy $\approx n \cdot S + \lambda \cdot \log n$ and the generalization error $\approx \lambda/n$. Neural networks are almost always singular, so λ , not the parameter count, governs generalization. The proof uses Hironaka's resolution of singularities. [data]

20.1 Proven vs conjectured

- **Proven:** the asymptotic formulas above (for real-analytic models on compact parameter spaces, etc.); exact λ for a few architectures such as deep linear networks (Aoyagi).
- **Unproven:** that SGD-trained networks match the local structure of the Bayesian posterior; that training "stages" correspond to Bayesian phase transitions; that λ describes finite-sample, ReLU, non-realizable real LLMs.

20.2 Evidence

WORK	SCALE	RESULT
Lau et al. (2308.12108)	deep linear networks, etc.	the local learning coefficient (LLC) estimator recovers known exact λ
Hoogland et al. (2402.02364)	~3M-parameter transformer	plateaus and turns in the LLC mark discrete developmental stages of in-context learning
Kreer et al. (ICLR 2026)	billions of parameters	Hessian-free data attribution at SOTA
Spectroscopy at Scale (2026-04)	Pythia-1.4B	57,236 structural clusters, 4,800 H200-hours
Patterning (2601.13548)	small models	reweighting data along susceptibility directions can speed up or delay the emergence of induction circuits

[data] The largest scale is about 1–3B; **there are no frontier-model results, and no case where SLT found a safety problem that other methods could not**. The LLC is

estimated by SGLD sampling, which the authors concede loses unbiasedness and is hyperparameter-sensitive.

20.3 What AI can do

- **Now:** (a) symbolic computation (Newton polyhedra, blow-up algorithms) to produce exact λ for many small architectures, giving estimators ground truth; (b) agents that sweep hyperparameters and replicate across seeds, fixing the estimator-robustness pain point.
- **Medium term:** running LLC/susceptibility pipelines on large models is mostly compute-limited; human effort can be driven very low.
- **Hard:** Lean formalization (Mathlib lacks Hironaka's theorem; a huge undertaking); a theory of the SGD–Bayes gap (type B).

The decisive experiment: on a model organism with an implanted backdoor or reward hacking, can λ or a susceptibility give a warning **before the behavior shows**? This decides whether SLT becomes a standard tool for monitoring training or remains a scientific tool for understanding learning dynamics. [judgment] Agents will bring the cost of this experiment down to a few weeks for a small team.

PART II · LOOKING AHEAD · AREAS 3 AND 4

21. Guaranteed Safe AI and neurosymbolic systems

The GSAI idea (Dalrymple, Skalse, Bengio, Russell, Tegmark et al., arXiv 2405.06624): instead of understanding an AI's internals, require each output to carry a **certificate** proving it satisfies a **safety specification** in a **world model**, checked by a trusted **verifier**.

Example A: verifying a sort function

Spec: "the output is sorted and is a permutation of the input." The world model is the programming language's semantics. Lean/Dafny proves it for all inputs. **Pitfall:** drop "is a permutation of the input" and returning an empty list also passes.

Example B: verifying a braking controller

World model: position and velocity update equations plus "road disturbance at most 0.1." Spec: never hit the obstacle. The verifier does reachability analysis. **Pitfall:** the certificate holds only inside the model; on ice the disturbance exceeds 0.1 and the guarantee fails, so runtime monitoring and a fallback controller are needed.

Replace example B with "a chatbot never gives harmful advice": the world model becomes "human society" and the spec becomes "harm," and neither can be written down. That is GSAI's hardest end.

21.1 The flagship institution has pivoted (ARIA Safeguarded AI) [data]

DATE	EVENT
2024-02	Thesis v1: TA1 mathematical framework / TA2 ML (a planned £18m new organization) / TA3 applications; £59m total
2025-11	davidad halts TA2 Phase 2: "every frontier model has more capability than I expected"; money moves to TA1
2026-02	Applications narrowed to cybersecurity; cyber-physical Phase 2 cancelled
2026-04	Nora Ammann becomes programme director; davidad becomes technical advisor
2026-05	Thesis v2: keeps only "mathematical-assurance tooling + cybersecurity applications"; reasons given are that frontier progress was faster than expected and that timely international consensus on a gatekeeper workflow has become less likely, hence a turn to "accelerating improvements to societal resilience"

[judgment] The original thesis was "AI capability first, safety certificates as the gate"; v2 effectively retreats to "use AI to make formal methods cheap and harden critical software." The gatekeeper is now a long-term option only.

21.2 AI is taking over the verifier leg [data]

BENCHMARK / CASE	EARLY	LATEST
DafnyBench (782 programs)	~68% (2024)	92.7%–96% (2025–2026)
CLEVER (161 problems, Lean, end-to-end)	1/161 (2025-05)	~87.5% (2026-05)
VERINA (189 problems, Lean)	o3 proof success 4.9% (2025-05)	an 8B specialized model averages 62% on three Lean benchmarks (2026-03)
seL4 (8,700 lines of C)	by hand: ~20 person-years, ~\$350/line (2009)	a 7B model proves 51.7% of the lemmas in seL4's important theories (2026, statements given)

Pattern: the same benchmark goes from single digits to 80–90% in about a year, the same shape as HumanEval in 2023. Dafny is easier than Lean because it hands part of the reasoning to an SMT solver and the model only writes hints — which is itself neurosymbolic.

21.3 Order of arrival [judgment]

DOMAIN	WORLD MODEL	SPEC	EXPECTED
Crypto libraries, protocols, kernels, C→Rust equivalence	language semantics, exists	often exists (RFCs, reference implementations)	now to 2027
Hardware, RTL, firmware, distributed protocols	ISA, memory model; engineering-limited	writable, needs human review	2027–2029
Control systems with reliable physics (grid, braking)	physical laws + disturbance bounds; math-limited	safe sets; conservative but writable	later; "finite-horizon certificate + runtime monitoring"
Open-world social and biological domains	no auditable model	"harm" cannot be written down	last, perhaps only weak guarantees forever

A counterintuitive consequence: AI makes finding vulnerabilities cheaper, but also makes proving their absence cheaper. **In software, the offense–defense balance may tilt toward defense** — provided defenders actually do formal verification. Biology and social domains lack this symmetry, because no spec can be written.

21.4 Neurosymbolic systems

A neural network "guesses" (proof steps, spec drafts, programs, world-model code) and a symbolic system "judges" (Lean kernel, SMT solver, reachability analysis). The judge is reliable; the guesser can be a black box — the same structure as GSAI's "black-box policy, trusted verifier."

- **Status:** it is already the de facto default architecture in every verifiable domain. Programmatic world models (PoE-World, NeurIPS 2025: an LLM writes 4,000+ lines of Python to model the Montezuma game world and generalizes zero-shot to new levels) show the route works, but only in Atari-like settings. AWS's Automated Reasoning checks translate text into logic and decide it with a solver. [data]
- **The weakest link is the translation layer:** every step from natural language to formal language can produce a vacuous spec or a mis-formalization, and this layer has **no sound check** — only cross-consistency of independent translations, testing, and human review.
- **Another attack surface:** DeepSeek-Prover-V2 exploited a toolchain bug in Lean 4.9.0 to get false proofs to compile. [data] "Only the checker needs to be trusted" must itself be maintained by engineering; whether RL-trained provers systematically seek such holes is a reward-hacking question.

22. Timeline: T0 → T4

Each node has four parts: **capability** (central scenario of the three curves), **what unlocks**, **new opportunities**, and **signals that would falsify it**. Dates carry roughly ± 1 year of uncertainty: the optimistic scenario shifts everything half a year to a year earlier, the pessimistic one a year or two later.

T0 · Now (2026 Q4)

- **Capability:** AI completes about 1–2 workdays of work on its own; single-attempt open-problem solve rate about 3–5%; formalization about \$0.01–0.02 per line; competition math saturated.
- **Unlocks:** theorems about debate protocols can be handed to AI to formalize; SLT's symbolic λ computations and estimator tuning can be handed to agents; the proof step of software verification is largely automated; neurosymbolic is the default in verifiable domains.
- **Opportunities:** the most valuable work is **what cannot yet be automated** — methods for reviewing statements and specs; reward hacking against provers and verifiers; designing oversight experiments with real capability gaps.
- **Falsifier:** a new METR long-task suite shows progress stalling at 1–2 days.

T1 · Mid 2027

- **Capability** [model]: task length about 150 hours ($\approx$ one work-month); open-problem solve rate about 9%; proof costs down another order of magnitude.
- **Unlocks: type-A theory problems start falling in bulk** — debate-protocol variants, λ for specific architectures, all kinds of "remove this assumption" theorems. One theorist plus AI can do in a month the proof work a group used to do in a year. SLT estimator robustness is essentially solved. Crypto and protocol libraries ship with proofs by default.
- **Opportunities:** the "AI-assisted theorist" becomes a role: you pose statements; AI proves them or finds counterexamples. The scarcest skill is **knowing which statement is worth proving**. Superhuman-oversight experiments built on formal math become practical.
- **Falsifiers:** the next round of FrontierMath Erdős is still below 5%; Anthropic's "AI-led, human-supervised" R&D share stalls around 30%.

T2 · End 2027

- **Capability** [model]: task length about 400 hours (2–3 work-months); open-problem solve rate about 18%; a Fermat-scale formalization costs a few thousand dollars.
- **Unlocks**: the idealized-model theory of debate largely converges; what remains — "do real arguments satisfy stability?", "does training reach the honest equilibrium?" — is empirical and ML work. SLT's decisive early-warning experiment becomes feasible. Hardware and RTL verification begins to scale.
- **Opportunities**: "**auditing statements and specs**" becomes a field: how to judge quickly whether a formal statement is faithful, how to design cross-consistency checks, how to measure human reviewers' throughput. Almost nobody works on this today.
- **Falsifier**: by end 2027 there is still no field-level important problem solved mainly by AI and confirmed by human experts, meaning curve ②'s cap was set too high.

T3 · 2028

- **Capability** [model]: task length about 2,800 hours (more than a work-year); open-problem solve rate about 45%; proof costs negligible.
- **Unlocks**: **most type-A problems are solved or proven impossible**. In all four areas only type-B and conceptual problems remain. The value of debate and SLT depends on one thing: whether they give guarantees outside the formalizable boundary. "Critical software ships with proofs" could become an industry norm (if regulation or insurance pushes it).
- **Opportunities**: **conceptual work has the highest return**: people who can turn "deception," "reward hacking" or "harm" into measurable, provable statements are the bottleneck itself. If AI R&D is highly automated, AI-safety theory itself is automated at scale, and **can humans review fast enough?** becomes the central question.
- **Falsifier**: task length stalls at a few hundred hours (success depends more and more on the spec), turning curve ① into an S-curve.

T4 · 2029–2030

- **The scenarios diverge**:
 - **Full automation of AI R&D** (the CASP scenario): every date compresses; theory, proofs and experiments are almost fully automated, and the human role concentrates on setting goals, reviewing statements and deciding deployment.
 - **A hard bottleneck appears** (compute, data, or spec review): progress returns to exponential or slower; type-A problems are solved while type-B and conceptual problems move slowly.

- **True in both:** safety guarantees for software and protocols are far stronger than today, and the verification asymmetry favors defense; guarantees in the open world (biology, society) remain weak and rely mainly on monitoring, access control and governance; **who reviews the statements and specs** is the ceiling of the whole system.

22.1 Where this projection is most likely wrong

1. **Curve ①'s measurement is breaking down:** unreliable above 16 hours, and cheating treatment can shift it by 4.5 doublings. Post-2027 numbers rest mainly on "other signals are rising in step."
2. **Curve ② has a single anchor** (3%, 68 problems), and the slope is a conservative choice with wide error.
3. **The type-A/type-B boundary will move:** AI may become good at "proposing good statements" sooner than expected. There is no evidence this has happened yet — OpenAI's 719 papers mostly solve existing problems and propose almost no new concepts (compare Tao's 21 dimensions of "good mathematics"). If it happens, every node moves earlier and the human role shrinks further.
4. **Variables left out:** pacing or pause agreements, major safety incidents, geopolitics. These can only slow the timeline.

22.2 For researchers: what is worth doing now

Rule: **don't do by hand what AI will do within a year, unless it is infrastructure or helps you judge "what to ask."**

OPPORTUNITY	WHY NOW
Reward hacking against provers and verifiers	A prover has already exploited a Lean toolchain hole; most valuable to study before AI proving is deployed at scale
Auditing the faithfulness of statements and specs	The shared bottleneck of every area from T1 to T3; the line-by-line reading of Lean statements in this deck is an early version
Superhuman-oversight experiments on formal math	Existing experiments manufacture their capability gaps; formal math gives a real gap plus perfect ground truth
SLT "early warning" experiment	The experiment that decides SLT's safety value; nobody has run it
Manipulation-resistance of AI R&D automation indicators	CASP's policy proposals rely on these indicators, and reward hacking will inflate them

Key numbers in this part are sourced inline. The four underlying research drafts (debate, SLT, GSAI and neurosymbolic, trajectory data; each item tagged [P]/[R]/△ with links) are kept in the author's research notes and are not published with this deck.

23. Explainer 01: The quasi-Riemann hypothesis

This is the "explained from zero" version of §4 of the [notes/openai_math_explained/](#) deck. All you need going in: you can work with powers and logarithms, you know the summation sign Σ , and you know what a function is. Every new concept is explained before it is used. Every number in this chapter was either computed by script or has a source. The **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Estimated reading time: 60–90 minutes. You can read it in three sittings: Sections 1–3 (primes and the error), Sections 4–6 (the ζ function and its zeros), Sections 7–10 (OpenAI's result and what it means).

0. The bottom line: the one thing this chapter explains

Prime numbers (2, 3, 5, 7, 11, ...) look randomly scattered, but taken as a whole they are very regular. Mathematicians can write down a formula that predicts, accurately, "roughly how many primes there are up to x ." The truly hard question is: **how wrong can that prediction be, at most?**

This chapter explains one causal chain:

How big the prediction error is $\Leftarrow$ is determined by a set of "waves" $\Leftarrow$ each wave corresponds to a "zero" of the ζ function $\Leftarrow$ the farther a zero sits from a certain line, the bigger the error.

The Riemann hypothesis says that all the zeros line up neatly on one line, so the error is as small as it can be. Nobody can prove it.

The **quasi-Riemann hypothesis** is a much weaker version. It does not require the zeros to lie on that line; it only requires that they **not sit too far to the right**. Nobody has proved even this weak version in 125 years. OpenAI claims to have proved it.

After reading this chapter, you should be able to explain in your own words:

- what exactly the relationship is between "zeros" and "the error in counting primes";
- why there is a qualitative difference between "a zero-free region that gets narrower and narrower" and "a zero-free region of fixed width";
- what OpenAI's claimed result, if true, concretely means, and what it does not mean.

1. Primes: count them, look for a pattern

1.1 What primes are and why they matter

A prime is an integer greater than 1 that is divisible only by 1 and itself: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, ...

They matter because **every integer can be split into a product of primes in exactly one way** (the fundamental theorem of arithmetic):

- $12 = 2 \times 2 \times 3$
- $360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$

So primes are the "atoms" of the integers. The HTTPS encryption (RSA) you use every day rests on one fact: multiplying two large primes is easy, but splitting the product back apart is extremely hard.

1.2 Counting

Write $\pi(x)$ for "the number of primes that are at most x ." This π is just a name. It has nothing to do with the circle constant; it is a historical notation.

x	$\pi(x)$ (actual count)
10	4 (2, 3, 5, 7)
100	25
1,000	168
1,000,000	78,498
1,000,000,000	50,847,534

Primes get sparser as you go further out. Below 100, one number in four is prime. Near 1 million, only about 72 in every 1000 numbers are prime (that is the estimate from the $1/\ln x$ rule in the next section; counted by script, the 1000 numbers from 999,001 to 1,000,000 actually contain 65 primes).

1.3 The first pattern: the density is about $1/\ln x$

As a teenager, Gauss stared at prime tables and noticed: **near x , about one number in every $\ln x$ is prime.**

Here $\ln$ is the natural logarithm, with base $e \approx 2.718$. $\ln(1,000,000) \approx 13.8$, which means that near 1 million about one number in 14 is prime. The script count above was 65 per 1000, or about one in 15. Very close.

[Check 1] $\ln(1000) \approx 6.9$. By this rule, about how many primes are there per 1000 numbers near 1000?

1.4 The prime number theorem

Integrate the "density $1/\ln t$ " from 2 to x and you get a prediction for the number of primes up to x :

$$\pi(x) \approx \text{Li}(x) = \int_2^x \frac{dt}{\ln t}$$

$\text{Li}(x)$ is called the "logarithmic integral." If you are not comfortable with integrals, that is fine: think of it as "add up the density $1/\ln t$ at every position."

x	actual $\pi(x)$	rough prediction $x/\ln x$	precise prediction $\text{Li}(x)$	error of Li
1,000	168	145	177	+9
1,000,000	78,498	72,382	78,627	+129
1,000,000,000	50,847,534	48,254,942	50,849,235	+1,701

($\text{Li}(10^9)$ uses the standard value from the literature; the rest were computed by script.)

The **prime number theorem** (proved independently by Hadamard and de la Vallée Poussin in 1896) says that the **ratio** of $\pi(x)$ to $\text{Li}(x)$ tends to 1. You can see this in the table: the absolute error is growing ($9 \rightarrow 129 \rightarrow 1,701$), but relative to the total it becomes more and more negligible (at 10^9 it is off by only 0.3 parts in 10,000).

2. The real question: how big is the error?

2.1 Two ways of saying "the error is getting smaller," with very different strength

The prime number theorem only says "error / $x \rightarrow 0$," that is, the relative error goes to 0. It does not say **how fast** it goes to 0.

An intuitive way to measure this: roughly what power of x is the error?

- If the error $\approx x^{0.5} = \sqrt{x}$: at 10^9 , $\sqrt{x} \approx 31,623$. The actual error in the table above, 1,701, is even smaller than that.
- If the error $\approx x^{0.875} = x^{(7/8)}$: at 10^9 this is about 75 million, far bigger than the actual error.

Note: the **actual error** and the **error bound you can prove** are two different things.

Numerically, the actual error is very small (roughly on the order of $\sqrt{x}$). But mathematicians

need a **proof**: "for all x , the error is definitely no more than such-and-such." Everything in this chapter is about the **bound you can prove**.

2.2 "Power saving": why $x^{7/8}$ is a qualitative change

Since 1958, the best provable bound has roughly had the form:

$$\text{error} \leq x \cdot \exp(-c \cdot (\ln x)^{0.6}) \quad (c \text{ is some small constant})$$

Compare this with $x^{7/8}$ by looking at how "error / x " (the relative error) changes with x . To compare the trends, take $c = 1$ below (the real c is smaller, so the old bound is actually even worse):

x	relative error of the old bound $\exp(-(\ln x)^{0.6})$	relative error of the new bound $x^{-1/8}$
10^{100}	$10^{-11.4}$	$10^{-12.5}$
10^{1000}	$10^{-45.2}$	10^{-125}
10^{10000}	10^{-180}	10^{-1250}

(Computed by script.)

The old bound also shrinks, but **more slowly than any "negative power of x ."** The new bound is " x to the power $-1/8$," and the gap widens the further out you go. Mathematicians call this a **power saving**: the error is smaller than x by a fixed power.

Why this matters: many harder problems in number theory plug the error term of the prime number theorem into further calculations. If the error is only "a little" smaller than x (like the old bound), it often gets swamped by other error terms once you plug it in. If it is smaller by a fixed power, you have room to spare for all kinds of consequences.

3. The key turn: the error is made of "waves"

This was Riemann's stroke of genius in 1859, and it is the single most important step in the chapter.

3.1 First, an analogy with sound

A sound (say, a chord) can be broken down into a sum of pure tones (sine waves). Each pure tone has two properties:

- **frequency**: how fast it vibrates
- **amplitude**: how loud it is

This is called **Fourier decomposition**. The EEG spectra you saw in neuroscience (alpha waves at 8–12 Hz, beta waves, ...) are the same idea: break a complicated signal into waves of different frequencies.

3.2 Riemann's discovery

Riemann discovered that the error in counting primes can be broken down the same way:

$$(\text{weighted count of primes}) = (\text{smooth main term}) - \sum(\text{individual "waves"}) - (\text{some small corrections})$$

(To keep the formula clean, mathematicians usually use a weighted version of the prime count, $\psi(x)$, in which each prime p gets weight $\ln p$. It is essentially the same thing as $\pi(x)$.)

The main term is smooth, roughly just x . All of the error comes from those "waves," and:

- **Each wave corresponds to one zero of the ζ function** (Section 5 explains what a zero is).
- A zero is written in the form $\rho = \beta + i\gamma$, with two coordinates β and γ .
- **γ determines the wave's frequency** (how fast it oscillates).
- **β determines how fast the wave's amplitude grows with x :** the size of this wave is about x^β .

So:

- If every zero has β equal to $1/2$, each wave has size about $\sqrt{x}$, and the error is about $\sqrt{x}$. This is exactly the **Riemann hypothesis**.
- If some zero has β close to 1, its wave has size close to x , almost as big as the main term, and the error will be huge.
- If you can prove that **every zero has β at most θ** , then the error is at most about x^θ .

In one sentence: the β coordinate of the zeros (how far "to the right" they sit) directly determines how big the error in counting primes is. That is why number theorists care so much about "where the zeros are."

3.3 Why complex numbers show up

To describe these "waves" we need complex numbers. The next section explains them first.

4. Complex numbers: the 5-minute version

4.1 A complex number is a point in the plane

Real numbers sit on a line. **Complex numbers are points in a plane:** the horizontal coordinate is called the real part, the vertical coordinate the imaginary part.

$s = \sigma + i \cdot t$ σ is the horizontal coordinate (real part, written $\text{Re } s$), t is the vertical coordinate (imaginary part, written $\text{Im } s$)

i is a symbol satisfying $i^2 = -1$. Don't worry about whether it "exists"; treat it as a marker for "one step in the vertical direction" and that is enough.

4.2 Multiplying by a complex number = scaling + rotation

The most useful property of complex numbers: **multiplying by a complex number rotates a point around the origin and scales it**. For example, multiplying by i rotates 90 degrees counterclockwise: $1 \rightarrow i \rightarrow -1 \rightarrow -i \rightarrow 1$, one full turn.

4.3 x to a complex power = "size part" $\times$ "rotation part"

For a positive real number x and a complex number $\beta + i\gamma$:

$$x^{(\beta+i\gamma)} = x^\beta \times (\cos(\gamma \cdot \ln x) + i \cdot \sin(\gamma \cdot \ln x))$$

- x^β is the size part: it grows with x , and how fast is set by β .
- $\cos(\gamma \cdot \ln x) + i \cdot \sin(\gamma \cdot \ln x)$ is the rotation part: its length is always 1; it just goes around the unit circle. How fast it turns is set by γ .

These are the "waves" from Section 3: **β controls the amplitude (x^β), γ controls the frequency (oscillation on the scale of $\ln x$).**

5. The ζ function and its zeros

5.1 Definition of the ζ function

$$\zeta(s) = 1/1^s + 1/2^s + 1/3^s + 1/4^s + \dots$$

At $s = 2$: $\zeta(2) = 1 + 1/4 + 1/9 + 1/16 + \dots = \pi^2/6 \approx 1.6449$ (computed by Euler in 1734; here π is the circle constant).

At $s = 1$: $\zeta(1) = 1 + 1/2 + 1/3 + 1/4 + \dots$, the "harmonic series," which **diverges to infinity**.
Computed by script: the sum of the first 1000 terms ≈ 7.49 , the first 1 million terms ≈ 14.39 .
It grows very slowly but never stops.

5.2 The Euler product: how ζ "knows" about primes

Euler discovered:

$$\zeta(s) = 1/(1-2^{-s}) \times 1/(1-3^{-s}) \times 1/(1-5^{-s}) \times 1/(1-7^{-s}) \times \dots \quad (\text{product over all primes})$$

Why it holds: each factor $1/(1-p^{-s})$ expands into $1 + 1/p^s + 1/p^{2s} + \dots$ (a geometric series). Multiply these expansions together over all primes, and every term has the form $1/(p_1^a \cdot p_2^b \cdot \dots)^s$. Because every integer has a **unique** prime factorization, each $1/n^s$ appears exactly once, and the product is exactly $\zeta(s)$.

By hand for $s = 2$ (checked by script):

- Multiplying only the three factors for $p = 2, 3, 5$: $(4/3)(9/8)(25/24) = 1.5625$
- Multiplying the factors for all primes below 100: 1.6419
- True value $\pi^2/6 = 1.6449$

So **the ζ function ties "all integers" to "all primes"**, and this is the bridge for the whole story.

A beautiful consequence (Euler's proof that "there are infinitely many primes"): at $s = 1$ the left side is the harmonic series, which diverges to infinity. If there were only finitely many primes, the right side would be a product of finitely many finite numbers and could not be infinite. So there must be infinitely many primes.

5.3 "Analytic continuation": extending the definition to the whole plane

The sum above only converges (adds up to a finite number) when the real part of s is > 1 . But Riemann needed to talk about $\zeta(s)$ on the whole complex plane.

An analogy for how to extend it:

$$1 + x + x^2 + x^3 + \dots = 1/(1-x)$$

The left side converges only when $|x| < 1$ (for example, at $x = 2$ the left side is $1+2+4+8+\dots$, which diverges), but the right side $1/(1-x)$ makes sense everywhere except $x = 1$ (at $x = 2$ it equals -1). We can **use the right side to define** something that originally only made sense for $|x| < 1$. This is called **analytic continuation**.

One can prove mathematically that such a continuation, if it exists, is unique. Riemann used a similar but more elaborate technique to continue $\zeta(s)$ to the whole complex plane (except the single point $s = 1$).

5.4 Zeros

A **zero** is a complex number s that makes $\zeta(s) = 0$.

- **Trivial zeros:** $s = -2, -4, -6, \dots$, which come from a simple factor in the continuation formula. Nothing mysterious.
- **Non-trivial zeros:** all the other zeros. They all lie in the vertical strip $0 < \operatorname{Re} s < 1$, which is called the **critical strip**.

The first three non-trivial zeros (in the upper half-plane) are approximately:

$$\frac{1}{2} + 14.13i, \quad \frac{1}{2} + 21.02i, \quad \frac{1}{2} + 25.01i$$

Their real parts are all exactly $1/2$.

5.5 The Riemann hypothesis

Riemann hypothesis (1859): every non-trivial zero has real part equal to $1/2$.

Numerical checks have examined **more than ten trillion zeros**, all on the line $\operatorname{Re} s = 1/2$ (Gourdon in 2004 verified the first 10^{13} zeros $\triangle$ defer to the original paper for the exact figure). But no amount of checking is a proof. It is one of the seven Clay Millennium Prize Problems, with a \$1 million prize.

Connecting Section 3 to here: **Riemann hypothesis $\Leftrightarrow$ every wave has amplitude $\sqrt{x} \Leftrightarrow$ the error in counting primes is about $\sqrt{x}$.**

6. Zero-free regions: what can be proved when the Riemann hypothesis can't

Since nobody can prove "all zeros are on the line $1/2$," mathematicians take a step back and ask: **can we at least prove where the zeros are not?** A region proved to contain no zeros is called a **zero-free region**.

6.1 Step one: no zeros on the line $\operatorname{Re} s = 1$ (1896)

The proof of the prime number theorem essentially amounts to proving that **there are no zeros on the vertical line $\operatorname{Re} s = 1$.**

The key trick is a trigonometric identity:

$$3 + 4\cos \theta + \cos 2\theta \geq 0 \quad (\text{holds for all } \theta)$$

[Check 2] Use $\cos 2\theta = 2\cos^2\theta - 1$ to prove that $3 + 4\cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2$, so it is always ≥ 0 . (The script swept 6,284 points on $[0, 2\pi]$; the minimum was 1.4×10^{-14} , i.e. 0 plus rounding error.)

Combining this identity with the Euler product, one can show that if $1 + iy$ were a zero, it would lead to a contradiction.

6.2 Step two: a strip to the left of $\text{Re } s = 1$ that "keeps getting narrower" (1899, 1958)

de la Vallée Poussin (1899) pushed the result a little further: not only are there no zeros on $\text{Re } s = 1$, there are none in **a narrow strip to its left** either:

zero-free region: $\text{Re } s > 1 - c / \ln(t)$ (t is the height, i.e. the imaginary part)

Note the $\ln t$ in the denominator: **the bigger the height t , the narrower this strip.**

- At $t = 100$, the width is about $c / 4.6$
- At $t = 10^{10}$, the width is about $c / 23$
- As $t \rightarrow \infty$, the width $\rightarrow 0$

Vinogradov–Korobov (1958) widened the strip a bit (the denominator changes from $\ln t$ to $(\ln t)^{2/3}$ times an even smaller factor), but **it still shrinks toward 0 with height**. This was the best result for the following 60-plus years.

6.3 Why "keeps getting narrower" and "fixed width" are qualitatively different

Back to Section 3: the error is the sum of the waves from all zeros, and **the rightmost zero** sets the size of the error.

- If the zero-free region keeps getting narrower: in principle, higher and higher up there could be zeros whose β gets closer and closer to 1. You cannot rule them out, so the error bound can only be "a little" smaller than x . That is the $\exp(-c(\ln x)^{0.6})$ from Section 2.
- If there is a zero-free region of **fixed width**, say no zeros at all in $\text{Re } s > 7/8$: then every zero has $\beta \leq 7/8$, every wave is $\leq x^{7/8}$, and the error bound becomes of order $x^{7/8}$. That is the **power saving** from Section 2.

The **quasi-Riemann hypothesis** is exactly such a statement:

There is some $\theta < 1$ such that ζ has no zeros anywhere in the region $\text{Re } s > \theta$.

It is much weaker than the Riemann hypothesis ($\theta = 1/2$), but before 2026 **nobody could prove it for any $\theta < 1$.**

6.4 Why the classical methods can't do it

The " $3 + 4\cos \theta + \cos 2\theta \geq 0$ "-type argument from Section 6.1 starts from properties of the Euler product in $\text{Re } s > 1$ and "seeps" a little way to the left. But this seeping power decays

with the height t , so all you can get is a strip that keeps getting narrower. To get a fixed width, you need **a fundamentally different mechanism**.

Another line of attack is called "zero-density estimates" (for example, the important breakthrough of Guth–Maynard 2024). These can prove that "there are **few** zeros with $\text{Re } s > \sigma$," but not that "there are **none at all**." And for the error bound, even a single zero with β close to 1 is enough to make the error large.

7. Dirichlet L-functions and Siegel zeros

OpenAI's result is not only about ζ ; it also covers all **Dirichlet L-functions**.

7.1 Primes in arithmetic progressions

Apart from 2, all primes are odd, so each has the form $4k+1$ (5, 13, 17, ...) or $4k+3$ (3, 7, 11, 19, ...).

Computed by script:

- Below 100: 11 of type $4k+1$, 13 of type $4k+3$
- Below 1000: 80 of type $4k+1$, 87 of type $4k+3$

The two types are roughly half and half (type $4k+3$ is usually slightly ahead, but not always: at $x = 26,861$ type $4k+1$ takes the lead for the first time; this phenomenon is called Chebyshev's bias). In 1837 Dirichlet proved that for any modulus q and any a coprime to q , there are infinitely many primes of the form $qk + a$, and that in the long run the classes split evenly.

7.2 Dirichlet L-functions

To study each class separately, Dirichlet built variants of ζ :

$$L(s, \chi) = \chi(1)/1^s + \chi(2)/2^s + \chi(3)/3^s + \dots$$

χ (pronounced "chi") is called a "character." It is a function that maps integers to certain complex numbers, and its job is to "pick out" and "tell apart" the different residue classes. For example, the non-trivial character mod 4: $\chi(\text{odd numbers } 4k+1) = 1$, $\chi(\text{odd numbers } 4k+3) = -1$, $\chi(\text{even numbers}) = 0$.

Just like ζ , the zeros of L-functions control the error in the distribution of primes across the classes mod q .

7.3 Siegel zeros: a ghost that has haunted the field for 90 years

For certain special L-functions, classical methods cannot rule out one possibility: **that it has a real zero very, very close to 1**. Such a hypothetical zero is called a **Siegel zero** (or Landau–Siegel zero).

If one exists, the consequences are strange: primes for the corresponding modulus would be heavily skewed toward certain residue classes. Although people widely believe it does not exist, nobody can rule it out.

Worse, Siegel's related theorem from 1935 is "ineffective": it proves that a certain constant **exists** but tells you nothing at all about **how big** it is. As a result, a whole set of problems, such as "how large is the class number of an imaginary quadratic field at least," can only get "there exists a bound," not an actual number.

If the quasi-Riemann hypothesis holds for all Dirichlet L-functions, Siegel zeros are automatically ruled out: there are no zeros anywhere in $\text{Re } s > 7/8$, so of course there is no zero close to 1.

8. What OpenAI claims

8.1 The main result

Abstract of the paper *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\Re s > 7/8$* (2026-09-30, 199 pages):

"We prove that all finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ and all Dirichlet L-functions are zero-free in the half-plane $\Re s > 7/8$... In particular, the Riemann zeta function is zero-free in this half-plane, proving the quasi-Riemann hypothesis."

In plain terms: **ζ and all Dirichlet L-functions have no zeros anywhere in the half-plane $\Re s > 7/8$.**

The paper itself makes a point of stressing: "The Riemann hypothesis remains open." The Riemann hypothesis says the zeros are all at $1/2$; this only proves they are all $\leq 7/8$. Between $7/8$ and $1/2$ there is still a large unknown territory.

There are two more papers in the same family: one proves an $11/12$ version by a different method (49 pages; the README says its writing was edited by humans), and a 9-page paper separately proves that Siegel zeros are uniformly ruled out.

8.2 Intuition for the method (we only read the introduction; this is a simplified version)

1. **Switch number systems.** Instead of working in the ordinary integers, work in the "Eisenstein integers" $\mathbb{Z}[\omega]$, where ω is a cube root of unity: $\omega^3 = 1$, $\omega \neq 1$. You can picture them as a triangular lattice of points in the plane (the ordinary integers are points on a line,

the Gaussian integers a square lattice, the Eisenstein integers a triangular lattice). The field corresponding to this number system is written $\mathbb{Q}(\sqrt{-3})$.

2. **Build one quantity and compute it two ways.** The authors construct a sum made from the coefficients of a "cubic theta series," which can be computed along two independent routes: one uses a "reflection formula," the other uses "Poisson summation." The result of the second route contains terms like $1/L(s, \eta)$, i.e. the reciprocal of an L-function.
3. **Contradiction.** If some L-function had a zero in $\text{Re } s > 7/8$, $1/L$ would blow up there, and the two computations could not agree. Think of it as weighing the same object on two independent scales: if an assumption would make the two scales read differently, the assumption is wrong.
4. **Transfer to ordinary Dirichlet L-functions.** Using a factorization formula $L_F(s, \chi \circ N) = L(s, \chi) \cdot L(s, \chi\chi_{-3})$, the conclusion over $\mathbb{Q}(\sqrt{-3})$ can be "split" back into L-functions over the ordinary integers, including ζ .

Technically there are two stages: first a simpler version gives $11/12$, then extra correction terms push it to $7/8$. The tools used include the sextic large sieve, Heath-Brown's estimates for cubic sums, and Dunn–Radziwiłł's 2024 Annals result on the Patterson conjecture.

9. How far it has been verified

9.1 The Lean statement: one line, and you can read it yourself

```
theorem riemannZeta_ne_zero_of_seven_eighths_lt_re {s : ℂ} (hs : (7/8 : ℝ) < s.re) :  
  riemannZeta s ≠ 0
```

Word by word:

- `theorem ...`: the name of the theorem, which literally means "ζ is nonzero when the real part is greater than 7/8."
- `{s : ℂ}`: s is a complex number (ℂ is the set of complex numbers).
- `(hs : (7/8 : ℝ) < s.re)`: the hypothesis hs is "the real part of s > 7/8."
- `riemannZeta s ≠ 0`: the conclusion is "ζ(s) is not equal to 0."

So this one line says: **for any complex number s, if its real part is greater than 7/8, then $\zeta(s) \neq 0$** . That is exactly the quasi-Riemann hypothesis.

The `riemannZeta` here is the **standard definition** that has long been maintained in Mathlib (Lean's community mathematics library), not something OpenAI wrote itself. This matters: if the definition were home-made, it could quietly "change the subject."

9.2 The proof itself

- The proof is about 2,926 Lean files and 487,000 lines.
- We grepped it: no `sorry` ("leave this step blank for now"), no added axioms, no `native_decide` (a backdoor that trusts the compiler).
- Independent re-check: Dave Goldblatt used the Lean FRO's Comparator tool, plus an independently implemented checking kernel (nanoda), and got the proof to go through, in about 63 minutes.
- **Remaining risk:** a patch to one dependency library is about 40,000 lines, and in it two upstream `sorry`s were removed and replaced with proofs. In other words, part of the so-called "dependency library" is OpenAI's own code. The Lean kernel checks these proofs, but if definitions were changed, that could affect what the statement means. Nobody has reviewed this line by line yet.

9.3 Human experts

- Alex Kontorovich (Rutgers; he himself started the Lean formalization project for the prime number theorem): "Quasi-RH?!?!?! Are you kidding me? If a human did this, it would be an instant Fields Medal..."
- **As of 2026-10-08, no human expert has publicly said they have read and verified the 199-page paper.**

9.4 How to think about "proved by machine, but not understood by people"

If you believe two things, (1) Mathlib's definition of ζ is correct and (2) the Lean kernel has no bugs, then this theorem has been proved in the machine sense, and it is far more trustworthy than a paper nobody has refereed.

But the mathematical community usually also asks for **understanding**: why is it true? Does the method generalize? This is what Tao means by "the end of Math 1.0": a conclusion can be confirmed by machine first, and understanding may only catch up much later.

10. If true, what it means (and what it does not mean)

10.1 For mathematics

1. **For the first time, the prime number theorem has an error bound with a power saving:**
 $\pi(x) - \text{Li}(x) = O(x^{7/8} \cdot (\ln x)^2)$. This is one of the biggest single steps forward in the theory of the distribution of primes in 125 years.
2. **Siegel zeros are ruled out:** a large batch of constants that "exist but can't be computed" become computable.
3. **Pushing from 1 to 7/8 is a qualitative change, but it is still far from 1/2.** The Riemann hypothesis remains open.

10.2 For computer science: square roots mod p

One of the paper's corollaries is an upper bound on the "least quadratic non-residue." First, what that means.

Quadratic residue: a number that "is the square of something" mod p . Take $p = 7$ as an example:

- $1^2 = 1$, $2^2 = 4$, $3^2 = 9 \equiv 2$, $4^2 = 16 \equiv 2$, $5^2 = 25 \equiv 4$, $6^2 = 36 \equiv 1 \pmod{7}$
- So the squares mod 7 are $\{1, 2, 4\}$; these are called quadratic residues. The rest, $\{3, 5, 6\}$, are called **quadratic non-residues**.
- The **least quadratic non-residue** is 3.

[Check 3] Which numbers are quadratic residues mod 23? What is the least quadratic non-residue? (Answer at the end of the chapter.)

Why it matters: to take a square root mod p (given a , find x with $x^2 \equiv a$), there is a classical algorithm (Tonelli–Shanks) that first needs to find **any one** quadratic non-residue. Pick a number at random and there is a one-in-two chance it is a non-residue, so the randomized algorithm is fast. But a **deterministic** algorithm (no dice) has to guarantee that "if you try numbers upward from 2, you will hit a non-residue quickly," and for that you need to know how small the least non-residue is.

The paper derives: least quadratic non-residue $\leq C \cdot (\ln p)^{32}$. That is, trying upward from 2, you are guaranteed to find one within on the order of $(\ln p)^{32}$ tries, so taking square roots gets an **unconditional** deterministic polynomial-time algorithm. Previously this conclusion required assuming the "generalized Riemann hypothesis."

No impact on practical cryptography:

- In practice people have always used the randomized algorithm, which is already fast.
- A deterministic polynomial-time algorithm for primality testing has existed since 2002 (AKS).
- The security of RSA and elliptic-curve cryptography rests on "factoring large numbers is hard" and "discrete logarithms are hard," which have nothing to do with this result.

10.3 For AI

This is one of the strongest cases of "AI gets a result on an open problem that humans agree is hard-core, and the result is machine-verified." It also creates a new situation: **a 199-page paper + 480,000 lines of Lean**, where trust rests mainly on "are the kernel and the definitions correct," not on "has anyone understood it." Human reviewing bandwidth has become the new bottleneck.

11. Self-test

Try to answer these after reading. If you can't, go back and reread the corresponding section.

1. In one sentence: why does the "real part" of the zeros affect the error in counting primes? (Section 3)
2. What is the fundamental difference in shape between de la Vallée Poussin's zero-free region and the quasi-Riemann hypothesis's zero-free region? Why does that difference matter? (Sections 6.2–6.3)
3. Why can't zero-density estimates, however good, replace a zero-free region? (Section 6.4)
4. What is a Siegel zero? Why does the quasi-Riemann hypothesis rule it out? (Section 7.3)
5. In the Lean statement, what does the hypothesis `hs` in front of `riemannZeta s ≠ 0` say? (Section 9.1)
6. Why does this result not affect the security of RSA? (Section 10.2)

Appendix: answers to the checks

- **Check 1:** about $1000/6.9 \approx 145$ primes per 1000 numbers. In reality there are 168 primes below 1000, so the order of magnitude matches.
- **Check 2:** $\cos 2\theta = 2\cos^2\theta - 1$; substituting gives $3 + 4\cos \theta + 2\cos^2\theta - 1 = 2\cos^2\theta + 4\cos \theta + 2 = 2(\cos \theta + 1)^2 \geq 0$.
- **Check 3:** the quadratic residues mod 23 are $\{1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18\}$ (11 of them, exactly half of 22), and the least quadratic non-residue is **5** (checked by script).

Appendix: Glossary

Term	One-line explanation
Prime number theorem	The number of primes up to x is about $\text{Li}(x)$; the ratio tends to 1
Power saving	The error is smaller than x by a fixed power, e.g. $x^{7/8}$
Complex number	A point in the plane; multiplication = scaling + rotation
ζ function	$1/1^s + 1/2^s + \dots$, connected to the primes through the Euler product
Analytic continuation	Uniquely extending a function defined only on part of a region to a larger region
Zero	A point where the function's value is 0
Critical strip	The vertical strip $0 < \text{Re } s < 1$, where all non-trivial zeros lie
Riemann hypothesis	Every non-trivial zero has real part $1/2$
Zero-free region	A region proved to contain no zeros
Quasi-Riemann hypothesis	There is a fixed $\theta < 1$ such that there are no zeros where $\text{Re } s > \theta$
Dirichlet L-function	A variant of ζ used to study primes in arithmetic progressions
Siegel zero	A hypothetical real zero of certain L-functions, very close to 1
Quadratic residue	A number that is the square of something mod p

Appendix: Further reading

- **3Blue1Brown**, *Visualizing the Riemann zeta function and analytic continuation* (YouTube): an animated version of Section 5.3, "analytic continuation." Highly recommended.
- **3Blue1Brown**, *But what is the Fourier Transform?: the intuition behind Section 3*, "breaking a signal into waves."
- **Barry Mazur & William Stein**, *Prime Numbers and the Riemann Hypothesis* (Cambridge University Press, 2016): a book written specifically for readers who are not math specialists; the first half needs only high-school math.
- Original paper: <https://github.com/openai/math/blob/main/preprints/The-Quasi-Riemann-Hypothesis-September-30-2026/paper.pdf>

PART III · EXPLAINED FROM ZERO · GOES WITH §5

24. Explainer 02: The irrationality exponent of π , $\mu(\pi) = 2$

This is the "explained from zero" version of §5 of the [notes/openai_math_explained/](#) deck. All you need going in: you can work with fractions, powers and logarithms, and you

know what it means for an infinite sum to "converge." Every new concept is explained before it is used. Every number in this chapter was either computed by script (Python + mpmath, with π to 3000 digits of precision) or has a source. The **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Estimated reading time: 60–80 minutes. You can read it in three sittings: Sections 1–3 (approximating by fractions, continued fractions, Dirichlet), Sections 4–6 (the irrationality exponent, the history for π , the Flint Hills series), Sections 7–9 (OpenAI's result, its verification and what it means).

0. The bottom line: the one thing this chapter explains

π is irrational: it cannot be written as a fraction. But it can be **approximated** by fractions: $3, 22/7, 355/113, \dots$ The bigger the denominator, the more accurate the approximation, which is nothing special. The real question is:

Relative to the size of the denominator, how "well" can π be approximated?

This chapter explains one chain:

Any irrational number can be approximated to within "error $\approx 1/q^2$ " (Dirichlet; everyone gets this) $\Rightarrow$ the question becomes "can you do **much better than $1/q^2$** , and do so infinitely often" $\Rightarrow$ a number μ (the irrationality exponent) measures this capacity for "exceptional approximation" $\Rightarrow$ almost every real number has $\mu = 2$, but for the **specific** number π , for 70 years all anyone could prove was $\mu \leq 7.1$.

OpenAI claims to have proved $\mu(\pi) = 2$. That is: in this respect π is as ordinary as "a real number picked at random"; there are not infinitely many "absurdly good" fractional approximations. As a side effect, it settles a popular puzzle: the Flint Hills series $\sum 1/(n^3 \sin^2 n)$ converges.

After reading this chapter, you should be able to explain in your own words:

- how to measure precisely "how well a number can be approximated by fractions," and why the baseline is $1/q^2$;
- why "almost every number has $\mu = 2$ " is easy to prove, while " π has $\mu = 2$ " is extremely hard;
- what playbook OpenAI's proof uses (in one sentence: "a nonzero integer is at least 1"), and how far it has been verified;
- what this result does **not** mean (it is not transcendence, not "normality," not "bounded continued fraction").

1. Approximating π by fractions: from 3 to 355/113

1.1 A table

$\pi = 3.14159265358979\dots$. Some historically famous fractions:

Fraction	Origin	Error $ \pi - p/q $	$1/q^2$	$q^2 \times \text{error}$
3/1	Antiquity	1.42×10^{-1}	1	0.142
314/100	Decimal truncation	1.59×10^{-3}	1×10^{-4}	15.9
22/7	Archimedes' "yuelü" ("approximate ratio")	1.26×10^{-3}	2.04×10^{-2}	0.062
333/106		8.32×10^{-5}	8.90×10^{-5}	0.935
355/113	Zu Chongzhi's "milü" ("close ratio," 5th century)	2.67×10^{-7}	7.83×10^{-5}	0.0034
103993/33102		5.78×10^{-10}	9.13×10^{-10}	0.633

(Computed by script.)

1.2 Why divide by the "cost of the denominator"

A smaller error is not remarkable in itself: if you allow denominators up to 10^6 , you can just take 3141593/1000000 and push the error down to around 10^{-7} . So looking only at the error is unfair; you have to look at the error **relative to the size of the denominator**.

Compare 314/100 and 22/7: their errors are about the same (both around 10^{-3}), but 22/7 uses only the denominator 7, while 314/100 uses the denominator 100. 22/7 is much better "value for money."

A natural measure: **the error is one over what power of the denominator?** If $|\pi - p/q| \approx q^{-v}$, we say this approximation has "effective exponent" v . The bigger v , the more "exceptional" the approximation.

- 314/100: error $\approx 100^{-1.4}$, $v \approx 1.4$. Decimal truncation only reaches about the $v \approx 1$ level (error $\sim 1/q$).
- 22/7: $v \approx 3.43$ (computed by script).
- 355/113: $v \approx 3.20$ (computed by script). This is what is magical about the "milü": a three-digit denominator gives accuracy of about 10^{-7} (the first 6 decimal places are all correct).

The last column of the table, $q^2 \times \text{error}$, is another angle: if it is < 1 , the error is smaller than $1/q^2$. For 355/113 this column is 0.0034, i.e. about **290 times** better than $1/q^2$.

[Check 1] Verify with a calculator: $355/113 = 3.14159292\dots$; how far is it from π ? What is 113^2 ? Roughly what is the error times 113^2 ?

2. Continued fractions: a machine for finding "best approximations"

The fractions 3, 22/7, 333/106, 355/113 in the table were not cobbled together; they are the output of a machine. The machine is called a **continued fraction**.

2.1 The algorithm: take the integer part, take the reciprocal, repeat

For a number x :

1. Write down the integer part a_0 ; what remains is the fractional part r .
2. Take the reciprocal $1/r$, write down its integer part a_1 , and keep the new fractional part.
3. Repeat.

The first two steps for π by hand:

- $\pi = 3 + 0.14159\dots$, so $a_0 = 3$.
- $1/0.14159\dots = 7.0625\dots$, so $a_1 = 7$.
- $1/0.0625\dots = 15.996\dots$, so $a_2 = 15$.

Continuing (computed by script):

```
 $\pi = [3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, \dots]$ 
```

This means $\pi = 3 + 1/(7 + 1/(15 + 1/(1 + 1/(292 + \dots))))$.

2.2 Truncation = good fractions

Cut it off at some position and you get a fraction, called a **convergent**:

- Cut at [3] $\rightarrow 3$
- Cut at [3; 7] $\rightarrow 3 + 1/7 = 22/7$
- Cut at [3; 7, 15] $\rightarrow 333/106$
- Cut at [3; 7, 15, 1] $\rightarrow 355/113$
- Cut at [3; 7, 15, 1, 292] $\rightarrow 103993/33102$

A classical theorem (Lagrange and others, 18th century): convergents are "best approximations." Among all fractions with denominator at most q , none is closer to x .

2.3 The bigger the next number, the more accurate the current fraction

This is the most useful rule of thumb in the chapter:

The error of a convergent p/q is about $1 / (a_{\text{next}} \times q^2)$, where a_{next} is the number in the continued fraction that **comes right after the cut** (called a "partial quotient").

- 22/7 is followed by 15: error $\approx 1/(15 \times 49) \approx 1.4 \times 10^{-3}$; actual 1.26×10^{-3} .
- 355/113 is followed by **292**: error $\approx 1/(292 \times 12769) \approx 2.7 \times 10^{-7}$; actual 2.67×10^{-7} .
- 333/106 is followed by 1: error $\approx 1/(1 \times 106^2)$, the same order as $1/q^2$, not "exceptional" at all ($q^2 \times \text{error} = 0.935$).

So asking "can π be approximated exceptionally well" is equivalent to asking "do especially huge partial quotients show up in π 's continued fraction." 292 is one. Will there be even more outrageous ones later?

2.4 Control group: some "well-behaved-looking" numbers

Number	Continued fraction (computed by script)	Pattern
Golden ratio $\phi = (1+\sqrt{5})/2$	[1; 1, 1, 1, 1, ...]	All 1s; the hardest number to approximate
$\sqrt{2}$	[1; 2, 2, 2, 2, ...]	Always 2
e	[2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, ...]	1, 1, 2k pattern; grows slowly
π	[3; 7, 15, 1, 292, 1, 1, 1, 2, ...]	No known pattern

For patterned numbers like $\sqrt{2}$ and e, the size of the partial quotients is obvious at a glance, so "how well it can be approximated" can be computed directly. π 's continued fraction has **no known pattern at all**, and that is the root of the difficulty: you can't just stare at its continued fraction and prove "a super-large number will never appear later."

[Check 2] Compute the continued fraction of 355/113 by hand. (Hint: $355 = 3 \times 113 + 16$; then continue with 113/16.)

3. Dirichlet's theorem: exponent 2 comes free for everyone

3.1 The statement

Dirichlet's approximation theorem (1842): for any irrational number x , there are infinitely many fractions p/q with $|x - p/q| < 1/q^2$.

In other words, effective exponent 2 is "free": every irrational number can achieve it. That is why the baseline is $1/q^2$ and not something else.

3.2 Proof: the pigeonhole principle, doable by hand

Take an integer Q , say $Q = 10$. Look at the **fractional parts** of the 11 numbers $0 \cdot \pi, 1 \cdot \pi, 2 \cdot \pi, \dots, 10 \cdot \pi$ (computed by script):

q	fractional part of $q\pi$
0	0.000
1	0.142
2	0.283
3	0.425
4	0.566
5	0.708
6	0.850
7	0.991
8	0.133
9	0.274
10	0.416

Cut $[0, 1)$ into 10 drawers of length 0.1. Put 11 numbers into 10 drawers and **at least two land in the same drawer**. From the table: $q = 1$ (0.142) and $q = 8$ (0.133) are both in $[0.1, 0.2)$.

Subtract: $8\pi - 1\pi = 7\pi$, and the fractional parts differ by only 0.009. This says 7π is very close to an integer: $7\pi \approx 21.991$, only 0.0089 away from 22. So

$$|7\pi - 22| < 1/10 \quad \Rightarrow \quad |\pi - 22/7| < 1/(7 \times 10) \leq 1/7^2$$

In general, with Q drawers and $Q+1$ numbers, you can always find $q \leq Q$ with $|qx - p| < 1/Q$, so $|x - p/q| < 1/(qQ) \leq 1/q^2$. Letting Q grow larger and larger gives infinitely many such fractions. QED.

[Check 3] Find another pair of q values in the table that land in the same drawer. What is their difference? Which fraction does it give?

3.3 Can you do better than $1/q^2$?

Computed by script: among all reduced fractions with denominators $2 \leq q \leq 100,000$, count how many satisfy $|\pi - p/q| < q^{-(\nu)}$:

v	Number of fractions that satisfy it	Their denominators
2	8	6, 7, 106, 113, 33102, 33215, 66317, 99532
2.2	2	7, 113
2.5	2	7, 113
3	2	7, 113

At $v = 2$ the count keeps growing (Dirichlet guarantees infinitely many). Just above 2 a few remain (computed by script: denominators 6, 106, 33102, 33215, 66317 and 99532 have effective exponents between 2.01 and 2.11); once $v \geq 2.2$, within 100,000 only two "early miracles" remain: 22/7 and 355/113.

$\mu(\pi) = 2$ says exactly this: for any $v > 2$ (even 2.0001), there are only **finitely many** such fractions. There can be a few miracles early on, but miracles cannot keep appearing forever.

4. The irrationality exponent μ : definition and a "zoo"

4.1 Definition

Irrationality exponent (also called the irrationality measure) $\mu(x)$ = the supremum of all v such that " $|x - p/q| < q^{-(v)}$ has infinitely many solutions."

In plain words: $\mu(x)$ is the highest exponent at which x can be "exceptionally approximated infinitely many times." The bigger μ is, the easier it is to approximate x very well by fractions, and in that sense the "closer to rational" x is.

Continued-fraction version (a standard result): $\mu(x) = 2 + \limsup (\ln a_{n+1} / \ln q_n)$. That is, **$\mu = 2$ if and only if the partial quotients a_{n+1} grow more slowly than any positive power of q_n .**

An example computed by script: among the first 1000 terms of π 's continued fraction, the largest partial quotient is 20776 (term 431), and the denominator of the convergent just before it has 216 digits. The effective exponent of this approximation is only **2.020**. With a denominator that big, even a partial quotient of more than 20,000 contributes almost nothing to the exponent. By contrast, 355/113 has effective exponent 3.20; that was "luck while the denominator was small."

4.2 The zoo

Number	μ	Who proved it
Rational numbers	1	Trivial
Any irrational number	≥ 2	Dirichlet 1842 (Section 3)
Quadratic irrationals such as $\sqrt{2}$	2	Liouville-type argument, see 4.3
Any algebraic irrational (e.g. $\sqrt[3]{2}$)	2	Roth 1955 , Fields Medal 1958
e	2	Computed directly from its patterned continued fraction
Almost every real number (in the Lebesgue-measure sense)	2	Khinchin (1920s), see 4.5
Liouville number $\sum 10^{-(k!)}$	∞	Liouville, mid-19th century
π	Known ≤ 7.10 ; OpenAI claims = 2	See Sections 5 and 7

4.3 A proof you can do by hand: why $\sqrt{2}$ can't be approximated exceptionally well

This small proof contains the core playbook of the whole chapter (including OpenAI's proof), so it is worth reading slowly.

For any fraction p/q , look at the integer $p^2 - 2q^2$:

- It is **not 0** (because $\sqrt{2}$ is irrational).
- It is an **integer**.
- So $|p^2 - 2q^2| \geq 1$.

On the other hand, $p^2 - 2q^2 = q^2 (p/q - \sqrt{2})(p/q + \sqrt{2})$. So

$$|\sqrt{2} - p/q| = |p^2 - 2q^2| / (q^2 \cdot (p/q + \sqrt{2})) \geq 1 / (q^2 \cdot (p/q + \sqrt{2})) \approx 1/(2.83 q^2)$$

The error is never smaller than about $0.35/q^2$, so the effective exponent can't exceed 2, and $\mu(\sqrt{2}) = 2$.

By hand for $p/q = 7/5$: $p^2 - 2q^2 = 49 - 50 = -1$, absolute value exactly 1. Predicted error = $1/(25 \times (1.4 + 1.41421)) = 0.01421$; actual $|1.41421 - 1.4| = 0.01421$. (The script also checked that for the convergents of $\sqrt{2}$, $q^2 \times \text{error}$ stays steady at around 0.35.)

The core playbook:

Build a quantity tied to the approximation. **Arithmetically**: it is a nonzero integer, so it is ≥ 1 . **Analytically**: if the approximation were too good, it would be < 1 . The two contradict each other.

Remember this sentence; you will meet it again in Section 7.

4.4 Liouville numbers: the monsters with $\mu = \infty$

$L = \sum 10^{-k!} = 0.1100010000000000000000001000\dots$ (the 1s are at decimal places 1, 2, 6, 24, 120, ...).

Cut it off after one of the 1s, and the resulting fraction has a tiny error (computed by script):

- Cut at place 2: $q = 10^2$, error $\approx 10^{-6} = q^{-3}$
- Cut at place 6: $q = 10^6$, error $\approx 10^{-24} = q^{-4}$
- Cut at place 24: $q = 10^{24}$, error $\approx 10^{-120} = q^{-5}$

The effective exponents 3, 4, 5, ... grow without bound, so $\mu(L) = \infty$. Liouville used numbers of this kind to construct the first numbers in history that were **proved to be transcendental** (since algebraic numbers have finite μ , a number with $\mu = \infty$ cannot be algebraic).

This point is crucial for understanding the difficulty: Liouville numbers are transcendental, with $\mu = \infty$. π is also transcendental (Lindemann 1882). So **the fact that " π is transcendental" is of no help at all for $\mu(\pi)$:** any argument that uses only " π is transcendental" would be refuted by Liouville numbers. To prove $\mu(\pi) = 2$, you have to use structure that only π has.

4.5 Why "almost every number has $\mu = 2$ " is easy to prove

Throw a random point x into $[0, 1]$. For a fixed q , the region where " x is within q^{-v} of some p/q " has total length about q small intervals $\times$ length $2q^{-v}$ each $= 2q^{1-v}$.

Add this up over all q : $\sum 2q^{1-v}$. When $v > 2$, the exponent $1 - v < -1$, and the sum **converges** (it is just 2 times $\zeta(v-1)$ from Explainer 01; $\zeta(s)$ is finite for $s > 1$). Probability theory has a result called the Borel–Cantelli lemma: if the probabilities of a sequence of events have a finite sum, then almost surely only finitely many of the events happen. So almost every x has only finitely many exceptional approximations with exponent v , and $\mu(x) \leq v$. v can be taken arbitrarily close to 2, so almost every x has $\mu = 2$.

This argument cannot be applied to π at all: it is about "a random number," and π is one specific number. It's like how "almost nobody is left-handed" can't tell you whether a particular person is left-handed. This is the typical shape of problems like this: **typical behavior is obvious at a glance; a specific instance is extremely hard to prove** (" π is normal" has the same shape, and is still unsolved).

5. The history of upper bounds for π : 70 years from 42 to 7.1

5.1 Timeline

Year	Author	Proved upper bound on $\mu(\pi)$	Source
1953	Mahler	42 (and said it could eventually be brought to 30)	<i>On the approximation of π</i> , Indag. Math. 15
1974	Mignotte	21 (all denominators) / 20 (eventually)	Bull. SMF Mém. 37
1993	Hata	8.01604539...	Acta Arith. 63
2008	Salikhov	7.606308...	Russ. Math. Surveys 63(3)
2020	Zeilberger–Zudilin	7.103205334137...	Moscow J. Comb. Number Theory 9(4); arXiv:1912.06345
2026-09	Bai (preprint)	7.101862832357	arXiv:2609.11276 $\triangle$ not peer-reviewed
2026-09-24	OpenAI (claimed)	= 2	Section 7 of this chapter

(Values in the table follow the survey in §1.1 of the OpenAI paper, cross-checked against the deck author's research notes.)

Note the size of the last jump: the first 70 years went $42 \rightarrow 7.10$, each step a large amount of delicate work; then a single jump to 2, and it is an **exact value**, not yet another upper bound.

5.2 The ceiling of the old methods

All previous upper bounds came from the same framework (Zudilin's survey arXiv:math/0404523 explains it clearly):

1. Build a sequence of "linear forms" with integer coefficients, $a_n\pi - b_n$, that are **very small but not 0**. Usually these are built from carefully designed integrals.
2. Two quantities pull against each other:
 - how fast the linear form decays (faster is better);
 - how fast the coefficients a_n grow (slower is better). To make the coefficients integers, you usually have to multiply by a large number like $\text{lcm}(1, 2, \dots, n)$ to "clear denominators," and this cost is high.
3. A standard lemma: if the decay rate is $e^{(-\beta n)}$ and the coefficient growth is $e^{(\alpha n)}$, you get $\mu \leq 1 + \alpha/\beta$.

To get $\mu = 2$ you need α and β to be almost equal, i.e. the "cost of clearing denominators" has to be almost zero. For every known integral construction for π , this ratio is far from 1. The Zeilberger–Zudilin generation of work was tightening screws inside this framework (optimizing the parameters of the integrals), and each time could only move the bound a little.

5.3 Why Roth's method can't be used directly

Roth's theorem gives $\mu = 2$ for all algebraic numbers, and its method is **multivariable**: take several approximations $p_1/q_1, p_2/q_2, \dots$ at once, with wildly different denominators, build a multivariable polynomial, and derive a contradiction using the "nonzero integer ≥ 1 " playbook from Section 4.3.

But Roth's proof fundamentally uses the fact that x satisfies a polynomial equation with integer coefficients (x is algebraic). π is not algebraic, so there is no such equation to use. And as Section 4.4 said, you also can't use only " π is transcendental." OpenAI's reasoning summary happens to record the moment the model ran into this wall (verbatim):

"Wait this would work for any transcend, contradiction with Liouville. Examine."

Meaning: it noticed that one of its arguments, if valid, would work for **all transcendental numbers**, but Liouville numbers are a counterexample, so the argument must have a hole.

6. A concrete consequence: the Flint Hills series

6.1 What the series is

$$\sum 1/(n^3 \sin^2 n) = 1/(1^3 \sin^2 1) + 1/(2^3 \sin^2 2) + 1/(3^3 \sin^2 3) + \dots \quad (n \text{ in radians})$$

The question: does it converge? Pickover popularized the problem, Alekseyev 2011 (arXiv:1104.5100) discussed it systematically, and it remained unsolved.

6.2 Why it is related to approximating π

The denominator contains $\sin n$. When an integer n is very close to some integer multiple $k\pi$ of π , $\sin n$ is very small, and that term blows up. And " $n \approx k\pi$ " means " $n/k \approx \pi$," i.e. **n/k is a good fractional approximation of π .**

Partial sums computed by script:

n	This term	Partial sum
1	1.412	1.412
3	1.860	3.423
21	0.00015	3.555
22 (22/7)	1.199	4.754
354	3.2×10^{-8}	4.807
355 (355/113)	24.60	29.406
1,000	1.5×10^{-9}	30.175
103,993 (103993/33102)	2.4×10^{-6}	30.3145
1,000,000	8.2×10^{-18}	30.3145

The whole curve is a staircase: most of the time it barely moves, and every time n hits the numerator of an exceptionally good approximation of π , it jumps. The $n = 355$ term alone contributes 24.6, which is 81% of the sum of the first 1 million terms.

[Check 4] Explain why the $n = 355$ term is about 24.6. (Hint: $\sin(355) = \sin(355 - 113\pi)$, and $355 - 113\pi \approx 3.01 \times 10^{-5}$; the sine of a very small angle $\approx$ the angle itself.)

6.3 Whether it converges is decided entirely by $\mu(\pi)$

- If π has infinitely many "too good" approximations, there will be infinitely many large enough jumps, and the series diverges.
- If exceptional approximations are rare enough, the jumps add up to something finite, and the series converges.

Two published criteria tie this precisely to $\mu(\pi)$:

- Alekseyev 2011: if the series converges, then $\mu(\pi) \leq 5/2$.
- Meiburg 2022 (arXiv:2208.13356): if $\mu(\pi) < 5/2$, then the series converges.

So you only need to prove $\mu(\pi) < 2.5$ to settle the problem. But the best known bound was 7.10, far away. OpenAI's reasoning summary shows that the model's initial goal was exactly $\mu < 5/2$, and only later did it find it could push all the way to 2 (see Section 7.3).

§5 of the paper also gives its own short proof (group the integers n by the nearest $q\pi$, and control each group with $|\sin n| \geq (2/\pi) \cdot |n - q\pi|$), and derives a more general result: $\sum 1/(n^a |\sin n|^b)$ converges if and only if $a > \max\{1, b\}$.

7. What OpenAI claims

7.1 The main result

Abstract of the paper *The irrationality exponent of π is 2* (OpenAI, 2026-09-24, 23 pages):

"We prove the conjecture that the irrationality exponent of π is 2. As a consequence, the classical Flint–Hills series $\sum_{n \geq 1} 1/(n^3 \sin^2 n)$ converges, with angles in radians."

The precise statement of the main theorem (Theorem 1.1):

For every real number $v > 2$, there exists an integer $Q(v)$ such that for all integers p and all $q \geq Q(v)$, $|\pi - p/q| \geq q^{-v}$.

The paper itself spells out three points (all important):

1. p/q is not required to be in lowest terms.
2. **$Q(v)$ is not effectively computable:** "the argument does not give it effectively". That is, it proves "from some denominator on, there are no more exceptional approximations with exponent v ," but does not tell you where that starts.
3. **$\mu = 2$ is not the same as "bounded partial quotients":** "an exponent of 2 is different from a uniform positive lower bound of the form c/q^2 , or equivalently from bounded continued-fraction partial quotients". The latter (whether the partial quotients of π 's continued fraction are bounded) is still unsolved.

The paper also records other claims in the same area: Mantzakouras and López Zapata's Flint Hills convergence claim was withdrawn in 2026-09; Carella claims $\mu(\pi) = 2$ and the stronger "bounded partial quotients," and the OpenAI paper points out in a footnote that his proof of the stronger claim has a sign error (this is OpenAI's side of the story $\triangle$, with no third-party confirmation).

The paper says that Waldschmidt recorded this statement as a conjecture in *Open Diophantine problems* (Moscow Math. J. 4(1), 2004, p. 265).

7.2 Intuition for the method (we only read the introduction and §5; this is a simplified version)

Recall the playbook from Section 4.3: "a nonzero integer is ≥ 1 , but an approximation that is too good would make it < 1 ." OpenAI's proof is a very heavy-duty version of this playbook.

1. **Proof by contradiction.** Fix some $v > 2$, and assume π has infinitely many exceptional approximations with exponent v .
2. **Pick a few, with widely separated denominators.** From these infinitely many, pick m of them, each denominator much bigger than the last: $q_1 \ll q_2 \ll \dots \ll q_m$. This step is the same as Roth (multivariable, separated denominators).
3. **Use structure only π has: it is a period of the exponential function.** $e^{2\pi i} = 1$, so $2\pi i \cdot j$ is a "period" of the exponential function, and also the source of the multivaluedness of the

logarithm. The authors turn the exceptional approximations p_i/q_i into $2i \cdot j \cdot p_i/q_i$, which approximate the true periods $2\pi i \cdot j$. This step answers the question from Section 4.4: the proof does not use " π is transcendental," but "this specific relationship between π and $\exp/\log$," which Liouville numbers do not have.

4. **Build a big determinant.** Take a multivariable polynomial $P(Y, X_1, \dots, X_m)$, expand it along the "logarithmic curve" $Y = 1 + t$, $X_i = c_i + \log(1 + t)$, take the Taylor coefficients, and arrange them into a big matrix. A new theorem in Section 2 of the paper ("separated-weight interpolation," using tools from algebraic geometry such as blowups and ampleness) guarantees that this matrix has full rank, so a **nonzero** determinant can be picked out.
5. **Arithmetic lower bound.** The matrix entries are all in $\mathbb{Q}(i)$ (of the form $a + bi$, with a and b fractions). After multiplying out the denominators, the determinant is a nonzero "Gaussian integer," with absolute value ≥ 1 .
6. **Analytic upper bound.** Swap the approximations $2i \cdot j \cdot p_i/q_i$ back for the true periods $2\pi i \cdot j$, and many rows become Taylor coefficients of the same functions, nearly dependent on one another, so the determinant cancels heavily (the paper calls this "quadratic saving"); the remaining terms carry the tiny approximation errors $|\pi - p_i/q_i|$ as factors. Result: the determinant has absolute value < 1 .
7. **Contradiction.** ≥ 1 and < 1 at the same time is impossible, so the assumption is false, and there are only finitely many exceptional approximations with exponent ν .

Why it reaches any $\nu > 2$: the dimension m (how many approximations you pick) can grow with ν . The higher the dimension, the faster the "cancellation saving" grows, until it eventually beats the cost of any $\nu > 2$. Why $Q(\nu)$ can't be computed: the argument starts from "assume there are infinitely many," and the approximations it picks have no concrete location. This is the same source as the "ineffectivity" of Roth's theorem.

7.3 The reasoning summary: how the AI got here

OpenAI released a 42-page "abridged chain-of-thought summary"

([reasoning_traces/irrationality-exponent-of-pi.pdf](#)). The script read its full table of contents; the structure is:

- **Part I (sections 1–44):** the goal is only the $\mu < 5/2$ needed for Flint Hills. The model tried dozens of routes, including the BBP formula, modular forms, Padé approximation, the Chudnovsky series, p -adic logarithms and elliptic functions, and kept running into the wall it repeatedly called "height vs denominator cost" (height barrier), i.e. the "cost of clearing denominators" from Section 5.2. Along the way it repeatedly drew on the determinant idea from OpenAI's own Catalan's-constant manuscript. In the end it used a weighted interpolation determinant to get $\mu \leq 62/25 = 2.48$, and hence Flint Hills convergence.

Almost all the titles of sections 32–44 are "audit" or "stress-testing," i.e. repeated self-checking of this step.

- **Part II (section 45):** "takes the earlier report's interpolation-and-determinant technique as a starting point and claims exponent two, explicitly without assuming the earlier approximation bound". On the same technique, it tuned parameters and raised the dimension, pushing to $\mu = 2$.

This document is itself a good case study of the research process: the goal was set conservatively (2.5), and the audit revealed that the technique's actual limit was 2.

8. How far it has been verified

8.1 The Lean statement: you can read it yourself

The "challenge statement" file in the repo,

`lean/ComparatorChallenges/PiExponent.lean` (it imports only Mathlib):

```
theorem main :
  (∀ v : ℝ, 2 < v → ∃ Q : ℤ, 2 ≤ Q ∧
    ∀ p q : ℤ, Q ≤ q →
      (q : ℝ) ^ (-v) ≤ |Real.pi - (p : ℝ) / (q : ℝ)|) ∧
  sSup {v : ℝ | 0 < v ∧
    Set.Infinite {r : ℚ | 2 ≤ r.den ∧
      0 < |Real.pi - (r : ℝ)| ∧
      |Real.pi - (r : ℝ)| < (r.den : ℝ) ^ (-v)}} = 2
```

Piece by piece:

- **First part** (before the `∧`):
 - `∀ v : ℝ, 2 < v →`: for any real number v greater than 2,
 - `∃ Q : ℤ, 2 ≤ Q ∧`: there exists an integer threshold $Q \geq 2$,
 - `∀ p q : ℤ, Q ≤ q →`: such that for all integers p and all integers $q \geq Q$,
 - `(q : ℝ) ^ (-v) ≤ |Real.pi - p/q|`: the distance between π and p/q is at least q^{-v} .
 - This is exactly Theorem 1.1. Note that it only says "Q exists"; it does not give Q , consistent with the paper's "not effective."
- **Second part** (after the `∧`):
 - `{r : ℚ | 2 ≤ r.den ∧ 0 < |π - r| ∧ |π - r| < r.den^(-v)}`: the set of all rationals with denominator ≥ 2 that are exceptional approximations with exponent v (Lean's $\mathbb{Q}$ is automatically in lowest terms; `r.den` is the denominator).

- `Set.Infinite ...`: this set is infinite.
- `sSup {v | 0 < v ∧ ...} = 2`: the supremum of all v that "have infinitely many solutions" is 2. This is the definition of μ from Section 4.1, matched word for word.

Two details worth noting:

- The statement **uses only Mathlib's standard definitions** (`Real.pi`, `sSup`, `Set.Infinite`, `ℚ`), and no concepts defined by OpenAI. This is much more trustworthy than "writing the statement with custom definitions."
- Lean's `sSup` has a "junk value" convention: for an empty set or a set with no upper bound, it returns 0. Here the conclusion is $= 2$, not $= 0$, so it does not fall into this trap.

8.2 The proof itself (counted by script on a local sparse clone, snapshot fd4aeeb)

- Solution directory `lean/OAI/NumberTheory/PiExponent/`: **869 .lean files, 94,948 lines**.
- `grep`: 0 occurrences of `sorry`, 0 of `native_decide`, 0 custom `axiom`s.
- The comparator config `PiExponent.json` allows only the three standard axioms (`propext`, `Quot.sound`, `Classical.choice`).
- The solution imports only Mathlib and files in its own directory, and **does not depend on any patched external library in `lean/patches/`** (the script checked every import). Compare Explainer 01: the proof of the quasi-Riemann hypothesis depends on a patch library of about 40,000 lines, which is one of its main remaining risks; the π proof does not have this problem.
- The entry file `Main.lean` also contains a `flint_hills_summable`: it uses Mathlib's `Summable` to state that $\sum 1/(n^3 \sin^2 n)$ converges, and gives a proof. But the official scope note `lean/docs/017.md` says Flint Hills is "outside this selected statement," meaning it is **not in the statement the comparator checks**. The accurate way to put it: a Lean proof of Flint Hills exists in the repo, but it has not gone through the same "statement-proof" comparison process as the main theorem.

8.3 What has not been done (△)

- △ This formalization is **not registered** in `lean/formalization.yaml` (we grepped for `PiExponent` and `017` and found neither). It may have been missed in the 10-07 update.
- △ We **did not compile and run** the comparator ourselves, and we **found no third-party re-check** (Dave Goldblatt's independent check in Explainer 01 covered only the ζ proof).
- On 2026-10-07, an issue appeared in Google DeepMind's formal-conjectures repository (#6909, by mo271) proposing to mark its `flint_hills_series_converges` as solved, but the issue itself notes it was "prepared with AI assistance and may contain mistakes" and asks people to check it first. When we looked, it had no comments.

- **⚠ As of 2026-10-08, we found no named expert in Diophantine approximation / transcendental number theory publicly commenting on the correctness of this paper.**
John D. Cook's 2026-10-07 blog post is only a popular explainer and does not assess the proof.

8.4 How to think about it

Compared with the quasi-Riemann hypothesis, this one is cleaner on the "machine verification" dimension: the statement uses only standard Mathlib concepts and reads like the textbook definition; the proof does not depend on a patch library; and the size (about 95,000 lines) is much smaller. If you trust that Mathlib's definitions of π , real powers and suprema are sound, and that the Lean kernel has no bugs, then " $\mu(\pi) = 2$ " is very credible in the machine sense. The remaining gaps are procedural (nobody has run a third-party comparator) and about understanding (no human expert has publicly said they understand the proof).

9. If true, what it means (and what it does not mean)

9.1 For mathematics: what it means

1. **A formally recorded conjecture is solved:** π is "typical" in its approximation properties, like almost every real number. The 70-year race of upper bounds ($42 \rightarrow 7.10$) ends with an "exact value."
2. **The Flint Hills series converges**, along with a complete convergence criterion for the more general $\sum 1/(n^a |\sin n|^b)$.
3. **The method may matter more than the result.** If "separated-weight interpolation along the logarithmic curve + determinants" generalizes, the next targets would be the irrationality exponents of $\log 2$, $\zeta(3)$ and other logarithm values (section 43 of the reasoning summary mentions the model "considered an extension to algebraic logarithms," but this is a speculative direction with no conclusion ⚠).

9.2 For mathematics: what it does not mean

Commonly confused claim	Relation to $\mu(\pi) = 2$
π is irrational	Long known (Lambert 1761). $\mu = 2$ is far stronger than "irrational"
π is transcendental	Long known (Lindemann 1882). And transcendental numbers can have $\mu = \infty$ (Liouville numbers), so neither implies the other
The partial quotients of π 's continued fraction are bounded	Still unsolved. The paper says explicitly that $\mu = 2$ is not the same thing
π is normal (every digit string appears with the same frequency)	Still unsolved; a completely different problem
You can compute "from which denominator on there are no more exceptional approximations"	No; $Q(v)$ is not effectively computable
Good approximations like $22/7$ and $355/113$ don't exist	Of course they exist. $\mu = 2$ only says there are finitely many "miracles with exponent > 2 "

9.3 For computer science

Almost no direct impact.

- Floating-point trig functions need "argument reduction" (reducing a large x into the interval $[-\pi/4, \pi/4]$), and the worst case depends on how close some floating-point number gets to a multiple of $\pi/2$. But this is a **finite-range** problem (the range of a double is finite), and engineers long ago computed the worst cases directly with continued fractions (see J.-M. Muller, *Elementary Functions*, the chapter on argument reduction [△](#) check the original book for the exact chapter). An "asymptotic theorem that gives no threshold" will not change these concrete numbers.
- No connection to cryptography or complexity theory.

9.4 For AI

- This is a relatively **easy-to-check** case of "AI solves a classical conjecture formally recorded in the literature": the statement is short, uses only standard definitions, and the Lean proof does not depend on a patch library. If you want to show someone "which results in this batch are the most trustworthy," it is a better example than the quasi-Riemann hypothesis.
- The reasoning summary shows a process feature worth studying: a conservative goal (2.5) + a long stretch of failed exploration (40-plus sections) + extensive self-auditing, ending with the discovery of the technique's true limit. This is useful for studying the research process of AI itself (and for eval design).
- At the ecosystem level: in the same month there is Bai's 7.1018 preprint (the deck author's research notes record that Bai's acknowledgments mention help from OpenAI models [△](#) not checked against the original), a withdrawn Flint Hills claim, and an AI-drafted issue on

formal-conjectures. The number of "AI-assisted math claims" is surging, and human expert reviewing bandwidth is still the bottleneck, consistent with the conclusion of Explainer 01.

10. Self-test

Try to answer these after reading. If you can't, go back and reread the corresponding section.

1. Why can't you judge a fractional approximation by the size of its error alone? How is the "effective exponent ν " defined? (Section 1.2)
2. Why is $355/113$ especially accurate? Explain in the language of continued fractions. (Section 2.3)
3. Use the pigeonhole principle to explain why every irrational number has $\mu \geq 2$. (Section 3.2)
4. In the "nonzero integer ≥ 1 " playbook, what quantity plays that role in the $\sqrt{2}$ proof, and in OpenAI's proof? (Sections 4.3, 7.2)
5. Why doesn't " π is transcendental" help prove $\mu(\pi) = 2$? What structure of π does OpenAI's proof use? (Sections 4.4, 7.2)
6. Why are the partial sums of the Flint Hills series a staircase? Why is $\mu(\pi) < 5/2$ enough to make it converge? (Section 6)
7. Which words in the definition of μ does the second part of the Lean statement, `sSup {...} = 2`, correspond to? (Section 8.1)
8. Does $\mu(\pi) = 2$ imply "the partial quotients of π 's continued fraction are bounded"? Does it imply " π is normal"? (Section 9.2)

Appendix: answers to the checks

- **Check 1:** $355/113 = 3.14159292\dots$, which differs from π by about 2.67×10^{-7} ; $113^2 = 12769$; the product is about 0.0034, i.e. about 290 times better than $1/q^2$.
- **Check 2:** $355 = 3 \times 113 + 16$, so $355/113 = 3 + 16/113 = 3 + 1/(113/16)$; $113 = 7 \times 16 + 1$, so $113/16 = 7 + 1/16$. Putting it together, $355/113 = [3; 7, 16]$. (π itself is $[3; 7, 15, 1, \dots]$, and $[3; 7, 15, 1] = [3; 7, 16]$; the two forms are the same fraction.)
- **Check 3:** $q = 2$ (0.283) and $q = 9$ (0.274) are both in $[0.2, 0.3]$; the difference is 7, giving $22/7$ again. ($q = 3$ and $q = 10$ are both in $[0.4, 0.5]$, also with difference 7. With $Q = 10$ the pigeonhole principle can only "see" fractions with denominator ≤ 10 , and the best of those is $22/7$.)
- **Check 4:** $\sin(355) = \sin(355 - 113\pi) \approx 3.01 \times 10^{-5}$ (computed by script: $355 - 113\pi = 3.0144 \times 10^{-5}$). $\sin^2 \approx 9.09 \times 10^{-10}$; $355^3 \approx 4.47 \times 10^7$; the product ≈ 0.0407 , and its reciprocal ≈ 24.6 .

Appendix: Glossary

Term	One-line explanation
Effective exponent ν	If $ x - p/q \approx q^{(-\nu)}$, then ν is how "exceptional" this approximation is
Continued fraction	The expansion $[a_0; a_1, a_2, \dots]$ obtained by repeatedly "taking the integer part, taking the reciprocal"
Convergent	A fraction obtained by truncating a continued fraction; the best approximation among fractions with comparable denominators
Partial quotient	Each a_i in a continued fraction; the bigger it is, the more accurate the convergent just before it
Dirichlet's approximation theorem	Every irrational number has infinitely many approximations with $ x - p/q < 1/q^2$
Irrationality exponent $\mu(x)$	The largest ν at which x can be approximated "infinitely many times" to accuracy $q^{(-\nu)}$
Algebraic number	A root of some polynomial with integer coefficients, such as $\sqrt{2}$ or $\sqrt[3]{2}$
Transcendental number	A number that is not algebraic, such as π , e , or Liouville numbers
Roth's theorem	Every algebraic irrational has $\mu = 2$ (1955, Fields Medal)
Liouville number	A number with $\mu = \infty$, e.g. $\sum 10^{(-k!)}$
Linear form $a\pi - b$	The "very small but nonzero" quantity built by the old methods to put an upper bound on μ
Ineffective	A proof shows a constant exists but cannot compute its value
Flint Hills series	$\sum 1/(n^3 \sin^2 n)$; whether it converges is decided by $\mu(\pi)$
Bounded partial quotients (badly approximable)	$ x - p/q \geq c/q^2$ holds for all fractions; stronger than $\mu = 2$
Normal number	A number in whose expansion every digit string appears with the same frequency; unrelated to μ

Appendix: Further reading

- **Numberphile**, *The Golden Ratio (why it is so irrational)* (YouTube, Ben Sparks): an intuitive version of Section 2.4's "the golden ratio is the hardest number to approximate," which is about its continued fraction being all 1s.
- **Mathologer**, *Infinite fractions and the most irrational number* (YouTube): continued fractions and approximation, with clear animations.
- **A. Ya. Khinchin**, *Continued Fractions* (Dover reprint): a slim book, the classical source for Sections 2–4 and Section 4.5; readable with undergraduate math.
- **Wadim Zudilin**, *An essay on irrationality measures of π and other logarithms*, arXiv:math/0404523: the authoritative survey of Section 5.2's "linear forms framework."

- **Michel Waldschmidt**, *Open Diophantine problems*, Moscow Math. J. 4(1), 2004: <https://webusers.imj-prg.fr/~michel.waldschmidt/articles/pdf/odp.pdf> (p. 265 records this conjecture)
- Zeilberger–Zudilin, arXiv:1912.06345; Alekseyev, arXiv:1104.5100; Meiburg, arXiv:2208.13356.
- Original paper: <https://github.com/openai/math/blob/main/preprints/The-irrationality-exponent-of-pi-is-2-September-24-2026/paper.pdf>
- Reasoning summary: https://github.com/openai/math/blob/main/reasoning_traces/irrationality-exponent-of-pi.pdf
- Lean scope note: <https://github.com/openai/math/blob/main/lean/docs/017.md> ; challenge statement: <https://github.com/openai/math/blob/main/lean/ComparatorChallenges/PiExponent.lean>
- formal-conjectures issue #6909: <https://github.com/google-deepmind/formal-conjectures/issues/6909>
- John D. Cook popular explainer: <https://www.johndcook.com/blog/2026/10/07/irrationality-exponent-of-pi/>

PART III · EXPLAINED FROM ZERO · GOES WITH §6

25. Explainer 03: The matrix multiplication exponent $\omega \leq 9/4$

This is the "explained from zero" version of §6 of the `notes/openai_math_explained/` deck. You only need to know matrix multiplication, how to take a logarithm (things like $\log_2 8 = 3$), and what "complexity $O(n^3)$ " means. New ideas such as tensors, rank and "tensor characters" are all explained before they are used. Every number here is either computed by script (script: `精讲/scripts/03_matmul.py`, with symbolic checks in sympy) or has a source. **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. $\triangle$ = I could not fully verify this. Estimated reading time: 60–90 minutes. You can split it into three sittings: §1–3 (Strassen and ω), §4–7 (tensors, approximation, the asymptotic spectrum), §8–10 (OpenAI's result and what it means). One difference from Explainer 01: this paper is **only 13 pages, and I read all of it** (not just the introduction). So §8 can lay out the full skeleton of the proof.

0. The bottom line: the one thing this chapter explains

Multiplying two $n \times n$ matrices the textbook way takes n^3 multiplications. In 1969 Strassen found that you can do it with fewer. Ever since, theoretical computer science has been

asking: **what is the smallest power of n you need?** That power is called ω (the matrix multiplication exponent).

This chapter explains one chain of cause and effect:

How small ω is $\Leftarrow$ how fast the rank of the matrix multiplication "tensor" grows with $n \Leftarrow$ (Strassen's duality theorem) how large every "tensor character" is on matrix multiplication $\Leftarrow$ OpenAI: using a simpler tensor, **polynomial multiplication**, they pinned down the value of every character.

Result: $\omega \leq 9/4 = 2.25$. The previous best human result was about 2.3712 (2026-08), and for 35 years progress had only moved the third decimal place.

After this chapter you should be able to explain in your own words:

- why "one fewer multiplication for 2×2 " can rewrite the whole n^3 ;
- what ω means exactly, and why it is the same thing as "the rank of a tensor";
- roughly what steps OpenAI's 13-page proof takes, and why it is not the same road as the "laser method" of the past 35 years;
- why this result **gives no direct speedup at all to GPUs or LLM training**.

1. Matrix multiplication: counting the multiplications

1.1 The textbook algorithm

$C = A \cdot B$. Each entry of C is a row times a column:

$$c_{ij} = a_{i1} \cdot b_{1j} + a_{i2} \cdot b_{2j} + \dots + a_{in} \cdot b_{nj}$$

C has n^2 entries, and each takes n multiplications and $n-1$ additions. In total: **n^3**

multiplications + $(n^3 - n^2)$ additions. For 2×2 : 8 multiplications, 4 additions.

1.2 Why theorists only count "multiplications"

You may find this odd: additions also take time, so why count only multiplications? The reason is recursion (you'll see it in §2 shortly). When a and b are themselves matrix blocks, "adding blocks" takes only $(n/2)^2$ operations, while "multiplying blocks" has to recurse, which costs far more. **In a recursion, the number of multiplications sets the exponent; additions only affect the constant.**

1.3 How this connects to your daily work

Most of the FLOPs in LLM training are GEMMs (general matrix multiplies). The GEMMs in cuBLAS and CUTLASS use the n^3 algorithm, just cut into small tiles that fit Tensor Cores and

shared memory (tiling). That will still be true at the end of this chapter; §10.2 explains why.

2. Strassen 1969: 7 is enough

2.1 Seven products

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$. Use only 7 multiplications:

$$\begin{array}{llll} M_1 = (a+d)(e+h) & M_2 = (c+d)e & M_3 = a(f-h) & M_4 = d(g-e) \\ M_5 = (a+b)h & M_6 = (c-a)(e+f) & & M_7 = (b-d)(g+h) \\ \\ C_{11} = M_1 + M_4 - M_5 + M_7 & C_{12} = M_3 + M_5 & & \\ C_{21} = M_2 + M_4 & C_{22} = M_1 - M_2 + M_3 + M_6 & & \end{array}$$

Check C_{12} : $M_3 + M_5 = af - ah + ah + bh = af + bh \checkmark$. (The script expanded all four entries with sympy and checked them; all four are correct.)

[Check 1] Verify by hand that $C_{21} = M_2 + M_4 = ce + dg$.

The cost: multiplications go down from 8 to 7, but additions/subtractions go up from 4 to 18. Doesn't look like a good deal, does it?

2.2 The key: the formulas never use $ab = ba$

Notice that in these 7 products, every multiplication has "only entries of A on the left, only entries of B on the right", and the order of multiplication is never swapped. So $a\dots h$ can **themselves be matrix blocks** (matrix multiplication is not commutative, but we never need it to be). So:

- cut the $n \times n$ matrix into 4 blocks of size $(n/2) \times (n/2)$;
- do 7 "block multiplications" (each one a recursive call) + 18 "block additions";
- recurrence: $T(n) = 7 \cdot T(n/2) + 18 \cdot (n/2)^2$.

Unroll the recursion: at each level down, the number of multiplications goes $\times 7$ and the size goes $\div 2$. There are $\log_2 n$ levels, so the total number of multiplications $= 7^{(\log_2 n)} = n^{(\log_2 7)}$.

$$\log_2 7 = 2.807354\dots \text{ (computed by script)}$$

The one saved multiplication compounds at every level of the recursion. The table below compares exact counts for $n = 2^k$, recursing all the way down to 1×1 :

n	Naive multiplications n^3	Strassen multiplications $7^{\log_2 n}$	Ratio	Naive total ops	Strassen total ops (incl. 18 block additions)	Ratio
2	8	7	0.875	12	25	2.08
16	4,096	2,401	0.586	7,936	15,271	1.92
128	2,097,152	823,543	0.393	4,177,920	5,666,497	1.36
1024	1,073,741,824	282,475,249	0.263	2,146,435,072	1,971,035,287	0.918
4096	6.87×10^{10}	1.38×10^{10}	0.201	1.37×10^{11}	9.68×10^{10}	0.704

(Computed by script.) Two things are worth noticing:

1. Counting only multiplications, Strassen wins right away. Counting additions too, **it only starts winning at $n = 1024$** (if you recurse all the way to 1×1). Real implementations switch to the naive algorithm once blocks get small enough (say 64), which moves the crossover earlier. But the pattern "asymptotically faster $\neq$ faster at small sizes" will come up again and again in this chapter.
2. This is only the operation count. Real hardware also has memory traffic and numerical stability; more on that in §10.2.

3. ω : the matrix multiplication exponent

3.1 Definition

ω = the infimum of all τ such that, for every $\varepsilon > 0$, $n \times n$ matrix multiplication can be done with $O(n^{(\tau+\varepsilon)})$ arithmetic operations (+, −, ×).

Unpacking this:

- "Infimum": ω itself might not be achievable, but you can get arbitrarily close to it.
- "+ ε ": allows a tiny bit of slack. That way something like $n^2 \cdot \log n$ still counts as exponent 2.
- The constant inside $O(\dots)$ **can grow arbitrarily large as ε shrinks**. This is the root of the "galactic algorithms" in §10.
- The computation is exact, over the complex numbers (or any fixed field). Floating-point error and the bit length of numbers are ignored.

3.2 Why $2 \leq \omega \leq 3$

- $\omega \leq 3$: the textbook algorithm.
- $\omega \geq 2$: the output has n^2 numbers, and each has to be computed at least once. (Reading in the $2n^2$ inputs also takes that many steps.)
- Strassen proved $\omega \leq 2.807$.

The mainstream conjecture: $\omega = 2$. That is, matrix multiplication can be almost as fast as "reading the data once". Nobody can prove it, and nobody can prove $\omega > 2$ either.

3.3 Why it matters

For many basic problems, the best known algorithm's complexity can be written directly as n^ω : solving linear systems, matrix inversion, determinants, LU decomposition, transitive closure of a graph, maximum matching in bipartite graphs, and more [P] (standard results; see the Bürgisser–Clausen–Shokrollahi textbook in "Further reading"). When ω goes down, their theoretical complexity goes down with it.

4. Tensors: turning "an algorithm" into "a number"

This section is the foundation of the whole chapter. It replaces the task of "designing an algorithm" with the task of "finding the rank of a three-dimensional array".

4.1 Starting from the rank of a matrix

The rank of a matrix = the smallest number of "outer products" (column vector $\times$ row vector) it can be written as a sum of. For example, $[[1,2],[2,4]] = [1,2]^T \cdot [1,2]$, which has rank 1.

4.2 The matrix multiplication tensor

Write matrix multiplication as a big "polynomial" in three groups of variables (a trilinear form):

$$T_n = \sum_{\{i,j,k\}} x_{ij} \cdot y_{jk} \cdot z_{ki}$$

x stands for the entries of A , y for the entries of B , and z is a placeholder for "position (i,k) of C ". The expression means: "entry (i,j) of A times entry (j,k) of B contributes to position (i,k) of C ." Arrange the coefficients into a three-dimensional array and you have a **tensor** (think of it as a three-dimensional matrix). Its three groups of variables x, y, z are called its three **legs**.

For 2×2 , T_2 has $2^3 = 8$ terms, matching the 8 multiplications of the textbook algorithm.

4.3 Tensor rank = the minimum number of multiplications

The **rank of a tensor $R(T)$** = the smallest number of "rank-1 terms" it can be written as a sum of, where each rank-1 term looks like:

$$(\text{some linear combination of } x) \cdot (\text{some linear combination of } y) \cdot (\text{some linear combination of } z)$$

Compare with Strassen's $M1 = (a+d)(e+h)$: the first two brackets are exactly "a linear combination of x" and "a linear combination of y", and the third bracket records which positions of C M1 gets added to (C11 and C22). **A rank-r decomposition = an algorithm that uses only r multiplications**, and vice versa (such algorithms are called "bilinear algorithms"). So:

Strassen 1969 $\Leftrightarrow R(T_2) \leq 7$. (Winograd 1971 proved that 7 is optimal: $R(T_2) = 7$ [P].)

4.4 Tensor product = recursion

"Multiply" T_n and T_m (the tensor product, which pairs up the indices on each leg) and you get exactly $T_{\{nm\}}$. This is the algebraic version of the "blocks inside blocks" recursion from §2.2. So $R(T_{\{nm\}}) \leq R(T_n) \cdot R(T_m)$, and:

As soon as you find a $k \times k$ decomposition using r multiplications, you get $\omega \leq \log_k r$.

Size k	Multiplications r	Source	$\log_k r$ (computed by script)
2	8	Naive	3.0000
2	7	Strassen 1969	2.8074
3	23	Laderman 1976	2.8540
4	49	Strassen applied twice	2.8074
4	48	AlphaEvolve 2025 (over the complex numbers)	2.7925
4	47	AlphaTensor 2022 (GF(2) only, i.e. arithmetic mod 2)	2.7773

Note: by "finding one good decomposition of a fixed small size", the best anyone has reached is about 2.77. Every upper bound on ω below that number comes from the "asymptotic" methods of §5–7.

[Check 2] If someone found a 4×4 decomposition with 46 multiplications, what bound on ω would follow?

4.5 Another tensor: polynomial multiplication (the star of OpenAI's proof)

Multiply two polynomials: $(x_0 + x_1t) \cdot (y_0 + y_1t) = x_0y_0 + (x_0y_1 + x_1y_0)t + x_1y_1t^2$. Written as a tensor:

$$C(a, b) = \sum_{\{i < a, j < b\}} x_i \cdot y_j \cdot z_{\{i+j\}}$$

a and b are the numbers of coefficients of the two input polynomials, and the output has $a + b - 1$ coefficients. The naive method takes $a \cdot b$ multiplications. But its rank is only **$a + b - 1$** :

- evaluate both polynomials at $a + b - 1$ different points and multiply the values ($a + b - 1$ multiplications);

- then recover the coefficients of the product polynomial by interpolation (interpolation uses only additions and multiplication by constants, which don't count as "variable × variable" multiplications).

The case $a = b = 2$ is the **Karatsuba trick**: 3 multiplications $P_0 = x_0y_0$, $P_1 = (x_0+x_1)(y_0+y_1)$, $P_2 = x_1y_1$, and the middle coefficient $= P_1 - P_0 - P_2$ (checked by script: it expands to $x_0y_1 + x_1y_0$ ✓). This is also where the idea behind fast FFT multiplication comes from.

Key point: the rank of polynomial multiplication is known exactly ($a + b - 1$); the rank of matrix multiplication is unknown. OpenAI's entire proof is "use the known thing to pin down the unknown thing".

[Check 3] Using "evaluate at the three points $t = 0, 1, -1$ ", design a different 3-multiplication algorithm for $(x_0 + x_1t)(y_0 + y_1t)$, and write down how to recover the middle coefficient.

5. Approximations count too: border rank and "degeneration"

5.1 A strange example

Look at this small tensor (it has nothing to do with matrix multiplication; it's just a demonstration):

$$W = x_0y_0z_1 + x_0y_1z_0 + x_1y_0z_0$$

One can prove its rank is 3. But look at this expression (ϵ is a small parameter):

$$\begin{aligned} & [(x_0+\epsilon x_1)(y_0+\epsilon y_1)(z_0+\epsilon z_1) - x_0y_0z_0] / \epsilon \\ &= W + \epsilon \cdot (x_0y_1z_1 + x_1y_0z_1 + x_1y_1z_0) + \epsilon^2 \cdot x_1y_1z_1 \end{aligned}$$

(Expanded and checked by script with sympy.) The left side has only **2** rank-1 terms, and as $\epsilon \rightarrow 0$ it tends to W . So W can be approximated arbitrarily well by "something of rank 2". We say W has **border rank 2**.

5.2 Bini's interpolation: approximate algorithms can become exact ones

Bini et al. found in 1979 that a variant of 2×2 matrix multiplication shows this kind of "approximately save a multiplication" behavior, and Bini 1980 proved a key fact [P]:

asymptotically, approximate algorithms are as good as exact ones. The method: treat the expression containing ϵ as a polynomial in ϵ , compute it at several different values of ϵ , then interpolate out the ϵ^0 term. The cost is only a factor independent of n , which disappears in the exponent.

5.3 "Degeneration": giving variables weights

The ε trick above has a more systematic version, which OpenAI's proof uses over and over:

1. give each variable an integer **weight** w_i ;
2. multiply each variable by ε^{w_i} ;
3. the total weight of a term = the sum of the weights of its three variables; **keep only the terms with the smallest total weight**. All other terms carry higher powers of ε and vanish in the limit.

Getting B from A this way is written $A \rightsquigarrow B$. Bini's interpolation guarantees that B is no "harder" than A. This is a powerful tool: **by choosing weights cleverly, you can "cut" the small tensor you want out of a big tensor, while throwing away every term you don't want.**

6. Fifty years of history: a war over the third decimal place

6.1 The asymptotic sum inequality and the laser method

- **Schönhage 1981:** if you can compute several **mutually independent** matrix multiplications **at the same time** more cheaply than computing them separately, you also get an upper bound on ω (the "asymptotic sum inequality"). Note that "independent" means no variables are shared on any of the three legs. This is called a **full direct sum**.
- **Strassen 1986–87: the laser method.** Find a tensor that is not itself matrix multiplication but has very low rank, take a very high power of it (tensor it with itself many times), and then "cut" a large number of mutually independent matrix multiplications out of it. Like a laser, it filters the in-phase part out of messy light.
- **Coppersmith–Winograd 1990:** found a particularly good starting tensor (the CW tensor) and, combined with the combinatorics of "sets with no three-term arithmetic progression" for the cutting, got 2.3755.

6.2 Timeline

Year	Authors	Upper bound on ω	Drop from the previous row (computed by script)
1969	Strassen	2.8074	—
1978	Pan	2.796	0.011
1979	Bini et al. (border rank)	2.780	0.016
1981	Schönhage (asymptotic sum inequality)	2.522	0.258
1986	Strassen (laser method)	2.479	0.043
1990	Coppersmith–Winograd	2.3755	0.104
2014	Le Gall (preceded by Stothers 2010, Vassilevska Williams 2012)	2.3728639	0.0026
2021	Alman–Vassilevska Williams	2.3728596	0.0000043
2023	Duan–Wu–Zhou	2.371866	0.00099
2024	Vassilevska Williams–Xu–Xu–Zhou	2.371552	0.00031
2024	Alman–Duan–Vassilevska Williams–Xu–Xu–Zhou	2.371339	0.00021
2026-08	Dupont et al. (incl. AlphaEvolve)	2.371177	0.00016
2026-09-24	OpenAI (any field, still within the laser-method framework) [R]	2.371054886	0.00012
2026-09-24	OpenAI (complex numbers) [R]	2.258	0.113
2026-10-02	OpenAI (complex numbers, the subject of this chapter) [R]	2.25 = 9/4	0.008

(Values before 2026-08 are all peer-reviewed results [P]; Dupont et al.'s 2.371177 is the arXiv preprint 2608.16884, and OpenAI's 9/4 paper itself cites it as the previous record.)

A few comparisons computed by script:

- The 34 years from 1990 → 2024: a total drop of 0.0041.
- The 12 years from 2014 → 2026-08: a total drop of 0.0017.
- 2026-08 → 9/4: a single drop of **0.121**.

6.3 The wall the laser method hit

Ambainis–Filmus–Le Gall (STOC 2015) [P] proved that this kind of laser-method analysis, applied to powers of the CW tensor, **cannot go below about 2.3078**. Later, Alman–Vassilevska Williams and others gave more general barriers [P] $\triangle$ (I did not check the specific values one by one). So 2.25 cannot be "the laser method optimized a bit more"; it has to come from a different road. This is exactly what Aaronson means by "via a completely different approach".

7. Strassen's asymptotic spectrum: "measuring" instead of "constructing"

This section is the key to understanding OpenAI's proof, and also the most abstract part of the chapter. I'll use engineering analogies as much as I can, and flag where an analogy breaks down.

7.1 "Which tensor is stronger"

$A \geq B$ (B is a "restriction" of A) means: you can get B by applying a linear substitution to each of A 's three legs. Intuitively, **if you can compute A , you can compute B** . For example:

- $R(T) \leq r$ is equivalent to " r independent scalar multiplications $\geq T$ " (where r denotes "a sum of r mutually unrelated xyz terms").
- Drop all terms of T_n with $i \neq j$, and what's left is n independent scalar multiplications, so $T_n \geq n$.

7.2 Tensor characters: a "consistent measuring device"

A **tensor character** (also called a "point" of the asymptotic spectrum) is a function λ that maps tensors to non-negative real numbers and satisfies four properties:

Property	Meaning	Engineering analogy
$\lambda(1) = 1$	A single scalar multiplication has value 1	Unit calibration
$\lambda(A \oplus B) = \lambda(A) + \lambda(B)$	Independent tasks run in parallel; values add	Throughput adds up
$\lambda(A \otimes B) = \lambda(A) \cdot \lambda(B)$	Nesting (recursion); values multiply	Recursive costs multiply
$A \geq B \Rightarrow \lambda(A) \geq \lambda(B)$	If you can compute A you can compute B , so A 's value is no smaller	Monotone

These four immediately imply: **for any nonzero tensor, $1 \leq \lambda(A) \leq R(A)$** . So every character gives a **lower bound** on rank.

Example: flattening rank. Stand one leg of the three-dimensional array up vertically and lay the other two legs out together as a row. You get an ordinary matrix; take its matrix rank. This satisfies all four properties, and these are the three simplest characters. For T_n , the flattening rank is n^2 .

7.3 Every character's value on matrix multiplication is $n^{(3t)}$

§2.1 of the paper uses a small trick to prove that for any character λ there is a number $t > 0$ such that

$$\lambda(T_n) = n^{(3t)} \quad \text{for all } n$$

(Proof idea: T_n can be split into a tensor product of three "dot product" tensors, and each dot product has value some power of m under λ ; multiplying gives $n^{(p_X + p_Y + p_Z)}$, and we write $3t = p_X + p_Y + p_Z$.)

For example, flattening rank: $\lambda(T_n) = n^2 \Rightarrow 3t = 2, t = 2/3$.

7.4 Strassen's duality theorem: measurements decide everything

Strassen 1986–88 [P] proved a deep theorem (Appendix A of the OpenAI paper re-proves the special case it needs):

The rank exponent of matrix multiplication $\omega \leq 3 \cdot (\text{the largest } t \text{ over all characters})$. More precisely: if $R(T_d)$ is large, then there must be some character λ that "detects" it, i.e. $\lambda(T_d)$ is also large.

Analogy: this is a lot like duality in linear programming, or Farkas' lemma: either a good algorithm exists, or there is a "certificate" (a character) proving it doesn't. So to prove "a good algorithm exists" (ω is small), you **don't need to construct the algorithm**. You only need to prove that "**no** certificate can show it doesn't exist", i.e. **every character has $t \leq 3/4$** .

This is the most important turn in the chapter:

- Laser method: **construct** one specific large tensor decomposition.
- Asymptotic spectrum: **bound every possible measurement**, then use the duality theorem to conclude that an algorithm exists.

An honest note: Strassen proposed the asymptotic-spectrum route back in the 1980s. The hard part is that nobody knows what all the characters look like. Over the complex numbers, the only known infinite family is the "quantum functionals" of Christandl–Vrana–Zuiddam (JAMS 2023) [P]. OpenAI's proof **sidesteps** this difficulty: it doesn't need to know what the characters look like, and derives its inequalities from the four properties alone.

If the $\omega = 2$ conjecture is true, then every character has $t = 2/3$ on matrix multiplication (the same as flattening rank). The $9/4$ result says: $t \leq 3/4$.

8. What OpenAI claims

8.1 The main result

The abstract of the paper *An Upper Bound of $9/4$ for the Matrix Multiplication Exponent* (2026-10-02, 13 pages) is a single sentence:

"We prove that the exponent of matrix multiplication over the complex numbers is at most $9/4$."

Theorem 1.1: "For every $\varepsilon > 0$, two $n \times n$ complex matrices can be multiplied using $O_\varepsilon(n^{(9/4+\varepsilon)})$ arithmetic operations. In particular, $\omega \leq 9/4$."

The paper itself also says: "Theorem 1.1 concerns the asymptotic arithmetic exponent; its proof does not specify a competitive finite matrix size."

The same family (number 107) has two more manuscripts:

- *Complex Matrix Multiplication Below 2.258 and Rectangular Bounds* (2026-09-24, 83 pages): over fields of characteristic 0, $\omega < 2.258$, the "dual exponent" $\alpha > 0.465$, and $\omega(1, 0.709, 1) < 2.092$. Once $9/4$ came out, the 2.258 was superseded by it.
- *Staggered extraction for exact matrix multiplication over every field* (2026-09-24, 35 pages): over **every field** (including finite fields and positive characteristic), $\omega < 2.371054886006746$. Still within the laser-method framework; about 1.2×10^{-4} better than Dupont et al.

8.2 The skeleton of the proof (I read the whole paper; this is a faithful simplification)

The whole proof uses only the tools from §4–7. It has five steps.

Step 1: separate blocks that "share a leg" (§3, the separation lemma).

An old problem for the laser method is that the small blocks you cut out often share variables, so they are not fully independent. OpenAI's core construction handles this case: M blocks that are independent of each other on the y and z legs but **share** the x leg. The method:

- make $5M$ copies, and give each one a phase from a different power of an $L = 5M$ -th root of unity ζ (this is a finite Fourier transform), tagging each x variable with a "tentative label" g ;
- then use the "weighting" from §5.3: x gets weight g^2 , y gets weight $h_u - h^2$, z gets weight $-h_v$. On the terms that survive the Fourier transform, the total weight is exactly $(g - h)^2$;
- keep only the terms with weight 0, i.e. $g = h$: each block gets its own private copy of x , and the blocks become fully independent. As a side effect, each block also gains a "dot product" of length M .

The intuition for this step: **use the square $(g-h)^2 \geq 0$ as a "penalty", so that every term whose labels don't match vanishes as $\varepsilon \rightarrow 0$** . Combined with the standard "count by frequency" technique, this gives an entropy inequality (Corollary 3.2).

Step 2: look only at polynomial multiplication. For any character λ , define the "symmetrized value" $P(a, b)$ of the polynomial multiplication tensor $C(a, b)$ (evaluate all 6

permutations of the three legs, multiply the 6 values together, then take the $1/(6t)$ -th power; see the paper's eq. 4.4). It satisfies three simple properties (Eq. 4.5):

- $P(a, b) = P(b, a)$;
- $P(1, b) = b$ ($C(1, b)$ is just a dot product);
- **$P(a, b) \leq (a + b - 1)^{1/t}$** : because the rank of polynomial multiplication is $a + b - 1$ (§4.5), and a character is $\leq$ rank.

Here t is exactly the "exponent of λ on matrix multiplication" from §7.3.

Step 3: two new inequalities (§4).

- **Discrete concavity**: $2P(a, b) \geq P(a, b+1) + P(a, b-1)$. (A "determinantal filtration" degenerates $C(a, b) \otimes$ dot product into two blocks, $C(a, b+1)$ and $C(a, b-1)$; then apply Step 1.)
- **Shifted tripling**: $P(a, 3h + a - 1) \geq 3 \cdot P(a, h)$. (Cut the y and z indices of $C(a, 3h+a-1)$ into left, middle and right segments; after weighting, exactly three copies of $C(a, h)$ remain.)

You can work out the smallest case of shifted tripling by hand: $a = 2, h = 1$, i.e. $C(2, 4)$, which has 8 terms $x_i \cdot y_j \cdot z_{i+j}$. Cut y into left $\{0\}$, middle $\{1, 2\}$, right $\{3\}$; cut z into left $\{0, 1\}$, middle $\{2\}$, right $\{3, 4\}$. Middle y gets weight $+1$, middle z gets weight -1 , everything else 0:

Term (x, y, z)	Weight	Result
(0,0,0), (1,0,1)	0	Left block kept
(0,2,2), (1,1,2)	$+1-1 = 0$	Middle block kept
(0,3,3), (1,3,4)	0	Right block kept
(0,1,1), (1,2,3)	+1	Vanishes

(Checked term by term by script; matches Figure 1 of the paper.) The three blocks don't overlap on y or z , and they share x : exactly the shape Step 1 can handle.

[Check 4] Without looking at the table, work out why the term $(0,1,1)$ has weight $+1$.

Step 4: a pure sequence argument (§5, Lemma 5.1). Using only "symmetry + $P(1,b) = b$ + concavity + shifted tripling", you can derive growth along the diagonal:

$$P(a, a) \geq a^{4/3}$$

This step is entirely elementary: concavity says the increments decrease, tripling says the increments can't decrease too fast, and together they squeeze out a recurrence $H_a \geq (1 + 1/(3(a-1))) \cdot H_{a-1}$, finished off with $(1 + 1/(3m))^3 \geq 1 + 1/m$. The script checked this chain of lower bounds (for example, at $a = 1000$ the paper's intermediate lower bound gives $P(a,a) \geq 16,790$, while $a^{(4/3)} = 10,000$).

Step 5: squeeze from both sides. Step 2 gives an upper bound and Step 4 gives a lower bound:

$$a^{(4/3)} \leq P(a, a) \leq (2a - 1)^{(1/t)}$$

As $a \rightarrow \infty$, the left side behaves like $a^{1.333}$ and the right side like $a^{(1/t)}$. For this to hold for all a , we need $1/t \geq 4/3$, i.e. $t \leq 3/4$. Computed by script: if some character had $t = 0.8$, this inequality would already be violated at $a = 32,761$; with $t = 0.76$ it would take until $a \approx 3.8 \times 10^{22}$.

So **every** character has $\lambda(T_d) = d^{(3t)} \leq d^{(9/4)}$. Then by the duality theorem of §7.4, the rank exponent is $\leq 9/4$, so $\omega \leq 9/4$.

8.3 Why the final answer is a rational number

Laser-method bounds come from the optimal value of a numerical optimization problem, so they are "computed" irrational-looking numbers like 2.3728639. Here $9/4 = 3 \times 3/4$; the $3/4$ comes from "growth like the $4/3$ power", and the $4/3$ in turn comes from the 3 in the "tripling" construction. Hence Aaronson's line: "a *rational* exponent for once (!)" [R].

8.4 What "does not specify a size" actually means

The last paragraph of the paper turns " $\omega \leq 9/4$ " into an algorithm: by the definition of the rank exponent, there **exists** some fixed u and a decomposition of T_u with $r < u^{(9/4+\delta)}$ multiplications; then you recurse as Strassen did. But the proof **doesn't tell you what u is, and doesn't give the decomposition**, because the duality theorem of §7.4 uses compactness and a fixed-point theorem (Schauder–Tychonoff). It is essentially an existence proof.

What this means for engineering can be estimated with known lower bounds (computed by script):

- $u = 2$: $R(T_2) = 7 > 2^{2.25} \approx 4.76$. Impossible.
- $u = 4$: you'd need $r \leq 22$, but Bläser 2003 proved $R(T_n) \geq 2.5n^2 - 3n$ [P] $\triangle$ (details of the source not re-checked), which gives a lower bound of 28 at $n = 4$. Impossible.

- By this lower bound, **u must be at least 34** for a decomposition meeting the requirement to possibly exist. The real u may be far larger; nobody knows.

Remark 5.2 of the paper adds that the coefficients of this decomposition can be taken to be algebraic numbers (in some number field). So in principle it could be found by a finite search, but there is no practical feasibility at all. (This is my inference $\triangle$.)

A comment on Aaronson's blog (Bram Cohen, #42) says the result "isn't constructive". That judgment is right, but his description "ramsey type thing" is inaccurate: the non-constructiveness comes from the compactness + fixed-point argument in the duality theorem, not from a Ramsey-type combinatorial argument.

9. How far it has been verified

9.1 The Lean statement: read the model first, then the theorem

Unlike Explainer 01, ω here is not a ready-made Mathlib definition; it is written by the challenge file `lean/ComparatorChallenges/MatrixMultiplication.lean` itself. So **reviewing the definitions** matters more than reviewing the theorem. Translating block by block:

- `Gate`: there are only five kinds of "gate": `constant` (a constant), `input` (read one input), `add`, `sub`, `mul` (operate on two existing registers).
- `Gate.cost`: constants and inputs cost 0; **add, subtract and multiply each cost 1**. So this is the standard "number of arithmetic operations" model. (Multiplying by a constant also uses a `mul` gate, so it also costs 1.)
- `Program`: a sequence of gates, where each gate can only refer to earlier registers. This is a "straight-line program": no loops, no branches, no division.
- `MatrixAlgorithm F a b c`: a program, plus "which register each position of the output matrix takes".
- `Correct`: for **all** input matrices A, B, the output equals `A * B` (Mathlib's standard matrix multiplication).
- `AdmissibleExponent F  $\tau$` : for every $\varepsilon > 0$ there is a constant C such that for **every** $n \geq 1$ there is a correct program with cost $\leq C \cdot n^{(\tau+\varepsilon)}$.
- `omega F := sInf { $\tau$  | AdmissibleExponent F  $\tau$ }`: the infimum.

Then the theorem itself:

```
theorem complex_omega_le_nine_quarters :
  Arithmetic.omega  $\mathbb{C} \leq (9 : \mathbb{R}) / 4$ 
```

Translation: **over the complex numbers, the matrix multiplication exponent, defined by the model above, is $\leq 9/4$** . This matches the definition in §3.1. Note that it is "non-uniform" (each n may have a different program; there is no requirement for one uniform algorithm that generates them), which is also the standard definition of ω in the literature.

A specification trap worth noting (especially interesting for AI safety people): in Lean/Mathlib, **the `snf` of the empty set is defined to be 0**, and the `snf` of a set with no lower bound is also defined to be 0. So if some definition were written wrong so that "no program is ever judged correct", then $\omega = 0$ and " $\omega \leq 9/4$ " would hold **trivially**. I checked this: the solution proves `omega_le_three` (the naive algorithm works, so the set is non-empty) and `admissibleExponent_two_le` (every admissible exponent is ≥ 2 , so the set is bounded below), and the $9/4$ derivation goes through `csInf_le`, which requires actually exhibiting an admissible exponent. So this trap is ruled out.

The same challenge file has two more theorems: $\alpha > 93/200$ and $\omega(\mathbb{C}; 1, 0.709, 1) < 523/250$.

9.2 The proof itself

- The Lean proof of the $9/4$ theorem is in `0AI/LinearAlgebra/MatrixMultiplication/AuxiliarySeparation/` (73 files, 10,895 lines). Computing the import closure from its `Main.lean`, the project-internal total is **129 files, about 20,700 lines** (computed by script, including shared directories such as Arithmetic, Tensor and Polynomial). The whole matrix multiplication directory (including 2.258 and the rectangular bounds) is 509 files, about 83,000 lines.
- I grepped: no `sorry`, no custom `axiom`, no `native_decide` (the three hits for `admit` are all the English word "admits" in comments).
- The Comparator config allows only three standard axioms: `propext`, `Quot.sound`, `Classical.choice`. Note that the config has `"enable_nanoda": false`, i.e. **this challenge did not turn on a second, independent kernel**.
- **External dependency**: the fixed-point theorem in Appendix A uses the third-party library `fixed-point-theorems-lean4` (harfe, the Brouwer fixed-point theorem). OpenAI applied a porting patch to it (about 77 lines added, 63 deleted). I looked at it: it mainly adapts imports and syntax to Lean 4.34, and I saw no `sorry` or new axioms.
- A small documentation oddity: the scope note `lean/docs/107.md` lists only the 2.258 and staggered papers, not the $9/4$ paper; but the $9/4$ theorem is indeed proved in `AuxiliarySeparation/Main.lean` (the file comment says "Theorem 1.1").

9.3 Independent re-checks

- **erenciracioglu-dotcom/openai-math-9-4-lean-check** [R]: compiled these three theorems with Lean v4.34.1 on Windows 11, used `pp.all` to compare the challenge statement and the solution statement and found them identical, and reports dependence on only the three standard axioms. But it says itself: "It does **not** replace Comparator: there was no sandbox and no second, independent kernel replay." Also, it used openai/math commit `adc7f12`, not the `fd4aeeb` used in these notes $\triangle$ (I did not compare whether this part changed between the two commits).
- I **did not compile** it myself.

9.4 Human experts

- Scott Aaronson ("The Mathocalypse", 2026-10-07): "Matrix multiplication in $O(n^{9/4})$ time—a rational exponent for once (!), and via a completely different approach than was used for $O(n^{2.373})$ and so forth" [R]. This is a comment, not a verification.
- **As of 2026-10-08, I found no laser-method / tensor expert (Vassilevska Williams, Alman, Le Gall, Zuiddam, Christandl, etc.) publicly saying they have read and verified this paper** $\triangle$. The vibemathed.com entry also says "No independent mathematician has checked this yet" [R].
- One favorable factor (my judgment): the paper is only 13 pages, uses only standard tools, and Appendix A fully re-proves the special case of the duality theorem it needs. It is **one of the headline results of the whole openai/math release most likely to be fully understood by human experts within a few weeks.**

9.5 Remaining risks

1. The custom computational model: I read it and think it is faithful (§9.1). But that is the level of "one person read it once".
2. No second-kernel re-check.
3. Correctness of the third-party fixed-point library: the Lean kernel checks its proofs, so the risk is mainly at the level of definitions, and it is low.
4. Risks beyond correctness: the conclusion is right, but humans don't yet understand "why nobody made this road work before".

10. What it means if true (and what it doesn't)

10.1 For mathematics and theoretical computer science

1. **The biggest single step in 35 years:** from 2.3712 to 2.25, a drop about 70 times the total of the previous 12 years (computed by script: 0.121 vs 0.0017).
2. **It gets around the laser-method barrier** (§6.3), showing that Strassen's asymptotic-spectrum route is workable, and can be used in a quite "elementary" way.

3. The theoretical upper bounds of all algorithms whose complexity is written as n^ω (§3.3) go down together.
4. **Is $\omega = 2$ now closer?** This is speculation. Where this method's bottleneck lies (whether the $4/3$ growth rate is a limit of the method itself), and whether it can be pushed to 2, the paper doesn't discuss $\triangle$.
5. **Only for the complex numbers** (and, via the algebraic-coefficient argument in the paper's Remark 5.2). Over fields of positive characteristic, OpenAI only claims 2.371054886.

10.2 For GPUs and LLM training: no direct impact

This is where engineers most easily get it wrong, so let me be concrete:

1. **We barely use even Strassen.** The table in §2.2 shows that, counting additions, plain Strassen only wins on operation count at $n \approx 1024$. On real hardware there is also: (a) the extra block additions need extra memory reads and writes, and GEMM is often bandwidth-bound; (b) Tensor Cores only accelerate "multiply-add", so block additions don't get accelerated; (c) numerical stability is worse than the naive algorithm (Higham's textbook analyzes this specifically). Some work has achieved practical Strassen speedups on CPUs (Huang et al., "Strassen's Algorithm Reloaded", SC16), but mainstream deep learning libraries don't use it.
2. **The 9/4 algorithm is a "galactic algorithm":** asymptotically faster, but only wins on absurdly large inputs. The paper gives no constant, so what follows is a **hypothetical** calculation (computed by script): if the 9/4 algorithm's constant is C and the naive algorithm's is $2n^3$, the crossover is around $n \approx (C/2)^{4/3}$. With $C = 10^6$, $n \approx 4 \times 10^7$, and a single such fp16 matrix takes about 3×10^{15} bytes (3 PB). And nobody knows the real constant; as §8.4 said, even the size u of the base decomposition is at least 34, and the actual decomposition is unknown.
3. **The 2.37 laser-method algorithms are galactic algorithms too.** So this is not a problem specific to 9/4: **none of the results with ω below about 2.77 has ever been usable in practice.**
4. **Compare AlphaTensor / AlphaEvolve:** AlphaTensor (Fawzi et al., *Nature* 610, 2022) [P] and AlphaEvolve (2025, 48 multiplications for 4×4 complex matrices) [R] look for low-rank decompositions of a **fixed small size**. That kind of thing can be written directly into a kernel and gives constant-factor speedups (the AlphaTensor paper even optimized decompositions for specific GPUs). ω is about the exponent as $n \rightarrow \infty$. The two are almost different problems. Dupont et al. (2026-08) used AlphaEvolve to optimize parameters inside the laser method; that is AI progress on ω itself, but the size of the improvement is 1.6×10^{-4} .

10.3 For AI

- **This is a "new idea", not a "bigger optimization".** Compared with Dupont et al. "using AI to tune parameters and squeeze out 1.6×10^{-4} ", the 9/4 proof switched to a different route, and the final argument is short enough to write out by hand. If it holds up, it is one of the clearest examples of "AI producing a mathematical idea that humans didn't find in 35 years".
- **13 pages + about 20,000 lines of Lean make a good touchstone for human verification.** Unlike Explainer 01's 199 pages + 480,000 lines of Lean, here human experts can read the whole thing independently. How experts assess it over the next few weeks is a good signal for judging "the credibility of the whole openai/math release".
- **Spec review matters more than proof review** (§9.1): a Lean proof is only trustworthy if "the computational model is written correctly". This is the math version of the "spec gaming" problem in AI safety, and here the review only requires reading 100 lines of definitions.

11. Self-test

Try answering these after reading. If you can't, go back and reread the relevant section.

1. Why can Strassen's 7-multiplication formula be used recursively? Which property does it crucially not use? (§2.2)
2. What is the relationship between "a rank- r tensor decomposition" and "an algorithm using r multiplications"? (§4.3)
3. Why is the rank of the polynomial multiplication tensor $C(a, b)$ equal to $a + b - 1$? How does this relate to Karatsuba? (§4.5)
4. What four properties must a tensor character satisfy? Why is every character $\leq$ rank? (§7.2)
5. Between the laser method and the asymptotic-spectrum route, which is "constructing" and which is "measuring"? Why can the latter get around the known barriers? (§6.3, §7.4)
6. In the last step of OpenAI's proof, which two inequalities squeeze out $t \leq 3/4$? (§8.2, step 5)
7. In Lean, why does " $\text{slnf of the empty set} = 0$ " create a specification risk? How is it ruled out here? (§9.1)
8. Why won't $\omega \leq 9/4$ make your GPU training faster? Give at least three reasons. (§10.2)

Appendix: answers to the checks

- **Check 1:** $M_2 + M_4 = (c+d)e + d(g-e) = ce + de + dg - de = ce + dg \checkmark$.
- **Check 2:** $\omega \leq \log_4 46 \approx 2.7618$ (computed by script). Better than Strassen's 2.8074, but far from the laser method's 2.37.
- **Check 3:** Evaluate at $t = 0$: $P_0 = x_0 y_0$; $t = 1$: $P_1 = (x_0 + x_1)(y_0 + y_1)$; $t = -1$: $P_2 = (x_0 - x_1)(y_0 - y_1)$. The product polynomial is $c_0 + c_1 t + c_2 t^2$, so $P_1 = c_0 + c_1 + c_2$ and $P_2 = c_0 - c_1 + c_2$. Hence $c_0 = P_0$,

$c_1 = (P_1 - P_2)/2$, $c_2 = (P_1 + P_2)/2 - P_0$. Dividing by 2 is multiplication by a constant, not a "variable $\times$ variable" multiplication.

- **Check 4:** In $(0,1,1)$, the y index 1 is in the middle segment, weight +1; the z index 1 is in the left segment $\{0, 1\}$, weight 0. The total weight is $+1 > 0$, so it vanishes as $\varepsilon \rightarrow 0$.

Appendix: Glossary

Term	One-line explanation
ω (matrix multiplication exponent)	The optimal exponent for the number of arithmetic operations needed for $n \times n$ matrix multiplication; $2 \leq \omega \leq 3$
Bilinear algorithm	An algorithm in which every multiplication is "a linear combination of A's entries $\times$ a linear combination of B's entries"
Tensor	A three-dimensional array; here equivalent to a trilinear form $\sum c \cdot x \cdot y \cdot z$
Leg	One of the tensor's three groups of variables x, y, z
Tensor rank	The smallest number of rank-1 terms it can be written as a sum of; equals the minimum number of multiplications of a bilinear algorithm
Border rank	The smallest rank when approximation by expressions containing ε is allowed
Degeneration	Give variables weights as powers of ε and keep only the lowest-weight terms
Restriction	Get another tensor by linear substitutions on the three legs; $A \geq B$ means "if you can compute A you can compute B"
Full direct sum	A sum of several tensors that share no variables on any of the three legs
Laser method	Cutting many independent matrix multiplications out of a high power of some low-rank tensor (Strassen 1986)
Asymptotic spectrum / tensor character	The set of all "measurement functions" that are normalized, additive, multiplicative and monotone; Strassen proved they determine asymptotic complexity
Flattening rank	The rank of the tensor after laying it out as a matrix; the simplest tensor character
Galactic algorithm	An algorithm that is asymptotically faster but only wins on unrealistically large inputs
Straight-line program	A sequence of operations with no loops or branches; the Lean formalization uses it to model algorithms

Appendix: Further reading

- **The original paper** (13 pages, readable in full):
<https://github.com/openai/math/blob/main/preprints/Matrix-Multiplication-Nine-Fourths-October-2-2026/paper.pdf>
- **Lean statement** (about 100 lines; I recommend reading the definitions yourself):
<https://github.com/openai/math/blob/main/lean/ComparatorChallenges/MatrixMultiplication.lean>

- **Independent Lean re-check:** <https://github.com/erenciracioglu-dotcom/openai-math-9-4-lean-check>
- **Quanta Magazine**, *New Breakthrough Brings Matrix Multiplication Closer to Ideal* (2024-03-07): popular coverage of the Duan–Wu–Zhou step, with intuitive pictures of the laser method. <https://www.quantamagazine.org/new-breakthrough-brings-matrix-multiplication-closer-to-ideal-20240307/>
- **Markus Bläser**, *Fast Matrix Multiplication* (Theory of Computing Graduate Surveys 5, 2013): the best introductory survey, free. <https://theoryofcomputing.org/articles/gs005/>
- **J. M. Landsberg**, *The complexity of matrix multiplication: developments since 2014* (arXiv 1811.11667): a survey from a more geometric point of view.
- **Bürgisser, Clausen, Shokrollahi**, *Algebraic Complexity Theory* (Springer 1997): Chapters 14–15 cover tensor rank, the laser method and the asymptotic spectrum; the standard textbook.
- **Christandl, Vrana, Zuiddam**, *Universal points in the asymptotic spectrum of tensors* (JAMS 36, 2023; arXiv 1709.07851): "what the known characters look like" from §7.
- **Ambainis, Filmus, Le Gall**, the laser-method barrier (STOC 2015): <https://arxiv.org/abs/1411.5414>
- **Alman et al.**, *More Asymmetry Yields Faster Matrix Multiplication* (SODA 2025): <https://arxiv.org/abs/2404.16349>
- **Dupont et al.**, *Improving the matrix multiplication exponent with modern optimization and AlphaEvolve* (2026-08): <https://arxiv.org/abs/2608.16884>
- **Fawzi et al.**, *Discovering faster matrix multiplication algorithms with reinforcement learning* (AlphaTensor, *Nature* 610, 2022)
- **Huang, Smith, Henry, van de Geijn**, *Strassen's Algorithm Reloaded* (SC16): when Strassen is worth using on real hardware.
- **Scott Aaronson**, *The Mathocalypse* (2026-10-07): <https://scottaaronson.blog/?p=10169>
- **3Blue1Brown**, *Linear transformations and matrices* (YouTube, Essence of Linear Algebra episodes 3–4): if you want to shore up the intuition that "matrix multiplication = composition of linear transformations" first.

PART III · EXPLAINED FROM ZERO · GOES WITH §7

26. Explainer 04: The Unique Games Conjecture

This is the "explained from zero" version of §7 of the [notes/openai_math_explained/](#) deck. You only need to know what a graph is (vertices and edges), how to do modular

arithmetic ($7 \bmod 3 = 1$), that "polynomial time" roughly means "grows mildly with input size", and to have a vague sense of P vs NP. Every new idea is explained before it is used. Every number here is either computed by script or has a source. **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Estimated reading time: 60–90 minutes. You can split it into three sittings: §1–4 (NP-hardness, approximation, 0.878, PCP), §5–7 (what a unique game is, why it matters, the 24-year tug-of-war), §8–10 (OpenAI's result, verification, what it means).

0. The bottom line: the one thing this chapter explains

Many important optimization problems (cutting a network into two halves, placing the fewest guards to watch every road...) are believed to have **no fast exact algorithm**. As a fallback, we ask: **what percentage of the optimum can a fast algorithm guarantee at best?**

For the most classic of these, the Max-Cut problem, someone gave an algorithm in 1995 that guarantees **87.8%**. On the other side, in 1997–2001 it was proved that guaranteeing more than **94.1%** is NP-hard. For 30 years, nobody knew where the truth lay in the 6 percentage points in between.

This chapter explains one chain of cause and effect:

How well you can approximate $\Leftarrow$ is decided by how hard it is to "tell nearly satisfiable apart from far from satisfiable" $\Leftarrow$ the most important such constraint system is called Unique Games $\Leftarrow$ the Unique Games Conjecture (UGC) says: telling these apart is NP-hard.

If **UGC is true**, then 87.8% is **exactly** the limit for Max-Cut, and the same argument settles the approximation limits of a whole class of problems at once. Khot proposed it in 2002, and for 24 years nobody could prove or refute it. **OpenAI claims to have proved it**, with a Lean formal proof attached.

After this chapter you should be able to explain in your own words:

- why a "fully satisfiable" unique game is easy, while a "nearly satisfiable" one is conjectured to be hard;
- exactly how UGC relates to the number 0.878;
- what OpenAI's result means if true, and what it does **not** mean (it has nothing to do with P vs NP).

1. P vs NP and NP-hardness: the 5-minute version

1.1 "Easy to check" is not "easy to find"

Sudoku: given a filled-in grid, checking whether it's right takes seconds; filling it in from an empty grid can take a long time of trial and error.

- **P**: problems where you can **find** the answer in polynomial time (say n^2 or n^3 steps).
- **NP**: problems where, given a candidate answer, you can **check** whether it is right in polynomial time.

P vs NP asks: is everything that is easy to check also easy to find? Almost everyone believes **P $\neq$ NP**, but nobody has proved it (it is one of the Clay Millennium Problems).

1.2 Reductions and NP-hardness

Reduction: quickly translate any instance of problem A into an instance of problem B, so that B's answer is A's answer. If you can do this, then "B has a fast algorithm" implies "A has a fast algorithm", so B is at least as hard as A.

3SAT: given a bunch of clauses like " $(x_1 \text{ OR NOT } x_2 \text{ OR } x_5)$ ", each with 3 variables, can you assign true/false to the variables so that all clauses hold at once? The Cook–Levin theorem (1971) says every problem in NP can be reduced to 3SAT.

NP-hard: if 3SAT can be reduced to problem B, B is called NP-hard. Meaning: **unless P = NP, B has no polynomial-time algorithm.**

Every "X is NP-hard" in this chapter should be read as "unless P = NP, X cannot be done". None of them **proves** P $\neq$ NP; they just tie X's fate to P vs NP.

2. The fallback: approximation algorithms

2.1 Max-Cut

Given a graph, split the vertices into two groups (say, color them black and white). An edge is "cut" if its two ends have different colors. **Max-Cut** asks: what is the largest number of edges you can cut? Deciding "can you cut $\geq k$ edges" is one of Karp's 21 NP-complete problems from 1972.

Computed by script (brute-force enumeration of all colorings):

Graph	Edges	Max cut	Expected cut of random coloring	Random / optimal
Triangle	3	2	1.5	0.750
5-cycle (pentagon)	5	4	2.5	0.625
Petersen graph	15	12	7.5	0.625

Why a triangle can cut at most 2: three vertices and only two colors means two vertices must share a color, and the edge between them can't be cut.

2.2 Approximation ratio, and "1/2 for free"

An **α -approximation algorithm**: runs in polynomial time and always returns a solution worth at least " $\alpha \times$ the optimum".

Random coloring: each vertex flips an independent coin to pick its color. The two ends of an edge get different colors with probability $1/2$, so the expected cut is "half the edges". Since the optimum can't exceed the number of edges, random coloring is at least a **1/2-approximation**.

From 1976 (Sahni–Gonzalez) until 1995, $1/2$ was the best guarantee known.

[Check 1] What is the max cut of a 4-cycle (a square, 4 edges)? How many edges does a random coloring cut in expectation?

2.3 Another lead character: Vertex Cover

Pick the fewest vertices in a graph so that every edge has at least one endpoint picked (imagine posting guards at intersections to watch every road).

A simple algorithm known since the 1970s: repeatedly find an edge with neither end picked, and **pick both ends**. The optimal solution has to pick at least one end of that edge, and we picked both, so we use at most **2 times** the optimum. That is a **2-approximation**.

For 50 years, nobody has managed "1.99 times".

3. Goemans–Williamson: turning vertices into vectors

3.1 Relaxation: first solve a looser problem

Max-Cut is hard because each vertex can take only two values (black = $+1$, white = -1). An edge (i, j) is cut exactly when $x_i \cdot x_j = -1$, so the number of cut edges = $\sum (1 - x_i \cdot x_j)/2$.

The idea of Goemans–Williamson (1995): **replace each ± 1 with a unit vector v_i** , and replace the product $x_i \cdot x_j$ with the inner product $v_i \cdot v_j$:

maximize $\sum_{\text{edges}} (1 - v_i \cdot v_j)/2$ subject to: each v_i is a vector of length 1

This is called a **semidefinite program (SDP)**, and it can be solved (approximately) in polynomial time. Since ± 1 are "one-dimensional unit vectors", every solution of the original problem is also a solution of this one, so **the SDP optimum $\geq$ the true max cut**. Intuitively: the more "opposite" the two ends of an edge are pushed (the closer their angle is to 180°), the better.

3.2 Random hyperplane: turning vectors back into black and white

Once you have the vectors, pick a random plane through the origin, splitting space in two: color the points on one side black and the other side white.

Key fact: **two vectors at angle θ are separated by a random plane with probability exactly θ/π** . It's easiest to see in 2D: a random line through the origin falls "between" the two vectors with probability $\theta/180^\circ$.

Verified by script with 1,000,000 random planes:

Angle	Theory θ/π	Simulation
90°	0.5000	0.4994
120°	0.6667	0.6659
150°	0.8333	0.8337

3.3 Where 0.878 comes from

For each edge, compare two quantities:

- the "score" the SDP gives it: $(1 - \cos \theta)/2$;
- the probability it is actually cut after rounding: θ/π .

As long as, for **every** angle, the second is at least α times the first, the whole graph gets at least α times the SDP value, and so at least α times the optimum. So the guarantee = the minimum of this ratio over all angles:

$$\alpha_{\text{GW}} = \min \text{ over } \theta \text{ of } (\theta/\pi) / ((1 - \cos \theta)/2) = \min 2\theta / (\pi(1 - \cos \theta))$$

Computed by script (numerical minimization):

Angle θ	SDP score	Probability cut	Ratio
60°	0.250	0.333	1.333
90°	0.500	0.500	1.000
120°	0.750	0.667	0.889
133.56°	0.845	0.742	0.8786 (minimum)
150°	0.933	0.833	0.893
180°	1.000	1.000	1.000

$\alpha_{\text{GW}} = 0.878567\dots$, attained at $\theta \approx 133.56^\circ$ (inner product $p \approx -0.689$). This matches the 0.878567... given in Eq. (1.1) of OpenAI's Max-Cut paper.

3.4 An example: the pentagon

The optimal SDP solution for the 5-cycle places the 5 vectors in a plane, one every $144^\circ (= 4\pi/5)$. Computed by script:

- SDP value = $5 \times (1 - \cos 144^\circ)/2 = \mathbf{4.5225}$;
- true max cut = 4;
- expected cut of a random plane = $5 \times 144/180 = \mathbf{4.000}$.

So on the pentagon the SDP value is noticeably higher than the true optimum ($4/4.5225 = 0.8845$). This kind of "SDP is too optimistic" gap is called the **integrality gap**. Feige–Schechtman (2002) built graphs where this ratio gets arbitrarily close to 0.878, so **this SDP plus any rounding method can never beat 0.878**. But that only says "this one algorithm" has hit its limit, not that "all algorithms" have.

4. Approximation can be hard too: the PCP theorem

4.1 The intuition behind the PCP theorem

The **PCP theorem** (Arora–Safra; Arora–Lund–Motwani–Sudan–Szegedy, 1992 conference, 1998 journal; Dinur 2007 gave a combinatorial proof) says something that sounds like magic:

Any mathematical proof can be rewritten in a special format such that a referee who **randomly reads only a constant number of bits** of it can tell, with high confidence, whether the proof is right: a correct proof always passes; a wrong proof, no matter how it is written, gets caught with probability at least one half.

4.2 Why it has to do with "approximation"

Restate PCP: there is a reduction starting from 3SAT that produces a constraint system such that

- formula satisfiable $\Rightarrow$ the constraint system can be **fully** satisfied;
- formula unsatisfiable $\Rightarrow$ the constraint system can satisfy **at most** a fixed fraction $s < 1$.

This is called a **gap**. Now suppose some algorithm guarantees an approximation ratio $\alpha > s$. Then you could use it to tell the two cases apart, and so solve 3SAT. Therefore: **a gap $\Rightarrow$ approximating above the gap is NP-hard**.

The game for the following decades was to make this gap "sharper and sharper", until it sits exactly at the position of a known algorithm.

4.3 The gap for Max-Cut

Håstad (2001, *JACM*), together with the "gadgets" of Trevisan–Sorkin–Sudan–Williamson, proved: **approximating Max-Cut above $16/17 \approx 0.941$ is NP-hard.**

So:



Who owns the region between 0.878 and 0.941? A better algorithm, or a stronger hardness proof? This is what UGC is meant to answer. (FIG_MAXCUT in the deck draws exactly this number line.)

Vertex cover has a similar gap: the algorithm achieves 2, and the NP-hardness lower bound was Dinur–Safra's **1.3606** (2005, *Annals*), raised in 2018 to $\sqrt{2} \approx 1.414$ (Khot–Minzer–Safra, see §7.2).

5. What Unique Games are

5.1 Labeling games

A **label constraint system** has:

- a graph;
- an "alphabet": each vertex picks a label from $\{0, 1, \dots, k-1\}$;
- a constraint on each edge, saying which labels at the two ends count as "compatible".

The goal: pick labels so that the fraction of satisfied constraints (called the instance's **value**) is as high as possible.

"Label Cover", the standard output of the PCP theorem, has exactly this form, except that its constraints are "many-to-one": once one end is fixed, the other end has several choices.

5.2 "Unique": fix one end and the other is determined

Unique Game: every edge constraint is a **permutation** (a one-to-one correspondence). Once one end picks a label, there is **exactly one** label at the other end that satisfies the edge.

The most common example, and the one this chapter uses: **difference equations mod k**

$$x_i - x_j \equiv c \pmod{k}$$

Given x_j , x_i is uniquely determined as $x_j + c$. (OpenAI's paper uses "translation constraints" $a(v) = a(u) + c$, which just replace numbers mod k with bitwise XOR on 0/1 vectors; it's essentially the same.)

5.3 By hand: a triangle, mod 3

Three variables a, b, c , labels $\{0, 1, 2\}$:

Instance A: $b - a \equiv 1, c - b \equiv 1, a - c \equiv 1 \pmod{3}$. Add them up around the loop: $0 \equiv 3 \equiv 0$, no contradiction. Take $a = 0$; propagating gives $b = 1, c = 2$; check $a - c = -2 \equiv 1 \checkmark$.

Everything is satisfied; value = 1.

Instance B: change the last one to $a - c \equiv 2$. Add them up around the loop: $0 \equiv 1 + 1 + 2 = 4 \equiv 1$, a contradiction. So the three can't all hold at once, but dropping any one of them makes the rest satisfiable. **Value = 2/3.** (Confirmed by the script by brute-force enumeration of all 27 labelings.)

The general rule: **on a cycle, add up the constraint constants around the loop; if the sum is not 0 (mod k), at least one constraint on that cycle must be violated.**

[Check 2] Four variables $x_0 \dots x_3$, mod 5, constraints: $x_1 - x_0 \equiv 2, x_2 - x_1 \equiv 3, x_3 - x_2 \equiv 1, x_0 - x_3 \equiv 4, x_2 - x_0 \equiv 0$. Can they all be satisfied? What if the last one is changed to $x_2 - x_0 \equiv 1$?

5.4 Why "fully satisfiable" is easy

If a unique game can be **fully** satisfied, there is a simple polynomial-time algorithm:

1. in each connected component, pick any vertex and try one of its k labels;
2. propagate along the edges: each edge uniquely determines the neighbor's label;
3. on hitting a contradiction, move to the next starting label; if all k fail, it can't be fully satisfied.

Each attempt is one pass over the graph, k attempts in total. Compare brute-force enumeration: 100 variables mod 3 means trying $3^{100} \approx 10^{47.7}$ possibilities (computed by script).

This is completely different from 3SAT: even when a 3SAT formula is "fully satisfiable", finding a solution is hard. **So UGC can't be about "satisfiable vs unsatisfiable"; it has to be about "nearly satisfiable vs far from satisfiable".** That is where the $1 - \epsilon$ in UGC comes from.

5.5 Why "nearly satisfiable" might be hard

Once a few constraints are "wrong", propagation carries the error a long way. The script ran an experiment: a random graph with 2000 vertices and on average 6 edges per vertex, mod

7; first plant a reference answer, then randomly tamper with a fraction of the constraints; then assign labels by propagation (breadth-first) from random starting points, trying 20 starting points:

Fraction tampered	Value of planted answer	Propagation, average value	Propagation, best value
0%	1.000	1.000	1.000
1%	0.990	0.958	0.982
5%	0.947	0.728	0.845
10%	0.901	0.610	0.709

(Computed by script; the random graph has a very small number of isolated vertices, whose label defaults to 0; this doesn't affect the conclusion.)

The reason: a vertex's label is "passed along" a path from the starting point, and as soon as the path crosses one bad edge, everything downstream is shifted.

To be honest: this experiment only shows that "propagation" fails, not that the problem itself is hard. On **random** noisy instances like these, smarter algorithms (such as SDP) actually do very well. What UGC says is: there **exist** carefully constructed instances where no polynomial-time algorithm can tell "it is 99% satisfiable" from "it is only 1% satisfiable".

5.6 The precise statement of UGC

Unique Games Conjecture (Khot 2002): for every $\epsilon > 0$ and $\delta > 0$, there is an alphabet size k such that, for unique games with alphabet k , it is NP-hard to tell apart the following two cases:

- YES: value $\geq 1 - \epsilon$ (nearly all satisfiable);
- NO: value $\leq \delta$ (nearly all unsatisfiable).

Note the **order of quantifiers**: first ϵ and δ are given, **then** the alphabet size k is chosen. If you flip it and let k grow with the instance, there are known algorithms that solve it (the SDP algorithm of Charikar–Makarychev–Makarychev 2006 gets about $1 - O(\sqrt{\epsilon \log k})$, which is very good when k is small). §1.2 of OpenAI's paper discusses this point specifically.

6. Why UGC matters: one conjecture governs a whole territory

The appeal of UGC: assuming it holds, the **exact** approximation limits of many problems can be computed, and they often **exactly equal some known algorithm**.

Problem	Best algorithm	Unconditional NP-hardness (before 2026)	Assuming UGC
Max-Cut	0.878 (GW 1995)	16/17 $\approx$ 0.941 (Håstad 2001)	0.878 is the limit (KKMO 2004/07)
Vertex cover	2	$\sqrt{2} \approx 1.414$ (KMS 2018)	2 is the limit (Khot–Regev 2003/08)
All constraint satisfaction problems (CSPs)	Basic SDP + rounding	Varies	Basic SDP is the optimal algorithm (Raghavendra 2008)

A few key results:

- **KKMO** (Khot–Kindler–Mossel–O'Donnell, FOCS 2004, *SICOMP* 2007): UGC $\Rightarrow$ beating α_{GW} on Max-Cut is NP-hard. A probability statement they needed, "Majority Is Stablest", was later proved by Mossel–O'Donnell–Oleszkiewicz (2005, *Annals* 2010). The odd 133.56° angle in the GW algorithm corresponds exactly to the "noise" parameter in the hardness reduction: the algorithm and the hardness meet at the same number.
- **Khot–Regev** (CCC 2003, *JCSS* 2008): UGC $\Rightarrow$ vertex cover can't achieve $2 - \epsilon$. The clumsy 1970s "pick both ends" algorithm is optimal.
- **Raghavendra** (STOC 2008): UGC $\Rightarrow$ for **every** CSP, the same generic SDP algorithm is the optimal polynomial-time approximation algorithm. One algorithm plus one conjecture settles an entire class of problems.

This is why UGC is called "the map of the field of approximation algorithms". Khot received the 2014 Nevanlinna Prize for it.

7. A 24-year tug-of-war: support and doubt

7.1 Doubt: a subexponential-time algorithm (2010)

Arora–Barak–Steurer (FOCS 2010, *JACM* 2015) gave an algorithm that, for a unique game with alphabet k , n variables and value $\geq 1 - \epsilon^c$ (c a fixed constant), finds a labeling satisfying $1 - \epsilon$ in **$\exp(k \cdot n^\epsilon)$** time.

Why did this make people doubt UGC? Because 3SAT is generally believed to need close to 2^n time, while UG needs only 2 to "a small power of n ". Comparison computed by script:

n	3^n (brute force)	$2^{(n^{0.1})}$	$2^{(n^{1/3})}$	n^3
100	$10^{47.7}$	3	$10^{1.4}$	10^6
1,000	10^{477}	4.0	$10^{3.0}$	10^9
1,000,000	10^{477121}	15.8	$10^{30.1}$	10^{18}

If UGC holds, the reduction from 3SAT to UG must **blow up** the instance to a very large polynomial size (otherwise 3SAT would also have a subexponential algorithm). That is not a contradiction, but it shows that even if UG is hard, it is "not thoroughly hard". Some researchers leaned toward believing UGC is false because of this.

7.2 Support: the 2-to-2 theorem (2018)

2-to-2 constraints: a bit looser than unique; once one end is fixed, the other end has 2 allowed labels. A series of works by Khot–Minzer–Safra and Dinur–Khot–Kindler–Minzer–Safra in 2017–2018 (the last piece being the KMS FOCs 2018 theorem on expansion in Grassmann graphs) proved: **nearly satisfiable 2-to-2 games are NP-hard.**

Corollaries:

- Unique games with **completeness about $1/2$** are NP-hard: telling " $\text{value} \geq 1/2 - \epsilon$ " from " $\text{value} \leq \delta$ ". This has been called "half of UGC".
- Vertex cover to within $\sqrt{2} - \epsilon$ is NP-hard.

What remained hard: pushing completeness from $1/2$ to $1 - \epsilon$. §1.3 of OpenAI's paper also says: "Obtaining Unique Games completeness arbitrarily close to one was a separate problem."

7.3 September 2026: a race

According to Quanta (2026-10-07), MIT's Dor Minzer and his students Yumou Fei and Shuo Wang, after hearing rumors that OpenAI was about to release a proof of UGC, rushed out a proof of a 4-to-1 variant of Khot's 2-to-1 conjecture on 2026-09-14. O'Donnell's comment: "They solved the problem in the old-fashioned way, with their minds, and wrote it with their own fingers." [R]

8. What OpenAI claims

8.1 The main result

The abstract of the paper *The Unique Games Theorem* (2026-09-23, 58 pages, author listed as "OpenAI"):

"We prove the Unique Games Conjecture. For every fixed $\epsilon, \delta \in (0, 1/2)$, we give a deterministic polynomial-time reduction from 3SAT to Unique Games over a fixed finite alphabet, with completeness at least $1 - \epsilon$ and soundness at most δ ."

In plain words: **for any fixed ϵ, δ , there is a deterministic polynomial-time reduction that translates 3SAT into unique games with a fixed alphabet: satisfiable formulas become instances with value $\geq 1 - \epsilon$, and unsatisfiable ones become instances with value $\leq \delta$.** This is exactly the statement of UGC from §5.6.

Theorem 1.1 also states a stronger form: the alphabet is the 0/1 vector space F_2^s , every constraint is a translation, and the graph is simple, bipartite and unweighted. The paper says this is a known equivalent form of UGC (citing KKM0), so "stronger" is not "swapping in a different problem".

8.2 Other papers in the same family (family 102)

Paper	Pages	Conclusion
A Direct Proof of Optimal Max-Cut Hardness	38	Beating α_{GW} on Max-Cut is NP-hard
The Factor-Two Hardness Threshold for Vertex Cover	25	Vertex cover can't be approximated to any constant factor < 2
Constant-factor hardness of Min-UnCut	—	Every constant factor is NP-hard
Constant-factor hardness of directed feedback vertex set	—	Every constant factor is NP-hard

The repo's CONTENTS.md says these 4 papers are **independent direct reductions** that "use established PCP and Label Cover hardness results" and don't depend on the UGC paper. In other words, even if the UGC paper has an error, its two most famous consequences, Max-Cut 0.878 and vertex cover 2, have a separate independent path. This is what Aaronson's blog means when it says "UGC isn't actually necessary for its most famous consequences" [R].

In addition, **family 105** (*Perfect completeness for 2-to-1 games*) claims a proof of Khot's 2-to-1 conjecture (the perfect-completeness version). Footnote 1 of the UGC paper says it is "not an input to this proof".

8.3 Intuition for the method (I read only the introduction; this is a simplified version)

1. **Starting point:** Håstad's hardness result for "nearly satisfiable systems of parity equations" (each equation $x + y + z \equiv b \pmod{2}$).
2. **Why the 2-to-2 road got stuck at 1/2:** in the "matrix short code" of Barak–Kothari–Steurer, the probability that an honest proof's answer stays unchanged under one random rank-one perturbation is $(1 + 2^{(-\ell)})/2$, i.e. about 1/2. So completeness could only reach 1/2.
3. **The key new idea: a different kind of noise.** The paper constructs a nonlinear "encoder" C and a special noise μ : honest answers barely change under this noise (change probability $\leq$ an arbitrarily small p), but any "cheating", high-rank linear observation still notices the noise with probability at least 1/8. An analogy: design a jitter that barely affects correct readings but exposes forged ones. This lets completeness be pushed to $1 - \epsilon$ while keeping the structural tools that the 2-to-2 approach uses for soundness.

4. **Soundness (unsatisfiable formulas have value $\leq \delta$):** reuses the Grassmann expansion theorem of KMS 2018 (in the form of an "inverse short code theorem") to extract a strategy, then uses parallel repetition (Dinur–Steurer 2014) to push the success rate down to $\exp(-c \cdot k^{1/3})$, giving a contradiction.
5. **Finishing up:** remove weights and repeated edges to get a simple, unweighted, bipartite instance.

9. How far it has been verified

9.1 The Lean statement: much longer than the quasi-Riemann hypothesis, but still readable

The Lean statement of the quasi-Riemann hypothesis is a single line, because Mathlib already has the ζ function. UGC is different: Mathlib has **no** ready-made definitions of "unique game", "3SAT encoding" or "polynomial-time reduction to a graph", so the challenge file `ComparatorChallenges/UniqueGamesTheorem.lean` (239 lines) has to define all of these itself. The core theorem:

```
theorem theorem11 (ε δ : ℝ)
  (hε : 0 < ε) (hεhalf : ε < 1 / 2)
  (hδ : 0 < δ) (hδhalf : δ < 1 / 2) :
  Nonempty (Explicit.MachineOutputContract.BinaryGapReduction ε δ)
```

Word by word:

- `(ε δ : ℝ)`: ε and δ are real numbers.
- `hε`, `hδ`: assume $0 < \varepsilon < 1/2$ and $0 < \delta < 1/2$.
- `Nonempty (BinaryGapReduction ε δ)`: the conclusion is "there **exists** a reduction satisfying all of the conditions below".

`BinaryGapReduction ε δ` is a structure that requires all of the following at once:

Field	Meaning
<code>alphabet</code> , <code>dimension</code> , <code>coordinates</code>	Alphabet size $k \geq 2$, identified with F_2^s (0/1 vectors)
<code>construct : List Bool → Instance alphabet</code>	Takes a string of bits (the encoding of a 3SAT formula) and outputs a unique game
<code>simpleBipartite</code>	The output graph is a simple bipartite graph
<code>translations</code>	Every constraint is a translation $a(v) = a(u) + c$
<code>computation : Turing.TM2ComputableInPolyTime ...</code>	<code>construct</code> can be computed in polynomial time by Mathlib's multi-tape Turing machine
<code>finiteAlphabet</code>	The Turing machine's tape alphabet is finite
<code>completeness</code>	Input is a satisfiable formula $\Rightarrow$ there is a labeling satisfying a fraction $\geq 1 - \epsilon$
<code>soundness</code>	Input is not a satisfiable formula $\Rightarrow$ every labeling satisfies a fraction $\leq \delta$

A few details worth noting:

- Permutations are represented as "forward table + inverse table + two proofs that they are inverses", so they really are one-to-one.
- Repeated edges are counted by multiplicity; there are no hidden weights.
- If the input string can't even be parsed as a formula, it counts as "unsatisfiable", and the reduction must still output an instance with value $\leq \delta$. This is a stricter requirement, not a loophole.
- Restricting $\epsilon, \delta < 1/2$ doesn't affect the conclusion: UGC is precisely about small ϵ, δ .
- Completeness is written as $\geq 1 - \epsilon$ rather than $= 1$, which is correct: as §5.4 said, the fully satisfiable case is easy.

My judgment: this is a faithful encoding of UGC, in a stronger form. One outside fact the reader has to supply: the theorem says "reduces from 3SAT"; "therefore NP-hard" also needs the Cook–Levin theorem (3SAT is NP-complete). That step is not in this statement, but it is a textbook result and uncontroversial.

9.2 The proof itself

- Solution directory `OAI/Computability/UniqueGames/`: **478 Lean files, 186,782 lines** (counted by script).
- grep for `sorry`, `axiom`, `admit`, `native_decide`, `implemented_by`, `unsafe`: 0 hits. The Comparator config allows only three standard axioms (`propext`, `Quot.sound`, `Classical.choice`).

- Apart from Mathlib, the proof imports nothing from other directories in the repo. Since Mathlib has no PCP theorem, **the PCP theorem, Håstad's result, KMS's Grassmann expansion theorem, parallel repetition, and the polynomial-time Turing machine implementation** should all have been formalized from scratch within these 478 files. The subdirectory names (`PCP/` with 104 files, `Inverse/` with 39 files including `KMS...`, `Repetition/`, `Machines/` with 144 files) are consistent with this. $\triangle$ This is inferred from the directory structure and import relations; I did not open each one to check.
- The direct proofs for Max-Cut and vertex cover also have Lean: `MaxCut/` has 372 files, 186,763 lines; `VertexCover/` has 354 files, 66,335 lines. In the Max-Cut statement, α_{GW} is defined directly as the `sInf` of $2 \cdot \arccos \rho / (\pi(1 - \rho))$, which is exactly the formula from §3.3 ($\arccos \rho = \theta$).
- **What wasn't done:** I did not compile it. The Comparator config for the UGC family has `enable_nanoda: false`, meaning no second, independent kernel re-check. Unlike the quasi-Riemann hypothesis, I found no third party who has publicly re-run this proof $\triangle$. The repo's `formalization.yaml` itself says `review: status: unchecked`.

9.3 Human experts

- Lance Fortnow (blog post "Open No More", 2026-10-07): "if they hold up, we've seen more progress in TCS in the last 24 hours than in the previous three decades combined." [R]
- Mark Braverman (Princeton, Quanta 2026-10-07): "Math by press release is not that healthy for math." [R]
- Quanta's report says this proof "didn't undergo any human editing or review by independent experts". [R]
- Dana Moshkovitz criticized the paper as extremely hard to read (as relayed by Aaronson's blog and Implicator) $\triangle$ secondhand quote; original wording not verified.
- Khot himself: as of 2026-10-08 I found no public statement $\triangle$.
- **As of 2026-10-08, no human expert has publicly said they have read through and verified these 58 pages.**

9.4 Remaining risks

1. **Statement level:** I read it line by line and found no problems, but I am not a complexity theory expert. The things most worth an expert's re-check are the use of `TM2ComputableInPolyTime` (whether the input and output encodings are reasonable, and whether the polynomial really is independent of the input), and whether the 3SAT encoding has a loophole that makes the problem easier.
2. **Compilation level:** depends on third parties or re-running it yourself. No nanoda re-check.

3. **Understanding level:** this is the same situation as the quasi-Riemann hypothesis chapter. It may already be proved in the machine sense, but nobody can explain "why that nonlinear noise works". The paper is very hard to read, which will slow down human understanding.

10. What it means if true (and what it doesn't)

10.1 For theoretical computer science

1. **A large batch of conditional results become theorems:** Max-Cut's 0.878, vertex cover's 2, Raghavendra's "SDP is optimal for every CSP", and so on. The premise is removed from the many papers of the past 20 years that say "assuming UGC, then...".
2. **The map of approximation algorithms is basically complete:** for many problems, the gap between "achievable" and "not achievable" is closed. The 1995 GW algorithm is proved optimal.
3. **The subexponential algorithm is no longer a counterexample:** UGC being true and ABS's $\exp(n^\epsilon)$ algorithm coexist. This means the reduction from 3SAT to UG must produce an extremely large polynomial (the paper says the degree and constants of the polynomial depend only on ϵ , δ , but gives no concrete numbers).

10.2 What it does not mean

- **It does not resolve P vs NP.** Every conclusion is "X is NP-hard", i.e. "unless $P = NP$, X cannot be done". If someone proved $P = NP$ tomorrow, all these hardness results would become void. Fortnow says in the same blog post that these results don't bring us closer to resolving P vs NP [R].
- **It does not affect practical engineering.** Industry uses heuristics for graph partitioning and clustering, and on real data they often do far better than the worst-case 0.878. UGC is about **guarantees** in the **worst case**.
- **It does not overturn any known algorithm.** It only proves that "existing algorithms can't be improved further".

10.3 For AI

- This is the result in OpenAI's batch **closest to AI researchers' daily work:** its core objects (SDP relaxations, randomized rounding, PCP-style verification) are basic tools of computer science.
- An association (**my speculation, not an established result**): the complexity-theoretic foundation of debate-style scalable oversight (Irving–Christiano–Amodei 2018, arXiv:1805.00899) is interactive proofs and PCP: "can a weak verifier, looking at only a little information, judge a strong prover's argument?" UGC characterizes the boundary where "a little noise makes efficient distinction impossible". For designing "weak overseer + strong

model" protocols, it suggests there may be an essential computational-hardness difference between **perfect agreement** and **near agreement**.

- As with the quasi-Riemann hypothesis, we again have the combination "a hard-to-read 58-page paper + 187,000 lines of Lean". The bottleneck of trust moves from "is the proof right" to "**was the statement encoded correctly**", which is exactly the spec review that §9.1 does.

11. Self-test

Try answering these after reading. If you can't, go back and reread the relevant section.

1. What exactly does "X is NP-hard" mean? Does it prove $P \neq NP$? (§1.2)
2. Why is random coloring at least a $1/2$ -approximation? (§2.2)
3. In the GW algorithm, 0.878 is the minimum of the ratio of which two quantities, and where is it attained? (§3.3)
4. Why does the "gap" in the PCP theorem imply hardness of approximation? (§4.2)
5. Why can a fully satisfiable unique game be solved in polynomial time, while 3SAT can't? (§5.4)
6. Why does the order "first ϵ , δ , then the alphabet size" in UGC matter? (§5.6)
7. What has the 2-to-2 theorem already proved, and what was still missing? (§7.2)
8. What does the **soundness** field in the Lean statement say? What happens if the input string isn't a valid formula encoding at all? (§9.1)
9. Why might the optimality of Max-Cut 0.878 still hold even if the UGC paper has an error? (§8.2)

Appendix: answers to the checks

- **Check 1:** A 4-cycle is bipartite; color it alternately black and white and all 4 edges are cut, so $\max \text{cut} = 4$. A random coloring cuts $4 \times 1/2 = 2$ edges in expectation.
- **Check 2:** The first four constraints form the cycle $x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_0$, with constant sum $2 + 3 + 1 + 4 = 10 \equiv 0 \pmod{5}$, so no contradiction. The fifth, $x_2 - x_0 \equiv 0$: along the cycle, $x_2 - x_0 = 2 + 3 = 5 \equiv 0 \pmod{5}$. So everything can be satisfied, e.g. $(x_0, x_1, x_2, x_3) = (0, 2, 0, 1)$. After changing it to $x_2 - x_0 \equiv 1$, the cycle $x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow x_0$ sums to $2 + 3 - 1 = 4 \not\equiv 0 \pmod{5}$, so at least one constraint is violated, and at most $4/5$ can be satisfied (confirmed by the script by brute-force enumeration of all 625 labelings).

Appendix: Glossary

Term	One-line explanation
P / NP	Problems where you can quickly find the answer / problems where you can quickly check an answer
Reduction	Quickly translating problem A into problem B, so that a fast algorithm for B solves A
NP-hard	3SAT reduces to it; unless $P = NP$, it has no polynomial-time algorithm
3SAT	Boolean satisfiability with 3 variables per clause
Max-Cut	Split the vertices into two groups to maximize the edges crossing between groups
Vertex cover	Pick the fewest vertices that cover every edge
α -approximation	Always reaches α times the optimum in polynomial time
Semidefinite program (SDP)	An optimization problem that relaxes ± 1 variables to unit vectors; solvable in polynomial time
Random hyperplane rounding	Make one random cut to turn vectors back into black and white
α_{GW}	0.878567..., the guarantee of the GW algorithm
Integrality gap	The ratio gap between the relaxation's optimum and the true optimum
PCP theorem	Proofs can be rewritten in a form that can be verified by randomly spot-checking a constant number of bits
Gap	YES instances have value $\geq c$, NO instances have value $\leq s$, with $c > s$
Completeness / soundness	The fraction satisfiable in the YES case / the most that can be satisfied in the NO case
Label Cover	The standard output of the PCP theorem: a labeling problem with "many-to-one" constraints on each edge
Unique Game	A labeling problem where each edge constraint is a permutation (fix one end and the other is unique)
UGC	Telling apart unique games with value $\geq 1 - \epsilon$ from those with value $\leq \delta$ is NP-hard
2-to-2 theorem	"Half of UGC" (completeness about 1/2), proved in 2018
Subexponential time	Faster than any $2^{(n^c)}$ (for fixed $c > 0$), slower than polynomial

Appendix: Further reading

- **Ryan O'Donnell**, *Analysis of Boolean Functions* (Cambridge University Press, 2014; free online version <https://analysisofbooleanfunctions.net/>): Chapter 7 covers PCP and Håstad, Chapter 11 covers "Majority Is Stablest" and KKMO. The best textbook for understanding the UGC hardness proofs. ⚠ Chapter numbers from memory; go by the table of contents.

- **Williamson & Shmoys**, *The Design of Approximation Algorithms* (Cambridge, 2011; free PDF <http://www.designofapproxalgs.com/>): Chapter 6 covers the GW algorithm, Chapter 16 covers hardness of approximation and UGC. ⚠ Chapter numbers from memory.
- **Subhash Khot**, "On the Unique Games Conjecture" (survey, CCC 2010): a survey by the person who proposed it. <https://cs.nyu.edu/~khot/papers/UGCSurvey.pdf>
- **Erica Klarreich**, "Approximately Hard: The Unique Games Conjecture" (Simons Foundation, 2011): popular science for a general audience.
- **Quanta**, "As AI Closed In on 'Unique Games' Proof, Researchers Raced to Beat the Machines" (2026-10-07): <https://www.quantamagazine.org/as-ai-closed-in-on-unique-games-proof-researchers-raced-to-beat-the-machines-20261007/>
- **Lance Fortnow**, "Open No More" (2026-10-07): <https://blog.computationalcomplexity.org/>
- **Scott Aaronson**, "The Mathocalypse" (2026-10-07): <https://scottaaronson.blog/?p=10169>
- Classic original references: Goemans–Williamson, *JACM* 42 (1995); Håstad, *JACM* 48 (2001); Khot, STOC 2002; KKMO, *SICOMP* 37 (2007); Khot–Regev, *JCSS* 74 (2008); Raghavendra, STOC 2008; Arora–Barak–Steurer, FOCS 2010 / *JACM* 62 (2015); Khot–Minzer–Safra, FOCS 2018.
- Original paper: <https://github.com/openai/math/blob/main/preprints/The-Unique-Games-Theorem-September-23-2026/paper.pdf>
- Direct Max-Cut proof: <https://github.com/openai/math/blob/main/preprints/A-Direct-Proof-of-Optimal-Max-Cut-Hardness-September-23-2026/paper.pdf>
- Lean statement:
<https://github.com/openai/math/blob/main/lean/ComparatorChallenges/UniqueGamesTheorem.lean>

PART III · EXPLAINED FROM ZERO · GOES WITH §8

27. Explainer 05: $L = RL = BPL$ (in small-memory computation, randomness doesn't help)

This is the "explained from zero" version of §8 of the deck in

[notes/openai_math_explained/](#). You only need these prerequisites: you know what $\log_2$ is, you know the phrase "an algorithm's running time grows with the input size n ", and you have seen matrix multiplication. Every new concept is explained before it is used. Every number in the text is either computed by script or has a source (scripts: [精讲/scripts/05_bpl_toy.py](#), [05_bpl_spec.py](#), implemented in numpy). The **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Tags: **[Established]** = a published, peer-reviewed result; **[OpenAI claim]** = a manuscript in

the openai/math repository; **[Speculation]** = my own judgment; $\triangle$ = something I could not fully verify. Estimated reading time: 60–90 minutes. You can read it in three sittings: Sections 1–3 (memory complexity and randomness), Sections 4–6 (random walks, Reingold, the history of derandomization), Sections 7–9 (OpenAI's result, its verification status, and what it means).

0. The bottom line: the one thing this chapter explains

Picture a machine with tiny memory: it can read the input as much as it likes, but its working memory only holds **a few pointers**. If this machine can flip coins, can it solve more problems?

This chapter explains the following causal chain:

The whole computation of a small-memory machine $\Leftrightarrow$ a random walk on a "configuration graph" $\Leftrightarrow$ acceptance probability = a sum of powers of a large matrix $\Rightarrow$ derandomization = computing this probability accurately with very little memory $\Rightarrow$ the difficulty is that the precision and the recursion depth needed to "compute it accurately" eat up the memory.

L is the set of problems that deterministic small-memory computation can solve. **BPL** is the set of problems solvable with "small memory + coin flips + a small allowed probability of error". Almost everyone believes $L = BPL$, but since 1999 the best unconditional result has only reached "BPL can be simulated deterministically with $(\log n)^{1.5}$ memory". That has always been a factor of $\sqrt{\log n}$ away from $\log n$.

OpenAI claims to have closed this gap completely: **$L = RL = BPL$** . Few people ever doubted the conclusion itself; the hard part is the proof. It comes with a Lean formalization, but as of 2026-10-08 no human expert has publicly said they have read and verified this 108-page paper.

After reading this chapter, you should be able to explain in your own words:

- why " $O(\log n)$ bits of memory" roughly equals "can store a constant number of pointers", and why such a machine must halt in polynomial time;
- why a random walk can solve undirected graph connectivity with tiny memory, and why the same method fails on directed graphs;
- why "seed length" is the central measure in this field, and what separates Nisan's $\log^2 n$ from the target $\log n$;
- what OpenAI's result would mean if true, and what it does **not** mean (it does not mean $P = BPP$, and it does not mean $L = NL$).

1. Space complexity: memory is a resource

1.1 Time and space

There are two rulers for measuring an algorithm:

- **Time:** how many steps it takes.
- **Space:** the maximum number of bits of **working memory** it uses at any one moment.

There is a key convention here: **the input does not count toward space**. The input sits on a "read-only tape"; the machine can read it back and forth but cannot write to it. Only the "work tape" (read-write scratch paper) counts as space. This is what lets us talk about "memory smaller than the input".

The engineering analogy is **streaming plus random access**: the data sits on a big read-only disk, you can seek to any position and read it, but your local RAM is only a few dozen bytes.

1.2 Logarithmic space L: room for a constant number of pointers

L (logspace) = the problems that a deterministic machine can decide with $O(\log n)$ bits of working memory.

Why $\log n$? Because a pointer that can point to any position in the input needs $\lceil \log_2 n \rceil$ bits. $O(\log n)$ bits means "a constant number of pointers or counters".

Input length n	$\log_2 n$ (one pointer)	$(\log_2 n)^{1.5}$	$(\log_2 n)^2$
10^3	10	31	99
10^6	20	89	397
10^9	30	163	894
10^{12}	40	252	1,589

(Computed by script; the unit is bits.) With an input of 10^9 bytes, **L** allows only tens to hundreds of bits of memory, about the size of a few registers. The $(\log n)^2$ and $(\log n)^{1.5}$ columns will come up again and again; they are the two milestones in the history of derandomization.

[Check 1] The input is a graph with 10^9 vertices. How many bits does it take to store one vertex ID? How many bits in total, roughly, to store "current vertex + target vertex + a step counter (counting up to at most n^3)"?

1.3 An important fact: a deterministic logspace machine must halt in polynomial time

The machine's complete state at a given moment is called its **configuration**: control state + the positions of all read/write heads + the contents of the work tape. The work tape has only

$c \cdot \log n$ bits, so it has at most $2^{(c \cdot \log n)} = n^c$ possible contents; the read head has n possible positions; the control state has a constant number of possibilities. So **the total number of configurations is polynomial in n** .

If a deterministic machine halts, it can never pass through the same configuration twice (otherwise it would loop forever). So its number of steps is at most the number of configurations, which means polynomial time.

This fact that "the number of configurations is polynomial" is the cornerstone of the whole chapter. Everything that follows can be seen as working on a polynomial-size "configuration graph".

2. What randomness can buy

2.1 Example: polynomial identity testing

You are given two polynomials written as formulas and asked whether they are identical. For example, $(x+y)^2$ and $x^2 + 2xy + y^2$.

You can settle this example by expanding by hand. The hard case is when the polynomial is written as a very compact formula or circuit that would have exponentially many terms once expanded (for example, an $n \times n$ determinant has $n!$ terms when expanded). In that case no fast **deterministic** algorithm is known to this day. This is the **PIT problem** (polynomial identity testing). [Established]

The randomized algorithm is simple: **plug in a few random numbers and check whether the two sides are equal.**

- If the polynomials are identical, any input gives equal values.
- If they are not identical, the difference of the two sides is a nonzero polynomial, and it has few roots. The Schwartz–Zippel lemma says: if you pick values at random from a set of size $|S|$, the probability that a nonzero polynomial of degree d evaluates to 0 is $\leq d/|S|$.

[Established]

Worked example: the false "identity" $(x+y)^2 = x^2 + y^2$. The difference of the two sides is $2xy$, a degree-2 polynomial. Pick x and y at random from $\{0, \dots, 99\}$: 10,000 combinations in total.

[Check 2] How many combinations make the two sides exactly equal (that is, "fool" the random test)?

Computed by script: 199, or 1.99%. The Schwartz–Zippel bound is $d/|S| = 2/100 = 2\%$, which matches. With one random substitution, the chance of being fooled is under 2%; with 10 independent substitutions, it is under $2\%^{10} \approx 10^{-17}$.

2.2 P and BPP

- **P**: problems solvable in deterministic polynomial time.
- **BPP**: problems solvable in polynomial time + random bits, with error probability $\leq 1/3$ on every input.

The number $1/3$ does not matter, because you can **amplify**: run independently k times and take a majority vote.

Repetitions k	Majority-vote error probability (single-run success $2/3$)	Bits needed for the counter
1	3.3×10^{-1}	1
11	1.2×10^{-1}	4
51	6.9×10^{-3}	6
101	2.7×10^{-4}	7
201	5.5×10^{-7}	8

(Computed by script, summing binomial tails.) Note the last column: amplification only needs a counter, **so it works for small-memory machines too**. That is why it doesn't matter whether the definition of BPL uses a threshold of $2/3$ or 0.51 .

The mainstream conjecture is $P = BPP$, meaning randomness gives no fundamental advantage. The evidence is strong: Impagliazzo–Wigderson 1997 proved that as long as certain problems require exponential-size circuits (a widely believed "hardness assumption"), $P = BPP$. [Established] But proving $P = BPP$ **unconditionally** is considered extremely hard: Kabanets–Impagliazzo 2004 proved that even just derandomizing PIT would imply certain circuit lower bounds that we currently have no idea how to prove. [Established]

This is the "hardness vs. randomness" framework: **derandomization and proving lower bounds are almost the same thing**. The time version is blocked by this wall. The space version has no known barrier of the same kind, which is why $L = BPL$ has long been seen as "the derandomization conjecture most likely to be proved unconditionally".

3. RL and BPL: small memory + coin flips

3.1 Definitions

The machine gains one ability: at each step it can get a fresh fair coin. The constraints are:

- $O(\log n)$ bits of working memory;
- it halts in polynomial time for **all** coin sequences;

- **coins are single-use:** once read, a coin is gone. To use it again, the machine must store it on its work tape itself, and the work tape has only $\log n$ bits.

The last constraint is crucial. The time version, BPP, can store the random string and reread it as often as it likes; a small-memory machine cannot. It can only "use the coins as it flips them", like an online, single-pass stream.

Two ways of making errors:

- **RL** (one-sided error): when the answer is "no", it always rejects; when the answer is "yes", it accepts with probability at least $1/2$.
- **BPL** (two-sided error): when the answer is "yes", it accepts with probability at least $2/3$; when the answer is "no", it accepts with probability at most $1/3$.

Known inclusions [Established]:

$L \subseteq RL \subseteq BPL \subseteq L^{(3/2)}$ (Saks–Zhou 1999)
 $RL \subseteq NL \subseteq L^2$ (Savitch 1970)

NL is "nondeterministic" logspace (the machine can "guess"). Reachability in directed graphs is the representative NL problem.

3.2 The whole computation = a random walk on a graph

Use the fact from Section 1.3: a randomized machine has polynomially many configurations on input x . Draw each configuration as a node; if flipping one coin can take configuration u to configuration v , draw an edge from u to v with weight $1/2$.

So **the machine's run on x = a random walk on this configuration graph, starting from the start node; the acceptance probability = the probability that the walk reaches an "accept" configuration within the time limit.**

Worked example: a toy machine with only 3 states, flipping one coin per step:

- State 0: heads goes to 0, tails goes to 1
- State 1: heads goes to 2, tails goes to 0
- State 2: absorbing (stays at 2 forever); it represents "accept"

Write this as a transition matrix T , start from $v = (1, 0, 0)$, and at each step compute $v \leftarrow vT$:

Step	P(at 0)	P(at 1)	P(at 2)
1	0.5	0.5	0
2	0.5	0.25	0.25
3	0.375	0.25	0.375
4	0.3125	0.1875	0.5

(Computed by script; it also enumerated all $2^4 = 16$ coin sequences, and exactly 8 of them reach state 2, matching the 0.5 from the matrix.)

[Check 3] By hand only, verify the distribution after step 2, (0.5, 0.25, 0.25).

This turns the problem into linear algebra: **acceptance probability = some powers of the transition matrix applied to the start vector**. The OpenAI paper starts the same way: it turns the machine into a matrix S , and the acceptance probability is one component of $(I - S)^{-1}e$, which is equivalent to $\sum_j S^j e$ (Section 7.2).

3.3 Why "just compute it" is not in L

If it is just matrix powers, why not compute it deterministically? The problem is **memory**:

- The matrix is $n^c \times n^c$. It cannot be stored, so each entry has to be computed on demand.
- When computing T^t by "repeated squaring", each entry of T^t depends on entries of $T^{(t/2)}$, each entry of $T^{(t/2)}$ depends on entries of $T^{(t/4)}$, and so on. The recursion has $\log t \approx O(\log n)$ levels, and each level has to remember where it is ($O(\log n)$ bits), plus the bits of numerical precision.
- Total space $\approx$ levels $\times$ per level $= O(\log n) \times O(\log n) = O(\log^2 n)$.

This is the intuition behind Borodin–Cook–Pippenger 1983's result that "randomized logspace $\subseteq$ deterministic $\log^2$ space", and it has the same structure as the "recursion stack" in Savitch's theorem. [Established] **Forty years of work in this field have been about pushing down this "levels $\times$ per level"**.

4. Undirected connectivity: the classic win for random walks

4.1 The problem

USTCON (undirected s-t connectivity): given an undirected graph and two vertices s and t , are they connected?

BFS/DFS makes this easy, but they need a "visited" array, which takes n bits. What do you do with only $O(\log n)$ bits?

4.2 The random walk algorithm (AKLLR 1979)

In 1979, Aleliunas, Karp, Lipton, Lovász and Rackoff gave a minimal algorithm [Established]:

```
v ← s
repeat N steps:
  if v = t, answer "connected"
  v ← a random neighbor of v
answer "not connected"
```

It only needs to remember "the current vertex v " and "a step counter": $O(\log n)$ bits. If s and t are not connected, it never reaches t , so it **never turns a "no" into a "yes"**; if they are connected, then for large enough N it reaches t with high probability. This is exactly the one-sided error form of RL.

The key bound they proved: on a connected graph, the expected number of steps for a random walk to visit every vertex (the **cover time**) is $\leq 2|E|(n-1)$, where $|E|$ is the number of edges. Since $|E| \leq n^2$, this is on the order of $\leq n^3$. Taking $N = 4n^3$ and applying Markov's inequality pushes the failure probability below $1/2$.

4.3 Measured by script

Small example: 6 vertices, edges 0–1, 0–2, 1–2, 2–3, 3–4, 4–5 (a triangle with a tail attached). Walk from 0 to 5.

One actual simulated trajectory (script, random seed 1):

```
0 1 0 2 0 2 1 2 3 4 3 2 1 0 2 1 0 2 1 0 1 2 0 1 0 1 2 3 2 1 0 1 2 1 0 2 0 1 2 1 0 2 3 2
0 2 0 2 3 4 3 4 5
```

It took 52 steps. It circled around the triangle for a long time, because from 2 it has only a $1/3$ chance of heading into the tail, and once at 3 it has a $1/2$ chance of going back.

Averaged over 20,000 simulations: about 29.1 steps from 0 to 5, and about 30.4 steps to cover the whole graph. The upper bound $2|E|(n-1) = 2 \cdot 6 \cdot 5 = 60$; the measurements are within the bound.

Hitting times on several kinds of graphs:

Graph	n	Measured / exact expected steps	Comparison
Path (from one end to the other)	10	81.7 (simulated)	exact value $(n-1)^2 = 81$
Path	40	1,523.7 (simulated)	$(n-1)^2 = 1,521$
"Lollipop" (complete graph on $n/2$ vertices + a tail of $n/2$ vertices)	40	8,019 (solving a linear system, exact)	$n^3 = 64,000$, ratio 0.125
Lollipop	80	64,039 (exact)	$n^3 = 512,000$, ratio 0.125

(Computed by script.) The lollipop graph is the worst case for random walks: the walk spins around inside the complete graph, rarely enters the tail, and easily falls back out once it does. Brightwell–Winkler 1990 proved that the maximum hitting time on an n -vertex graph is about $(4/27)n^3 \approx 0.148n^3$ (attained when the clique takes $2/3$ of the vertices).

[Established; $\triangle$ constant quoted from memory of the literature, not checked against the original] Our "half and half" version gives $0.125n^3$, the same order of magnitude. So **n^3 steps are generally enough, and you basically can't do with fewer.**

4.4 Why the same method fails on directed graphs

Consider a directed graph with vertices $0, 1, \dots, n$. From vertex i there are two edges: one to $i+1$ and one back to 0 . The target is n .

A random walk from 0 must choose "forward" n times in a row to get there. The expected number of steps is $2^{(n+1)} - 2$:

n	Expected steps (exact, solving a linear system)
5	62
10	2,046
20	2,097,150
30	2,147,483,646

(Computed by script; matches the formula exactly.) **Exponential.** So random walks cannot solve directed reachability. Directed reachability is NL-complete; whether it is in L ($L = NL?$) is still open, and it is **unrelated to this chapter's result**.

[Check 4] Compute this expected value by hand for $n = 2$. Hint: let h_i be the expected number of steps to reach n starting from i ; $h_n = 0$, and $h_i = 1 + \frac{1}{2}h_{i+1} + \frac{1}{2}h_0$.

Why undirected graphs are nice: an undirected edge can be walked back along, so the random walk cannot get stuck in a "one-way trap". In linear-algebra terms, the transition

matrix of an undirected graph is symmetric (after normalization), its eigenvalues are all real, and its "mixing" speed is controlled by the spectral gap, which is the star of Section 5.

5. Reingold 2005: undirected connectivity is in L (SL = L)

5.1 History

After AKLLR, people tried to remove the randomness from USTCON:

- Nisan–Szemerédi–Wigderson 1992: $O(\log^{1.5} n)$ space [Established]
- Armoni–Ta-Shma–Wigderson–Zhou 2000: $O(\log^{4/3} n)$ space [Established]
- **Reingold 2005**: $O(\log n)$ space, i.e. $USTCON \in L$. [Established] STOC 2005, JACM 2008, Gödel Prize.

SL (symmetric logspace) is the class for which USTCON is complete, so this result is also written **SL = L**. It was a milestone in "removing the randomness from a specific problem", but only for one structure: undirected graphs.

5.2 Expanders: graphs that are "very well" connected

An **expander** is a graph with very small degree (each vertex has only a constant number of edges) that is nonetheless extremely well connected: any small set of vertices has many edges going out of it.

The measure is the **spectral gap**: the largest eigenvalue of the random-walk transition matrix is 1, the second largest is λ_2 , and the gap = $1 - \lambda_2$. The larger the gap, the faster the random walk "forgets" its starting point. Roughly, the number of steps needed to mix $\propto (\log n) / \text{gap}$.

n	Gap of the cycle (degree 2)	1/gap	Gap of a random 4-regular graph	1/gap
64	0.00482	208	0.162	6.2
256	0.00030	3,320	0.138	7.2
1024	0.00002	53,122	0.134	7.4

(Computed by script, eigenvalues via numpy; the random 4-regular graph is built as the union of two random permutations.) The cycle's gap falls to 0 like $1/n^2$, while the random 4-regular graph's gap stays at a constant (the theoretical optimum is $1 - 2\sqrt{3}/4 \approx 0.134$, the Ramanujan bound, and the measurement is already right up against it).

An expander has diameter only $O(\log n)$. So if the graph is a constant-degree expander, you can **enumerate all paths of length $O(\log n)$** : each step has only a constant number of choices, a path can be described in $O(\log n)$ bits, and there are $d^{O(\log n)}$ = polynomially many paths in total. A counter can try them one by one, with no randomness needed.

5.3 Reingold's idea: "rebuild" any graph into an expander

The difficulty is that the input graph is not necessarily an expander. Reingold repeatedly rebuilds it with two operations (each connected component is rebuilt separately):

1. **Squaring** (powering): in the new graph, two vertices are adjacent if and only if there is a path of length 2 between them in the original graph. The spectral gap roughly doubles (λ_2 becomes λ_2^2), at the cost of squaring the degree.
2. **The zig-zag product** (Reingold–Vadhan–Wigderson 2000): "replace" each vertex with a small fixed expander, which brings the degree back down to a constant while losing only a constant factor in the gap. [Established]

Alternating these for $O(\log n)$ rounds grows the gap from $1/\text{poly}(n)$ to a constant.

Why it doesn't eat memory: the rebuilt graph is **never stored explicitly**. To look up "the i -th neighbor of vertex v in the new graph", you recursively send queries to the graph one level up. The key is that the zig-zag design makes each level cost only a **constant** number of extra bits (rather than $\log n$ bits). $O(\log n)$ levels $\times$ $O(1)$ bits = $O(\log n)$.

Compare with Section 3.3: in "levels $\times$ per level", Reingold pushed "per level" from $\log n$ down to a constant. Engineering analogy: multi-level virtual address translation where each level only has to remember an offset of a few bits rather than a full address.

Limitation: both zig-zag and squaring rely on the graph's symmetry (undirectedness). The configuration graph of a general BPL computation is directed. Reingold–Trevisan–Vadhan 2006 extended the approach to special cases such as "regular directed graphs", but did not cover the general case. [Established]

6. Derandomizing general BPL: stuck at $3/2$ for 25 years

6.1 Pseudorandom generators and "seed length"

The most naive way to derandomize: **try every random string and count the fraction that accept**. A BPL machine reads $\text{poly}(n)$ random bits, so there are $2^{\text{poly}(n)}$ random strings. Far too many.

The idea of a **pseudorandom generator** (PRG): use a short **seed** to generate a long string of bits that "looks random". "Looks random" is relative to some class of observers: for any small-memory machine, the acceptance probability with the pseudorandom string is almost the same as with a truly random string.

Given a PRG with seed length s , derandomization is: enumerate all 2^s seeds, run once on each, and tally. The space needed is about $s + O(\log n)$.

Seed length	Number of seeds to enumerate	Derandomization result
$\text{poly}(n)$ (truly random)	$2^{\text{poly}(n)}$	useless
$O(\log^2 n)$ (Nisan 1992)	$n^{O(\log n)}$ (quasi-polynomial)	$\text{BPL} \subseteq L^2$
$O(\log n)$ (the dream)	$\text{poly}(n)$	$L = \text{BPL}$

So **the whole problem boils down to: for small-memory machines, can we build a PRG with seed length $O(\log n)$?** This question went unsolved for more than 30 years.

6.2 Nisan's generator (1992): why $\log^2$

Nisan's core observation: **a small-memory machine cannot remember which random bits it has used, so they can be safely "reused".** [Established]

The construction (simplified):

- Seed = a random string x of $O(\log n)$ bits, plus k random hash functions $h_1, \dots, h_k$, each of $O(\log n)$ bits.
- Level 1: output $x, h_1(x)$.
- Level 2: output $x, h_1(x), h_2(x), h_2(h_1(x))$.
- Each added level doubles the output length.

Intuition: after the machine has read the first half, its state is only $O(\log n)$ bits, so it has almost "forgotten" the specific contents of the first half. So the second half can be replaced by "the first half passed through a hash", and the machine can't tell.

To output $\text{poly}(n)$ bits takes $k = O(\log n)$ levels, each with an $O(\log n)$ -bit hash function, so the total seed length is $O(\log n) \times O(\log n) = \mathbf{O(\log^2 n)}$. "Levels $\times$ per level" again.

Nisan 1994 then proved that this simulation can achieve polynomial time and $O(\log^2 n)$ space at the same time ($\text{BPL} \subseteq \text{SC}$). [Established] (Note: Borodin–Cook–Pippenger 1983 had already obtained $\log^2$ space with the matrix method, but not polynomial time.)

6.3 Saks–Zhou 1999: 3/2

Saks and Zhou cut Nisan's $\log n$ levels into $\sqrt{\log n}$ segments of $\sqrt{\log n}$ levels each. Each segment uses Nisan's method to approximately compute a matrix power, and then the low-order bits of the result are **randomly rounded** away. This makes each segment's result almost independent of which particular hash functions were used, so **all segments can share the same set of hash functions**. [Established; this description is heavily simplified, [△](#) see the original for details]

Space: one set of hash functions, $\sqrt{\log n} \times \log n$, plus the recursion, $\sqrt{\log n}$ segments $\times$ $\log n$ per segment, for a total of $\mathbf{O(\log^{1.5} n)}$.

6.4 Later progress, and the stall

Year	Result	Deterministic space
1983	Borodin–Cook–Pippenger	$\log^2 n$
1992	Nisan PRG	$\log^2 n$
1999	Saks–Zhou	$\log^{1.5} n$
2021	Hoza (weighted pseudorandomness)	$\log^{1.5} n / \sqrt{(\log \log n)}$
2026	OpenAI (claimed)	$\log n$

(The Hoza bound is based on [19, Theorem 1.6 and Corollary A.5] as cited in §1.1 of the OpenAI paper, and on my memory; the deck author's research notes originally flagged it $\triangle$, and this time I checked it against the paper's citation; the two agree.)

Hoza 2021 saved only a factor of $\sqrt{(\log \log n)}$, which for $n = 10^9$ is only about 2.2×. The exponent for general BPL has been stuck near 3/2 for 27 years.

Along the way there were many wins on "special structures": Reingold (undirected graphs), Reingold–Trevisan–Vadhan (regular directed graphs), Ahmadinejad et al. (computing random-walk probabilities to high precision on Eulerian graphs), Cohen–Doron–Sberlo–Ta-Shma (long products of random matrices). [Established] In addition, Cheng–Hoza 2022 proved that a logspace-enumerable "hitting set" (which only guarantees finding one accepted input, and is weaker than a PRG) is enough to imply $L = BPL$. [Established] All of this shows that the "shape" of the problem had been mapped out very clearly; only the last step was missing.

7. What OpenAI claims

7.1 The main result

The paper is *Exact Derandomization of Logarithmic Space: $L = RL = BPL$* (2026-09-23, 108 pages, authorship credited to "OpenAI"). The abstract is a single sentence:

"We prove $L = RL = BPL$, resolving the derandomization problem for polynomial-time randomized logarithmic space."

In plain words: every randomized computation that is "polynomial time + logarithmic space + bounded error probability" can be simulated in deterministic logarithmic space.

The paper also gives a **stronger quantitative version** (Theorem 1.2): for any such randomized machine M and input x , the acceptance probability can be estimated deterministically to precision $2^{(-q)}$, using $O(\log n + q)$ space and $\text{poly}(n) \cdot 2^{O(q)}$ time. When $q = O(\log n)$, this means precision of any inverse polynomial can be computed in

logarithmic space. This is much stronger than "deciding whether the probability is above $2/3$ or below $1/3$ ": it can even estimate probabilities that have "no gap".

In addition, Section 13 gives a **compiler**: given the source program of a randomized machine and its time/space bounds, it outputs a deterministic decider together with explicit resource bounds.

7.2 Intuition for the method (I only read the introduction, §1–§2; what follows is a simplified version)

1. **Turn it into matrix inversion** (§2). Write the configuration graph of the machine on input x as a matrix S (each configuration carries a timestamp, and edges only point to later times, so S is "strictly forward" and nilpotent). The acceptance probability is $p_0 =$ the value of $(I - S)^{-1}e$ at the start node, where e marks the accepting configurations. This is the same thing as the toy example in Section 3.2.
2. **"Correction and copy hierarchy"** (§4). The authors use a sequence of transformations to replace S with new matrices $C_1, C_2, \dots$. Each step "moves" part of the transition probability into a reward vector (using the identity $I - D = (I + E)(I - C)$), and copies some vertices to spread out large matrix entries. After $O(\log n)$ stages, reading off the accumulated reward along a single "main copy" path gives the acceptance probability with inverse-polynomial error.
3. **Estimate using only $O(\log n)$ random bits** (§5–§9). These matrices are not stored explicitly; instead they are estimated by sampling with a **shared finite random environment** σ (an encoding of length $O(\log n)$). The paper proves that on more than $3/4$ of the σ 's, the estimate is accurate; and for **every** σ (including the ones where the estimate is inaccurate), the estimation procedure halts within $O(\log n)$ space.
4. **Enumerate + take the median** (§12). σ has only $O(\log n)$ bits, so there are only polynomially many σ 's in total. Enumerate them one by one, take the median, and compare it with $1/2$. This step is the "enumerate the seeds" idea from Section 6.1, except that instead of building a general-purpose PRG, it squeezes the randomness down to $O(\log n)$ bits for this one specific estimator.
5. **Keep the recursion from eating memory** (§9–§11). Use short **fingerprints** (polynomial evaluation + affine hashing) instead of full vertex addresses for comparisons; use a shared "catalytic" bit vector; use Cook–McKenzie's "walk around the tree's contour" trick to recover incoming edges without a stack. The data kept by each recursive call is held within $A(w - w') + D(t - j)$ bits, and this **telescopes** along the call chain, for a total of $O(\log n)$.

Compare step 5 with Sections 3.3 and 5.3: it is still a "levels $\times$ per level" problem, and the solution is to tie the number of bits each level spends to "how much the recursion budget drops", so the whole call chain adds up to no more than one $O(\log n)$ total budget.

Engineering analogy: a recursive call stack in which the size of each frame is bounded by a global quota, and the quota shrinks as you go down the stack.

Two details that surprised me:

- The mixing bound for "conditional averaging" in §6 relies on a result of Shalom 1999 about **Kazhdan's property (T)** (a deep property from group theory). [OpenAI claim] Group-theoretic tools showing up in small-memory derandomization is uncommon in the existing literature. [Speculation]
- The "shared catalytic bit vector" is reminiscent of recent work on **catalytic computing**: borrow a piece of memory that is already full of data, use it, and give it back exactly as it was. Cook–Mertz used it to bring the space for the tree evaluation problem down to $O(\log n \cdot \log \log n)$, and Ryan Williams 2025 built on this to prove that "a computation running in time t can be simulated in about $\sqrt{t \log t}$ space". [Established] If OpenAI's proof holds, catalytic tricks may once again be a key component. [Speculation, $\triangle$ I did not read the details of §11]

8. How far it has been verified

8.1 The Lean statement: you can read it yourself

The last line of `lean/ComparatorChallenges/LogspaceEquality.lean` is:

```
theorem exact_logarithmic_space_derandomization :  
  L = RL  $\wedge$  RL = BPL
```

Word-by-word translation:

- `L = RL  $\wedge$  RL = BPL`: L equals RL, **and** RL equals BPL.

This line is very short; the real content is all in the **definitions**. Unlike the quasi-Riemann hypothesis, the L, RL and BPL here are **not** standard Mathlib definitions (Mathlib has no such complexity classes). OpenAI wrote them itself in the challenge file. So the definitions are what need reviewing. Here is what I found, reading them one by one:

Definition	What it says	Matches the textbook?
Machine $q w h$	A finite-state machine with h read-only input heads and w binary work tapes; reads one coin b per step	Yes, a standard multi-tape Turing machine
readInput	The input has end markers at both ends	Yes
step	A configuration that has already produced output (halted) no longer changes	Yes
spaceThrough	Up to time t , the sum over work tapes of the number of distinct cells visited	Yes, and fairly strict (blank cells that were visited also count)
LogSpace	There exists $c > 0$ such that for all inputs, all coin sequences and all times, $\text{space} \leq c \cdot \lceil \log_2(n+2) \rceil$	Yes
Deterministic	The transition function does not depend on the coin	Yes
acceptanceProbability	At clock time t , the fraction of the 2^t coin strings that stop in "accept"	Yes
L	Deterministic + logspace + gives the correct answer at some time on every input	Yes (no time bound is written, but Section 1.3 shows that a halting deterministic logspace machine is automatically polynomial-time)
RL	Logspace + halts within $c \cdot (n+2)^k$ steps for all coin sequences + if yes, probability $\geq 1/2$, otherwise probability = 0	Yes
BPL	Same as above, with thresholds $2/3$ and $1/3$	Yes

My judgment: they match the standard definitions. One choice worth noting is "halts in polynomial time for all coin sequences". This is standard BPL; if you drop the time bound you get a different class (§1.1 of the paper also specifically stresses the difference between a "space bound" and "simultaneous time and space bounds"). This agrees with the conclusion in the author's research notes.

8.2 The Comparator configuration

LogspaceEquality.json locks **20 definitions** (from Word to BPL; I counted), which means the solution file cannot quietly swap out any of them. The only allowed axioms are Lean's three standard axioms (propext, Quot.sound, Classical.choice).

Note: this configuration has "enable_nanoda": false, meaning there is **no requirement for re-checking by a second, independent kernel**. (Only 2 of the 416 configurations in the

whole repository turn on nanoda; this is from the author's count over all configurations in the repository.)

8.3 The proof itself (what I found)

- Solution directory `lean/OAI/Computability/Logspace/`: **39 files, about 43,000 lines** (commit fd4aeeb, counted by script with `wc -l`).
- grep for `sorry | axiom | admit | native_decide | implemented_by | unsafe`: **0 hits**.
- Apart from Mathlib, it imports nothing from other `OAI.*` directories in the repository. So there is no "custom code hidden in a dependency library" problem; in this respect it is cleaner than the quasi-Riemann hypothesis.
- The top-level proof is only three lines: `L ⊆ RL` (trivial), `RL ⊆ BPL` (trivial, amplify a bit), `BPL ⊆ L` (all the difficulty is here).
- **What I didn't do:** I did not compile it, and I did not run Comparator. $\triangle$ **I found no public record of any third party rerunning Comparator on this family** (the quasi-Riemann hypothesis had an independent re-check by Goldblatt; this family has none).
- The paper's **compiler** and **explicit running-time bounds** are not in the Lean statement. The scope document `lean/docs/103.md` says verbatim: "The paper's explicit compiler and numerical running-time bounds are not separately stated in this Comparator target." Theorem 1.2 (arbitrary-precision estimation) is not formalized either.

8.4 Human experts (as of 2026-10-08)

- **Scott Aaronson** (blog post "The Mathocalypse", 2026-10-07): "L=BPL (i.e., probabilistic logspace and deterministic logspace are the same thing), one of the great derandomization conjectures short of P=BPP." He also said: "its truth was never in serious doubt, there was a whole subcommunity focused on proving this." [R]
- **Lance Fortnow** (blog post "Open No More", 2026-10-07) lists it as "A full derandomization of randomized log space", and says: "And none of these results get us any closer to settling $P \vee NP$." [R]
- **Ran Raz** (Princeton, as reported by *Scientific American* on 2026-10-08): "We have strong reasons to believe that, but we don't know how to prove it", and, earlier, "I didn't see any directions that were promising". [R, $\triangle$ the quotes were relayed through a web-fetch tool; check the original for context (whether he meant $L = BPL$ itself or derandomization more generally)]
- **No human expert has publicly said they have read and verified the 108 pages.**
Aaronson's comment on the whole batch of results: "it also appears that no human has understood just about any of these proofs yet." [R]

8.5 An inconsistency worth flagging

△ Fortnow's "OpenAI Math: TCS Selection" page (lance.fortnow.com, dated 10-07) **shows no Lean marker** for number 103, while the repository at commit fd4aeeb (10-07 22:20 PT) contains the full Lean challenge and solution. history.md says 6 formalizations were "added" on 10-07, so one possible explanation is that 103's Lean was added that day and Fortnow's page was based on an earlier snapshot. I could not confirm which 6 they were (the repository has only one merge commit, so per-file history is not visible). This judgment is based only on a web-fetch tool's summary; I did not see the page myself.

8.6 How to think about the trust structure here

Compared with the quasi-Riemann hypothesis: **the definitions are self-written**, but they are only 142 lines, and I read them one by one and they match the standard; **the proof is self-contained** (it depends only on Mathlib), with no large-scale patches; **the conclusion is widely believed**, so "the conclusion is right" cannot be turned around into evidence that "the proof is right". Nearly all the credibility rests on the Lean kernel and on "whether anyone has actually compiled it successfully" (△ no third-party report yet).

9. What it would mean if true (and what it would not)

9.1 For complexity theory

1. The first natural computational resource class to be "fully derandomized"

unconditionally. Within "logarithmic space + polynomial time", flipping coins is of no use at all. This answers a question stated explicitly back in AKLLR 1979.

2. **It does not imply $P = BPP$.** The time version is blocked by the "derandomization $\Rightarrow$ circuit lower bounds" wall (Section 2.2). The proof of the space version goes around that wall but does not tear it down. A deterministic polynomial-time algorithm for PIT is still unknown.

3. **It does not imply $L = NL$, and it is even further from P vs NP .** It is still unknown whether directed reachability (Section 4.4) can be solved in logarithmic space. The new picture of inclusions is:

$$L = RL = BPL \subseteq NL \subseteq L^2$$

1. **Theorem 1.2 may be more useful than 1.1.** Being able to estimate random-walk probabilities to high precision in logarithmic space means that many small-space linear algebra problems (for example, Laplacian solving, or stationary distributions of Markov chains) may get new deterministic algorithms. [Speculation]

2. **The technique may matter more than the conclusion.** Aaronson says the conclusion was "never in serious doubt", so what the field really wants is the **proof method**. Unfortunately, nobody understands it yet. If the proof holds but nobody can extract reusable ideas from it,

that would be a very strange situation: the problem is closed, but the field learned nothing.
[Speculation]

9.2 For engineering

Almost no direct impact on real systems: the randomized algorithms used in practice (hashing, sampling, Bloom filters) are already fast enough, and most of them are not in the extreme "logarithmic space" setting anyway. The theorem only guarantees that a deterministic replacement algorithm **exists**; the polynomial's degree and constants are likely astronomical. [Speculation] The implications for streaming algorithms are also limited: the streaming model can read the input only in a single pass, while L allows repeated reads.

9.3 For AI

- Like the other headline results, this is a case of "AI produces a proof of a recognized hard open problem, with a machine check attached".
- What sets it apart is that **everyone believes the conclusion; the hard part is the proof**. Problems of this kind may suit AI especially well: the direction is clear, there's no need to "guess the right conclusion", and what's needed is to get every step of a 108-page complex construction right, which Lean can check step by step. [Speculation]
- The lesson for AI safety research is the same as in §16 of this deck: once you have Lean, the bottleneck of trust moves from "is the proof right?" to "**were the statement and definitions written correctly?**" This family's definitions are self-written, so the table in Section 8.1 is a spec review. It is the same kind of work as auditing the behavioral spec of an AI agent.

10. Self-test

Try answering these after reading. If you can't, go back and reread the corresponding section.

1. Why does " $O(\log n)$ bits of working memory" roughly equal "can store a constant number of pointers"? Why must such a deterministic machine halt in polynomial time? (Sections 1.2, 1.3)
2. What is the difference between RL and BPL? Why does the restriction "once a coin is read, it's gone" matter? (Section 3.1)
3. Why does the random walk algorithm work on undirected graphs and fail on directed graphs? Give a number for each. (Sections 4.3–4.4)
4. In Reingold's algorithm, what are "levels" and "per level" in "levels $\times$ per level"? How does this differ from "levels $\times$ per level" in Nisan's generator? (Sections 5.3, 6.2)
5. Why is "a PRG with seed length $O(\log n)$ " enough to imply $L = BPL$? (Section 6.1)

6. In OpenAI's Lean statement, who defined L, RL and BPL? How is this different from the quasi-Riemann hypothesis case? Why does it matter? (Section 8.1)
7. Why does $L = BPL$ not imply $P = BPP$? (Sections 2.2, 9.1)

Appendix: answers to the checks

- **Check 1:** $\lceil \log_2 10^9 \rceil = 30$ bits. Current vertex 30 + target vertex 30 + counter ($n^3 = 10^{27}$, needing $\lceil \log_2 10^{27} \rceil = 90$ bits) = 150 bits, less than 20 bytes.
- **Check 2:** the two sides are equal $\Leftrightarrow 2xy = 0 \Leftrightarrow x = 0$ or $y = 0$. There are 100 combinations with $x = 0$ and 100 with $y = 0$; $(0, 0)$ is counted twice, so there are 199 in total (checked by script).
- **Check 3:** after step 1 the distribution is $(0.5, 0.5, 0)$. Step 2: of state 0's 0.5, half stays at 0 and half goes to 1; of state 1's 0.5, half goes to 2 and half goes back to 0. So $P(0) = 0.25 + 0.25 = 0.5$, $P(1) = 0.25$, $P(2) = 0.25$.
- **Check 4:** $h_2 = 0$; $h_1 = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}h_0$; $h_0 = 1 + \frac{1}{2}h_1 + \frac{1}{2}h_0$, i.e. $h_0 = 2 + h_1$. Substituting gives $h_1 = 1 + \frac{1}{2}(2 + h_1) = 2 + \frac{1}{2}h_1$, so $h_1 = 4$ and $h_0 = 6 = 2^3 - 2$.

Appendix: Glossary

Term	One-line explanation
Space complexity	The maximum number of bits of working memory used at any one moment (the input doesn't count)
L	Problems decidable deterministically with $O(\log n)$ bits of working memory
NL	Nondeterministic logspace; directed reachability is its representative problem
RL	Logspace + polynomial time + coins, one-sided error ("no" is never accepted)
BPL	Logspace + polynomial time + coins, two-sided error ($\geq 2/3$ vs $\leq 1/3$)
BPP	Polynomial time + coins, two-sided error
Configuration	The machine's complete state at a given moment; a logspace machine has only polynomially many configurations
PIT	Polynomial identity testing; has a simple randomized algorithm, but no known deterministic polynomial-time algorithm
USTCON	Undirected s-t connectivity
Cover time	The expected number of steps for a random walk to visit every vertex
Expander	A graph with small degree but excellent connectivity; its spectral gap is a constant
Spectral gap	$1 - \lambda_2$; measures how fast a random walk "forgets its starting point"
Zig-zag product	A graph operation that reduces a high-degree expander to constant degree while mostly preserving the gap
Pseudorandom generator (PRG)	Stretches a short seed into a bit string that "looks random to a certain class of observers"
Seed length	The number of truly random bits a PRG needs; $O(\log n)$ means the seeds can be enumerated
Derandomization	Turning a randomized algorithm into a deterministic one
Hardness vs. randomness	A family of theorems of the form "hard functions exist" $\Rightarrow$ "derandomization is possible"
Catalytic computing	Computing with borrowed memory that is already full, and returning it exactly as it was
Comparator	A Lean FRO tool that checks that a solution proves exactly the statement in the challenge file, using only allowed axioms

Appendix: Further reading

- **Sanjeev Arora & Boaz Barak**, *Computational Complexity: A Modern Approach* (Cambridge University Press, 2009): Chapter 4 (space complexity), Chapter 7 (randomized computation), Chapter 21 (expanders and Reingold's algorithm). Free draft: <https://theory.cs.princeton.edu/complexity/>

- **Salil Vadhan**, *Pseudorandomness* (Foundations and Trends in TCS, 2012): Chapter 2 (random walks and USTCON), Chapter 4 (expanders), Chapters 7–8 (PRGs). Free: <https://people.seas.harvard.edu/~salil/pseudorandomness/>
- **William Hoza**, *Recent Progress on Derandomizing Space-Bounded Computation* (BEATCS survey, 2022): the best introductory survey of the current state of the L vs BPL field. [△](#) Please search for and confirm the link yourself.
- Original literature: AKLLR, FOCS 1979; Nisan, *Combinatorica* 12(4), 1992; Saks & Zhou, *JCSS* 58(2), 1999; Reingold, *JACM* 55(4), 2008; Hoza, APPROX/RANDOM 2021, LIPIcs 207:28.
- The original paper: <https://github.com/openai/math/blob/main/preprints/Exact-Derandomization-of-Logarithmic-Space-L-equals-RL-equals-BPL-September-23-2026/paper.pdf>
- Lean statement: <https://github.com/openai/math/blob/main/lean/ComparatorChallenges/LogspaceEquality.lean>; scope: `lean/docs/103.md`
- Reactions: Aaronson <https://scottaaronson.blog/?p=10169>; Fortnow <https://blog.computationalcomplexity.org/2026/10/open-no-more.html>; Fortnow's selection page <https://lance.fortnow.com/openai-math-tcs.htm>; *Scientific American* <https://www.scientificamerican.com/article/the-most-exciting-claims-from-openais-heap-of-new-proofs/>

PART III · EXPLAINED FROM ZERO · GOES WITH §9

28. Explainer 06: The chromatic number of the plane $\chi \geq 6$ (the Hadwiger–Nelson problem)

This is the "explained from zero" version of §9 of the deck in

`notes/openai_math_explained/` (`<section id="color">`). The only prerequisites: you know what a graph is (vertices + edges), you can use the Pythagorean theorem to compute the distance between two points, and you know cos/sin. Words like measure and ergodic theory are explained the first time they appear. Every number in the text is either computed by script (script: `精讲/scripts/06_coloring.py`, with results copied into the text) or has a source. The **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Estimated reading time: 50–80 minutes. You can read it in three sittings: Sections 1–4 (graph coloring, lower bound 4, upper bound 7), Sections 5–7 (the 2018 breakthrough, "weird sets" and the axiom of choice, why 6 is hard), Sections 8–10 (OpenAI's result, verification, what it means).

0. The bottom line: the one thing this chapter explains

The problem itself is something a grade-schooler can understand: **give every point in the plane a color, so that any two points at distance exactly 1 have different colors. What is the minimum number of colors?** This minimum is called the "chromatic number of the plane", written $\chi(\mathbb{R}^2)$.

Since 1950 it has been known that the answer is between 4 and 7, and then it stayed stuck for 68 years. In 2018 the lower bound was pushed to 5. OpenAI claims to have pushed it to 6, so the answer can only be 6 or 7.

This chapter explains a causal chain:

Lower bounds come from "finding a finite point set that can't be colored" (the Moser spindle $\rightarrow$ de Grey's 1581-point graph) $\Leftarrow$ in principle you can always find one this way (the de Bruijn–Erdős compactness theorem) $\Leftarrow$ but the point set needed may be too large to find by search $\Leftarrow$ OpenAI took a different road: first turn "any weird coloring" into "a measurable coloring" (ergodic theory), then use geometry to rule out measurable 5-colorings, and the last step comes back to the 7-point Moser spindle.

After reading this chapter, you should be able to explain in your own words:

- why 3 colors are not enough (you can prove it by drawing 7 points by hand), and why 7 colors are enough (a hexagonal tiling);
- why "are the color classes measurable or not" is the real difficulty in this problem, and how it relates to the axiom of choice;
- what OpenAI's proof does structurally, why its Lean statement can be checked at a glance, and what it does **not** settle (6 or 7).

1. Graph coloring: an old friend from CS

1.1 Definitions

A **graph** is a set of vertices plus a set of edges. A **proper coloring**: give each vertex a color so that any two vertices joined by an edge have different colors. The minimum number of colors a graph needs is its **chromatic number** $\chi(G)$.

You've seen this in CS:

- **Register allocation**: variables are vertices, two variables that are "live at the same time" are joined by an edge, and colors are registers. Chromatic number = the minimum number of registers needed.
- **Course/exam scheduling**: courses are vertices, two courses that share a student are joined by an edge, and colors are time slots.

Deciding whether a general graph can be 3-colored is NP-complete. This will matter later: with brute-force computer search, once the graph gets big, the search becomes infeasible.

1.2 A few toys to work by hand

- One edge: needs 2 colors.
- **Triangle** (3 points, all pairwise connected): 2 colors aren't enough. The script enumerated all $2^3 = 8$ colorings and found **0** proper ones. So a triangle needs 3 colors.
- In general, if a graph has k points all pairwise connected (called a k -clique), it needs at least k colors. The converse doesn't hold: the chromatic number can be larger than the largest clique. The Moser spindle below is an example: it has no 4-clique, yet it needs 4 colors.

[Check 1] A ring of 5 points (a 5-cycle, where each point is connected only to its left and right neighbors): what is the minimum number of colors? Why aren't 2 enough?

2. Turning the whole plane into a graph

2.1 The unit-distance graph

Now treat **every point of the plane** as a vertex (there are uncountably infinitely many), and join two points by an edge if and only if their distance is **exactly 1**. This enormous graph is called the **unit-distance graph** of the plane, and its chromatic number is $\chi(\mathbb{R}^2)$.

An equivalent way to say it: split the plane into k pieces (one color per piece) such that **no** piece contains two points at distance exactly 1. What is the minimum number of pieces?

Two things to note:

- The distance is "exactly 1", not "less than 1". Two points at distance 0.999 or 1.001 may have the same color.
- **There is no requirement at all on the shape of the color pieces.** They can be nice hexagons, or weird sets built with the axiom of choice that can't be drawn at all. This is the core difficulty of the whole problem; Section 6 is devoted to it.

2.2 The first few steps

- $\chi \geq 2$: any two points at distance 1 already need 2 colors.
- $\chi \geq 3$: draw an equilateral triangle with side 1. Its three points are pairwise at distance 1; it's the triangle from Section 1.2, and 2 colors aren't enough.

The recipe for lower bounds is set right here: **find a finite set of points in the plane whose unit-distance edges form a graph, prove that this finite graph can't be colored with k colors, and then the plane can't be colored with k colors either** (because any

proper coloring of the plane, restricted to these few points, is a proper coloring of this small graph).

3. Lower bound 4: the Moser spindle (1961)

3.1 Step one: a rhombus forces its two tips to share a color

Put two equilateral triangles of side 1 together to form a rhombus: $O=(0,0)$, $B=(1/2, \sqrt{3}/2)$, $C=(1,0)$, $T=(3/2, \sqrt{3}/2)$. $O-B-C$ is one triangle, $B-C-T$ is the other, and they share the edge BC .

With only 3 colors: O , B and C are pairwise adjacent, so each takes one of the three colors; T is adjacent to both B and C , so T can only use the remaining color, which is O 's color.

Conclusion: with 3 colors, O and T must have the same color. Script check: this rhombus has 6 proper 3-colorings in total, and in every one of them O and T have the same color.

The distance between O and T is $\sqrt{3} \approx 1.732$ (computed by script), not 1, so their sharing a color does not by itself break any rule.

3.2 Step two: add a second rhombus and rotate it

Rotate the whole rhombus about O by an angle θ to get a second rhombus O, B_2, C_2, T_2 . By the same argument, T_2 must also have the same color as O . Now choose θ so that the distance between T and T_2 is exactly 1.

Both tips lie on the circle centered at O with radius $\sqrt{3}$, separated by angle θ . The square of their distance is:

$$|T - T_2|^2 = (\sqrt{3})^2 + (\sqrt{3})^2 - 2 \cdot 3 \cdot \cos \theta = 6 - 6 \cos \theta$$

Setting it equal to 1 gives **$\cos \theta = 5/6$** , $\theta \approx 33.56^\circ$ (computed by script).

Now T and T_2 must both have the same color as O , but they are at distance 1 from each other, so they must have different colors. **Contradiction. So 3 colors aren't enough, and $\chi \geq 4$.**

3.3 Script check

The script builds 7 points from the coordinates above:

Point	Coordinates (four decimal places)
O	(0, 0)
B ₁	(0.5, 0.866)
C ₁	(1, 0)
T ₁	(1.5, 0.866)
B ₂	(−0.062, 0.9981)
C ₂	(0.8333, 0.5528)
T ₂	(0.7713, 1.5508)

It then checks the distances of all 21 pairs of points; exactly **11 pairs** are at distance 1: 5 edges from each rhombus, plus T₁–T₂. No unexpected extra unit distances.

Then brute-force enumeration:

- 3 colors: $3^7 = 2187$ colorings, **0** proper.
- 4 colors: **384** proper ($= 4! \times 16$, i.e. if you ignore the names of the colors there are 16 essentially different colorings).

So this 7-point, 11-edge graph has chromatic number exactly 4. It is called the **Moser spindle** (published by the Moser brothers in 1961).

[Check 2] Above we used $|T_1 - T_2|^2 = 6 - 6\cos \theta$. Derive it yourself using the law of cosines, and verify that the distance really is 1 when $\cos \theta = 5/6$.

Remember this 7-point graph: the last step of OpenAI's 62-page proof still uses it.

4. Upper bound 7: a hexagonal tiling

Lower bounds come from "finding a small graph that can't be colored". An upper bound requires **giving a coloring that is proper for the entire plane**.

4.1 The construction

1. Tile the plane with regular hexagons (a honeycomb). The paper takes the hexagons' circumradius to be $r = 2/5$.
2. Any two points in the same hexagon are at distance at most its diameter, $2r = 0.8 < 1$. **So a whole hexagon can be given one color**, and no pair of points inside it is at distance 1.
3. Color the hexagons periodically with 7 colors, so that two different hexagons of the same color are far enough apart. In the paper, the centers of same-colored hexagons are at least $\sqrt{21} \cdot r \approx 1.833$ apart, so points in the two pieces are at least $(\sqrt{21} - 2) \cdot r \approx \mathbf{1.033 > 1}$ apart (computed by script).

4. Points on the boundaries can be assigned to any one of the adjacent pieces (this is how the paper handles them, and the Lean version also covers the boundaries).

4.2 Where "7 colors" comes from

The centers of the hexagons form a triangular lattice. The paper writes it with complex numbers: lattice points have the form $a \cdot (m + n \cdot \omega)$, where $\omega = e^{(2\pi i/3)}$ is a cube root of unity (the Eisenstein integers from Explainer 01 §8.2 show up again).

Multiplying by $(2 - \omega)$ gives a sublattice that is "scaled up and rotated a bit". It makes up $1/7$ of the original lattice, because $|2 - \omega|^2 = 7$ (computed by script: 7.0000000000000001, where the last digit is floating-point error). The original lattice splits into 7 classes according to "which translated copy of the sublattice a point belongs to", and each class gets one color. In the script this corresponds to a very simple rule: the center $a \cdot (m + n\omega)$ gets color $(m + 2n) \bmod 7$.

4.3 Script check

The script randomly scattered **200,000** pairs of points at distance exactly 1 (for each pair, pick a random point, then walk distance 1 in a random direction), colored them with the rule above, and found **0 pairs with the same color**.

[Check 3] r can't be just anything. For the construction to work, we need $2r < 1$ and $(\sqrt{21} - 2) \cdot r > 1$. What is the allowed range of r ?

So $4 \leq \chi(\mathbb{R}^2) \leq 7$. This interval held from 1950 (Nelson got 4 and Isbell got 7, neither published at the time; Soifer later pieced together the historical record; Hadwiger published it formally in 1961) until 2018.

5. 2018: de Grey pushes the lower bound to 5

5.1 A breakthrough by an amateur mathematician

In April 2018, Aubrey de Grey (known for his research on the biology of aging; mathematics is a hobby for him) posted *The chromatic number of the plane is at least 5* on arXiv (arXiv:1804.02385). The method was still the recipe from Section 2.2: assemble a huge finite unit-distance graph out of many parts like the Moser spindle, and prove that it **can't be 4-colored**. The corrected graph has **1581 vertices**, and the "can't be 4-colored" step was verified by computer.

5.2 Follow-ups: making the evidence smaller and readable

Year	Who	What they did	Source
2018	Heule	Used a SAT solver plus "unsatisfiable core" extraction to get a 553-vertex graph needing 5 colors	arXiv:1805.12181
2018/2020	Exoo, Ismailescu	Independently gave another proof of $\chi \geq 5$	arXiv:1805.00157
2020	Parts	Shrank the graph to 509 vertices (graph minimization)	arXiv:2010.12665
2020	Parts	Gave a proof of $\chi \geq 5$ that a human can check by hand	arXiv:2010.12661

This phase also included the crowdsourced project Polymath16 (organized by Dustin Mixon on his blog), and the work above was intertwined with it.

Where the 509 points come from: according to the introduction of the OpenAI paper, SAT plus unsatisfiable cores gave Heule's **553** points; the **509** points came from Parts's graph minimization. Both are connected to Polymath16.

5.3 Why this road keeps getting harder

A graph with n vertices and k colors has k^n possible colorings (orders of magnitude computed by script):

Graph	Vertices	Colors	$k^n \approx$
Moser spindle	7	3	$10^{3.3}$
Parts	509	4	10^{306}
Heule	553	4	10^{333}
de Grey	1581	4	10^{952}

A SAT solver of course doesn't enumerate these one by one; it relies on pruning. But to prove "there are none at all", the solver has to rule out the whole space, and the graph for 5 colors is expected to be much larger still. Going from 4 to 5 already needed a graph with over a thousand points; going from 5 to 6, nobody knows how big the graph has to be, and nobody found one during Polymath16.

5.4 A safety-net theorem: de Bruijn–Erdős compactness

If the finite graph we're looking for might be absurdly large, **does it necessarily exist?** Yes. This is the **de Bruijn–Erdős theorem (1951)**:

A graph (possibly infinite) can be colored with k colors if and only if **every finite subgraph** of it can be colored with k colors.

So "the plane can't be 5-colored" is equivalent to "some finite unit-distance graph can't be 5-colored". But the proof of this theorem is **non-constructive**: it guarantees that such a

finite graph exists, but doesn't tell you what it looks like or how big it is.

Here is an analogy that comes naturally to CS people: it's like "if every finite subset of the constraints is satisfiable, then the whole infinite constraint system is satisfiable". The proof uses compactness of an infinite product space (Tychonoff's theorem), which requires the **axiom of choice** (explained in the next section). §4 of the OpenAI paper states this step explicitly: "using product compactness in ZFC".

6. The real difficulty: weird sets, measurability and the axiom of choice

6.1 What "measurable" means

Roughly speaking, a set in the plane is **measurable** if it has a sensible "area". Every shape you can draw (polygons, disks, countable unions of them, plus some more complicated but "well-behaved" sets) is measurable.

But in the usual set theory (ZFC, the axiom system that includes the **axiom of choice**), you can construct **non-measurable** sets: they have no sensible area, you can never draw them, and you can only prove they exist. The most famous examples are the Vitali set and the Banach–Tarski paradox (a ball is cut into finitely many pieces and reassembled into two balls of the same size; the pieces are non-measurable).

The axiom of choice says: given infinitely many nonempty boxes, you can always pick one element from each box, even if you can't write down a rule for picking. The vast majority of mathematicians accept it by default.

6.2 The measurable chromatic number: a different problem

If you require each color class to be measurable, the minimum number of colors is called the **measurable chromatic number** $\chi_m(\mathbb{R}^2)$. Clearly $\chi_m \geq \chi$ (an extra restriction can only make coloring harder).

- **Falconer (1981):** $\chi_m(\mathbb{R}^2) \geq 5$. The tool is "density points": near almost every point of a measurable set, the set fills almost 100% of the area. Around such a point, it's hard for the unit circle to avoid points of the same color; combined with a rotation argument, this gives a contradiction.

In other words, **under the "measurable" restriction, 5 was proved back in 1981**, 37 years before de Grey. The hard part is removing the "measurable" condition.

6.3 A cautionary example

The introduction of the OpenAI paper cites an example by Payne (2009) that illustrates the issue well: consider a subgraph of the plane in which two points are joined only when their difference is exactly "a unit vector with rational coordinates" (for example, $(3/5, 4/5)$).

- If any coloring is allowed, **2 colors** are enough for this graph;
- If the coloring must be measurable, it needs **at least 5 colors**.

So "measurable colorings" and "arbitrary colorings" can't be assumed to be the same, even when their distance constraints look identical. (△ I didn't read Payne's original; this restates what §1 of the OpenAI paper says.)

6.4 Shelah–Soifer: the answer might depend on the axioms

In *Axiom of choice and chromatic number of the plane* (J. Combin. Theory Ser. A 103 (2003) 387–391), Shelah and Soifer constructed some "distance graphs" whose chromatic number **depends on which axioms you use**: it is 2 in ZFC, but uncountable in the axiom system that "drops the axiom of choice and replaces it with 'every set of reals is measurable'" (ZF + DC + LM; Solovay proved in 1970 that this system is consistent, provided you assume an "inaccessible cardinal" exists).

They also proposed a conditional theorem, roughly: if all finite unit-distance graphs can be 4-colored, then the chromatic number of the plane is 4 in ZFC, but at least 5 in ZF+DC+LM (because in that world every coloring is measurable, so Falconer's result applies). (△ I paraphrased the exact statement of the conditional theorem from Soifer's later survey and did not check it word for word against the original.)

The premise of this conditional theorem has been refuted by de Grey, but the possibility it reveals is still there: **"how many colors the plane really needs" might not have an answer independent of the axioms**. That is also why the OpenAI paper states at the very start that it is "working throughout in ZFC".

6.5 "Nice" colorings: the answer is already known to be 7

If you require the color pieces to be "map-like" (with regular curves as boundaries), the situation is different again:

- Townsend (announced 1981, details published 2005): map-like colorings need at least **6** colors.
- Sokolov–Voronov (2025, arXiv:2502.01958): in a "polygonal, locally finite" framework, at least **7** colors are needed.

In other words, **for colorings "you can draw", the answer is 7**. The remaining question is: can weird sets save colors compared with nice colorings?

7. Why ≥ 6 is hard: both roads are blocked

Before 2026, the situation looked like this:

Approach	What it can achieve	Where it gets stuck
Finite graphs + computers (de Grey, Heule, Parts)	$\chi \geq 5$	The finite graph needed for 6 colors may be enormous; nobody has found one
Measurability / density points (Falconer)	$\chi_m \geq 5$	Holds only for measurable colorings; says nothing about weird sets
Regular boundaries (Townsend, Sokolov–Voronov)	map-like ≥ 6 , polygonal = 7	Requires regular boundaries; says nothing about general measurable sets, let alone weird sets

To get to 6, you need to clear two hurdles at once:

1. **From arbitrary colorings to measurable colorings:** how do you make weird sets "analyzable"?
2. **From measurable colorings to a geometric contradiction:** the boundaries of measurable sets can be extremely irregular, so you can't just draw them as a map.

OpenAI's paper corresponds exactly to these two steps.

8. What OpenAI claims

8.1 The main result

The abstract of the paper *The Euclidean plane is not five-colorable* (OpenAI, 2026-09-23, 62 pages, Family 158):

"We prove that every coloring of the Euclidean plane with five colors has a monochromatic unit-distance pair, with no regularity assumption on the color classes. Consequently, the chromatic number of the plane is either six or seven."

In plain words: **every 5-coloring of the plane contains two points at distance 1 with the same color, and no regularity is required of the color pieces. So the chromatic number of the plane is 6 or 7.**

The paper's Theorem 1.1 writes this as $6 \leq \chi(\mathbb{R}^2) \leq 7$, and immediately adds: "The remaining alternatives six and seven are unresolved."

A common misreading: it is easy to assume this result is "find a bigger finite graph and verify it with SAT". **It isn't.** The paper never gives a single finite graph that can't be 5-colored.

8.2 The two halves of the proof

The paper splits the proof into two theorems:

- **Theorem 1.3 (transfer theorem):** for every k , "a proper k -coloring exists" $\Leftrightarrow$ "a **weakly measurable** k -coloring exists". This holds for all k , not just 5.

- **Theorem 1.4:** no weakly measurable 5-coloring exists.

Together: if there were a proper 5-coloring, then by 1.3 there would be a weakly measurable 5-coloring, contradicting 1.4.

What a "weakly measurable k-coloring" is (a plain-language version of Definition 1.2): every color piece is measurable, and the "same-colored unit-distance pairs" are null in the sense of measure. That is, exceptional same-colored pairs are allowed, but there are so few of them that they have zero area and can be ignored. This is weaker than a "proper coloring" (it allows measure-zero violations), and more regular than an "arbitrary coloring" (it must be measurable).

8.3 Intuition for the first half: "averaging" a weird coloring into a statistical regularity

What follows is my simplified paraphrase of §1.1 of the paper; the technical details go far beyond this.

1. **First shrink to a countable point set.** Take all points whose coordinates are **algebraic numbers** (roots of polynomial equations, like $\sqrt{2}$ or $(1+\sqrt{5})/2$), and call this set E . It has only countably many points, but it is dense in the plane, and it is closed under "multiplying by an algebraic complex number of absolute value 1" (that is, rotation by an algebraic angle).
2. **Average over translations and rotations.** Restrict the weird coloring to E , then "take the average" over all translations and rotations to get a **random coloring** whose statistics are invariant under any translation or rotation. Averaging like this relies on these groups being **amenable** (roughly, "you can take a fair average over them"; the chapter on Thompson's group, family 248, covers this in detail).

An analogy with something familiar: this is like turning a very irregular signal into a **stationary random process**. A single sample may be weird, but its statistics (for example, its autocorrelation function) are regular.

1. **Use the spectral theorem for a "Fourier decomposition".** A stationary process can be decomposed by frequency (compare the power spectrum of an EEG). Here the "frequencies" come in two kinds: ordinary continuous frequencies (corresponding to ordinary waves $\exp(i\xi \cdot z)$ on the plane), and a kind of "wild" frequency that only makes sense on the countable point set E .
2. **Rigidity theorem (Theorem 2.3):** any distribution that is invariant under rotations and does not sit on the ordinary frequencies **must be completely uniform noise** (Haar measure). Uniform noise has zero correlation under any nonzero translation, so throwing it away doesn't break the correlation condition "unit-distance pairs have different colors". The proof

of this step uses the **Furstenberg–Zimmer structure of compact extensions**, a tool from ergodic theory developed for proving Szemerédi's theorem.

3. **The remaining ordinary-frequency part can be extended to the whole plane**, giving a genuine measurable function. Then taking the most likely color at each point gives a weakly measurable coloring.

The reverse direction (weakly measurable $\Rightarrow$ proper) uses a Falconer-style density-point argument plus the de Bruijn–Erdős compactness from Section 5.4.

8.4 Intuition for the second half: ruling out 5 colors with geometry

Again a simplified paraphrase:

1. **"Palette"**: around each point x , look at the unit circle and record which colors appear in which directions on the circle. The paper proves that almost every such palette has **at most 2 colors** (§5).
2. **A "boundary graph" between colors**: treat the 5 colors as vertices and record which pairs of colors have a lot of transitions between them. The paper proves this graph contains a **cycle** (§6–7, using mollification, thresholding and a topological argument about coverings of the plane).
3. **A cycle on 5 colors can only have length 3, 4 or 5**, and §8 rules them out one by one:
 - 5-cycle: a combinatorial contradiction about a "6-letter angle word";
 - 4-cycle: a parity contradiction;
 - 3-cycle: it leaves an open region that uses only 3 colors, and then a "placement certificate" with rational coordinates puts **the 7 points of the Moser spindle** into this region. Section 3 already proved that 3 colors aren't enough for those 7 points: contradiction.

You can see the definition of this "three-color region" directly in the Lean code

([OAI/Geometry/PlaneColoring/Moser.lean](#)):

```
def ThreeLabelRegion (x y : ℝ) : Prop :=  
  0 < x^2 + y^2 ∧ x^2 + y^2 < 4 ∧  
  y^2 * (3 * x^2 - y^2)^2 < (x^2 + y^2)^3 * (1 - (x^2 + y^2) / 4) * (x^2 + y^2 - 1)^2
```

It's a region of the plane described by polynomial inequalities, and each point of the spindle has to be verified to lie inside it. **The whole 62-page argument starts from the 7-point toy of 1961 and comes back to it in the end.**

8.5 Two side results

- **Corollary 1.5 (the "bad fraction" has a positive lower bound):** there is a constant $\delta > 0$ such that in any 5-coloring, the fraction of "randomly placed needles of length 1 whose two ends have the same color" is at least δ . But the paper itself says: "The graph and the positive constant here are existential." That is, δ is **only known to exist; nobody knows how big it is**, because it comes from that finite graph needing 6 colors whose shape nobody knows.
- **Footnote 1 (Reed's manuscript, 2026-05):** Reed claims $\chi = 7$ and attaches a "formalized theorem". The OpenAI paper points out that this theorem put a **false** density upper bound in as a **hypothesis** (counterexample: the half-open arc $\{e^{it} : 0 \leq t < \pi/3\}$), so the formalization itself is not wrong; what's wrong is the proposition being proved. $\triangle$ I didn't read Reed's original manuscript; I only saw OpenAI's account of it. This example will be used in Section 9.

9. How far it has been verified

9.1 The Lean statement: two lines, you can read it yourself

File `lean/ComparatorChallenges/EuclideanFiveColor.lean` (snapshot fd4aeeb):

```
def ProperColoring (colorCount : ℕ) (coloring : ℂ → Fin colorCount) : Prop :=
  ∀ point otherPoint : ℂ, ‖point - otherPoint‖ = 1 → coloring point ≠ coloring otherPoint

theorem no_proper_five_coloring : ¬ ∃ coloring : ℂ → Fin 5, ProperColoring 5 coloring
```

Word-by-word translation:

- `coloring : ℂ → Fin colorCount`: a coloring is just a function that maps each complex number (i.e. a point in the plane, using $\mathbb{C}$ as the plane) to a color in $\{0, 1, \dots, \text{colorCount}-1\}$. **The function can be arbitrary**; there's no requirement like "measurable" or "continuous".
- `∀ point otherPoint : ℂ`: for any two points in the plane.
- `‖point - otherPoint‖ = 1 →`: if their distance (the modulus of the complex difference) is exactly 1,
- `coloring point ≠ coloring otherPoint`: then their colors differ. This is exactly "proper coloring".
- `¬ ∃ coloring : ℂ → Fin 5, ProperColoring 5 coloring`: there **does not exist** any proper 5-coloring.

This is exactly the original statement of "5 colors aren't enough" for the Hadwiger–Nelson problem, with no sneaked-in measurability and no extra premises. Compared with the

quasi-Riemann hypothesis in Explainer 01, it doesn't even depend on whether Mathlib's definition of some complicated function (like ζ) is correct; it only uses complex numbers, the modulus and finite sets. **This is one of the easiest statements for a human to audit in the entire OpenAI repository.**

The upper bound is formalized too: `properColoring_seven` in `ComparatorChallenges/PlaneColoring.lean`, where the plane is `EuclideanSpace  $\mathbb{R}$  (Fin 2)` and the condition is `dist x y = 1  $\rightarrow$  c x  $\neq$  c y`. The proof file `Seven.lean` is only 68 lines.

9.2 The proof itself (statistics we ran ourselves)

- The proof entry point is `OAI/Geometry/PlaneColoring/Five.lean`. A script followed its `import`s to collect all of its transitive dependencies inside the OAI directory: **70 files, 30,406 lines**.
- `grep` over these 70 files: **no** `sorry`, `admit`, custom `axiom` or `native_decide`.
- The final assembly corresponds exactly to the two halves of the paper:

```
theorem no_proper_five_coloring :  $\neg \exists c : \mathbb{C} \rightarrow \text{Fin } 5, \text{ ProperColoring } 5 c := \text{by}$ 
  rintro (c,hc)
  obtain (d,hd,hw) := proper_to_borel_weak hc          -- transfer theorem (Theorem 1.3)
  exact no_measurable_weak_five_coloring hd hw        -- geometric obstruction (Theorem 1.4)
```

- The Comparator configuration (`EuclideanFiveColor.json`) allows only the three standard axioms: `propext`, `Quot.sound` and `Classical.choice`. Note:
 - `Classical.choice` is the **axiom of choice** in Lean. So what this formalization proves is "in mathematics with the axiom of choice, the plane can't be 5-colored", which agrees with the paper's "in ZFC". The point in Section 6.4 that "with a different set of axioms the answer may differ" is not touched by this proof.
 - `"enable_nanoda": false`: an independent second checking kernel is **not** enabled. Someone independently re-checked the quasi-Riemann hypothesis of Explainer 01 with nanoda; for this one, as of 2026-10-08 we found no public record of an independent re-check $\triangle$.
- We did **not compile** this proof ourselves; the above comes from reading the source.

9.3 Human experts

- **Gil Kalai** (blog post, 2026-10-07): "The new manuscript raises the lower bound to six, leaving six and seven as the two possible answers." The disclaimer section of the same post says "even Lean verification may have issues".

- **As of 2026-10-08, we found no public statement from de Grey, Polymath16 members or other combinatorial geometry experts saying they had read and verified the 62 pages.**
- The 2026-10-07 history of the OpenAI repository lists 3 retractions and 14 revisions; **this paper is not among them.** Family 158 is also **not** among the 10 families whose reasoning summaries are published in the README.

9.4 How to think about the remaining risk

- **Statement risk: very low.** Two lines that anyone can check against the original problem; see 9.1.
- **Proof risk: depends on the Lean kernel and Mathlib.** The measure theory, spectral theorem and so on used in the proof come from Mathlib or from these 70 files; the kernel checks every step. As long as the kernel has no bug and the statement is right, the conclusion holds.
- **Understanding risk: high.** Ergodic theory plus geometric topology, 62 pages, and no human expert has publicly read it all yet.
- **Reed as a cautionary tale:** Reed's "formalized theorem" doesn't count because the error was hidden in a **hypothesis**. OpenAI's statement has no hypotheses; nothing at all comes before `no_proper_five_coloring`, so the same problem can't arise. This shows exactly what the human half should be auditing in the division of labor "humans audit the statement, machines audit the proof".

10. What it would mean if true (and what it would not)

10.1 For mathematics

1. $\chi(\mathbb{R}^2) \in \{6, 7\}$. This is the first progress since 2018, and the first time the lower bound has been pushed without a finite graph.
2. **The transfer theorem itself may matter more than the "6"**: it holds for all k , which amounts to saying that "arbitrary colorings" and "weakly measurable colorings" are the same thing for this problem. Future work on the problem can go straight to the measurable world and use analytic tools, without worrying about weird sets. (This is my judgment; it counts as speculation.)
3. **A new open problem**: by de Bruijn–Erdős, there must be a finite unit-distance graph that can't be 5-colored, but nobody knows what it looks like or how big it is. Finding it (for example, with a SAT solver) would give a completely independent proof that can be verified finitely. This is a natural follow-up target for "AI + SAT" (speculation).

10.2 What it does not mean

- **It doesn't decide between 6 and 7.** The paper says explicitly that both possibilities remain unresolved.
- Section 6.5 says "nice colorings need 7 colors", so if the answer is 6, the coloring must use weird sets or extremely irregular measurable sets. Some people will guess from this that the answer is 7, but that's only intuition, not evidence (speculation).
- **It says nothing about worlds without the axiom of choice.** The proof is carried out in ZFC (`Classical.choice` in Lean). The question from Section 6.4 of what happens in a world where "all sets are measurable" is still open. Intuitively, in that world every coloring is measurable, so the geometric part, Theorem 1.4, would seem to be enough; but I haven't checked whether its proof uses more choice than DC $\triangle$ (speculation).
- **It gives no numerical value for δ ,** and no concrete finite graph needing 6 colors.

10.3 For AI and AI safety

- This is verifiability in its ideal form: **a two-line statement, a 30,000-line proof**, and humans only need to audit the two lines. Compared with the quasi-Riemann hypothesis, it doesn't even need the layer of trust "is Mathlib's definition of ζ correct?".
- Reed's counterexample shows the other side: **Lean only guarantees that "the given statement has been proved"**; the statement and its hypotheses still need human checking. This is the same kind of problem as specification gaming in reward models: the optimizer satisfies the spec you wrote down, not necessarily what you wanted.
- The "surprising" method (ergodic theory + geometric topology, rather than a bigger finite graph) deserves attention: if it holds up, it's an example of "the model found a route nobody in the field had taken", rather than "computing faster along a known route". Whether nobody had really taken it needs a domain expert's judgment $\triangle$.

11. Self-test

Try answering these after reading. If you can't, go back and reread the corresponding section.

1. Why does finding a finite point set in the plane that can't be k -colored imply $\chi(\mathbb{R}^2) > k$? (Section 2.2)
2. In three sentences, restate why the Moser spindle proves that 3 colors aren't enough. (Section 3)
3. In the hexagonal 7-coloring, which two inequalities does $r = 2/5$ satisfy? What does each one guarantee? (Section 4)
4. What does the de Bruijn–Erdős theorem say? Why does it guarantee that a finite graph needing 6 colors exists, but not help you find it? (Section 5.4)

5. Falconer proved "5" back in 1981, so why was de Grey's "5" in 2018 still a breakthrough? (Section 6.2)
6. What are the two halves of OpenAI's proof? Which half holds for every number of colors k ? (Section 8.2)
7. What is the fundamental difference in trustworthiness between Reed's "formalized theorem" and OpenAI's `no_proper_five_coloring`? (Sections 8.5, 9.4)

Appendix: answers to the checks

- **Check 1:** 3 colors. With 2 colors, the colors must alternate around the ring (red, blue, red, blue, ...). On a ring of odd length, when you go all the way around back to the start, the starting point would have to share a color with its own neighbor: contradiction. In general, a graph can be 2-colored if and only if it has no cycle of odd length.
- **Check 2:** T and T_2 are both at distance $\sqrt{3}$ from O , with angle θ between them. Law of cosines: $|T - T_2|^2 = 3 + 3 - 2 \cdot \sqrt{3} \cdot \sqrt{3} \cdot \cos \theta = 6 - 6 \cos \theta$. When $\cos \theta = 5/6$ this equals $6 - 5 = 1$, so the distance is 1.
- **Check 3:** $2r < 1 \Rightarrow r < 1/2$; $(\sqrt{21} - 2)r > 1 \Rightarrow r > 1/(\sqrt{21} - 2) \approx 0.3872$ (computed by script). So $0.3872 < r < 0.5$, and the paper's choice $r = 0.4$ is in range.

Appendix: Glossary

Term	One-line explanation
Proper coloring	A coloring in which adjacent vertices have different colors
Chromatic number $\chi(G)$	The minimum number of colors a proper coloring needs
Unit-distance graph	Vertices are points in the plane; two points at distance exactly 1 are joined by an edge
$\chi(\mathbb{R}^2)$	The chromatic number of the plane's unit-distance graph, i.e. the answer to the Hadwiger–Nelson problem
Moser spindle	A 7-point, 11-edge unit-distance graph that can't be 3-colored; proves $\chi \geq 4$
de Bruijn–Erdős theorem	An infinite graph can be k -colored $\Leftrightarrow$ every finite subgraph can be k -colored (needs the axiom of choice)
SAT solver	A program that decides whether a set of Boolean constraints can be satisfied simultaneously; used to verify that a finite graph can't be k -colored
Measurable set	A set with a sensible "area"
Measurable chromatic number χ_m	The minimum number of colors when all color pieces must be measurable; Falconer proved ≥ 5
Axiom of choice	You can always pick one element from each of infinitely many nonempty sets; it lets you build non-measurable sets
Weakly measurable coloring	Color pieces are measurable and same-colored unit pairs have measure zero (measure-zero exceptions allowed)
Amenable	The property of a group that "you can take a fair average" over it
Ergodic theory	The branch of mathematics that studies "long-run averages" and invariant statistical regularities
Haar measure	The completely uniform distribution on a group; corresponds to "pure noise"
Palette	The set of colors appearing on the unit circle around a point

Appendix: Further reading

- **The original paper:** <https://github.com/openai/math/blob/main/preprints/The-Euclidean-plane-is-not-five-colorable-September-23-2026/paper.pdf> (read Section 1 and Figure 1 in §1.1 first; the rest is technical detail)
- **Lean statement:**
<https://github.com/openai/math/blob/main/lean/ComparatorChallenges/EuclideanFiveColor.lean>
; scope note [lean/docs/158.md](#)
- **Gil Kalai's 2026-10-07 blog post:** <https://gilkalai.wordpress.com/2026/10/07/updates-sharing-ai-progress-on-mathematics-amazing-and-my-lecture-plans/>
- **Gil Kalai's 2018 introduction to de Grey's result** (the best entry point):
<https://gilkalai.wordpress.com/2018/04/10/aubrey-de-grey-the-chromatic-number-of-the->

plane-is-at-least-5/

- **de Grey 2018**: <https://arxiv.org/abs/1804.02385> ; **Heule's 553 points**: <https://arxiv.org/abs/1805.12181> ; **Parts's hand-checkable proof**: <https://arxiv.org/abs/2010.12661>
- **Quanta Magazine**, *Decades-Old Graph Problem Yields to Amateur Mathematician* (2018–04, popular coverage of de Grey)
- **Alexander Soifer**, *The Mathematical Coloring Book* (Springer, 2009; second edition 2024): the full history of the problem, including the chapters on Shelah–Soifer and the axiom of choice ("What If We Had No Choice?").
- **Shelah & Soifer**, *Axiom of choice and chromatic number of the plane*, J. Combin. Theory Ser. A 103 (2003) 387–391.
- **Payne**, *Unit distance graphs with ambiguous chromatic number*: <https://arxiv.org/abs/0707.1177>
- **Sokolov & Voronov**, map-like colorings need at least 7 colors: <https://arxiv.org/abs/2502.01958>
- **Terence Tao**, *254A Lecture 9: Ergodicity* (the ergodic theory introduction cited by the paper): <https://terrytao.wordpress.com/2008/02/04/254a-lecture-9-ergodicity/>

PART III · EXPLAINED FROM ZERO · GOES WITH §10

29. Explainer 07: Thompson's group F is nonamenable

This is the "explained from zero" version of §10 of the [notes/openai_math_explained/](#) deck. All you need: fraction arithmetic, what a function and function composition $f(g(x))$ are, and the length and inner product of vectors. **No group theory is needed.** Every new concept is explained before it is used. Every number here is either computed by script ([精讲/scripts/07_thompson.py](#)), exact arithmetic with Python's `Fraction` or sourced. **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Estimated reading time: 60–90 minutes. You can split it into three sittings: Sections 1–3 (groups, free groups, "fair averages"), Sections 4–5 (Banach–Tarski, Thompson's group itself), Sections 6–9 (OpenAI's proof, its verification, and what it means).

0. The bottom line: the one thing this chapter explains

On some infinite "sets of operations" (mathematicians call them **groups**) you can define a **fair average**: shift everything over, and the average does not change. The integers allow this. On other groups such an average is **impossible**. The classic example is the "free

group". The reason it cannot be fairly averaged is that it can be cut up, shifted around, and reassembled into **two copies** of itself. This is the mechanism behind the famous Banach–Tarski paradox (cut a ball into a few pieces and reassemble them into two balls).

For a long time mathematicians thought that "cannot be fairly averaged" was always caused by a free group hidden inside. Counterexamples overturned this starting in 1980. But **Thompson's group F**, the most natural and most famous candidate, stayed undecided: it contains no free group, yet it does not belong to any type that "obviously can be averaged". In 1979 Geoghegan conjectured that it **cannot** be averaged. Over the next 47 years, people announced proofs in both directions, and one after another they were found to be wrong and withdrawn.

This chapter explains one causal chain:

F can be fairly averaged $\iff$ F contains finite sets that "barely leak" (Følner sets) $\implies$ averaging over such a set produces an approximate fixed point for a map that "pushes every point away by at least $\frac{1}{2}$ " $\implies$ contradiction. So F cannot be fairly averaged.

OpenAI gave this argument in a 13-page paper on 2026-09-23, together with a Lean formalization.

After this chapter, you should be able to explain in your own words:

- what "amenable" actually means, and why it is the same thing as the Banach–Tarski paradox;
- what the two generators of Thompson's group F look like, and how to compose them by hand;
- why F is the most famous "litmus test" in this field, and why its history makes Lean verification especially valuable;
- which "borrowed" phenomenon OpenAI's proof uses (maps with no fixed point in infinite-dimensional space), and what this result does and does not mean.

1. Groups: a set of operations you can compose and undo

1.1 A concrete example first: the symmetries of a square

Take a square card and label its four corners A, B, C, D. Which operations put it back into its original outline (the corner labels may move around)?

- Rotations by 0° , 90° , 180° , 270° (4 rotations);
- Flips about the vertical axis, the horizontal axis, and the two diagonals (4 flips).

That is 8 operations in total. Starting from "rotate 90° (call it r)" and "flip about the vertical axis (call it s)" and combining them repeatedly, the script generates exactly these 8, no more and no fewer.

These 8 operations have a few properties:

1. **They compose:** do one, then another, and the result is still one of the 8. Write $r \circ s$ for "first s, then r" (the same order as function composition $f(g(x))$: the one on the right happens first).
2. **There is a "do nothing":** rotate by 0°, written e (the identity).
3. **Every operation can be undone:** rotating 90° is undone by rotating 270°; each flip is undone by itself.
4. **Associativity:** $(a \circ b) \circ c = a \circ (b \circ c)$. Function composition satisfies this automatically.

But **order matters:** computed by script, $r \circ s$ and $s \circ r$ send the four corners to different places (one swaps A and C, the other swaps B and D). So groups are in general **not commutative**.

1.2 A second example: adding integers

Take all integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$; an "operation" is "add some number". Composition is addition, the identity is 0, and +3 is undone by -3. This is an **infinite** group, and it is commutative.

1.3 Definition

Group = a set G together with an operation that combines two elements into one, such that: there is an identity element, every element has an inverse, and the operation is associative.

You can always think of a group as "all the symmetry operations of some object". Every group in this chapter is like that: the symmetries of a square, the translations of the integers, certain stretchings of the interval [0,1] (Section 5).

1.4 Generators and relations

Many groups can be built entirely out of **a few operations**. These operations are called **generators**.

- The 8 symmetries of the square are generated by r and s.
- $\mathbb{Z}$ is generated by the single element "+1" (-1 is its inverse, and every integer is a string of +1's or -1's).

Generators often satisfy **relations**. For example, in the square, $r \circ r \circ r \circ r = e$ (four rotations bring you back) and $s \circ s = e$. Relations make different "spellings" give the same element.

2. The free group F_2 : a group with no relations at all

2.1 Reduced words

Take two generators a, b and their inverses a^{-1}, b^{-1} (also written A, B below). The elements of the **free group F_2** are "words" spelled with these four letters. The only simplification rule is that an adjacent a and a^{-1} (or b and b^{-1}) cancel. A word with everything cancelled is called a **reduced word**. Apart from that there are **no relations at all**: $abab^{-1}$ and $baba^{-1}$ are different elements, and so are ab and ba .

Let's count the reduced words of length n . The first letter has 4 choices; each later letter cannot be the inverse of the letter before it, so it has 3 choices. So there are $4 \cdot 3^{n-1}$ words of length n .

Length n	Words of length exactly n ("sphere")	Words of length $\leq n$ ("ball")	Sphere / ball
0	1	1	1.000
1	4	5	0.800
2	12	17	0.706
3	36	53	0.679
5	324	485	0.668
10	78,732	118,097	0.667

(The script counts by enumerating reduced words one by one. Ball size = $2 \cdot 3^n - 1$.)

Two key points:

- The number of elements **grows exponentially** with n (roughly $2 \cdot 3^n$). Compare: $\mathbb{Z}$ has only $2n+1$ elements of length $\leq n$; $\mathbb{Z}^2$ (with L1 distance) has only $2n^2+2n+1$, which is 221 for $n=10$.
- **The outermost layer holds about 2/3 of everything**, and this fraction does not shrink as n grows. The outermost layer of $\mathbb{Z}$ has only 2 points, a fraction of $2/(2n+1) \rightarrow 0$.

[Check 1] There are 12 reduced words of length exactly 2. Write them all out. (Hint: $aa, ab, aB, \dots$)

2.2 Cayley graphs: drawing a group

Draw each element of the group as a point. If multiplying g on the right by a generator gives g' , draw an edge between g and g' . This picture is called the **Cayley graph**.

- $\mathbb{Z}$ (generator $+1$): an infinite straight line.
- $\mathbb{Z}^2$ (generators $(1,0), (0,1)$): an infinite square grid.

- F_2 (generators a, b): an **infinite tree in which every point has 4 edges**. Because there are no relations, different paths from the origin never come back to the same point, so there are no cycles.

In network terms: $\mathbb{Z}^2$ is like a 2D mesh, where the number of nodes grows polynomially with the radius; F_2 is like a 4-way fan-out tree, where **most nodes are on the boundary**. This property is the key to everything below.

3. Amenability: can you "average fairly"?

3.1 Where the question comes from

Averaging over a finite group is easy: add up the values at every element and divide by the number of elements. This average is "fair": shift the whole group (multiply every element on the left by the same h), and you are just adding the same numbers in a different order, so the average does not change.

On an infinite group this gets hard. There is no "uniform distribution" on $\mathbb{Z}$: if every integer has the same probability, that probability must be 0, and then the total is not 1.

Mathematicians get around this by relaxing the requirement. They ask only for a **mean**: a rule that assigns a score $M(\phi)$ to every bounded function $\phi: G \rightarrow \mathbb{R}$, such that

1. **Linear**: $M(\phi + \psi) = M(\phi) + M(\psi)$, $M(c\phi) = c \cdot M(\phi)$;
2. **Positive**: if $\phi \geq 0$ everywhere, then $M(\phi) \geq 0$;
3. **Normalized**: the mean of the constant function 1 is 1;
4. **Left-invariant**: shifting the whole function (replacing ϕ by $g \mapsto \phi(hg)$) does not change the mean.

If such an M exists on a group, the group is called **amenable**; otherwise it is **nonamenable**. This is exactly the definition used in the OpenAI paper and in the Lean statement (see Section 7).

Intuitively, M is an idealized version of "sprinkling a handful of sand evenly over the whole group". It only needs to be **finitely additive** (not countably additive). That is what gives groups like $\mathbb{Z}$ a chance to qualify.

3.2 The Følner criterion: find a finite set that "barely leaks"

The definition above is very abstract (M usually cannot be written down; proving it exists requires the axiom of choice). In 1955 Følner gave an equivalent criterion with a combinatorial flavor:

A group G is amenable $\iff$ for any finite collection of "shifts" h and any small $\varepsilon > 0$, you can find a finite set A with $|hA \Delta A| / |A| < \varepsilon$.

Here $hA = \{h \cdot a : a \in A\}$ is the result of shifting all of A , and Δ is the "symmetric difference" (the elements that appear on only one side). $|hA \Delta A|/|A|$ measures "after one shift, what fraction of the elements changed". The intuition: if A barely changes when shifted, averaging over A approximates a fair average; make A bigger and less and less "leaky", take the limit, and you get M .

Computed by script:

Group	Set A	A	Shift h	$ hA \Delta A / A $
$\mathbb{Z}$	$\{-n, \dots, n\}$, $n=10$	21	+1	0.0952
$\mathbb{Z}$	$n=1000$	2,001	+1	0.0010
$\mathbb{Z}^2$	$(2n+1) \times (2n+1)$ square, $n=10$	441	(1,0)	0.0952
$\mathbb{Z}^2$	$n=100$	40,401	(1,0)	0.0100
F_2	ball of length ≤ 4	161	a	1.006
F_2	ball of length ≤ 10	118,097	a	1.000

For $\mathbb{Z}$ and $\mathbb{Z}^2$ the fraction $\rightarrow 0$, so they are amenable. Shift a ball in F_2 and the fraction **tends to 1**: about half the elements move out and the same number of new elements move in ($|hA \Delta A|$ counts both sides). The reason is what Section 2 said: "most elements are on the boundary".

[Check 2] In $\mathbb{Z}$, let $A = \{-n, \dots, n\}$ and $h = +1$. Which two elements make up $hA \Delta A$? Use this to verify that the fraction is $2/(2n+1)$.

3.3 But "balls fail" does not mean "every set fails"

The table above tried only balls. The Følner criterion asks whether **some** A exists. In principle F_2 could contain some oddly shaped set that does much better than a ball. To prove that F_2 is not amenable, you need an argument that works for **every** set. The cleanest one is the following.

F_2 can be reassembled into two copies of itself. Let $W(a) =$ "the reduced words starting with a ", and define $W(a^{-1})$, $W(b)$, $W(b^{-1})$ the same way. Then:

- $F_2 = \{e\} \cup W(a) \cup W(a^{-1}) \cup W(b) \cup W(b^{-1})$, five pieces with no overlap.
- But **$W(a) \cup a \cdot W(a^{-1})$ is already all of F_2** : for any word w that does not start with a , putting a^{-1} in front gives $a^{-1}w$, which starts with a^{-1} (nothing cancels), and $w = a \cdot (a^{-1}w)$.
- Likewise $W(b) \cup b \cdot W(b^{-1})$ is also all of F_2 .

(The script checked both statements for every word of length ≤ 7 , and both hold; in the ball of length ≤ 8 , each of the four W pieces has 3,280 words.)

Now suppose F_2 has a fair mean M . Treat "belongs to a set" as a function taking values 0/1; then M gives each piece a "share". By left-invariance, the share of $a \cdot W(a^{-1})$ equals the share of W . So:

- $M(W(a)) + M(W(a^{-1})) \geq M(\text{all of } F_2) = 1$;
- $M(W(b)) + M(W(b^{-1})) \geq 1$;
- Added together, the four pieces have total share ≥ 2 . But they are non-overlapping parts of F_2 , so their total share is ≤ 1 . Contradiction.

So **F_2 is not amenable**. This "make two copies of yourself out of a few pieces" is called a **paradoxical decomposition**. In 1938 Tarski proved that a group is nonamenable **if and only if** it has a paradoxical decomposition. So "cannot be fairly averaged" and "can duplicate itself" are two sides of the same thing.

Also: **if a group contains a nonamenable subgroup, the group itself is nonamenable** (a standard fact). So every group containing F_2 is nonamenable.

4. The Banach–Tarski paradox and von Neumann's question

4.1 The paradox

Banach–Tarski (1924): a solid ball in three-dimensional space can be cut into finitely many pieces which, using only rotations and translations, can be reassembled into **two** solid balls each the same size as the original.

This does not violate physics, because the pieces are non-measurable "weird sets" (they require the axiom of choice) and have no volume to speak of. Its mathematical root is exactly the paradoxical decomposition from Section 3.3: the 3D rotation group $SO(3)$ **hides a copy of F_2** (take two rotations about different axes with suitably chosen angles; there are no relations between them). Transfer F_2 's "one copy becomes two" onto the sphere, and you get the paradox.

4.2 von Neumann, 1929

von Neumann asked: why does this paradox exist in three dimensions but not in two (the plane)? His answer: **the key is not the space but the group of operations**. The group of rotations and translations of the plane is amenable, so it can be fairly averaged. That means the plane admits a "finitely additive, motion-invariant" notion of area, and the paradox cannot happen. The group of motions of 3D space contains F_2 and is nonamenable, so there

is room for the paradox. This is how the concept of "amenable" was born (German *messbar*; the English name "amenable" was coined later by Day).

4.3 The von Neumann–Day problem

Since "contains $F_2 \Rightarrow$ nonamenable", the natural question is the converse:

Must every nonamenable group contain F_2 ?

This is usually called the von Neumann problem or the von Neumann–Day problem (Day stated it explicitly in 1957). The answer is **no**, but the counterexamples were hard to come by:

Year	Author	Counterexample	Features
1980	Ol'shanskii	"Tarski monster" groups	Infinite, but every proper subgroup is a finite cyclic group; the construction is extremely complicated
1982	Adyan	Free Burnside groups with large odd exponent	Some fixed power of every element is the identity
2002	Ol'shanskii–Sapir	A finitely presented counterexample	The first counterexample that can be written down with finitely many generators and finitely many relations
2013	Monod	Groups of "piecewise projective" homeomorphisms of the real line	The first "natural" counterexamples; they look a lot like Thompson's groups
2016	Lodha–Moore	A finitely presented subgroup of Monod's group	Finitely presented and torsion-free (no elements of finite order)

(Monod: *Groups of piecewise projective homeomorphisms*, PNAS 2013; Lodha–Moore: *A nonamenable finitely presented group of piecewise projective homeomorphisms*, Groups Geom. Dyn. 10 (2016) 177–200.)

Note: a common claim is that "the counterexamples are all weird, and F would be the first natural counterexample". That was true before 2013; **since 2013, Monod and Lodha–Moore have given natural, finitely presented, torsion-free counterexamples**. So F 's significance is not "the first natural counterexample" but "**the most famous, most studied group in this field** finally has an answer".

5. Thompson's group F

5.1 Definition

F = all functions $f: [0,1] \rightarrow [0,1]$ such that:

1. f is continuous and strictly increasing, with $f(0) = 0$ and $f(1) = 1$;
2. f is **piecewise linear**, with finitely many breakpoints;

3. the x-coordinates of the breakpoints are all **dyadic rationals** (numbers of the form $k/2^n$, e.g. $1/2, 3/4, 5/8$);
4. the **slope of every piece is an integer power of 2** ($\dots, 1/4, 1/2, 1, 2, 4, \dots$).

The group operation is function composition.

You can check that composing two such functions gives another such function, and so does taking the inverse (reflecting the graph across the diagonal). So F really is a group. It is infinite: for example, you can squeeze $[0,1]$ to the left by arbitrarily large amounts.

F was introduced by Richard Thompson in 1965 (originally for a problem in logic); McKenzie–Thompson 1973 gave an early published construction. The 1996 survey by Cannon–Floyd–Parry, *Introductory notes on Richard Thompson's groups*, is the standard introduction.

5.2 The two generators, written out

x_0 : breakpoints $(0,0) \rightarrow (1/2, 1/4) \rightarrow (3/4, 1/2) \rightarrow (1,1)$.

Interval	Maps to	Slope
$[0, 1/2]$	$[0, 1/4]$	$1/2$
$[1/2, 3/4]$	$[1/4, 1/2]$	1
$[3/4, 1]$	$[1/2, 1]$	2

Intuitively: squeeze the left half to half its size, stretch the rightmost quarter to twice its size, and shift the middle. The overall effect is to "push points to the left".

x_1 : do nothing on $[0, 1/2]$; on $[1/2, 1]$, do a "scaled-down x_0 ". Breakpoints $(0,0) \rightarrow (1/2, 1/2) \rightarrow (3/4, 5/8) \rightarrow (7/8, 3/4) \rightarrow (1,1)$, with slopes 1, $1/2$, 1, 2 in order.

Cannon–Floyd–Parry proved: **every element of F can be built from x_0 , x_1 and their inverses.**

5.3 Composing by hand

Exact computation with fractions (checked by script):

- $x_0(3/8)$: $3/8$ lies in $[0, 1/2]$, where the slope is $1/2$, so the result is $3/16$.
- Breakpoints of $x_0 \circ x_1$ (first x_1 , then x_0): $(0,0) \rightarrow (3/4, 3/8) \rightarrow (7/8, 1/2) \rightarrow (1,1)$, slopes $1/2, 1, 4$.
- Breakpoints of $x_1 \circ x_0$ (first x_0 , then x_1): $(0,0) \rightarrow (1/2, 1/4) \rightarrow (7/8, 5/8) \rightarrow (15/16, 3/4) \rightarrow (1,1)$, slopes $1/2, 1, 2, 4$.
- The two differ, so F is **not commutative**. A concrete example: $x_1(x_0(7/8)) = 5/8$, while $x_0(x_1(7/8)) = 1/2$.

- Breakpoints of x_0^{-1} : $(0,0) \rightarrow (1/4, 1/2) \rightarrow (1/2, 3/4) \rightarrow (1,1)$, slopes 2, 1, $1/2$ (just swap the coordinates of each breakpoint of x_0).

[Check 3] Compute the breakpoints and slopes of $x_0 \circ x_0$. (Hint: first compute $x_0(1/2)$ and $x_0(3/4)$, then plug them into x_0 ; you also need to find which point the first x_0 sends to $3/4$.)

5.4 Relations: why F is not a free group

F has relations. The most intuitive one comes from "**functions with disjoint supports commute**": if f only moves points in $[0, 3/4]$ and g only moves points in $[3/4, 1]$, it does not matter which one you do first.

Computed by script:

- $u = x_0 \circ x_1^{-1}$ has breakpoints $(0,0) \rightarrow (1/2, 1/4) \rightarrow (3/4, 3/4) \rightarrow (1,1)$, and is the identity on $[3/4, 1]$.
- $v = x_0^{-1} \circ x_1 \circ x_0$ has breakpoints $(0,0) \rightarrow (3/4, 3/4) \rightarrow (7/8, 13/16) \rightarrow (15/16, 7/8) \rightarrow (1,1)$, and is the identity on $[0, 3/4]$.
- So u and v commute: $u \circ v = v \circ u$. The script checked that $u^{-1}v^{-1}uv = \text{identity}$, and it holds; it also holds with v replaced by $x_0^{-2} \circ x_1 \circ x_0^2$.

These two are **all** the defining relations of F (Cannon–Floyd–Parry): F is **finitely presented** with 2 generators and 2 relations.

Relations make the balls smaller. Using x_0 , x_1 and their inverses as generators, count the distinct elements of length $\leq n$ (the script deduplicates using exact breakpoints):

n	Ball in F	Ball in F_2
1	5	5
4	161	161
5	475	485
6	1,381	1,457
8	11,237	13,121

The first 4 layers are identical; from layer 5 on, there are 10 fewer. The reason: the shortest relation, $u^{-1}v^{-1}uv$, has length $2+3+2+3 = 10$, so length 5 is the first point at which two different words can represent the same element. F still grows exponentially (layer 8 still has 7,280 new elements), but exponential growth by itself does not decide amenability (some exponentially growing groups are amenable).

5.5 Why F is the litmus test of this field

Both standard tools **fail** on F :

- **"Contains $F_2 \Rightarrow$ nonamenable" does not apply:** Brin–Squier proved in 1985 that F contains no non-abelian free subgroup (Inventiones 79).
- **"Built from abelian and finite groups $\Rightarrow$ amenable" does not apply either:** F is not an "elementary amenable" group (Cannon–Floyd–Parry Theorem 4.10).

So F falls through the gap between the two standard methods of deciding the question. Add to this that its definition is simple and it has very nice properties (Brown–Geoghegan proved in 1984 that it has strong finiteness properties), and it became the single most watched group after the von Neumann–Day problem. Geoghegan conjectured in 1979 that F is **nonamenable** (recorded in Cannon–Floyd–Parry 1996, p. 227).

Historical claims and retractions (each row has a primary source):

Date	Claim	Direction	Outcome
2009-02	Akhmedov, arXiv:0902.3849	Nonamenable	Withdrawn by the author (v8, 2013-12, pointing to the new manuscript 1310.4395)
2009-06	Shavgulidze, arXiv:0906.0107 (a related paper was published in a journal)	Amenable	Moore 2011 (arXiv:1102.0747) lists specific errors and writes that those who studied it carefully agree: "the errors are serious and do not seem to be repairable"; he also notes that the Math Reviews review is inaccurate
2011-12	Beklaryan, arXiv:1112.1942	Nonamenable	Withdrawn by the author in 2012-11, citing "a crucial error in Theorem B*"
2012-09	Justin Moore , arXiv:1209.2063	Amenable	Withdrawn by the author on 2012-10-01: "Azer Akhmedov pointed out a serious error in Lemma 4.13. This error appears to be both serious and irreparable."
2013/2021	Akhmedov, arXiv:1310.4395v3	Nonamenable	The OpenAI paper only says it "claims"; $\triangle$ I found no evidence that the community accepted it, and before 2026 the problem was generally listed as open

There are also solid partial results. For example, Moore proved in 2013 that **if** F is amenable, its Følner sets must be astonishingly large ("tower-type" growth, *Groups Geom. Dyn.* 7).

Note Moore's retraction: he is one of the leading experts on this problem, and a fatal error was still found by peer checking within three weeks. This is the key background for understanding "why it matters that this result comes with Lean".

6. What OpenAI claims

6.1 The main result

The paper is *Thompson's group F is nonamenable* (authored by OpenAI, 2026-09-23, **13 pages in total**: the proof is on pages 1–7, pages 8–11 are appendices, pages 12–13 are

references). The full abstract:

"We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture and resolves the amenability problem for F ."

Main theorem: **Theorem 1.1. Thompson's group F is not amenable.**

Page 1 of the paper honestly lists the failed historical claims above, and specifically notes that what Moore withdrew was a different paper, not his published Ramsey characterization.

6.2 The borrowed tool: a map that "pushes everything away"

First recall a classical theorem.

Brouwer fixed-point theorem: on a closed ball of **finite dimension** (a segment, a disk, a solid ball, ...), every continuous map f has at least one fixed point, $f(x) = x$. Intuition: you cannot continuously crumple a disk and put it back inside the disk while moving every point.

This theorem fails in infinite-dimensional space. The infinite-dimensional example is ℓ^2 : all infinite sequences $(x_1, x_2, x_3, \dots)$ whose squares have a finite sum, with length $\sqrt{x_1^2 + x_2^2 + \dots}$. Kakutani's classic example:

$f(x) = (1 - \|x\|, x_1, x_2, x_3, \dots)$ (shift the sequence right by one slot, and put $1 - \|x\|$ in front)

It maps the unit ball into the unit ball, and it **has no fixed point** (a fixed point would need all coordinates to be equal and the sum of squares to be finite, so they would all be 0; but then the first slot is 1, a contradiction).

But this example is not strong enough: its "displacement" can be **arbitrarily small**.

Computed by script: take $x = (t, t, \dots, t)$ (N coordinates) with $t = 1/(1+\sqrt{N})$; then $f(x) - x$ is t in the $(N+1)$ -th coordinate only:

N	Displacement $\ f(x) - x\ $
1	0.500
100	0.091
10^4	0.0099
10^6	0.0010

So Kakutani's f has "approximate fixed points".

Benyamini–Sternfeld (1983) proved something stronger: on the unit ball of any infinite-dimensional normed space, there is a **Lipschitz** map f (it never stretches the distance between two points by more than a factor L) that **moves every point by at least $\delta > 0$** . Such an f does not even have approximate fixed points. Appendix A of the OpenAI paper builds its own version with $\delta = \frac{1}{2}$ on $L^2([0,1]; \mathbb{R}^2)$ (using a curve in infinite-dimensional space and a "tube" around it), and every Lipschitz estimate in the appendix is written out.

6.3 Skeleton of the proof (Section 2, about 5 pages)

The overall strategy in one sentence: **if F has finite sets A that barely leak, then averaging over A produces an approximate fixed point of the map f above.** This contradicts "every point is moved by at least δ ".

Five steps, explained as intuitively as possible:

1. **Color the "dyadic partitions" of $[0,1]$.** A partition cuts $[0,1]$ into small cells of the form $[k/2^r, (k+1)/2^r]$. Fix in advance D pairwise non-adjacent "parent intervals" $I_1 < \dots < I_D$. For a partition T , take its part inside each parent interval and stretch it back to $[0,1]$; this gives a smaller partition $T_{\{I_j\}}$. Define the "color" of T (a vector in the unit ball) recursively:

$$\begin{aligned} p(T) &= f((p(T_{\{I_1\}}) + \dots + p(T_{\{I_D\}})) / D) && \text{(if } T \text{ cuts at every } I_j) \\ p(T) &= 0 && \text{(otherwise)} \end{aligned}$$

Each restriction to a subinterval strictly reduces the number of cells, so the recursion always terminates.

1. **Let elements of F produce partitions.** An element g of F sends each cell of a uniform partition to a dyadic interval (Lemma 2.1 of the paper), so each g brings a family of colors: $X_i(g)$ is the color of the i -th parent interval, and $Y_{ij}(g)$ is the color of the j -th "grandchild interval" inside the i -th parent interval. The recursive definition directly gives **$X_i = f(z_i)$** , where z_i is the average of $Y_{i1}, \dots, Y_{iD}$. Let m = the average of all the X_i .
1. **F can move any pair of separated intervals to any other pair** (Lemma 2.2: refill the gaps with dyadic cells, and map them linearly one-to-one in order). So for any two separated intervals $I < J$ there is an element $h_{\{I,J\}}$ of F that moves them to the reference pair (I_1, I_2) , and the colors move along exactly. Let S be the **finite** set made of these $h_{\{I,J\}}$. It is fixed in advance and does not depend on A .
1. **Average over a Følner set.** Let η = the largest leakage fraction $|hA \Delta A|/|A|$ of A under S . When averaging over A , "replacing g by hg " barely changes the average, with error at most η . So: **for any two separated intervals, the average over A of the inner product of their**

colors is within η of the same constant α . It is like saying: on a sample that is almost translation-invariant, all "pairs of separated positions" look equally correlated.

1. **A variance estimate, then a contradiction.** Expand $\|z_i - m\|^2$. Among its inner-product terms, the vast majority are "separated interval pairs", whose averages are all about α , and positive and negative contributions cancel; only the "nested pairs" (a grandchild interval and its own parent), a fraction $1/D$, are not controlled, but their inner products have absolute value at most 1. Working it out (equation 2.16 of the paper):

$$\text{average over } A \text{ of } \|z_i - m\|^2 \leq 4/D + 4(1 - 1/D) \cdot \eta$$

On the other hand, m is also the average of the $f(z_i)$; using convexity and the Lipschitz property:

$$\delta^2 \leq \|m - f(m)\|^2 \leq (L^2/D) \cdot \sum \|z_i - m\|^2$$

Combining: $\delta^2 \leq L^2 \cdot (4/D + 4(1-1/D) \cdot \eta)$. Rearranging gives (Proposition 2.3):

$$\eta \geq (\delta^2/L^2 - 4/D) / (4(1 - 1/D))$$

As long as D is chosen in advance to be larger than $4L^2/\delta^2$, the right-hand side is a **positive number that does not depend on A** . In other words: **every finite set A in F leaks at least a fixed positive fraction under this fixed finite set S** . This directly violates the Følner criterion, so F is not amenable.

A feel for the numbers (computed by script; L is an assumed value, since the paper does not give a concrete number for the appendix map's L $\triangleq$): with $\delta = 1/2$ you need $D > 16L^2$. If $L = 10$, taking $D = 3,200$ gives a lower bound of about 3.1×10^{-4} ; taking $D = 1,601$ (just past the threshold) gives about 3.9×10^{-7} . The fraction is tiny, but the proof only needs it to be positive and independent of A .

Why this proof is surprising: most earlier attempts looked for something inside F 's own combinatorial structure (trees, graphs, Ramsey properties, height functions). Here the key "impossibility" comes from **infinite-dimensional geometry**. F only supplies the flexibility to "move any pair of intervals to any pair", plus the self-similar recursion of dyadic partitions.

6.4 Two corollaries (relying on companion papers)

Section 3 of the paper derives two corollaries: F has a representation that is "almost isometric but cannot be unitarized" (relying on another OpenAI paper, *Unitarizability implies*

amenability for discrete groups); and percolation on the Cayley graph of F has $p_c < p_u$ (relying on another OpenAI paper on percolation). The paper states explicitly that these two corollaries are **not used** in proving the main theorem. $\triangle$ We have not checked these two companion papers.

7. How far it has been verified

7.1 The Lean statement: F is defined from scratch, and you can read it line by line

Unlike Explainer 01, this statement **does not rely on a ready-made F in Mathlib** (Mathlib has none). Instead, the challenge file

`ComparatorChallenges/ThompsonNonamenability.lean` (89 lines, including imports)

defines everything from scratch. The core:

```
def IsDyadic (x : ℝ) : Prop := ∃ (k : ℤ) (n : ℕ), x = (k : ℝ) / (2 : ℝ) ^ n

structure DyadicPLWitness (f : IntervalHomeomorph) where
  pieceCount : ℕ -- number of pieces
  knots : Fin (pieceCount + 1) → UnitInterval -- list of breakpoints
  strictMono_knots : StrictMono knots -- breakpoints in increasing order
  first : knots 0 = 0; last : knots (last index) = 1
  dyadic : ∀ i, IsDyadic (knots i) -- breakpoints are dyadic rationals
  exponent : Fin pieceCount → ℤ -- one integer exponent per piece
  affine : ... (f x) = f(knot_i) + 2^(exponent i) * (x - knot_i) -- each piece has slope a power of 2
  ...

def F := {f : IntervalHomeomorph // StrictMono f ∧ HasDyadicPLPieces f}

structure InvariantMean (G) [Group G] where
  toLinearMap : BoundedReal G →1[ℝ] ℝ -- linear
  positive : ∀ f, (∀ g, 0 ≤ f g) → 0 ≤ toLinearMap f -- positive
  normalized : toLinearMap 1 = 1 -- normalized
  left_invariant : ∀ h f, toLinearMap (leftPull h f) = toLinearMap f -- left-invariant

theorem thompson_F_nonamenable_composition :
  ∃ group : Group F, let I := group;
  (∀ (h g : F) (x : UnitInterval),
    (h * g).val.toHomeomorph x = h.val.toHomeomorph (g.val.toHomeomorph x)) ∧
  ¬ Nonempty (InvariantMean F)
```

(A few technical fields in the middle of the definitions are omitted; the full file is in the repository.)

Word-by-word translation:

- `IntervalHomeomorph`: a homeomorphism from $[0,1]$ to itself (continuous, invertible, with a continuous inverse).
- `F`: the homeomorphisms that are "strictly increasing" and have "a witness of being dyadic piecewise linear". This is the definition from Section 5.1, with nothing missing.
- `BoundedReal G`: bounded real-valued functions on G (Mathlib's ℓ^∞). `leftPull h f` is the function $g \mapsto f(hg)$.
- `InvariantMean`: the four conditions from Section 3.1: linear, positive, normalized, left-invariant.
- The theorem says: **there exists** a group structure on F whose multiplication **is function composition** ($h * g$ acting on x equals $h(g(x))$), and under this group structure there **is no** invariant mean.

Why "there exists a group structure" is not a loophole: once the group multiplication is required to be composition, the identity and the inverses are uniquely determined. So this "exists" just means " F really is a group under composition", and leaves no room for cheating. The multiplication convention $hg = h \circ g$ also matches the paper.

Our assessment: the statement is faithful to Theorem 1.1 of the paper. In addition, the Lean proof includes a file `StandardCharacterization.lean` proving that the definitions in the challenge file are equivalent to the "textbook" formulation, and `MeanToFølner.lean`, which **proves in Lean itself** the direction "invariant mean $\implies$ Følner sets". So the Følner criterion that the paper cites is also machine-checked, not taken as an axiom.

7.2 The proof itself

- The proof lives under `lean/OAI/GroupTheory/Thompson/`: **68 files, 9,457 lines** (counted by script on snapshot fd4aeeb). Of these, 22 files (with names starting with `Analytic` or `Tube`) construct the map from Appendix A.
- We grepped it: this directory contains no `sorry`, no `axiom` declarations, and no `native_decide`.
- The Comparator configuration allows only Lean's three standard axioms (`propext`, `Quot.sound`, `Classical.choice`), and `enable_nanoda` is `false` (no second independent kernel check is turned on).
- **We did not compile it locally, and we did not run Comparator.** "Verified in Lean" is currently OpenAI's claim, plus our static inspection of the source.
- Unlike the quasi-Riemann hypothesis in Explainer 01: **as of 2026-10-08, we found no public record of any third party independently running this proof** (the quasi-Riemann hypothesis had an independent recheck by Dave Goldblatt using Comparator + nanoda). $\triangle$

7.3 Human experts

- Gil Kalai (blog post, 2026-10-07) cited his 2009 post introducing Thompson's group and wrote: "The new manuscript announces that Thompson's group (F) is nonamenable." He uses "announces", a neutral wording.
- Group theorist Xiaolei Wu (compiled on proofsandprompts.com): "So Thompson's group F is non-amenable. OK, I had tried it many times using ChatGPT, but I guess I didn't have the newest model or enough tokens."
- **As of 2026-10-08, we found no public comments from central figures such as Geoghegan, Moore, or Akhmedov, and no named expert has publicly said they have read and verified the paper.** ⚠

7.4 Remaining risks

1. **Statement risk is very low:** the challenge file is under 90 lines and relies only on standard Mathlib objects (real numbers, homeomorphisms, ℓ^∞ , linear maps), and the definitions can be checked line by line against a textbook. This is the ideal form of the division of labor "humans check the statement, machines check the proof".
2. **Compilation and toolchain risk:** we did not compile it ourselves, and there is no second-kernel check. Such risks are usually small, but not zero.
3. **Correspondence between the paper and Lean:** Lean proves Theorem 1.1 itself, so even if the paper's write-up has flaws, the main conclusion is unaffected. The two corollaries (Section 6.4) are outside what Lean covers.
4. **An unusual piece of good news:** the core proof in this paper is only about 5 pages, and the tools it uses (Hilbert-space inner products, convexity, Lipschitz estimates, dyadic intervals) are all standard. Compared with the 199-page quasi-Riemann hypothesis paper, it is one of the headline results that **human experts are most likely to read in full and independently confirm in the short term.** (This is speculation.)

8. What it means if true (and what it doesn't)

8.1 For mathematics

1. **A 47-year-old problem (1979→2026) has been solved, and the answer matches Geoghegan's conjecture:** F is not amenable.
2. **F becomes the most famous counterexample to the von Neumann–Day problem:** nonamenable, yet containing no F_2 . But as Section 4.3 said, it is **not the first natural counterexample** (Monod 2013 and Lodha–Moore 2016 came first).
3. **A methodological surprise:** translate "averaging over a group" into "a fixed-point problem on the infinite-dimensional unit ball", then borrow a counterintuitive phenomenon from functional analysis, like Benyamini–Sternfeld, to reach a contradiction. Whether this route

works for other groups (for example other groups of piecewise linear/piecewise projective homeomorphisms) is a natural open question. (Speculation.)

4. **What it does not mean:** the paper gives no explicit constant for a specific generating set (the Lean document `docs/248.md` says verbatim: "No explicit boundary constant or prescribed generating set is given."); nor does it answer F's other famous questions.

8.2 For AI and for "verification" itself

This is the best case for "why Lean matters", for concrete reasons:

- The human history of this problem is a string of "proof announced → error found → withdrawn", in both directions, and even the leading expert withdrew once (the table in Section 5.5). Going by past experience, the community might take a long time to reach consensus on a new 13-page claim.
- With Lean, the object of trust shifts from "the author" to **"the kernel + an 89-line statement that can be audited line by line"**.
- At the same time, it reminds us what Lean cannot do: it cannot judge the **significance** of the result, it cannot tell whether the method generalizes, and it cannot replace human understanding of "why it is true".

For you (an AI safety researcher), this case has one more layer: it is a sample of "AI gives a short proof of a problem humans failed at for a long time, and humans can recheck it in a short time". Together with "AI gives a 480,000-line proof that humans can barely recheck" (the quasi-Riemann hypothesis), it marks the two ends of a **scalable-oversight** spectrum. (Speculation, but worth writing down.)

9. Self-test

Try to answer these after reading. If you can't, go back and reread the corresponding section.

1. Explain in one sentence what "amenable" means. (Section 3.1)
2. Why are balls in F_2 not good Følner sets? Why is that not enough to prove F_2 is nonamenable? (Sections 3.2–3.3)
3. Why does $W(a) \cup a \cdot W(a^{-1}) = F_2$ imply that F_2 has no fair mean? (Section 3.3)
4. The Banach–Tarski paradox holds in three dimensions but not in two. What did von Neumann think was the root cause? (Section 4.2)
5. Why does F fall "through the gap between the two standard methods of deciding the question"? (Section 5.5)
6. Neither the Kakutani map nor the Benyamini–Sternfeld map has a fixed point. What is the difference? Why does OpenAI's proof need the latter? (Section 6.2)

7. Why is "there exists a group structure" in the Lean statement not a loophole? (Section 7.1)

Appendix: answers to the checks

- **Check 1:** **aa, ab, aB, AA, Ab, AB, bb, ba, bA, BB, Ba, BA.** **aA, Aa, bB, Bb** don't count, because they cancel to the empty word. Each first letter has 3 choices after it, $4 \times 3 = 12$ (checked by script).
- **Check 2:** $hA = \{-n+1, \dots, n+1\}$. Compared with A , it gains $n+1$ and loses $-n$, so $hA \Delta A = \{-n, n+1\}$, 2 elements, fraction $2/(2n+1)$. For $n=10$ this is $2/21 \approx 0.0952$, matching the table.
- **Check 3:** $x_0(1/2) = 1/4$, $x_0(3/4) = 1/2$. Apply x_0 again: $x_0(1/4) = 1/8$, $x_0(1/2) = 1/4$. You also need the point that the first x_0 sends to $3/4$: on $[3/4, 1]$, x_0 is $y = 1/2 + 2(x - 3/4)$; setting $y = 3/4$ gives $x = 7/8$. So the breakpoints of $x_0 \circ x_0$ are **$(0,0) \rightarrow (1/2, 1/8) \rightarrow (3/4, 1/4) \rightarrow (7/8, 1/2) \rightarrow (1,1)$** , with slopes **$1/4, 1/2, 2, 4$** (checked by script). The slopes are still all powers of 2 and the breakpoints are still dyadic rationals, so it is still in F .

Appendix: Glossary

Term	One-line explanation
Group	A set of operations that compose, have an identity, and can each be undone
Generators	A few elements from which the whole group can be built
Relations	Equations between generators that make different spellings represent the same element
Free group F_2	The group with two generators and no relations at all
Cayley graph	The graph with group elements as points and edges given by generators; for F_2 it is a 4-way tree
Invariant mean	A rule for scoring bounded functions: linear, positive, normalized, invariant under left shifts
Amenable	An invariant mean exists; equivalent to having Følner sets
Følner set	A finite set in which only a tiny fraction of elements change after a shift
Paradoxical decomposition	Cutting a group into a few pieces and shifting them to form two copies of itself; equivalent to nonamenability (Tarski)
von Neumann–Day problem	Must every nonamenable group contain F_2 ? Answer: no
Thompson's group F	The group, under composition, of piecewise linear increasing homeomorphisms of $[0, 1]$ with dyadic breakpoints and slopes that are powers of 2
Fixed point	A point with $f(x) = x$
Lipschitz map	A map that stretches the distance between two points by at most a factor L
Benyamini–Sternfeld	On the infinite-dimensional unit ball there is a Lipschitz map that moves every point by at least δ

Appendix: Further reading

- **The original paper** (13 pages; Section 2 is fully within your reach):
<https://github.com/openai/math/blob/main/preprints/Thompsons-group-F-is-nonamenable-September-23-2026/paper.pdf>
- **Lean statement:**
<https://github.com/openai/math/blob/main/lean/ComparatorChallenges/ThompsonNonamenability.lean>
; scope notes: <https://github.com/openai/math/blob/main/lean/docs/248.md>
- **Cannon, Floyd, Parry**, *Introductory notes on Richard Thompson's groups*, L'Enseignement Math. 42 (1996) 215–256: the standard introduction to F ; the generators and relations in Section 5 all come from here.
- **Terence Tao**, *The Banach–Tarski paradox* (lecture notes, UCLA): full derivations of the paradoxical decomposition and amenability from Sections 3.3 and 4. Search for "Tao Banach–Tarski paradox notes".
- **Juschenko**, *Amenability of discrete groups by examples* (AMS Mathematical Surveys and Monographs, 2022): a modern textbook on amenability, with a chapter on F . $\triangle$ Chapter number not checked.
- Primary historical sources: Moore's note on Shavgulidze <https://arxiv.org/abs/1102.0747> ; Moore's withdrawn manuscript <https://arxiv.org/abs/1209.2063> ; Akhmedov <https://arxiv.org/abs/0902.3849> and <https://arxiv.org/abs/1310.4395> ; Beklaryan <https://arxiv.org/abs/1112.1942>
- Natural counterexamples to von Neumann–Day: Monod PNAS 2013 (*Groups of piecewise projective homeomorphisms*); Lodha–Moore <https://arxiv.org/abs/1308.4250>
- Gil Kalai's 2009 blog post introducing F : <https://gilkalai.wordpress.com/2009/05/01/the-thompson-group/> ; the 2026 update: <https://gilkalai.wordpress.com/2026/10/07/updates-sharing-ai-progress-on-mathematics-amazing-and-my-lecture-plans/>
- Compilation of reactions: <https://proofsandprompts.com/2026/10/08/100-reactions-to-100-solutions/>

PART III · EXPLAINED FROM ZERO · GOES WITH §11

30. Explainer 08: Finite-time blowup for the Navier–Stokes equations (with external force)

This is the "explained from zero" version of §11 of the [notes/openai_math_explained/](#) deck. All you need: how to take derivatives, what a partial derivative $\partial/\partial x$ means, and vectors and dot products. You don't need to have studied partial differential equations (PDEs); PDEs are explained from the start. Every number here is either computed by script

or sourced. **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Tags: **[P]** = primary material I read myself (paper PDF, Lean file, official Clay text); **[R]** = secondhand reporting; Δ = unverified or disputed. As of 2026-10-08. Estimated reading time: 60–90 minutes. You can split it into three sittings: Sections 1–3 (PDEs, the fluid equations, what blowup means), Sections 4–6 (why it's hard, history, the Clay problem statement), Sections 7–9 (OpenAI's result, its verification, what it means).

0. The bottom line: the one thing this chapter explains

The Navier–Stokes (NS) equations describe how water and air flow, and engineers use them every day. But the most basic question about them has no answer: **starting from a smooth, gentle initial state, can the flow velocity become infinite in finite time?**

This chapter explains one causal chain:

The equations contain a nonlinear term where "the fluid pushes itself", and it keeps making the velocity steeper $\Leftarrow$ viscosity is smoothing it out $\Leftarrow$ which of the two wins depends on scale $\Leftarrow$ in 3 dimensions, the only "global ledger" we have (energy) controls less and less at smaller and smaller scales $\Leftarrow$ so nobody can prove that viscosity always wins, and nobody can build an example where it loses.

In 2000 Clay listed this as a Millennium Prize Problem, with **four alternative statements (A)–(D)**; proving any one counts as a solution. (A) and (B) say "with no external force, solutions stay smooth forever"; (C) and (D) say "**there exists** some smooth external force that makes the solution break down in finite time".

On 2026-09-08 OpenAI claimed proofs of (C) and (D): a smooth external force, acting only within a bounded region of space and time, makes a fluid that starts at rest develop infinite velocity at one point at $t = 1$, while total energy stays bounded throughout. It comes with about 640,000 lines of Lean. The same day they also released what may be the more important result: **unforced 3D Euler (inviscid fluid) blowup from smooth initial data**.

After this chapter, you should be able to explain in your own words:

- what a PDE is, and what "smooth solutions exist forever" and "finite-time blowup" each mean;
- why energy conservation cannot stop blowup in 3 dimensions (supercritical scaling, with numbers);
- what each of Clay's (A)–(D) says, and what role the external force f plays in (C)/(D);
- whether OpenAI's result **formally** counts as a solution under Clay's official text, which hurdles remain before "collecting the prize", and why many analysts think the real problem is still (A)/(B).

1. What is a PDE: starting with the heat equation

1.1 ODEs and PDEs

The differential equations you know, like $y'(t) = -y(t)$, have as their unknown **a single** number $y(t)$ that changes over time. These are called ordinary differential equations (ODEs).

In a partial differential equation (PDE), the unknown is **a whole field**: a value $u(x, t)$ at every position x and every time t . The equation involves both the time derivative $\partial u / \partial t$ and spatial derivatives $\partial u / \partial x$.

Cut space into small cells, one unknown per cell, with neighboring cells influencing each other: a PDE is "**infinitely many coupled ODEs**", like a distributed system in which each node only talks to its neighbors.

1.2 The simplest PDE: the heat equation

Take a thin iron rod, and let $u(x, t)$ be the temperature at position x at time t . The heat equation is:

$$\partial u / \partial t = k \cdot \partial^2 u / \partial x^2$$

The term $\partial^2 u / \partial x^2$ on the right is the "curvature": if a point is hotter than both neighbors (like a peak), it is negative and the temperature drops; if it is colder than both neighbors (like a valley), it is positive and the temperature rises. So **the heat equation says "each point moves toward the average of its neighbors"**.

It's easier to see on a grid. With grid spacing Δx and time step Δt , write $r = k\Delta t / \Delta x^2$; one update step is:

$$\text{new}_i = \text{old}_i + r \cdot (\text{old}_{i-1} - 2 \cdot \text{old}_i + \text{old}_{i+1})$$

One step by hand ($r = 0.25$, checked by script): 7 cells `[0, 0, 0, 1, 0, 0, 0]`, with a hot spot in the middle.

- Middle: $1 + 0.25 \cdot (0 - 2 + 0) = 0.5$
- The two neighbors: $0 + 0.25 \cdot (0 - 0 + 1) = 0.25$
- Result: `[0, 0, 0.25, 0.5, 0.25, 0, 0]`, and the sum is still 1.

The hot spot has spread out, but the total heat is unchanged.

[Check 1] Take one more step ($r = 0.25$) from `[0, 0, 0.25, 0.5, 0.25, 0, 0]`. What is the result? What is the sum?

1.3 Numerical simulation: the heat equation smooths everything out

Computed by script: $k = 1$, initial temperature 1 where $|x| < 0.5$ and 0 elsewhere (a "box" whose edges are infinitely steep steps), advanced with the scheme above on 1001 grid points.

t	Center temperature (numerical)	Center temperature (exact)	Total heat	Steepest slope
0	1.0000	1.0000	0.990	100 (the steepest step the grid allows)
0.01	0.9996	0.9996	0.990	2.82
0.1	0.7316	0.7364	0.990	0.83
0.5	0.3794	0.3829	0.990	0.22

(The total heat is 0.990 rather than 1 because the box on the grid is 0.99 wide.)

Remember: **the heat equation only makes things smoother**. An infinitely steep step instantly becomes a smooth curve, and the slope keeps falling. It is linear and cannot "blow up" on its own. The **viscosity term** in NS is mathematically "the heat equation applied to the velocity", and it is on the side of "smooth".

2. Fluids: the velocity field, and the NS equations term by term

2.1 The velocity field

The unknowns for a fluid are the **velocity field** $u(x, t)$ (a 3D vector at every point and every moment) and the **pressure** $p(x, t)$ (a number).

A 2D example: $u(x, y) = (-y, x)$. At the point (1, 0) the velocity is (0, 1), pointing up; at (0, 1) it is (-1, 0), pointing left. Drawn out, it is a rigid rotation **counterclockwise around the origin**, faster the farther from the center (speed = radius r).

2.2 Incompressibility: zero divergence

Water is nearly incompressible. Mathematically this is written as **zero divergence**:

$$\nabla \cdot u = \partial u_1 / \partial x + \partial u_2 / \partial y + \partial u_3 / \partial z = 0$$

Divergence measures "how much flows out of a small region, net". Zero divergence = as much flows in as flows out, so volume is preserved.

[Check 2] Compute the divergence of $u = (-y, x)$ and of $w = (x, y)$. Which one is incompressible? What does w look like when drawn?

2.3 The NS equations: Newton's second law, written for a small parcel of fluid

The official Clay problem statement (written by Fefferman) says: "Equation (1) is just Newton's law $f = ma$ for a fluid element" [P]. The equations are:

$$\underbrace{\partial u / \partial t + (u \cdot \nabla) u}_{\text{accel. } a} = \underbrace{\nu \Delta u}_{\text{visc}} - \underbrace{\nabla p}_{\text{pres}} + \underbrace{f}_{\text{force}} \quad \nabla \cdot u = 0$$

Term by term:

Left side: the acceleration of a small parcel of fluid. It has two parts:

- $\partial u / \partial t$: the change of velocity over time, as seen from a fixed point.
- $(u \cdot \nabla) u$: the change caused by the parcel **moving to a place where the velocity is different**. Even if the flow field doesn't change in time (steady flow), fluid moving along a streamline can still be turning and speeding up.

$(u \cdot \nabla) u$ is u times the derivative of u , so it is **quadratic in u** . This is where the **nonlinearity** of NS comes from, and it is the source of all the difficulty. Intuitively: the fluid "pushes itself along".

Computing this term by hand for the rotating flow. For $u = (-y, x)$, the first component of $(u \cdot \nabla) u$ is $u_1 \cdot \partial u_1 / \partial x + u_2 \cdot \partial u_1 / \partial y = (-y) \cdot 0 + x \cdot (-1) = -x$; the second component is $(-y) \cdot 1 + x \cdot 0 = -y$. So $(u \cdot \nabla) u = (-x, -y)$, **pointing toward the center, with magnitude r** . This is exactly the centripetal acceleration from high-school physics, $v^2/r = r^2/r = r$.

First term on the right: viscosity $\nu \Delta u$. $\Delta = \partial^2 / \partial x^2 + \partial^2 / \partial y^2 + \partial^2 / \partial z^2$, the same thing as on the right side of the heat equation. When neighboring layers of fluid move at different speeds (shear), they drag on each other and smooth out the velocity difference.

Second term on the right: pressure $-\nabla p$. It pushes from high pressure toward low pressure. In incompressible flow, pressure is "the force that adjusts itself automatically so that $\nabla \cdot u = 0$ holds at every moment", i.e. the Lagrange multiplier for the incompressibility constraint.

Third term on the right: external force f . For example gravity, a stirrer, or an electromagnetic force. **This term is the key to understanding OpenAI's result;** Section 6 is devoted to it.

[Check 3] For the rotating flow $u = (-y, x)$, take $\nu > 0$ and $f = 0$. (a) Compute Δu . (b) This flow is steady ($\partial u / \partial t = 0$), so the equation requires $(u \cdot \nabla) u = -\nabla p$. Find the pressure p . Is the pressure higher at the center or at the edge?

2.4 Reynolds number: nonlinearity versus viscosity

Consider a flow with length scale L and speed U :

- the time for the nonlinear term to carry velocity across one length scale is about L/U ;
- the time for viscosity to smooth out differences at this scale is about L^2/ν .

Their ratio is the **Reynolds number** $Re = UL/\nu$. Large Re : nonlinearity dominates (turbulence). Small Re : viscosity dominates (smooth, like honey).

Computed by script (water $\nu \approx 1.0 \times 10^{-6} \text{ m}^2/\text{s}$, air $\approx 1.5 \times 10^{-5} \text{ m}^2/\text{s}$): water flowing at 1 m/s through a 0.1 m pipe has $Re = 10^5$; air flowing at 30 m/s past a 4 m car has $Re = 8 \times 10^6$.

Keep this quantity in mind; Section 4 uses it: **the blowup question is whether, at smaller and smaller scales, the Reynolds number can go up instead of down.**

3. "Smooth solutions exist forever" versus "finite-time blowup"

3.1 First, an ODE toy: the equation is perfect, but the solution breaks

Compare two equations that look very similar, both with initial value $y(0) = 1$:

t	Solution of $y' = y$: e^t	Solution of $y' = y^2$: $1/(1-t)$
0.5	1.649	2
0.9	2.460	10
0.99	2.691	100

(Computed by script; integrating $y' = y^2$ numerically with RK4 up to $t = 0.99$ gives 100.0000000001, matching the exact solution.)

e^t is finite at every finite time; $1/(1-t)$ **becomes infinite at $t = 1$** . This is called **finite-time blowup**. The right-hand side y^2 could not be smoother; what breaks is the solution: "the bigger it is, the faster it grows", so the growth rate itself is growing.

3.2 Adding damping: who wins depends on the initial value

Now add a "viscosity-like" damping term: $y' = y^2 - y$. The substitution $z = 1/y$ solves it exactly: $z(t) = 1 + (z_0 - 1)e^t$, and y blows up if and only if z hits 0.

Initial value y_0	Outcome (computed by script)
0.5	Decays, $y(3) \approx 0.047$
1.0	Stays put, $= 1$ forever (equilibrium)
1.1	Blows up, $t^* = \ln 11 \approx 2.398$
2.0	Blows up, $t^* = \ln 2 \approx 0.693$

For small initial values damping wins; for large ones nonlinearity wins. The known theory of 3D NS has exactly this shape: Fefferman's statement says that in 3D "(A) and (B) hold provided the initial velocity u^0 satisfies a smallness condition", and that for large initial data, smooth solutions exist at least for a short time $[0, T)$ [P]. Whether large initial data stays smooth forever is the Millennium Problem.

3.3 A PDE toy: the Burgers equation

Cut NS down to 1 dimension and drop the pressure: $u_t + u \cdot u_x = \nu \cdot u_{xx}$. Initial value $u = -\sin x$. **Without viscosity**, the slope at $x = 0$ satisfies $(u_x)' = -(u_x)^2$, which has the same form as $y' = y^2$, giving $u_x(0, t) = -1/(1-t)$: at $t = 1$ the slope is infinite (a shock). **With viscosity**, the script simulates it with a spectral method, and the steepest slope is capped at about $1/(2\nu)$: 3.7 for $\nu = 0.1$ ($1/(2\nu) = 5$), 48.4 for $\nu = 0.01$ (50), 164.9 for $\nu = 0.003$ (166.7). The smaller ν , the higher the cap, but it is always finite (1D viscous Burgers can be turned into the heat equation by the Cole–Hopf transformation, and it is rigorously proven not to blow up).

The question for 3D NS is: can viscosity still cap things like this? 3D has one extra mechanism: **vortex stretching**. A vortex tube that gets stretched longer and thinner spins faster, like a figure skater pulling in their arms (angular momentum $r \cdot u_\theta$ is conserved: shrink the radius 10x, and the spin speed goes up 10x). This positive feedback does not exist in 1D or 2D.

3.4 Saying "blowup" precisely

"Smooth" means derivatives of all orders exist and are continuous (C^∞). **Global smooth solution**: smooth for all $t \geq 0$ with bounded energy. **Finite-time blowup**: the solution is smooth on $[0, T)$, but as $t \rightarrow T$, the velocity or one of its derivatives $\rightarrow \infty$ somewhere. Fefferman's statement records that for NS ($\nu > 0$), at blowup **the velocity itself** must be unbounded [P], which is why OpenAI's theorem is stated as " $\limsup \|u(t)\|_\infty = \infty$ ". Real water never has infinite velocity; the continuum model breaks down at molecular scales first. Blowup is a statement **about this mathematical model**.

4. Why 3D is so hard: energy and supercritical scaling

4.1 Energy: the only global ledger

With no external force, NS has a beautiful identity: the total kinetic energy $E(t) = \frac{1}{2} \int |u|^2 dx$ **can only decrease, never increase** (viscosity turns kinetic energy into heat). The nonlinear term $(u \cdot \nabla)u$ contributes exactly 0 to the total energy; it only **moves** energy between scales and never creates it.

So we have an upper bound valid for all time: $E(t) \leq E(0)$. The question is: **is this upper bound enough to prevent blowup?**

4.2 Scaling symmetry

NS has a symmetry: if $u(x, t)$ is a solution, then for any $\lambda > 0$,

$$u_\lambda(x, t) = \lambda \cdot u(\lambda x, \lambda^2 t)$$

is also a solution (with pressure and external force transformed accordingly). $\lambda > 1$ means "shrink the same flow to $1/\lambda$ of its size, play it λ^2 times faster, and the flow speed becomes λ times larger".

λ	Size	Time	Flow speed	3D energy	2D energy
1	1	1	$\times 1$	$\times 1$	$\times 1$
10	$1/10$	$1/100$	$\times 10$	$\times 0.1$	$\times 1$
100	$1/100$	$1/10^4$	$\times 100$	$\times 0.01$	$\times 1$
1000	$1/1000$	$1/10^6$	$\times 1000$	$\times 0.001$	$\times 1$

Where the energy column comes from: take $\int |\lambda u(\lambda x)|^2 dx$ and set $y = \lambda x$; in 3D, $dx = dy/\lambda^3$, so the energy is multiplied by $\lambda^2/\lambda^3 = 1/\lambda$.

[Check 4] Compute the 2D energy factor the same way, and verify that it is 1.

The 3D column is the key: shrink a "blowup structure" 1000 $\times$, speed its flow up 1000 $\times$, and the energy it needs is only one thousandth of the original. The energy bound places almost no constraint on small scales. This is called **supercritical**. In 2D the energy does not change under scaling, which is called **critical**; that is why 2D NS was proven long ago to be globally smooth (Ladyzhenskaya; see Fefferman's statement [P]).

4.3 Saying it again with the Reynolds number

Fix the total energy $E = 1$ and the viscosity $\nu = 1$. For a blob of flow of size L , how large can its speed be?

- 3D: energy $\approx U^2 L^3 \leq 1$, so $U \leq L^{-2/3}$, and the Reynolds number $Re = UL/\nu \leq L^{1/3}$.
- 2D: energy $\approx U^2 L^2 \leq 1$, so $U \leq 1/L$, and $Re \leq 1$.

Size L	Max Re allowed in 3D	Max Re allowed in 2D
1	1	1
10^{-2}	10	1
10^{-6}	1,000	1
10^{-8}	10,000	1

(Computed by script.)

2D: energy holds the Reynolds number at every scale below 1, so viscosity always has what it takes to win. 3D: the smaller the scale, the larger the Reynolds number the energy allows, so **viscosity might lose at small scales**. Energy neither rules out blowup nor proves it; it is simply "not enough". This is the core of the 3D NS problem.

4.4 Tao's "supercriticality barrier" (2016)

Tao (JAMS 29, 2016; arXiv:1402.0290) constructed an "averaged NS": it keeps the energy identity and the scaling, replaces only the nonlinear term with an averaged version, and he proved that it **does** blow up. Consequence: an argument that uses only "energy + generic harmonic-analysis estimates" cannot prove (A)/(B); it must use the fine structure of the real nonlinear term. He also proposed a **program**: build "logic gates" out of a real fluid, so that the fluid, like a self-replicating machine, passes its energy again and again to smaller, faster copies of itself, producing blowup in **unforced** NS. This program has not been carried out to this day.

5. 90 years of partial results

Year	Result	Significance
1934	Leray (Acta Math. 63): every finite-energy initial value has a global weak solution	Weak solutions satisfy the equation only "in an integral-average sense" and allow singularities; smoothness and uniqueness are unknown
1982	Caffarelli–Kohn–Nirenberg (CPAM 35): the singular set has parabolic 1-dimensional Hausdorff measure 0	Singularities "cannot form a curve in space-time" (Fefferman's wording [P]), but isolated points are not ruled out
2003	Escauriaza–Seregin–Šverák : if the L^3 norm stays bounded, there is no blowup	Blowup must make the L^3 norm diverge
2016	Tao : averaged NS blows up	See Section 4.4
2019	Buckmaster–Vicol (Annals 189): a class of weak solutions is non-unique	Weak solutions are too "soft"
2014 / 2022	Luo–Hou numerics (PNAS); Chen–Hou computer-assisted proof (arXiv:2210.07191): 3D axisymmetric Euler with a boundary blows up from smooth initial data	Needs a solid wall; not in all of space $\mathbb{R}^3$
2021	Elgindi (Annals 194): Euler on $\mathbb{R}^3$ blows up for $C^{(1,\alpha)}$ initial data	The initial data is not C^∞
2023 on	Córdoba–Martínez–Zorua (arXiv:2309.08495 and others): forced 3D Euler blows up, with a force of limited smoothness	Pioneers of the "forcing route": finer and finer vortex layers amplified stage by stage
2025	DeepMind et al. (arXiv:2509.14185): neural networks find unstable self-similar singularities for the Boussinesq, IPM and other equations, with accuracy close to machine precision	Paves the way for computer-assisted proofs in the boundaryless case
2026-08	Buckmaster–Alpöge : IPM, Boussinesq and 3D Euler blow up under smooth forcing , verified in Lean; made public September 7	Pushes the route in the previous row to smooth forcing [P, Buckmaster statement]

The pattern that keeps appearing in this table: **Euler (no viscosity) blows up more easily than NS; having a boundary, lower smoothness, or forcing is each easier than "all of space, C^∞ , no forcing"**. Every time one condition is relaxed, someone manages to produce blowup. Millennium Problem (A)/(B) is the version with no condition relaxed.

6. The Clay problem statement: four branches, and the role of the force f

6.1 The official text [P]

Fefferman's statement (<https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf>) first sets out "physically reasonable" conditions: all derivatives of the initial data and the external force decay rapidly (conditions 4, 5), the solution is C^∞ on $\mathbb{R}^3 \times [0, \infty)$ (6), and the energy is uniformly bounded (7); the periodic case uses conditions 8–11. Then:

"To give reasonable leeway to solvers while retaining the heart of the problem, we ask for a proof of one of the following four statements."

Branch	Domain	External force	What must be proven
(A)	$\mathbb{R}^3$	$f \equiv 0$	For every smooth decaying initial value $\rightarrow$ there is a global smooth solution with bounded energy
(B)	Periodic torus $\mathbb{R}^3/\mathbb{Z}^3$	$f \equiv 0$	Same, periodic version
(C)	$\mathbb{R}^3$	There exists a smooth decaying f	There exist some initial value and force for which no smooth, bounded-energy global solution exists
(D)	$\mathbb{R}^3/\mathbb{Z}^3$	There exists a smooth periodic f	Same, periodic version

Two points. (A)/(B) say "good for all initial values", while (C)/(D) say "there is one bad example": **opposite directions**. (A)/(B) require $f \equiv 0$, while (C)/(D) let you **choose the force yourself**, as long as it is smooth and decaying. Fefferman also writes that the corresponding problem for the Euler equations ($v = 0$) is "also open and very important", but it is not on the prize list [P].

6.2 Why "you can choose the force" changes the shape of the problem

Section 2 of the OpenAI paper says this itself [P]:

"For any incompressible flow u and pressure p , we can always define the external force f to be the residual in (1.1). The Navier–Stokes equations then hold by construction. The challenge is to choose a flow that blows up while this residual remains smooth."

In plain words: draw any incompressible flow that blows up, plug it into NS, subtract the right side from the left, and let f make up whatever the difference is. The equation then holds **automatically**. There is only one difficulty: **the force making up the difference must stay smooth**, and cannot break even at the blowup time.

An ODE toy makes the difficulty clear:

- **Linear equation $y' = -y + g(t)$:** to make $y = 1/(1-t)$ blow up, the needed force is $g = y' + y = 1/(1-t)^2 + 1/(1-t)$, which itself blows up and is not smooth. In fact, a linear equation with a smooth force **never** blows up.
- **Nonlinear equation $y' = y^2 + g(t)$:** for the same $y = 1/(1-t)$, $g = y' - y^2 = 1/(1-t)^2 - 1/(1-t)^2 = 0$. The force is 0, perfectly smooth.

So blowup must rely on the nonlinear term to cancel the singular part **by itself**, with the force only making up a smooth remainder. In NS, all divergent parts must cancel, to derivatives of every order; this is what those 166 pages are doing.

[Check 5] For $y' = -y + g(t)$, if g is smooth and bounded on $[0, 2]$ (say $|g| \leq 1$) and $y(0) = 0$, can $y(t)$ blow up at $t = 1$? Hint: compare with $y' \leq -y + 1$.

6.3 So is (C) easier than (A)?

In 2000 Clay put (C) side by side with (A), considering all four branches to be "retaining the heart of the problem". But the "forcing route" opened by Córdoba–Martínez–Zorua from 2023 on showed that **the force can do a great deal of "engineering" work**: inject energy where and when it is needed, and "plant" perturbations. In his statement Buckmaster says this route "is the route Luis and Diego opened", and calls the next goal "a path to unforced Euler" [P]. Even the people pushing this route see "unforced" as the deeper problem.

A secondhand summary: the forcing route is "a route the written problem permits but that most working mathematicians exclude from the question they care about" [R, Implicator]. $\triangle$ This is a journalist's summary, not a systematic survey of analysts.

7. What OpenAI claims

7.1 The main theorem [P]

The paper is *Finite time blowup for Navier–Stokes*, authored by "OpenAI", 166 pages, PDF generated 2026-09-08 12:06 PDT (<https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf>). Abstract:

"For every positive viscosity, we construct a solution of the three-dimensional incompressible Navier–Stokes equations that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy."

What Theorem 1.1 says: for every $\nu > 0$, there exist

- an external force $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty))$: smooth, and **nonzero only within a bounded region in both space and time** ("compactly supported");
- velocity and pressure u, p , smooth on $\mathbb{R}^3 \times [0, 1)$, with **initial value $u(\cdot, 0) = 0$** (starting from rest), and confined at all times to a fixed bounded region K ;
- with energy $\sup \|u(t)\|_{\{L^2\}} < \infty$ and velocity $\limsup_{t \uparrow 1} \|u(t)\|_{\{L^\infty\}} = \infty$.

The paper's own words: "This establishes alternative (C) in the Millennium problem statement ... Compact support also yields the corresponding construction on T^3 ... establishing alternative (D)" (Corollary 10.6).

Arbitrary viscosity, by hand. The paper first builds the construction at $\nu = 1$, then extends to any ν by the scaling $u_\nu(x, t) = \sqrt{\nu} \cdot u(x/\sqrt{\nu}, t)$. The blowup time does not change.

[Check 6] Verify: substituting $u_v = \sqrt{v} \cdot u(x/\sqrt{v}, t)$, each of the three terms $\partial u_v/\partial t$, $(u_v \cdot \nabla)u_v$ and $v\Delta u_v$ equals $\sqrt{v}$ times the corresponding original term.

7.2 Intuition for the mechanism (from reading Section 2 of the paper; a simplified version)

- 1. A vortex column that gets thinner and faster.** In the vortex core near the origin, fluid spirals inward and flows out up and down along the axis. Conservation of angular momentum makes the inflowing fluid spin faster; incompressibility makes the incoming fluid leave along the axis, so the inward flow can keep going.
- 2. Self-similar contraction.** Write $\tau = 1 - t$. The core radius $\ell_r \sim \tau^{(1/2)}$, the height $\ell_z \sim \tau^{(1/2-h)}$, where h is a fixed small number, $0 < h < 1/100$. The vortex column gets thinner and relatively longer.
- 3. The problem:** the transition annulus outside the core leaves a "residual", and the force needed to make it up diverges as $t \rightarrow 1$.
- 4. The fix: add oscillating pulses in the annulus.** The pulses have zero average velocity, but their **quadratic products** (momentum flux) do not average to zero. Two families of pulses are carefully arranged so that the nonlinear flux **exactly cancels** the divergent part of the residual. Each pulse is "planted" by an exponentially small force, and then grows by feeding on the background shear.
- 5. Outside:** a purely rotating flow that satisfies the heat equation and needs no force; then smoothly cut off to a bounded region.

This is exactly Section 6.2's "the nonlinear term cancels the singular part by itself", except that it must be done in 3D for derivatives of every order.

7.3 A feel for the numbers: how the vortex core shrinks

Take $h = 0.005$ (the paper only requires $0 < h < 1/100$; I chose this value for illustration), and compute with the scaling laws the paper gives (computed by script, $v = 1$, constant factors set to 1):

$\tau = 1 - t$	Core radius ℓ_r	Core height ℓ_z	Speed $\sim \tau^{(-1/2-h)}$	Core energy $\sim \tau^{(1/2-3h)}$	Swirl Reynolds number $\tau^{(-h)}$
10^{-2}	0.1	0.102	10.2	0.107	1.023
10^{-4}	0.01	0.0105	104.7	0.0115	1.047
10^{-8}	10^{-4}	1.10×10^{-4}	1.10×10^4	1.3×10^{-4}	1.096
10^{-12}	10^{-6}	1.15×10^{-6}	1.15×10^6	1.5×10^{-6}	1.148

- Speed $\rightarrow \infty$, core energy $\rightarrow 0$:** this is exactly the supercriticality from Section 4, reaching ever larger speeds at small scales with ever less energy.

- The speed carries an extra factor $\tau^{(-h)}$ beyond pure scaling $\tau^{(-1/2)}$; this is the **swirl Reynolds number**. It goes to infinity, but extremely slowly: as τ goes from 10^{-2} to 10^{-12} , it only rises from 1.02 to 1.15. The blowup is achieved right at the edge, "just barely beyond critical".
- Consistent with known theorems: the singularity is a **single point** (the origin, $t = 1$), which does not violate CKN; the cube of the L^3 norm $\sim \tau^{(-4h)}$, about 1.74 at $\tau = 10^{-12}$, and it is diverging, which fits the necessary condition from ESS ($\triangle$ ESS is about the unforced case; here it serves only as an intuition check).

7.4 The Euler result released at the same time

The same Lean repository also formalizes a second paper, *Finite time blowup for the Euler equation* (57 pages, <https://cdn.openai.com/pdf/315b36cd-ec98-4023-8342-93345194ece1/euler.pdf>) [P]:

"We exhibit finite-time blowup for the three-dimensional incompressible, unforced Euler equations from smooth, compactly supported, divergence-free initial data."

Compare with Section 5: Chen–Hou need a boundary, Elgindi's initial data is only $C^{(1,\alpha)}$, and Córdoba–Martínez-Zoroa and Buckmaster–Alpöge need a force. This result removes all of those relaxations; the only remaining difference from the Millennium Problem is "no viscosity". Fefferman says the corresponding problem for Euler is "also open and very important" [P]. My judgment (speculation): in terms of how close it is to the question people actually care about, it may carry more weight than the forced NS result. $\triangle$ I found no named expert's technical assessment of this paper.

Scale [R]: about 10,000 concurrent agents, 88 hours; Wikipedia cites 2.7 million messages and 130 billion output tokens, while other reports say "nearly 5 million" $\triangle$.

`formalization.yaml` records the model as "GPT-6 Astra" and the framework as Codex [P].

8. How far it has been verified

8.1 The Lean statement: OpenAI did not write it

The statement file `ComparatorChallenges/NavierStokes.lean` says at the top that it is copied from **Google DeepMind's Formal Conjectures project** (commit `8bf45ed`), with only the imports, namespace and notation changed [P]. If the solver writes the statement itself, it can quietly loosen conditions; here it comes from a third party. $\triangle$ I did not diff it line by line against upstream.

The statement of branch (C):

```

theorem navier_stokes_breakdown_R3 (nu : ℝ) (hnu : nu > 0) :
  ∃ (u₀ : ℝ³ → ℝ³) (f : ℝ³ → ℝ → ℝ³),
    InitialVelocityConditionDecay u₀ ∧ ForceConditionDecay f ∧
    ¬ (∃ v p, NavierStokesExistenceAndSmoothnessRn nu u₀ f v p)

```

Word-by-word translation:

- `(nu : ℝ) (hnu : nu > 0)`: for any positive viscosity ν ,
- `∃ (u₀ ...) (f ...)`: there exist an initial value u_0 and a force f ,
- `InitialVelocityConditionDecay u₀`: u_0 is smooth, divergence-free, and all its derivatives decay rapidly (Clay condition 4),
- `ForceConditionDecay f`: f is smooth, and all its space-time derivatives are $\leq C/(1 + |x| + t)^K$ (condition 5),
- `¬ (∃ v p, ...)`: such that there **do not exist** v, p satisfying all of NS (1), zero divergence (2), the initial value (3), C^∞ (6), and uniformly bounded energy (7).

I checked it item by item against the Clay text: (4)–(7) are all there; the periodic version (D) includes (8)–(11), and, as Clay's erratum requires, the pressure is also periodic [P]. **The structure of the statement matches Clay's (C)/(D)**. Note that (C) asks that "no global smooth solution exists at all", not just "the solution I built blows up"; this requires uniqueness of smooth solutions (the "Consequently..." sentence in Theorem 1.1), and that is proven in Lean too.

8.2 The proof itself

- Repository `openai/NavierStokesAndEuler` (snapshot `f9e8bc5`, 2026-09-10): **2,659 .lean files** (NS 816, Euler 1,839) [P], about 640,000 lines (a count from the author's research notes).
- `formalization.yaml`: sorry count 0; uses only the three standard axioms `propext`, `Classical.choice`, `Quot.sound`; Comparator has `enable_nanoda` turned on, i.e. a second, independently implemented kernel also checks it [P]. The review status reads "self-assessed".
- ⚠ I did not run `lake build` or Comparator, and found no public report of a third party running the NS Comparator successfully (the quasi-Riemann hypothesis had Goldblatt's independent recheck).

8.3 Human experts and institutions

- **Clay** (09-11) [R]: reportedly said "the Navier–Stokes problem has apparently been settled", and also "The process is deliberately unhurried, but we will provide updates". The problem page on Clay's website is still marked "Active" as of today (I checked on 10-08).

- **Gómez-Serrano** (NPR, 09-22) [R]: "The paper is not written for humans ... as of today, the paper doesn't teach us much."
- **Fefferman** [R]: "The heroes of the story ... are Córdoba and Martínez-Zoroa." ⚠ Original source not found.
- ⚠ **As of 2026-10-08, no human expert has publicly said they have read and verified the 166-page paper**, and there are no reports of journal submission.

8.4 The priority dispute (in brief)

- **Buckmaster's statement** (<https://cims.nyu.edu/~tristanb/statement.pdf>) [P]: he and Alpöge obtained Boussinesq and 3D Euler blowup under smooth forcing on August 15, verified it in Lean on August 22, and made it public on September 7 (dates per Wikipedia [R]). They also used LLMs heavily (Claude, Codex). He credits the route to Córdoba and Martínez-Zoroa. About OpenAI, he recounts that in a call on September 6, "Eventually it was agreed that [the first prompt] had been sent in the past few days, after information about our work had reached OpenAI", and also writes "I am not accusing anyone of anything."
- **OpenAI's response** [R]: "it is impossible for Dr. Buckmaster's Codex prompts over the last two months to have influenced the system in any way".
- **Citations** [P]: the PDF I read (generated 09-08 12:06 PDT) **already cites** three works by Córdoba–Martínez-Zoroa [6–8]; it **does not cite** Buckmaster–Alpöge.

8.5 Remaining risks

1. **Mathlib definitions**: the Laplacian, gradient, etc. come from Mathlib, and "junk value conventions" could make the statement drift from its intended meaning; but the statement requires C^∞ everywhere, so the effect is small.
2. **Toolchain**: Lean 4.34.0-rc2 is a pre-release version [P]; the nanoda dual-kernel check lowers the risk.
3. **Human understanding**: nobody can say whether this method extends to the unforced case.

9. What it means if true (and what it doesn't)

9.1 By Clay's text, does this count as "solving the Millennium Problem"?

The answer has three layers.

Layer one: the problem statement. Yes. The official text says "we ask for a proof of one of the following four statements", and (C) is one of them [P]. Rule 5(b) of the prize rules also states: "In the case of the P versus NP Problem and the Navier-Stokes Problem, a resolution in either direction will be evaluated by the standard evaluation procedure" [P, https://www.claymath.org/wp-content/uploads/2022/03/millennium_prize_rules_0.pdf]. Both directions are equally valid.

Layer two: the prize procedure. Still a long way off. Rule 4 requires all of the following [P]:

- (a) publication in a "qualifying outlet" (no reports of submission so far $\triangle$);
- (b) at least **2 years** have passed since publication;
- (c) "general acceptance" in the global mathematics community, as judged by CMI;
- (d) a satisfactory answer to the question posed in the official problem statement.

After that, CMI must also form a committee of at least 3 people to review it (Rule 7). Rule 8(b): CMI will "pay special attention to the question of whether a Prize solution depends crucially on insights published prior to the solution", and may add earlier contributors to the list of prizewinners; this is directly relevant to Córdoba–Martínez-Zoroa and Buckmaster–Alpöge. Rule 8(c): if no conclusion can be reached on correctness or **attribution**, the prize may not be awarded. Separately, OpenAI has reportedly said it will not claim the prize [R]. Speculation: even if it is published next year, with "2 years after publication", a conclusion would come around 2029 at the earliest.

Layer three: the question mathematicians care about. Most would say no. "Does a fluid blow up on its own?" is still completely open. OpenAI's result does not answer the unforced case, and does not carry out Tao's program.

Be careful with the "specification gaming" framing. §11.1 of the deck compares this to specification gaming in AI. But Fefferman included (C) **on purpose** in 2000, considering that it retains the heart of the problem. A more accurate way to put it: **the problem setters thought at the time that (C) was as hard as (A); progress since 2023 shows that the forcing route is easier than expected.** The gap between the text and the intent became visible in hindsight; it is not a case of the solver exploiting a loophole the setters never thought of.

9.2 For mathematics, physics and AI

1. **Mathematics:** if it holds, this is the first NS blowup in the setting of smooth forcing, all of space, and C^∞ : viscosity is not all-powerful. The unforced Euler blowup released at the same time (Section 7.4), if it holds, may be the bigger news. Methodologically, "synthesizing" the needed stress from the quadratic flux of oscillating pulses is related to convex integration for Euler (Daneri–Székelyhidi), as the paper itself says [P].
2. **Physics and engineering:** almost no direct impact. NS solvers never rely on global regularity theorems; in real fluids, molecular scales make the continuum model fail before a singularity is reached. The conclusion is "**the model itself** breaks under an artificially designed force", not "real water can reach infinite velocity".
3. **AI:** three independent gates: passing formalization $\neq$ a faithful statement $\neq$ acceptance by mathematicians. Here the second gate is fairly reliable because a third-party statement was

used; the third gate is far from passed. Buckmaster says LLMs mean one mathematician plus a model "can now do all this work in a month", and calls it "a Deep Blue-Kasparov moment" [P]. When an AI lab can quickly pour large amounts of compute into a direction after hearing it is making progress, how norms of priority and attribution should adapt is a real question.

10. Self-test

1. In one sentence: why does $y' = y^2$ blow up in finite time while $y' = y$ does not? (Section 3.1)
2. Which term in the NS equations is nonlinear? What is its physical meaning? (Section 2.3)
3. Why is energy for 3D NS called "supercritical"? Explain using the numbers for scaling $\lambda = 1000$. (Sections 4.2–4.3)
4. How do Clay's (A) and (C) differ in "external force" and in "quantifier (for all / there exists)"? (Section 6.1)
5. Why does "you can choose the force yourself" not make the problem trivial? What is the key difficulty? (Section 6.2)
6. Who wrote OpenAI's Lean problem statement? Why does that matter? (Section 8.1)
7. Under Clay's rules, what steps at minimum remain between now and a possible award? (Section 9.1)

Appendix: answers to the checks

- **Check 1:** `[0, 0.0625, 0.25, 0.375, 0.25, 0.0625, 0]`, sum 1 (checked by script). For example, the middle: $0.5 + 0.25 \cdot (0.25 - 1 + 0.25) = 0.375$.
- **Check 2:** $\nabla \cdot (-y, x) = \partial(-y)/\partial x + \partial x/\partial y = 0 + 0 = 0$, incompressible. $\nabla \cdot (x, y) = 1 + 1 = 2 \neq 0$: w is a flow "spraying" outward from the origin; every small region has net outflow, and volume is expanding.
- **Check 3:** (a) Each component of $u = (-y, x)$ is a linear function, so all second derivatives are 0 and $\Delta u = 0$: rigid rotation has no shear, so viscosity does nothing. (b) From Section 2.3, $(u \cdot \nabla)u = (-x, -y)$, so $\nabla p = (x, y)$ and $p = (x^2 + y^2)/2 + \text{constant}$. **Higher at the edge, lower at the center:** the pressure difference pushes inward from outside, supplying the centripetal force. When the OpenAI paper describes the vortex core as "Pressure decreases toward the axis, supplying the leading centripetal force", this is what it means.
- **Check 4:** In 2D, $dx = dy/\lambda^2$, so the energy factor is $\lambda^2/\lambda^2 = 1$.
- **Check 5:** No. From $y' \leq -y + 1$ and $y(0) = 0$ you can derive $y \leq 1 - e^{-t} < 1$; likewise $y \geq -1$. A linear equation with a bounded force has a solution that stays bounded.
- **Check 6:** $\partial u_v / \partial t = \sqrt{v} \cdot \partial u / \partial t$. $(u_v \cdot \nabla)u_v$: the velocity factors give $\sqrt{v} \times \sqrt{v}$, and the derivative ∇ acting on $x/\sqrt{v}$ brings $1/\sqrt{v}$, for $\sqrt{v}$ in total. $v \Delta u_v$: $v \times \sqrt{v} \times (1/\sqrt{v})^2 = \sqrt{v}$. All three

terms agree.

Appendix: Glossary

Term	One-line explanation
ODE / PDE	The unknown is a single number changing over time / a whole field
Heat equation	$\partial u / \partial t = k \Delta u$; each point moves toward the average of its neighbors, which only makes things smoother
Divergence $\nabla \cdot u$	Net outflow from a small region; $= 0$ means incompressible
Material derivative	$\partial u / \partial t + (u \cdot \nabla) u$, the acceleration of a small parcel of fluid itself
Viscosity $\nu \Delta u$	Neighboring fluid layers drag on each other; mathematically, the heat equation for the velocity
Reynolds number Re	UL/ν , the ratio of the strength of nonlinearity to viscosity
Weak solution	A solution that satisfies the equation only in an integral-average sense, allowing singularities (Leray 1934)
Scaling symmetry	$u_\lambda = \lambda u(\lambda x, \lambda^2 t)$ is still a solution
Supercritical	The conserved quantity (energy) controls less and less at small scales
Vortex stretching	A vortex tube stretched longer and thinner spins faster; only in 3D
Euler equations	NS with $\nu = 0$, an inviscid fluid
Clay (A)–(D)	The four alternative statements of the Millennium Problem; proving any one is enough

Appendix: Further reading

- **The official Clay problem statement** (Fefferman, 5 pages, strongly recommended to read in full): <https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf>
- **Clay prize rules**: https://www.claymath.org/wp-content/uploads/2022/03/millennium_prize_rules_0.pdf
- **OpenAI NS paper**: <https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf> (read Sections 1–2 first, about 6 pages, needing almost no PDE background)
- **OpenAI Euler paper**: <https://cdn.openai.com/pdf/315b36cd-ec98-4023-8342-93345194ece1/euler.pdf>
- **Lean repository**: <https://github.com/openai/NavierStokesAndEuler> ; upstream problem statement: <https://github.com/google-deepmind/formal-conjectures/blob/main/FormalConjectures/Millennium/NavierStokes.lean>
- **Buckmaster's statement**: <https://cims.nyu.edu/~tristanb/statement.pdf>

- **Terence Tao**, arXiv:1402.0290 (JAMS 2016): the introduction lays out the "supercriticality barrier" clearly; more accessible is his blog post *Why global regularity for Navier-Stokes is hard* (2007).
- **DeepMind et al.**, *Discovery of Unstable Singularities*, arXiv:2509.14185
- **Chen & Hou**, *Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: Analysis*, arXiv:2210.07191
- **Córdoba & Martínez-Zoroa**, *Blow-up for the incompressible 3D-Euler equations with uniform $C^{1,1/2-\varepsilon} \cap L^2$ force*, arXiv:2309.08495
- **Bertozzi & Majda**, *Vorticity and Incompressible Flow* (Cambridge, 2002): the textbook recommended in Fefferman's problem statement.
- Secondhand reporting (not checked word for word): Quanta 2026-09-08
<https://www.quantamagazine.org/ai-has-solved-one-of-maths-1-million-millennium-prize-problems-20260908/> ; Wikipedia "Navier–Stokes priority controversy"
https://en.wikipedia.org/wiki/Navier%E2%80%93Stokes_priority_controversy ; paraphrase of Clay's 09-11 statement <https://the-decoder.com/clay-mathematics-institute-says-the-navier-stokes-millennium-prize-problem-has-apparently-been-settled/> ; Implicator <https://www.implicator.ai/clay-institute-navier-stokes-openai-proof-claim/>

PART III · EXPLAINED FROM ZERO · GOES WITH §12

31. Explainer 09: The relativistic Vlasov–Maxwell system (can a plasma model break down on its own?)

This is the "explained from zero" version of §12 of the [notes/openai_math_explained/](#) deck. Prerequisites: you can take a derivative, you know the dot product and cross product of vectors, and you know that "a differential equation describes how something changes over time." You don't need to have studied partial differential equations (PDEs) or plasma physics. Every new concept is explained before it is used. Every number in the text was computed by script or has a source. **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Estimated reading time: 60–90 minutes. You can split it into three sittings: Sections 1–4 (plasma, phase space, the equations themselves), Sections 5–7 (relativity, what "the solution breaks down" means, 40 years of history), Sections 8–10 (OpenAI's result, how well it is verified, and what it means).

0. The bottom line: the one thing this chapter explains

Plasma (the ionized gas inside a fusion reactor or in the solar wind) has a standard mathematical model: the **relativistic Vlasov–Maxwell system**. It says: charged particles are

pushed around by the electromagnetic field, the particles' motion creates currents, and those currents in turn change the electromagnetic field. This loop is nonlinear.

The question mathematicians want answered is: **starting from a normal initial state, can a solution of this model "break down" in finite time?** That is, can some quantity suddenly become infinite, so that the equations stop making sense from then on?

This chapter explains one chain of reasoning:

Whether the solution breaks down $\iff$ (Glassey–Strauss 1986) depends only on "whether the momentum of the fastest particle can shoot to infinity in finite time" $\iff$ if you can prove "each doubling of momentum takes at least $c/\log P$ time" $\iff$ these times add up to a divergent series (like the harmonic series $1 + 1/2 + 1/3 + \dots$), so momentum cannot double infinitely many times in finite time.

OpenAI claims (2026-09-23, a 49-page paper + about 90,000 lines of Lean): in the **three-dimensional, single-species** case, for smooth initial data of any size and with no symmetry requirement, the solution **never breaks down** (it exists globally, is smooth, and is unique). This was an open problem for about 40 years.

After reading this chapter, you should be able to explain in your own words:

- why plasma has to be described by a "phase-space density" $f(t, x, v)$ rather than an ordinary density;
- why the whole problem reduces to "can a particle's momentum blow up in finite time," and why "the sum of the doubling times diverges" is enough;
- how well OpenAI's result has been verified, and what extra risks it carries compared with the ζ -function chapter;
- what it means for fusion engineering (almost nothing directly), and why.

1. Plasma: why we care

1.1 What a plasma is

Heat a gas to a high enough temperature and electrons get knocked out of the atoms. The gas turns into a soup of **positively charged ions + negatively charged electrons**. That is a plasma, often called the "fourth state" of matter.

Two examples you know:

- **Fusion reactors** (tokamaks, such as ITER): the core temperature is on the order of 10 keV. Converting to temperature: 1 eV corresponds to about 11,604.5 K, so 10 keV $\approx$ **116 million K** (computed by script).

- **The solar wind:** plasma the Sun continuously blows outward, with a typical speed of about 400 km/s (Δ this is the commonly quoted typical value; the real speed varies between roughly 300 and 800 km/s). As a fraction of the speed of light: $400 / 299,792 \approx \mathbf{0.0013}$ (computed by script).

1.2 "Collisionless": why fluid equations don't work

In an ordinary gas, molecules collide countless times per second. Collisions quickly turn the local velocity distribution into a "thermal equilibrium" bell curve, so the gas can be described by just a few quantities: density, mean velocity, and temperature. (That is fluid dynamics, for example Navier–Stokes in Explainer 08.)

A hot, thin plasma is different: a particle can travel a very long way between two collisions, in many cases farther than the size of the device (Δ the actual numbers vary a lot with density and temperature; only the order-of-magnitude relationship is stated here).

Collisions don't have time to "smooth out" the velocity distribution. At the same location there can be a group of fast particles and a group of slow ones at once, or even two streams of particles moving in opposite directions. **Recording only the mean velocity throws away key information.**

So we need a different description: record not only "how many particles are here," but also "at what velocity each of the particles here is moving." That is the phase space of the next section.

2. Phase space and the distribution function $f(t, x, v)$

2.1 Phase space: record position and velocity together

A particle's state is determined by two things: **where it is (x)** and **which way and how fast it is moving (v)**. Put the two together and you get a point (x, v) . The space of all possible (x, v) is called **phase space**.

- On a one-dimensional line, x is one number and v is one number, so phase space is a **two-dimensional plane**.
- In real three-dimensional space, x has 3 components and v has 3 components, so phase space is **6-dimensional**. Add time, and $f(t, x, v)$ is a function of 7 variables.

The **distribution function** $f(t, x, v)$ means: at time t , roughly how many particles there are per unit volume of phase space near (x, v) . Integrate f over all velocities and you get back the ordinary density:

$$\rho(t, x) = \int f(t, x, v) dv \quad (\text{total number of particles at position } x, \text{ regardless of velocity})$$

2.2 A toy example: why position alone is not enough

Four particles on a line, two scenarios:

	Particle positions	Particle velocities
Scenario A (two counter-streaming beams)	0, 0, 1, 1	+1, -1, +1, -1
Scenario B (at rest)	0, 0, 1, 1	0, 0, 0, 0

At $t = 0$, **looking only at positions**, the two scenarios are identical: 2 particles at position 0 and 2 at position 1.

But at $t = 2$ (each particle moves according to $x + v \cdot t$):

- Scenario A: the particles are at 2, -2, 3, -1. They have spread out.
- Scenario B: the particles are still at 0, 0, 1, 1.

So the ordinary density ρ cannot predict the future, but **the phase-space density f can**.

Physically, "counter-streaming beams" are exactly the typical configuration that triggers an instability in a plasma (the two-stream instability), and a fluid description can't see it.

2.3 Drawing a histogram in phase space (computed by script)

Here is a slightly larger example. 12 particles, positions x between 0 and 4, velocities ± 0.5 or ± 1 . Count them in bins of width 1 by position:

Position bin	$[-1, 0)$	$[0, 1)$	$[1, 2)$	$[2, 3)$	$[3, 4)$	$[4, 5)$
Particles per bin at $t = 0$	0	3	3	4	2	0
Particles per bin at $t = 1$ (each particle moves $v \times 1$)	1	0	5	3	2	1

(Computed by script; the particle data is in `精讲/scripts/09_vlasov_maxwell.py`.)

If you only have the 4 numbers in the $t = 0$ row, you cannot derive the $t = 1$ row. If you have "how many particles of each velocity are in each bin" (that is, a two-dimensional histogram in phase space), you can. **f is the limit of this two-dimensional histogram as the bins become infinitely small and the particles infinitely many.**

3. The Vlasov equation: particles follow trajectories, and f moves with them

3.1 First, no forces (free streaming)

With no forces at all, each particle moves in a straight line at constant speed: $x(t) = x_0 + v \cdot t$, and velocity doesn't change. A "clump of particles" in phase space just gets translated and sheared; particles don't appear or disappear. So

$$f(t, x, v) = f_0(x - v \cdot t, v)$$

This means: "the particle at (x, v) at time t is the particle that was at $(x - vt, v)$ at time 0."

Differentiate this expression with respect to t and you can verify that it satisfies

$$\partial f / \partial t + v \cdot \partial f / \partial x = 0$$

[Check 1] Let $f_0(x, v) = e^{-(x^2)} \cdot g(v)$, so under free streaming $f(t, x, v) = e^{-(x-vt)^2} \cdot g(v)$. Compute $\partial f / \partial t$ and $\partial f / \partial x$ directly and verify that $\partial f / \partial t + v \cdot \partial f / \partial x = 0$.

3.2 Adding a force: the Vlasov equation

When a force F acts, velocity changes too: $dv/dt = F$ (taking mass to be 1 here). f is still "constant along particle trajectories"; the trajectories are just curved now. As an equation:

$$\partial f / \partial t + v \cdot \nabla_x f + F \cdot \nabla_v f = 0$$

- $\nabla_x f$ is the vector of partial derivatives of f with respect to the three components of position; $\nabla_v f$ is the same with respect to the three components of velocity.
- The three terms mean: the change of f over time + the change due to particles moving in position + the change due to the force altering particle velocities = 0.

This is the **Vlasov equation** (1938), also called the "collisionless Boltzmann equation": add a collision term on the right-hand side and you get the Boltzmann equation.

An important consequence: since f is constant along trajectories, the **maximum of f can never grow** ($0 \leq f \leq$ the maximum of f_0). This is Lemma 3.1 in the paper. It means the bad scenario "particles crowd together in phase space to infinite density" cannot happen. Any danger has to come from somewhere else.

3.3 Where the force comes from: the Lorentz force

The force on a charged particle is the **Lorentz force** (charge and mass both normalized to 1):

$$F = E + v \times B$$

- E is the electric field, which pushes the particle directly;

- B is the magnetic field, which makes the particle **turn** through the cross product $\mathbf{v} \times \mathbf{B}$. $\mathbf{v} \times \mathbf{B}$ is always perpendicular to $\mathbf{v}$, so the magnetic field **only changes direction and does no work**. It doesn't change the particle's speed.

[Check 2] Take $\mathbf{v} = (1, 0, 0)$, $\mathbf{B} = (0, 0, 1)$. Compute $\mathbf{v} \times \mathbf{B}$ and verify that its dot product with $\mathbf{v}$ is 0.

4. Maxwell's equations: how particles create fields in return

4.1 The four equations in plain language

Set the speed of light $c = 1$ and absorb the various constants (this is the normalization the paper uses). Maxwell's equations are:

$\partial \mathbf{E} / \partial t - \nabla \times \mathbf{B} = -\mathbf{j}$	(the current $\mathbf{j}$ and a changing magnetic field produce a change in the electric field)
$\partial \mathbf{B} / \partial t + \nabla \times \mathbf{E} = 0$	(a changing electric field produces a change in the magnetic field; Faraday's law)
$\nabla \cdot \mathbf{E} = \rho$	(charge is the "source" of the electric field; Gauss's law)
$\nabla \cdot \mathbf{B} = 0$	(there is no "magnetic charge")

In plain language:

- **$\nabla \cdot$ (divergence)**: how much a field "flows outward" from a point. $\nabla \cdot \mathbf{E} = \rho$ says "wherever there is charge, the electric field flows out of it."
- **$\nabla \times$ (curl)**: how much a field "swirls" around a point. The first two equations say "the electric and magnetic fields drive each other by swirling." This mutual driving propagates outward at the **speed of light**; it is the electromagnetic wave (light, radio).

4.2 Self-consistent coupling: a nonlinear loop

Connect the particles and the fields:

$\rho(t, \mathbf{x}) = \int f \, d\mathbf{v}$	(charge density: add up the particles of all velocities)
$\mathbf{j}(t, \mathbf{x}) = \int (\text{particle velocity}) \cdot f \, d\mathbf{v}$	(current: particles carry their charge as they move)

This forms a loop:

f determines ρ and $\mathbf{j} \rightarrow \rho$ and $\mathbf{j}$ determine $\mathbf{E}$ and $\mathbf{B}$ through Maxwell's equations $\rightarrow \mathbf{E}$ and $\mathbf{B}$ determine how the particles move through the Lorentz force $\rightarrow$ the particles' motion changes f again.

Taken alone, the Vlasov equation is linear in f (for given E and B), and Maxwell's equations are linear in E and B . **The nonlinearity comes entirely from this loop:** the force $E + v \times B$ itself depends on f . This is the same kind of structure as "the velocity field transports itself" in Navier–Stokes in Explainer 08.

4.3 Two "conserved quantities"

Two things in this system never change (Section 3 of the paper uses exactly these):

- **the number of particles** (f is constant along trajectories, and phase-space volume is preserved);
- **the total energy** = the kinetic energy of all particles + the electromagnetic field energy $\frac{1}{2} \int (|E|^2 + |B|^2) dx$.

So "the total energy blows up" is impossible. If anything breaks, it can only be because **energy concentrates onto a very few particles, or onto extremely small scales**. This point is the key to understanding everything that follows.

5. Why relativity

5.1 The problem with the non-relativistic version

If you use Newtonian mechanics (velocity = momentum / mass, which can be arbitrarily large), particles can move faster than light. But in Maxwell's equations the fields propagate at the speed of light. Once a particle outruns the field it is producing, the model contradicts itself. So **the physically consistent version must be relativistic**: particle speeds are always below the speed of light.

5.2 The relation between momentum and velocity

Take $c = 1$ and mass = 1, and write the momentum as p (a three-dimensional vector).

In relativity:

$$\text{velocity } \hat{v} = p / \sqrt{1 + |p|^2}$$

⚠ **Notation warning:** in OpenAI's paper and Lean code, the letter **v** denotes momentum, and the velocity is written $u(v) = v / \sqrt{1+|v|^2}$ (written $\hat{v}$ in the reasoning summary). So the v in $f(t, x, v)$ is actually momentum. Below I consistently use p for momentum and $\hat{v}$ for velocity, to avoid confusion.

(Computed by script)

Momentum p	Velocity $\hat{v}$	Gap to light speed $1 - \hat{v}$
0.1	0.0995	0.90
0.5	0.447	0.55
1	0.707	0.29
2	0.894	0.11
10	0.995	0.0050
100	0.99995	0.00005
1000	0.9999995	0.0000005

At small momentum $\hat{v} \approx p$ (Newtonian mechanics). At large momentum $\hat{v}$ gets arbitrarily close to 1 but never reaches it. **Momentum can be arbitrarily large; velocity always stays below the speed of light.** This is why the "blowup" below has to be discussed in terms of momentum rather than velocity: velocity cannot possibly blow up.

[Check 3] What is $\hat{v}$ when $p = 3$? (Hint: $\sqrt{10} \approx 3.162$)

5.3 Where real particles sit in this table (computed by script)

The electron's rest energy is 511 keV. Given a kinetic energy K , we have $\gamma = 1 + K/511$, $p = \sqrt{(\gamma^2 - 1)}$, $\hat{v} = p/\gamma$:

Electron kinetic energy	Momentum p (in units of m·c)	Velocity $\hat{v}/c$
1 keV	0.063	0.062
10 keV (order of fusion core temperature)	0.20	0.19
100 keV	0.66	0.55
1 MeV	2.78	0.94
10 MeV	20.5	0.9988

Tokamaks have a dangerous phenomenon called **runaway electrons**: under a strong electric field, a small number of electrons are continuously accelerated to MeV or even tens of MeV ($\triangle$ the order of magnitude comes from fusion physics reviews, for example Breizman et al., *Nuclear Fusion* 59 (2019) 083001, not checked word for word). At that point relativistic effects cannot be ignored at all. Note: runaway electrons are a **physical phenomenon**, with very large but finite momentum. They are not the same thing as the "mathematically, momentum $\rightarrow \infty$ in finite time" discussed below.

5.4 The light cone: why Maxwell is harder than Poisson

If you ignore the magnetic field and assume the electric field responds "instantly" to the charge distribution ($\nabla \cdot \mathbf{E} = \rho$ and $\mathbf{E}$ is the gradient of some potential), you get the simpler

Vlasov–Poisson system. Its large-data problem in three dimensions was solved in 1991–92 (Pfaffelmoser 1992, Lions–Perthame 1991, Schaeffer 1991; the paper cites Pfaffelmoser, [20]).

Maxwell's fields don't respond instantly; they propagate at the speed of light. The field a particle feels at (t, x) comes from certain points in past spacetime: exactly those points that can send a signal to (t, x) at the speed of light. Together they form a **backward light cone**. The field produced this way is called the **retarded field**.

The difficulty: a source particle moving close to the speed of light is almost **running alongside the light it emits**. Its field piles up along the direction of motion (physicists call this "relativistic field compression"). When the source particle's velocity is almost parallel to the direction of the light ray, large factors appear in the retarded-field formula. The "loss" the OpenAI paper talks about (Section 8) happens right here.

6. What "the solution breaks down" actually means

6.1 An ODE toy: finite-time blowup

Look at the simplest equation $dp/dt = p^2$, $p(0) = 1$. The solution is $p(t) = 1/(1 - t)$: 2 at $t = 0.5$, 10 at $t = 0.9$, 100 at $t = 0.99$, and **it goes to infinity as $t \rightarrow 1$** . The equation itself is perfectly smooth, yet the solution breaks down at $t = 1$. This is called **finite-time blowup**.

Count it another way: p doubles from 1 to 2, from 2 to 4, from 4 to 8... Going from 2^k to 2^{k+1} ($k = 0, 1, 2, \dots$) takes time $1/2^k - 1/2^{k+1} = 1/2^{k+1}$.

Doubling number	Time taken by this doubling	Cumulative time
1st	1/2	0.5
2nd	1/4	0.75
3rd	1/8	0.875
60 doublings	...	$1 - 2^{-60}$ (computed by script ≈ 1.0)

The doubling times get shorter and shorter, and their sum converges (the geometric series sums to 1), so momentum can double infinitely many times in finite time. That is blowup.

6.2 The other way around: the sum of doubling times diverges $\Rightarrow$ no blowup

If you can prove "the k -th doubling takes at least c/k time," then the total time for N doublings is at least $c \cdot (1 + 1/2 + \dots + 1/N)$. This sum is called the **harmonic series**, and it **diverges**:

N	$1 + 1/2 + \dots + 1/N$ (computed by script)
10	2.93
60	4.68
100	5.19
1,000	7.49
1,000,000	14.39

It grows very slowly (roughly like $\ln N$), but without bound. So within any finite time T , momentum can only double finitely many times, and momentum stays bounded.

This contrast is the skeleton of the entire proof, and it is what the staircase figure in the deck (FIG_HARMONIC) shows:

Geometric series $1/2 + 1/4 + \dots$ (converges) $\rightarrow$ can blow up; harmonic series $1 + 1/2 + 1/3 + \dots$ (diverges) $\rightarrow$ cannot blow up.

[Check 4] If all you can prove is "the k -th doubling takes at least $1/k^2$ time," can you conclude there is no blowup? Why or why not?

(By the way: in Explainer 01, $\zeta(1) = 1 + 1/2 + 1/3 + \dots$ diverges; it is the same series.)

6.3 "Global regularity" for PDEs

For a PDE, "the solution breaks down" means some derivative or some quantity becomes infinite at a finite time, after which the equation can no longer hold pointwise. The opposite properties are:

- **Local existence:** starting from smooth initial data, a smooth solution exists at least for a short time interval $[0, T)$. For Vlasov–Maxwell this was proved by Wollman 1984 (paper [29]).
- **Global regularity / global classical solution:** this smooth solution can be continued for all time $t \geq 0$. This is what OpenAI claims.

There is another concept that is easy to confuse with these: **weak solutions**. DiPerna–Lions 1989 (paper [5]) proved that "weak solutions" exist globally for large data. A weak solution is allowed to be non-smooth, satisfies the equation only "on average," and is not known to be unique. So the existence of weak solutions **does not mean** the problem is solved. This is exactly the same situation as Leray's 1934 weak solutions for Navier–Stokes.

7. 40 years of history: how the problem was reduced, and where it got stuck

7.1 Glassey–Strauss 1986: you only need to control the fastest particle

Glassey & Strauss, "Singularity formation in a collisionless plasma could occur only at high velocities", ARMA 92 (1986) 59–90. The title is the conclusion:

If on $[0, T)$ the momenta of all particles stay below some finite number, then the solution can be continued smoothly past T .

Take the contrapositive: if the solution breaks down at a finite time T , then **some particle's momentum must have shot toward infinity before T** . The whole problem is reduced to a question about single particles:

Can a particle be accelerated to infinite momentum in finite time by the self-consistent electromagnetic field?

This criterion was later re-proved and strengthened many times: Klainerman–Staffilani 2002 with Fourier methods, Bouchut–Golse–Pallard 2003 with physical-space methods; Luk–Strain 2014 proved that it is enough to control the projection of momentum onto some fixed plane; Pallard, Sospedra-Alfonso–Illner and others gave various moment conditions (§1.1 of the paper lists the sources). But all of these are **conditional**: they tell you "it's enough for momentum to be bounded," not "momentum really is bounded."

7.2 Lower dimensions: Glassey–Schaeffer

If you make the problem "a few dimensions smaller," it can be done:

- **1½ dimensions** (1D position, 2D momentum): Glassey–Schaeffer 1990;
- **2½ dimensions**: Glassey–Schaeffer 1997;
- **2 dimensions** (2D position): Glassey–Schaeffer 1998, two papers (ARMA 141).

(All cited from the paper's references [6, 7, 9, 10].) In lower dimensions the singularity of the field is weaker, and energy conservation is enough to control acceleration. In three dimensions this "just barely enough" margin disappears.

7.3 Small data, symmetry

- **Small data / dilute plasma**: Glassey–Strauss 1987 (paper [12]), followed by Schaeffer 2004, Bigorgne 2020/2025, Wang 2022, Wei–Yang 2021, and others. The idea: if the particles are sparse enough, the field is weak and the acceleration is limited.
- **Symmetry**: Xuecheng Wang's large-data result under cylindrical symmetry (arXiv:2203.01199, v1 in 2022, v2 on 2026-07-16; the paper says "to appear in Annals of Mathematics"; a second paper is arXiv:2607.14685, 2026-07-16). This was the closest large-data result before OpenAI, but it requires the initial data to be rotationally symmetric about an axis.

Up to before 2026-09 (going by the survey in §1.1 of the paper), the three-dimensional large-data problem **with no size or symmetry restrictions** remained open.

7.4 Where the difficulty lies (intuitive version)

Conservation of total energy gives a "budget." To prove that a particle isn't accelerated to infinity, you need to estimate the total impulse it receives over a period of time. The most direct approach is to add up the **absolute values** of the force. But in three dimensions this estimate overshoots what you need by a factor of $\sqrt{w}$ (w is the energy of the receiving particle). It falls just short and can't be closed. The loss happens when the source particle's velocity is almost parallel to the direction of the light ray (Section 5.4).

So you have to exploit **cancellation**: the force is sometimes positive and sometimes negative, and its direction changes; if you integrate first and then take the absolute value, the result can be much smaller. The hard part is capturing this cancellation in a rigorous proof.

8. What OpenAI claims

8.1 The main result

The abstract of the paper *Global classical solutions of the three-dimensional relativistic Vlasov–Maxwell system* (OpenAI, 2026-09-23, 49 pages):

"We prove global existence and uniqueness for arbitrary smooth admissible initial data in the three-dimensional, one-species relativistic Vlasov–Maxwell system. The particle density is initially compactly supported, and the electromagnetic fields have finite energy and bounded derivatives of all orders. The solution remains smooth on every finite time interval. This resolves the large-data global classical regularity problem for this model, without size or symmetry restrictions on the data."

Breaking it down:

- **Dimension**: three-dimensional space + three-dimensional momentum ("3+3 dimensions").
- **Direction**: this is **global regularity** (no blowup), not blowup. That is the opposite direction from Navier–Stokes in Explainer 08.
- **One species**: there is only one kind of charged particle, with charge and mass both normalized to 1, and no background charge.
- **Initial data conditions** ("admissible"): f_0 is smooth, non-negative, and **compactly supported** (nonzero only in a bounded region of phase space, so the initial momentum is bounded); E_0 , B_0 are smooth, with all derivatives bounded and finite energy (they belong to L^2); and they satisfy the two Gauss constraints $\nabla \cdot E_0 = \rho_0$, $\nabla \cdot B_0 = 0$.

- **Conclusion:** there exists a unique global classical solution; it is smooth on every finite time interval $[0, T]$, and the particle support is compact (momentum is bounded, but **the bound may grow with T**).

The paper also points out specifically (after Theorem 1.1) that the total charge is allowed to be nonzero (so the electric field has a Coulomb tail $\sim 1/r^2$ far away), and it does not require all derivatives of the fields to be square-integrable.

[Check 5] The Coulomb field of total charge Q has magnitude about $Q/(4\pi r^2)$. Verify that outside radius R , the electric field energy $\frac{1}{2} \int |E|^2 dx$ is finite (you only need to check whether $\int_R^\infty (1/r^2)^2 \cdot r^2 dr$ converges). This shows that "allowing nonzero total charge" and "finite energy" don't contradict each other.

8.2 The skeleton of the proof: two propositions

Section 2 of the paper splits the whole proof into two pieces:

Proposition 2.2 (local theory + continuation criterion): the solution exists locally and is unique; if the maximal existence time $T_{\max}$ is finite, then the maximal momentum must be unbounded. This is the Glassey–Strauss criterion (in the formulation of Luk–Strain 2014), plus a technical fix: the Luk–Strain version requires the fields to be in H^5 (a Sobolev space, roughly "5 derivatives square-integrable"), while the initial data here only require bounded derivatives and finite energy, so the two don't quite match. The paper's approach is to keep the Coulomb part, cut off the rest of the field far away, and then add back the difference using "Maxwell solutions in vacuum," relying on "finite signal propagation speed" to guarantee that this doesn't affect the particles within the given time (§10). In the reasoning summary, the model found this gap itself and went back to check it several times.

Proposition 2.1 (signed momentum increment estimate): for a receiving particle with energy about w , over a time interval of length I ,

$$|\text{change in momentum}| \leq M \cdot P \cdot \sqrt{I} + A \cdot I \cdot P^2 \cdot \log(2+P) / w$$

where P is an upper bound on the current maximal momentum, and M, A are constants. The key word is "**signed**": first integrate the force over time (giving an impulse vector), then take its length, instead of first taking the magnitude of the force and then integrating.

8.3 From Proposition 2.1 to "no blowup": it's the staircase from Section 6

The paper's derivation (§2, "Proof of Theorem 1.1") goes like this: suppose $T_{\max}$ is finite; then the maximal momentum is unbounded. Let t_n be the first time the maximal energy reaches 2^n . During the n -th doubling, Proposition 2.1 gives

$$t_n - t_{n-1} \geq c / \log(2 + 1024 \cdot 2^n)$$

The denominator is about $n \cdot \log 2$, so the n -th doubling takes at least $\sim c/n$ time: **exactly the harmonic series**. Summing the paper's exact form (with $c = 1$) (computed by script):

Number of doublings N	$\sum 1/\log(2 + 1024 \cdot 2^n), n = 1 \dots N$
10	0.96
100	3.40
1,000	6.59
10,000	9.90
100,000	13.22

The sum grows without bound, which contradicts "doubling infinitely many times before a finite $T_{\max}$." So $T_{\max} = \infty$.

This step is the simplest in the whole paper. About 40 of the 49 pages go into proving Proposition 2.1.

8.4 How Proposition 2.1 is proved (I only read the introduction and Section 2; what follows is a simplified version)

- 1. Retarded fields + light-cone binning.** Follow the trajectories of the source particles to find where they intersect the backward light cone of the receiving particle, and sort the contributions into "bins" by momentum size, distance, and three angles (between the source velocity, the receiving velocity, and the direction of the light ray). Using the flux of energy through the light cone, plus estimates of "how long a source particle can stay in each bin" (occupation, i.e. occupation time), you get an upper bound on the **absolute value** of the force (§3–5).
- 2. Find the loss.** When the receiving energy is $\sim w$, the absolute-value estimate overshoots what is needed by a factor of $\sqrt{w}$. The loss is concentrated in a narrow strip of angles where "the source particle's velocity is closer to the light-ray direction than the receiving particle's is."
- 3. Signed cancellation** (§6). In this narrow strip, first integrate along the source particle's trajectory, then take the absolute value. An exact identity combines "the part without the source particle's acceleration" and "the part with the source particle's acceleration": the former exactly cancels a term produced by differentiating the light-cone geometry. The force is thereby written as "a total derivative + some less singular terms." One of these terms contains **the receiving particle's own acceleration**, which in turn equals the force on

the receiving particle, so it becomes "error $\propto$ the force on the receiving particle," and the coefficient in front of it has to be controlled.

4. **Count direction changes** (§7). Under the assumption that "all signed increments satisfy Proposition 2.1," the same identity is used to prove that the number of times a particle changes direction (turns through a given angle) is finite and countable. Between two turns, the relative motion is monotone, so the time a particle spends crossing a given angular region is compressed further, giving a better occupation estimate.
5. **Pick only the bins you need** (§8). The signed argument is used only on bins where the direct estimate exceeds a certain summable threshold. On those bins, after summing, the coefficient of the receiving particle's force is at most a constant $\times w^{(-1/2)}$, which **exactly cancels** the $\sqrt{w}$ loss.
6. **Bootstrap** (§9). All of the above is done under the premise "assume Proposition 2.1 holds on $[0, \tau]$." What comes out at the end is a **strictly better** estimate, and a continuity argument then pushes the assumption to the whole interval. Figure 1 in the paper draws this dependency graph.

An analogy in distributed-systems language: a bootstrap is a lot like "proving a protocol safe with an inductive invariant." Assume the invariant holds up to time τ , prove that it still holds a little beyond τ , and with margin to spare (a strict improvement), so it holds forever.

8.5 What the reasoning summary says

OpenAI published a 6-page "reasoning summary" ([reasoning_traces/relativistic-vlasov-maxwell.pdf](#)), released 2026-10-06). Some points worth noting:

- The model first tried a Pfaffelmoser-style "dwell time" strategy (exactly the route that solved Vlasov–Poisson in 1992), and also considered generalizing Wang's cylindrical-symmetry method to the general case.
- It hit a wall partway through and left concrete numbers: "the improved occupancy bound forced $H < 0.41z$, while the excessive coefficient required $H > 0.496z$ ". This rules out the "excessive coefficient" case through a contradiction between exponents: the two inequalities cannot both hold.
- The model found that the time window it was using to count direction changes was too long, and shortened it by a factor of $\sim M^2 + A \cdot \log P$.
- The last sentence is quite candid: "That assertion rests on the occupation bound, direction-count estimate, and cutoff cancellations needed to close the signed momentum bootstrap; the cited continuation theorem alone does not establish those estimates."

9. How well has it been verified

9.1 The Lean statement: much longer than the ζ chapter's, and the definitions are homemade

The Lean statement in Explainer 01 is one line, and the `riemannZeta` it uses is Mathlib's standard definition. **Here it's different:** Mathlib has no Vlasov–Maxwell equations, so the challenge file `lean/ComparatorChallenges/VlasovMaxwell.lean` (about 140 lines)

defines everything from scratch: momentum, velocity, cross product, curl, divergence, charge density, current, the equations themselves, and what a "classical solution" is.

The main theorem:

```
theorem global_classical_solution (d : Datum) (hd : Admissible d) :  
  ∃ s : Solution, Classical d s ∧ SmoothOnFiniteHorizons s ∧  
    ∀ s' : Solution, Classical d s' → SameNonnegativeTime s s'
```

Word-by-word translation:

- `d : Datum`: the initial data, a triple (f_0, E_0, B_0) .
- `hd : Admissible d`: the initial data satisfy `Admissible`, which unfolds to: f_0 is smooth (`ContDiff  $\mathbb{R}$   $\infty$`), compactly supported (`HasCompactSupport`), and non-negative; E_0, B_0 are smooth with all derivatives uniformly bounded (`BoundedSmooth`); E_0, B_0 are square-integrable (`MemLp ... 2`); $\nabla \cdot E_0 = \rho(f_0)$, $\nabla \cdot B_0 = 0$.
- `∃ s : Solution`: there exists a solution (f, E, B) .
- `Classical d s`: it is a classical solution: f, E, B are C^1 on $t \geq 0$; $f \geq 0$; E, B are continuous in time as L^2 -valued functions; on every finite time interval the support of f lies in a compact set; the equations hold pointwise; and at $t = 0$ it equals the initial data.
- `SmoothOnFiniteHorizons s`: on every $[0, T]$, f, E, B are infinitely differentiable.
- `∀ s', Classical d s' → SameNonnegativeTime s s'`: any other classical solution is identical to it on $t \geq 0$ (uniqueness).

This matches Theorem 1.1 of the paper, and the problem statement OpenAI gave the model (the prompt at the start of the reasoning summary).

I checked by hand the places in the definitions most likely to be written wrong:

- `velocity  $v = (q \ v)^{-1} \cdot v$` , `$q \ v = \sqrt{1 + \|v\|^2}$` : exactly $\hat{v} = p/\sqrt{1+|p|^2}$. ✓
- `cross`: $(a_1b_2 - a_2b_1, a_2b_0 - a_0b_2, a_0b_1 - a_1b_0)$, the standard cross product. ✓
- `curl`: first component $\partial_1 F_2 - \partial_2 F_1$, and so on; the standard curl. ✓

- **Equations**: $\partial_t f + \hat{v} \cdot \nabla_x f + (E + \hat{v} \times B) \cdot \nabla_v f = 0$; $\partial_t E - \nabla \times B = -j$; $\partial_t B + \nabla \times E = 0$; $\nabla \cdot E = \rho$; $\nabla \cdot B = 0$.
Matches the paper's (1.1) term by term, with the right signs. ✓

A trap specific to Lean: in Lean/Mathlib, the derivative of a non-differentiable function returns 0, and the integral of a non-integrable function also returns 0. If the statement didn't require smoothness, "the equations hold" could be satisfied by exploiting these default values. Here **Classical** requires C^1 and compact support, which closes that loophole (my judgment; no formal audit done $\triangle$).

9.2 The proof itself (local grep, snapshot fd4aeeb, 2026-10-07)

- Solution module **OAI/Analysis/VlasovMaxwell**: **207 .lean files, 89,993 lines** (computed by script). The largest subdirectories are **Population** (56 files, about 32,000 lines, corresponding to the paper's occupation and bin estimates) and **Retarded** (53 files, about 24,000 lines, corresponding to the retarded fields).
- grep finds no **sorry**, **admit**, **native_decide**, or custom **axiom**.
- **It imports only Mathlib** and doesn't depend on any patched external library. This is cleaner than Explainer 01 (which has a dependency patch of about 40,000 lines).
- In other words, the Glassey–Strauss continuation criterion **was not assumed as an axiom**; it was re-proved inside these 90,000 lines (**Continuation/GlobalContinuation.lean** contains theorems such as **bounded_momentum_uniform_restart**; I only looked at the theorem names and did not read the proofs $\triangle$).
- Comparator config **VlasovMaxwell.json**: only the three standard axioms are allowed (**propext**, **Quot.sound**, **Classical.choice**), and **enable_nanoda: false**, meaning **no re-check with a second, independent kernel**.

9.3 Independent checks and human experts

- **I found no public report of anyone successfully running Comparator on 362** (the Goldblatt re-check in Explainer 01 was for ζ). $\triangle$
- **As of 2026-10-08, I found no named public comment from any kinetic-theory / plasma PDE expert** (searched general media and the GitHub ecosystem around the release). $\triangle$ Media coverage just restates the abstract.
- A coincidence of timing worth noting: Wang's large-data cylindrical-symmetry paper had just been updated and accepted by Annals in 2026-07, and OpenAI's symmetry-free version appeared two months later. Experts in the field (Wang, Luk, Strain, Pallard, Bigorgne, and others) are the people best qualified to read these 49 pages.

9.4 Remaining risks

1. **Faithfulness of the statement**: 140 lines of homemade definitions; any subtle mistake could make the theorem "prove something else." I checked the equations and the key

definitions and found no problems, but this is not a formal audit.

2. **Scope:** single species only. Real plasmas have at least two species, electrons and ions, with opposite charges, so particles both repel and attract each other. Whether the proof generalizes directly is **unknown** $\triangle$. A caution: in the related relativistic Vlasov–Poisson system, the "repulsive" (plasma) and "attractive" (gravitational) cases can have completely different conclusions (Glassey–Schaeffer 1985, paper [8], proved global existence for the spherically symmetric repulsive case; the attractive case is known to be able to blow up $\triangle$ the latter point is from my memory, not rechecked against the original).
3. **No human has understood it:** the 49-page bootstrap and bin decomposition have so far only been checked by a machine.

10. What it would mean if true (and what it would not mean)

10.1 For mathematics

1. **A roughly 40-year-old flagship open problem (single-species case) is solved:** from the Glassey–Strauss 1986 reduction to "momentum really is bounded."
2. **Methodological novelty:** signed impulse estimates + counting direction changes + using cancellation only in the bins that need it. Whether this transfers to other kinetic systems (multi-species Vlasov–Maxwell, Einstein–Vlasov, etc.) is open (speculation).
3. **The conclusion is finer than "no blowup":** momentum is bounded on every finite time interval, but the bound may grow with time. It does **not** say what happens to particle momenta over long times (for example, whether they approach some stable state). Long-time behavior is a separate question.

10.2 For fusion and space physics: almost no direct impact

This deserves to be said clearly, because the media tend to write it up as "AI solves a hard math problem in fusion."

- **Engineering simulations don't depend on this theorem.** The fusion community uses particle-in-cell (PIC) methods, gyrokinetic codes, and fluid/MHD codes. They compute on discrete grids with finite resolution and finite time steps, and never need to assume that the continuum equations are globally smooth.
- **The model itself is far from a real device:** a real tokamak has at least two species of particles, collisions (important on long time scales), metal walls and boundary conditions, and strong magnetic fields from external coils. This theorem is proved for **all of unbounded three-dimensional space, with no external field and only one species of particle**.
- **Runaway electrons are not mathematical blowup:** the runaway electrons engineers worry about have very large but finite energy. The theorem says "nothing shoots to infinity in finite time"; it can't tell you how many MeV runaway electrons can reach at most in a real device.

A more accurate positioning: it is a **certificate of health for the model**. It tells physicists that this system of equations is self-consistent as a mathematical object: starting from reasonable initial data, it will not produce infinite momentum on its own in any finite time. (Note that what is controlled is each particle's **momentum**, not the total energy: the **total energy** is conserved anyway.) This has value for theorists (for example, it guarantees that the premises of certain derivations hold), but it won't change how anyone designs a fusion reactor today.

10.3 For AI

- This is **one of the weightiest mathematical-physics results in this batch**: a named open problem with a complete chain of literature, a clear conclusion, and **a Lean proof that depends only on Mathlib**.
- Compared with Explainer 01: the statement in the ζ chapter used community definitions (the risk was in the dependency patch); here the risk is in the **homemade statement**. For machine-proved PDE results, "is the statement faithful" is the new human bottleneck: reading 140 lines of Lean definitions is much easier than reading a 49-page paper, but it still needs someone who knows the field.
- The reasoning summary shows a process that looks fairly like a human researcher's: try known methods first, hit a wall, find a concrete contradiction between exponents, fix the time window, and repeatedly audit the order of dependencies. But what we see is a summary edited by OpenAI, not the raw reasoning.

11. Self-test

Try to answer these after reading. If you can't, go back and reread the corresponding section.

1. Why can't a plasma be described by the position density $\rho(t, x)$ alone? Give an example. (Section 2.2)
2. The Vlasov equation says "f is constant along particle trajectories." What property of the maximum of f follows from this? (Section 3.2)
3. Why, in the relativistic version, do we look at momentum rather than velocity when discussing "blowup"? (Section 5.2)
4. What did Glassey–Strauss 1986 reduce the whole problem to? (Section 7.1)
5. Why does "the k-th doubling takes at least c/k time" imply no blowup? What if it were $c/2^k$? (Sections 6.1–6.2)
6. Compared with the Lean statement for ζ in Explainer 01, where is the main extra risk in OpenAI's Lean statement? (Sections 9.1, 9.4)

7. Why is this result said to have almost no direct impact on fusion engineering? Give at least two reasons. (Section 10.2)

Appendix: answers to the checks

- **Check 1:** Let $s = x - vt$. $\partial f / \partial t = e^{(-s^2)} \cdot (-2s) \cdot (-v) \cdot g(v) = 2vs \cdot e^{(-s^2)} g(v)$; $\partial f / \partial x = e^{(-s^2)} \cdot (-2s) \cdot g(v) = -2s \cdot e^{(-s^2)} g(v)$. So $\partial f / \partial t + v \cdot \partial f / \partial x = 2vs \cdot e^{(-s^2)} g - 2vs \cdot e^{(-s^2)} g = 0$.
- **Check 2:** $\mathbf{v} \times \mathbf{B} = (0 \cdot 1 - 0 \cdot 0, 0 \cdot 0 - 1 \cdot 1, 1 \cdot 0 - 0 \cdot 0) = (0, -1, 0)$. Its dot product with $\mathbf{v} = (1, 0, 0)$ is 0. The particle is pushed in the $-y$ direction, but its speed doesn't change; it only turns.
- **Check 3:** $\hat{v} = 3/\sqrt{10} \approx 3/3.162 \approx \mathbf{0.949}$ (script: 0.9487).
- **Check 4:** No. $1/1^2 + 1/2^2 + 1/3^2 + \dots$ converges (to $\pi^2/6 \approx 1.645$, the $\zeta(2)$ of Explainer 01). The total time is finite, so momentum can double infinitely many times in finite time, and blowup can't be ruled out. You need a lower bound like $1/k$ whose "sum diverges."
- **Check 5:** $\int_{-R}^{\infty} (1/r^2)^2 \cdot r^2 \, dr = \int_{-R}^{\infty} r^{-2} \, dr = 1/R$, which is finite. With the constants included, $\frac{1}{2} \int_{\{|x|>R\}} |E|^2 dx = Q^2/(8\pi R)$; for $Q = 1$, the script computes $\int_{-R}^{\infty} (1/(4\pi r^2))^2 \cdot 4\pi r^2 \, dr = 1/(4\pi R)$, which is ≈ 0.0796 at $R = 1$ and ≈ 0.0080 at $R = 10$. In three dimensions a $1/r^2$ field is square-integrable (far away); the part near the origin doesn't diverge either, because the charge is smoothly distributed.

Appendix: Glossary

Term	One-line explanation
Plasma	An ionized gas, made of ions and electrons
Collisionless	Particles travel far between collisions, so collisions don't have time to smooth out the velocity distribution
Phase space	The space of position + velocity (or momentum); 6-dimensional in 3D
Distribution function $f(t, x, v)$	Number of particles per unit volume of phase space; in the paper v means momentum
Vlasov equation	f is constant along particle trajectories; the "collisionless Boltzmann equation"
Lorentz force	$E + \hat{v} \times B$; the magnetic field only changes direction and does no work
Maxwell's equations	How charges and currents produce electric and magnetic fields; fields propagate at the speed of light
Self-consistent coupling	The loop in which particles produce fields and fields push particles; the source of nonlinearity
Backward light cone / retarded field	The past spacetime points that can send a signal to the current point at the speed of light; the field coming from there
Vlasov–Poisson	Simplified version that ignores the magnetic field, with an instantly responding electric field; solved in 3D in 1991–92
Finite-time blowup	Some quantity of the solution goes to infinity at a finite time
Global classical solution / global regularity	A smooth solution exists for all $t \geq 0$
Weak solution	A solution that satisfies the equation only in an averaged sense, possibly non-smooth and non-unique
Continuation criterion	A theorem of the type "as long as some quantity is bounded, the solution can be continued"
Compact support	The function is nonzero only in a bounded region
Bootstrap	Assume an estimate holds, derive a stronger estimate, then use continuity to remove the assumption
Harmonic series	$1 + 1/2 + 1/3 + \dots$, which diverges

Appendix: Further reading

- **The original paper:** <https://github.com/openai/math/blob/main/preprints/Global-classical-solutions-of-the-three-dimensional-relativistic-Vlasov-Maxwell-system-September-23-2026/paper.pdf> (I suggest reading §1–2 first, 7 pages in total; the "Proof of Theorem 1.1" in Section 2 is the full version of Section 8.3 of this chapter)
- **The reasoning summary:**
https://github.com/openai/math/blob/main/reasoning_traces/relativistic-vlasov-maxwell.pdf

(6 pages)

- **The Lean statement:**

<https://github.com/openai/math/blob/main/lean/ComparatorChallenges/VlasovMaxwell.lean> ;
scope notes: <https://github.com/openai/math/blob/main/lean/docs/362.md>

- **Glasse & Strauss 1986**, ARMA 92:59–90, <https://doi.org/10.1007/BF00250732> (the reduction of the problem)
 - **Luk & Strain 2014**, *A new continuation criterion for the relativistic Vlasov–Maxwell system*, CMP 331:1005–1027, arXiv:1406.0165 (a modern, more readable proof of the continuation criterion; the OpenAI paper uses its formulation)
 - **Wang**, cylindrically symmetric large data: arXiv:2203.01199, arXiv:2607.14685 (the closest result before OpenAI)
 - **Robert Glassey**, *The Cauchy Problem in Kinetic Theory* (SIAM, 1996): the standard textbook of the field; the first few chapters cover the basic theory of Vlasov–Poisson and Vlasov–Maxwell.
 - **Introductory physics**: Francis F. Chen, *Introduction to Plasma Physics and Controlled Fusion* (its chapter on "kinetic theory" covers the Vlasov equation, $\triangle$ chapter number not checked); good for filling in plasma background.
-

PART III · EXPLAINED FROM ZERO · GOES WITH §13

32. Explainer 10: The Mézard–Parisi formula (diluted spin glasses)

This is the "explained from zero" version of §13 of the [notes/openai_math_explained/](#) deck. Prerequisites: you can work with exponentials and logarithms (e^x , $\ln x$), you know what an "expectation" (average) is, and you know the summation sign Σ . The physics concepts (energy, temperature, free energy) are all explained from scratch. The Hopfield network you met in neuroscience comes back in Section 10. Every number in the text was computed by script or has a source. **[Check]** items are small exercises you can work out yourself; answers are at the end of the chapter. Estimated reading time: 70–100 minutes. You can split it into three sittings: Sections 1–4 (spins, free energy, the replica method), Sections 5–7 (sparse graphs, the cavity method, the problem itself), Sections 8–10 (OpenAI's result, how well it is verified, and what it means).

0. The bottom line: the one thing this chapter explains

A magnet contains a huge number of tiny compass needles ("spins"), each of which can only point up or down. If the interactions between the needles are **random**, with some pairs

wanting to align and others wanting to point opposite ways, you get a situation where "no arrangement can satisfy everyone." Such a system is called a **spin glass**. The central quantity physicists care about is the **free energy**: once you know it, you can derive almost every macroscopic property of the system at every temperature.

Between 1979 and 2001, physicists (mainly Parisi and Mézard) wrote down formulas for the free energy using a set of **mathematically non-rigorous** methods (the replica method, the cavity method). Mathematicians spent decades turning parts of this into theorems. The chain of reasoning in this chapter is:

Free energy = the minimum of a "hierarchical trial functional" $\Leftarrow$ the upper bound was proved long ago (the interpolation method, 2003–04) $\Leftarrow$ the lower bound (equality) requires controlling "the spin patterns seen by several replicas at once" $\Leftarrow$ in the dense model (SK), pairwise overlaps are enough; in sparse models they are not, and this step was stuck for over twenty years.

OpenAI claims to have filled in the lower bound for **one class of sparse models** (Poisson-diluted, even-arity interactions, satisfying the Panchenko–Talagrand condition), thereby proving that equality holds, and has given a Lean formalization.

Note: the name "Mézard–Parisi" refers to at least two different things in the literature. What OpenAI proved is the **hierarchical cavity formula for diluted spin glasses** (Mézard–Parisi 2001), **not** the famous **random assignment problem** $\zeta(2) = \pi^2/6$ (Mézard–Parisi 1985–87, proved by Aldous in 2001). Section 6 covers the latter, because it is the best example for understanding "why physicists' non-rigorous methods are trustworthy."

After reading this chapter, you should be able to explain in your own words:

- what free energy is, why "averaging $\log Z$ " is hard, and what trick the replica method uses to get around it;
- the difference between dense (SK) and sparse (Viana–Bray / Bethe lattice) models, and why the cavity method is exact on trees;
- what exactly the theorem OpenAI proved covers and does not cover (3-SAT, the fixed-degree Bethe lattice, and Hopfield are all outside it);
- which of its connections to ML / neuroscience are real and which are only analogies.

1. Magnets and spins: the Ising model

1.1 Spins

Think of a magnet as a row of tiny compass needles, each with only two states: up is written $\sigma = +1$, down is written $\sigma = -1$. This variable that can only take ± 1 is called a **spin**. One

assignment of values to N spins (for example $+1, -1, -1, +1$) is called a **configuration**, and there are 2^N configurations in total.

You have seen the same thing in neuroscience: the binary "firing / not firing" model of a neuron, which is exactly how a Hopfield network encodes it.

1.2 Energy

Between two spins i, j there is a **coupling strength** J_{ij} . The energy of the system (physicists call it the Hamiltonian) is:

$$H(\sigma) = - \sum_{\text{connected } i,j} J_{ij} \cdot \sigma_i \cdot \sigma_j \quad (\text{sometimes with an extra term } - \sum h_i \sigma_i, \text{ where } h \text{ is an external magnetic field})$$

Look at a single pair of spins:

- $J_{ij} > 0$ (**ferromagnetic**): when $\sigma_i = \sigma_j$, the product $\sigma_i \sigma_j = +1$ and the energy contribution is $-J$, which is low; so they "want to align."
- $J_{ij} < 0$ (**antiferromagnetic**): they "want to point opposite ways."

Physical systems tend toward low-energy states. The configuration with the lowest energy is called the **ground state**.

1.3 The Boltzmann distribution: where temperature comes in

At temperature $T > 0$, the system doesn't stick to the ground state. Instead, it appears in every configuration with some probability, given by the **Boltzmann distribution**:

$$P(\sigma) = e^{(-\beta \cdot H(\sigma))} / Z, \quad Z = \sum_{\text{all configurations}} e^{(-\beta \cdot H(\sigma))}, \quad \beta = 1/T$$

β is called the **inverse temperature**. Z is the normalizing constant obtained by adding up all the weights, called the **partition function**. You see this formula in ML every day: it is just **softmax**, with the "logit" replaced by $-\beta \cdot \text{energy}$. β is the softmax's inverse-temperature parameter.

Smallest example: two spins, $J = 1$ (ferromagnetic), $H = -\sigma_1 \sigma_2$. Four configurations: $(+, +)$ and $(-, -)$ have energy -1 ; $(+, -)$ and $(-, +)$ have energy $+1$. Computed by script:

β	Z	P(a given aligned configuration)	P(a given anti-aligned configuration)	P(aligned, total)
0 (infinite temperature)	4.000	0.250	0.250	0.500
0.5	4.511	0.366	0.135	0.731
1	6.172	0.440	0.060	0.881
2	15.05	0.491	0.009	0.982
5	296.8	0.500	0.000	1.000

At high temperature all four configurations are equally likely (thermal motion smooths everything out); at low temperature essentially only the ground states remain. Temperature is the knob that sets the weighting between "energy" and "disorder."

[Check 1] By hand, compute $P(+,+) = e / (2e + 2e^{-1})$ at $\beta = 1$ and verify that it is about 0.440.

2. Frustration and spin glasses

2.1 The frustrated triangle

Put three spins on a triangle, with all three edges antiferromagnetic ($J = -1$): every pair wants to point opposite ways. The energy is $H = \sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3$.

The problem: three numbers that can only be ± 1 cannot all be pairwise different. So **no matter how you arrange them, at least one edge is unhappy**. This is called **frustration**.

The script enumerates all 8 configurations:

	Ground-state energy	Number of ground states	Zero-temperature entropy $\ln(\text{number of ground states})$
Ferromagnetic triangle ($J = +1$)	-3 (all three edges satisfied)	2 (all up, all down)	$\ln 2 \approx 0.693$
Antiferromagnetic triangle ($J = -1$)	-1 (only two edges can be satisfied)	6	$\ln 6 \approx 1.792$

The partition function of the antiferromagnetic triangle is $Z(\beta) = 2e^{(-3\beta)} + 6e^{\beta}$ (at $\beta = 1$ the script gets 16.409, matching the formula). Frustration has two consequences: (1) the ground-state energy is "raised" (-1 instead of -3); (2) the ground state is **highly degenerate** (many configurations tie for the lowest energy).

[Check 2] List the 8 configurations of the antiferromagnetic triangle with their energies, and confirm that exactly 2 have energy $+3$ and 6 have energy -1 .

2.2 Spin glasses: random couplings

If each edge's J_{ij} has a **random** sign (or is a random Gaussian number), frustration is everywhere: pick three random ± 1 signs, and the probability that the triangle is frustrated (the product of the three signs is negative) is exactly $1/2$ (the script enumerates the 8 sign combinations). Such a system is called a **spin glass**. The name comes from real materials (for example copper with a small amount of manganese mixed in): at low temperature their spins "freeze" in disordered directions, "orderly disordered" like glass.

There is a key distinction here that also shows up in ML: the couplings J are "**frozen randomness**" (quenched disorder), drawn once and then fixed; the spins σ are the variables that undergo "thermal motion" under the Boltzmann distribution. Analogy: J is a training set, and σ are the parameters on that training set.

2.3 A rugged energy landscape

"Frustration everywhere" produces a **rugged energy landscape**: many local minima, separated from each other by energy barriers. A script experiment: call a configuration where "flipping any single spin cannot lower the energy" a **single-flip stable configuration** (a local minimum), and count them:

System	N	Number of single-flip stable configurations
Fully connected ferromagnet (all $J = 1/N$)	12	2 (all up, all down)
SK spin glass (random Gaussian J)	12	13.1 on average (20 samples, min 6, max 24)
SK spin glass	16	23.2 on average (5 samples)

The literature result is that the number of single-flip stable configurations in SK grows **exponentially** with N , roughly as $e^{(0.199 \cdot N)}$ (Tanaka–Edwards 1980 / Bray–Moore 1980 [△](#) constant quoted from memory of the literature, not rechecked against the original). Plugging in $N = 12$ and 16 gives 10.9 and 24.2 , consistent in magnitude with the computed values in the table above.

3. Free energy: why physicists want it

3.1 Definition

Free energy $F = -(1/\beta) \cdot \ln Z$
 "Pressure" per spin, $(1/N) \cdot \ln Z$ (this is the quantity the OpenAI paper uses, called pressure)

Pressure = $-\beta F/N$: the two differ only by a factor of $-\beta/N$ ($-\beta$ from the sign and temperature, $1/N$ from "per spin"), so they are essentially the same thing.

3.2 Why they want it

Because $\ln Z$ is a "generating function": differentiating it gives almost every macroscopic quantity. For example, the mean energy $\langle E \rangle = -\partial(\ln Z)/\partial\beta$, and the entropy $S = \beta(\langle E \rangle - F)$. Computed by script on the frustrated triangle: at $\beta = 5$, the entropy of the antiferromagnetic triangle is $S = 1.7918$, almost exactly $\ln 6 = 1.7918$ (6 equally likely ground states); the entropy of the ferromagnetic triangle is $0.6931 = \ln 2$.

The ML counterpart: $\ln Z$ is just log-sum-exp; the ELBO is a lower bound on $\ln Z$.

3.3 The real difficulty: you have to average $\ln Z$

For a spin glass, Z itself is random (it depends on which J you drew). What is physically meaningful is the free energy of a **typical** sample, that is, $E[\ln Z]$ (the "quenched average").

$\ln E[Z]$ (the "annealed average") is much easier to compute: $E[Z]$ is the expectation of $\sum e^{...}$, so you can move the expectation inside the sum, and each term is a Gaussian integral. But the two are not equal. Since $\ln$ is concave, Jensen's inequality gives $E[\ln Z] \leq \ln E[Z]$.

Computed by script (SK model, $N = 10$, $\beta = 2$, 2000 random samples, exact enumeration of the 1024 configurations for each sample):

Quantity	Value per spin
Quenched $(1/N) \cdot E[\ln Z]$	1.387
Annealed $(1/N) \cdot \ln E[Z]$ (exact formula $\ln 2 + \beta^2(N-1)/(4N)$)	1.593

The gap is large. The reason is that $E[Z]$ is dominated by a very few "unusually lucky" samples and doesn't represent the typical case. For the SK model, as $N \rightarrow \infty$ the two are equal for $\beta \leq 1$ and separate for $\beta > 1$ (low temperature); this is exactly the spin glass phase transition (an established result).

All the difficulty in this chapter comes from "you have to compute $E[\ln Z]$, and $\ln$ cannot be swapped with the expectation."

4. The replica method and replica symmetry breaking: a tradition that is "not rigorous, but right"

4.1 The replica trick

Physicists use an identity:

$$\ln Z = \lim_{n \rightarrow 0} (Z^n - 1) / n$$

Check (script): $\ln 5 = 1.6094$; for $n = 1, 0.1, 0.01, 0.001$, $(5^n - 1)/n$ is 4.000, 1.746, 1.622, 1.611 in turn, which does approach $\ln 5$.

The benefit: when n is a **positive integer**, $Z^n = Z \cdot Z \cdot \dots \cdot Z$ is the partition function of **n independent copies** ("replicas"), and $E[Z^n]$ can be computed (again, you can move the expectation inside the sum). So the strategy is:

1. Compute $E[Z^n]$ for integer n ;
2. "Pretend" the resulting formula also holds for real n ;
3. Let $n \rightarrow 0$.

Step 2 has no mathematical justification at all (the values at the integers do not uniquely determine a function on the reals), and step 3 also requires swapping the two limits $n \rightarrow 0$ and $N \rightarrow \infty$. This is why it is called "non-rigorous."

4.2 Replica symmetry and its failure

When computing $E[Z^n]$, an $n \times n$ matrix Q_{ab} appears, called the **overlap**:

$$Q_{ab} = (1/N) \sum_i \sigma_i^a \cdot \sigma_i^b \quad (\text{how similar the configurations of replica } a \text{ and replica } b \text{ are, a value between } -1 \text{ and } 1)$$

The most natural guess is that all replicas are "equal," so all off-diagonal Q_{ab} are the same. This is called **replica symmetric** (RS). Sherrington and Kirkpatrick did exactly this for the SK model in 1975, and the result was clearly wrong at low temperature: the zero-temperature entropy came out as $-1/(2\pi) \approx -0.159$, while the entropy of an Ising system cannot be negative (there is at least 1 configuration, and $\ln 1 = 0$); the RS prediction for the ground-state energy, $-\sqrt{2/\pi} \approx -0.798$, also disagrees with numerical simulations.

4.3 Parisi's replica symmetry breaking

In 1979 Parisi proposed that the replicas are not equal but form a **hierarchy**. Split the n replicas into large groups, split each large group into smaller groups, those into even smaller groups... Overlaps within the same small group are large; overlaps between different large groups are small. Splitting into r levels is called **r -step replica symmetry breaking** (r -RSB), and $r \rightarrow \infty$ is called full RSB.

The physical interpretation (built up gradually starting with Mézard–Parisi–Sourlas–Toulouse–Virasoro 1984): at low temperature the Gibbs distribution splits into many "pure states" (roughly corresponding to the deep valleys of the energy landscape), and these valleys form a **tree**: the more "closely related" two valleys are, the deeper their common ancestor in the tree. This kind of distance structure is called **ultrametric**: among any three

points, the two largest distances are equal, just like the distances between species on a biological taxonomy tree.

The parameters $0 < m_1 < m_2 < \dots < m_r < 1$ that appear in the Parisi formula are the "branching strengths" at each level of the tree. The **Exponents m** in the OpenAI paper and Lean code later on refer to exactly these.

4.4 The physics was right; the math took 27 years

Year	Event	Nature
1975	Sherrington–Kirkpatrick propose the SK model (every pair of spins is coupled)	Definition
1979–80	Parisi writes down the SK free energy formula (RSB)	Physical derivation, non-rigorous
2003	Guerra uses the "interpolation method" to prove the Parisi formula is an upper bound on the free energy	Theorem
2006	Talagrand proves equality (Annals of Math. 163)	Theorem
2013	Panchenko proves ultrametricity (Annals of Math. 177)	Theorem
2021	Parisi wins the Nobel Prize in Physics (shared with Manabe and Hasselmann)	—
2024	Talagrand wins the Abel Prize (spin glass work is one of the main reasons)	—

The numerical value of the SK ground-state energy: the Parisi formula gives **−0.7632** per spin (high-precision value −0.763166..., Crisanti–Rizzo 2002 $\triangle$ as cited in the literature). The script enumerates small systems exactly ($H = -\sum_{\{i < j\}} J_{ij} \sigma_i \sigma_j$, $J_{ij} \sim N(0, 1/N)$):

N	Number of samples	Ground-state energy per spin
8	400	−0.571 ± 0.006
12	200	−0.609 ± 0.006
16	40	−0.655 ± 0.013
20	8	−0.683 ± 0.022

As N grows, the value moves toward −0.763, but slowly (finite-size corrections decay roughly like $N^{-(2/3)}$, $\triangle$ a result from the literature, not verified here by fitting). The point of this table: **brute-force enumeration of small systems cannot give you the limiting value; you need a formula.**

This is the most important background in this chapter: in the field of spin glasses, physicists' non-rigorous methods have repeatedly given the right answers, and

mathematicians have then spent anywhere from ten to several tens of years catching up. This OpenAI result is one link in that "catching up" chain.

5. Sparse graphs and the cavity method

5.1 Dense vs. sparse

In the SK model, **every pair** of spins is coupled, each spin has $N - 1$ neighbors, and each coupling is very weak (of order $1/\sqrt{N}$). This is called a **dense** (mean-field) model.

Real physics and many computational problems are **sparse**: each spin interacts with only **finitely many** other spins, so even when N is large, each spin has only a constant number of neighbors, like 3 or 5. Two standard random constructions:

- **Poisson dilution** (Viana–Bray 1985): scatter αN edges at random (in OpenAI's theorem the number of edges is itself drawn as $\text{Poisson}(\alpha N)$); each spin's degree is approximately Poisson-distributed.
- **Bethe lattice / random regular graph**: every spin has the same **fixed** degree. The original Mézard–Parisi 2001 paper uses this one.

5.2 Sparse random graphs "look locally like trees"

The key property: sparse random graphs have **very few short cycles**. Computed by script (Erdős–Rényi graphs with average degree 3):

N	Average number of triangles	Average degree	Fraction of isolated vertices
100	4.8	2.82	0.058
1,000	3.8	2.99	0.053
10,000	4.1	3.01	0.049
100,000	3.4	3.00	0.049

Theoretical values: the number of triangles tends to $c^3/6 = 4.5$ (**it does not grow with N**), and the fraction of isolated vertices tends to $e^{-3} \approx 0.0498$. In other words, the bigger the graph, the closer to zero the fraction of vertices with a short cycle nearby: looking a few steps out from any vertex, what you see is almost always a tree.

5.3 The cavity method: exact on trees

The idea of the **cavity method**: remove one spin and see how the rest of the system "treats" the empty slot.

On a **tree** this is completely exact. After removing spin i , its neighbors belong to separate subtrees that are not connected to each other, so they are independent. Each neighbor j

sends i a "message" (an effective field $u_{j \rightarrow i}$), and i adds up the messages it receives to get its own state. The update rule for messages is:

$$u_{j \rightarrow i} = (1/\beta) \cdot \operatorname{artanh}(\tanh(\beta \cdot J_{ij}) \cdot \tanh(\beta \cdot h_j^{\text{cavity}}))$$

where h_j^{cavity} is "the total field felt by j after i is removed." This is **belief propagation (BP)** from ML: the same algorithm, invented separately by physicists and by Pearl.

Smallest example ([Check 3] has you compute it by hand): two spins, $J = 1$, an external field $h = 0.5$ on spin 1, $\beta = 1$. The magnetization (mean value) of spin 2 is exactly $\tanh(1) \cdot \tanh(0.5) = 0.35195$. Exact enumeration by script gives 0.35195, which matches.

The script also does a tree with 5 vertices and random couplings and external fields: comparing the 5 magnetizations computed by exact enumeration of the 32 configurations against the BP result, the largest difference is 4.9×10^{-16} (machine precision). Add two more edges to create cycles, and the largest difference becomes **0.16**. **Cavity / BP is exact on trees and inexact with short cycles.** Sparse random graphs "look locally like trees," so there is reason to expect the cavity method to become exact as $N \rightarrow \infty$, but this needs proof.

5.4 The complication: with many pure states, messages become "distributions of distributions"

At high temperature, BP has a unique fixed point, with one message per edge; averaging over random graphs, the order parameter is "a distribution of messages" $\zeta \in P(\mathbb{R})$. This is the RS level ($r = 0$).

In the low-temperature spin glass phase there are many pure states (the tree from Section 4.3), and the message on an edge differs from one pure state to another. So you have to describe "how the message is distributed across pure states," then average over the randomness of the graph, and the order parameter becomes "a distribution of distributions of messages"; with r levels of RSB, there are $r + 1$ levels of nesting:

Level 0: $\mathbb{R}$ (one message)
 Level 1: $P(\mathbb{R})$ (a distribution of messages)
 Level 2: $P(P(\mathbb{R}))$ (a distribution of distributions of messages)

The **Hierarchy** in the Lean statement is exactly this definition: $H_0 = \mathbb{R}$, $H_{\{r+1\}} = P(H_r)$. This is what "hierarchical" means in "hierarchical cavity formula."

Mézard–Parisi 2001 (*The Bethe lattice spin glass revisited*, EPJ B 20:217) wrote down this framework and an explicit 1RSB functional for the fixed-degree Bethe lattice; the general form for any finite number of levels r , and the proof that it is an upper bound on the free energy, come from Panchenko–Talagrand 2004 (see Section 7).

6. The other "Mézard–Parisi": random assignment and $\zeta(2)$

This section is not OpenAI's result, but it is the most beautiful example of "physicists' non-rigorous method $\rightarrow$ mathematical theorem," and it is the same number as $\zeta(2) = \pi^2/6$ in Explainer 01.

6.1 The problem

n workers, n jobs. The cost $C_{i,j}$ for worker i to do job j is an independent random number, exponentially distributed with mean 1. Find a one-to-one assignment that minimizes the total cost. Question: what is the expected minimum total cost?

Intuition might say it grows with n (after all, you are adding n numbers), but the optimal assignment picks very small numbers from each row, and as n grows, the "very small numbers" get smaller and smaller.

6.2 Prediction and proof

- **Mézard–Parisi 1985/1987**: using the replica method, they predicted that as $n \rightarrow \infty$ the expected minimum cost $\rightarrow \zeta(2) = \pi^2/6 \approx 1.6449$.
- **Aldous 2001** (*The $\zeta(2)$ limit in the random assignment problem*, Random Structures & Algorithms 18): a rigorous proof, using "local weak convergence" (the same idea as the cavity method's "locally tree-like").
- **Parisi 1998** went further and conjectured an **exact** formula for finite n : $E[\text{minimum cost}] = 1 + 1/4 + 1/9 + \dots + 1/n^2$. This was proved independently by Linusson–Wästlund in 2004 and Nair–Prabhakar–Sharma in 2005.

Computed by script (optimal assignment via the Hungarian algorithm, `scipy.linear_sum_assignment`):

n	Number of trials	Simulated mean minimum cost	Parisi formula $\sum_{k \leq n} 1/k^2$
1	200,000	1.0000 $\pm$ 0.0022	1.0000
2	200,000	1.2498 $\pm$ 0.0019	1.2500
3	100,000	1.3628 $\pm$ 0.0023	1.3611
5	50,000	1.4614 $\pm$ 0.0026	1.4636
10	20,000	1.5500 $\pm$ 0.0029	1.5498
30	4,000	1.6093 $\pm$ 0.0038	1.6122
100	1,000	1.6373 $\pm$ 0.0042	1.6350
300	200	1.6435 $\pm$ 0.0055	1.6416
1000	40	1.6381 $\pm$ 0.0066	1.6439

Every row agrees within one or two standard errors, and for large n both approach $\pi^2/6 = 1.6449$. For comparison, a greedy method where "each worker in turn picks the cheapest remaining job" has an average cost of **5.14** at $n = 100$, more than three times the optimum.

[Check 4] For $n = 2$, with four costs a, b, c, d (matrix $[[a,b],[c,d]]$), the minimum cost is $\min(a+d, b+c)$. Prove that its expectation is $5/4$. (Hint: the sum X of two independent $\text{Exp}(1)$ variables satisfies $P(X > t) = (1+t)e^{-t}$; for a non-negative random variable, $E[Y] = \int_0^\infty P(Y > t) dt$. A direct script simulation with 2 million trials gives 1.24989.)

Lesson: the number the replica method gave ($\pi^2/6$) was right, and was later proved rigorously. Success stories like this, piling up, are the reason mathematicians are willing to spend decades proving the Parisi / Mézard–Parisi formulas.

7. What exactly is the problem: the upper bound is known, equality is open

7.1 What needs to be proved

For the diluted model, define the per-spin pressure $F_N = (1/N) \cdot E[\ln Z_N]$. The hierarchical cavity functional $B_r(\zeta, m)$ gives a number for any "trial law" ζ (the nested distributions from Section 5.4) and any branching parameters $0 < m_1 < \dots < m_r < 1$. The conjecture is:

$$\lim_{N \rightarrow \infty} F_N = \inf_{r \geq 0} \inf_{\{\zeta, m\}} B_r(\zeta, m)$$

The right-hand side takes the infimum over **all finite numbers of levels and all trial laws**.

7.2 Half of it is already proved: the upper bound (2003–04)

- **Franz–Leone 2003** (J. Stat. Phys. 111): carried Guerra's interpolation method over to diluted models and proved the RS and 1RSB upper bounds for even-arity models.

- **Panchenko–Talagrand 2004** (PTRF 130): proved the upper bound for any finite number of levels r : for every r, ζ, m , $\limsup F_N \leq B_r(\zeta, m)$. After Theorem 4 they **explicitly state equality as a conjecture**. The OpenAI paper says this is the conjecture it resolves (within the same model class).

The intuition for the interpolation method: build a path that continuously deforms "the real model" into "a trial model made of independent messages," and prove that the derivative of the pressure along the path has a fixed sign. Determining the sign uses a positivity condition; this is where $E[(-b)^n] \geq 0$ in the Panchenko–Talagrand condition below comes from, and also why only **even-arity** interactions can be handled.

7.3 Why the lower bound is hard

In SK, when Talagrand proved the lower bound, he only needed to control the **pairwise overlaps** Q_{ab} . In a sparse model, a newly inserted spin connects simultaneously to $p - 1$ old spins, and you need to know what pattern several replicas see **at the same time** at those positions. This depends on all the **multioverlaps**:

$$R_S = (1/N) \sum_i \prod_{a \in S} \sigma_i^a \quad (S \text{ is a set of replicas})$$

The ultrametric structure of pairwise overlaps does not determine all the multioverlaps (in the words of §1 of the paper: "pair overlaps alone do not determine all these patterns").

Existing partial progress (listed in the paper's introduction): Panchenko 2013 gave a different exact variational formula using "spin distributions," but not in hierarchical cavity form; Panchenko 2016 (arXiv:1406.4702) proved for "modified models" that finite-RSB Gibbs measures have the Mézard–Parisi structure, but was missing the step "a general state can be approximated by finite RSB"; Coja-Oghlan–Perkins 2019 gave another variational formula for random regular factor graphs under a positivity condition; Concetti 2025 (arXiv:2508.17049) gave a full-RSB upper bound for random regular graphs, with equality still a conjecture.

Neighboring known results (background):

- **Ferromagnetic** sparse Ising models (no frustration): Dembo–Montanari 2010 proved the Bethe formula.
- **Random k-SAT** (each clause is a k -ary interaction): Ding–Sly–Sun (STOC 2015, Annals 2022) proved the satisfiability threshold for **sufficiently large k** , which is exactly what the 1RSB formula predicts.

8. What OpenAI claims

8.1 The main result

Abstract of the paper *The Mézard–Parisi formula for diluted spin glasses* (OpenAI, 2026-09-23, 36 pages, family 221):

"We prove the Mézard–Parisi hierarchical cavity formula for diluted even-arity Ising models in the Panchenko–Talagrand class. This resolves the variational equality conjecture for that class: the limiting pressure equals the infimum of the trial functional over all finite hierarchy depths and trial laws. Only first moments of the interaction and external field are required."

Main theorem (Theorem 2.1): under assumptions (2.1)–(2.4), the thermodynamic limit exists, and $\lim F_N = \inf_r \Phi_r$.

8.2 What the model is (plain-language version)

- Fix an **even** $p \geq 2$ (how many spins each interaction involves) and a density $\alpha > 0$.
- Scatter $M_N \sim \text{Poisson}(\alpha N)$ interactions; each picks p spins at random (**with replacement**, so repeats are allowed) and applies a random function θ . Each spin also gets a random external field h .
- The only requirement is $E[\max|\theta|] + E[|h|] < \infty$ ("only first moments are needed": the interactions are not required to be bounded).

The Panchenko–Talagrand condition: e^θ can be written in the form $a \cdot (1 + b \cdot f_1(s_1) \dots \cdot f_p(s_p))$, where the f 's are i.i.d., b is independent of them, and **$E[(-b)^n] \geq 0$ holds for all $n \geq 1$.**

Check it on the Viana–Bray model ($p = 2$, $\theta = \beta J \cdot s_1 s_2$, with J symmetrically distributed): take $a = \cosh(\beta J)$, $b = \tanh(\beta J)$, $f(s) = s$.

[Check 5] Verify that $e^{(\beta J \cdot s_1 s_2)} = \cosh(\beta J) \cdot (1 + \tanh(\beta J) \cdot s_1 s_2)$, separately for $s_1 s_2 = +1$ and -1 . (The script, with $\beta J = 0.7$, verifies that both sides are 2.01375 and 0.49659.)

Because J is symmetric, the odd moments of $b = \tanh(\beta J)$ are 0 and the even moments are positive, so the positivity condition holds.

The coverage listed in Section 10 of the paper (OpenAI's claim):

- the **Viana–Bray model** ($p = 2$, symmetric random couplings);
- **diluted p -spin models with even p** (symmetric couplings);
- **soft random even K-SAT**, at every positive temperature ("soft": a violated clause is only penalized by energy βW , not strictly forbidden).

- Corollary 10.3: lowering the temperature to zero gives the limit of "maximum energy / minimum number of violated clauses," with error $\leq \ln 2 / \beta$.

8.3 What it does not cover (important)

- **Odd arity:** random **3-SAT** is not included. The paper's Remark 10.4 says so itself, and points to another OpenAI paper from the same batch, *Computing the Random 3-SAT Threshold* (2026-09-27), which uses a different trial functional (OpenAI's claim; not read for this chapter).
- **Fixed-degree Bethe lattice / random regular graphs:** the degrees in the theorem come from Poisson edge scattering. The setting of the original Mézard–Parisi 2001 paper is exactly the fixed-degree one, and that case is **not** covered by the theorem. In other words, the "Mézard–Parisi formula" in the title refers to its counterpart on Poisson-diluted models.
- **Ferromagnetic models** (my inference; I haven't seen the paper say it explicitly $\triangle$): for fixed $J > 0$, the standard decomposition gives $b = \tanh(\beta J) > 0$, so $E[(-b)^n]$ alternates in sign and the positivity condition fails. But the ferromagnetic case was already handled by Dembo–Montanari 2010, so it is not a gap.
- **The zero-temperature hard-constraint SAT threshold:** the theorem is a result at every positive temperature; the zero-temperature limit only gives the minimum number of violations, not the satisfiability threshold directly. So it does **not** re-prove or extend Ding–Sly–Sun.
- It **does not claim** that the infimum is attained by some trial law, and it does not construct an infinite-level limiting Gibbs state (the remarks after Theorem 2.1 in the paper).

8.4 Intuition for the method (I only read the introduction and Section 2.2; what follows is a simplified version)

The upper bound reuses the interpolation method (Section 4). The lower bound has four steps:

1. **Choose a "cavity reservoir":** compare the system with N spins and with $N + 1$ spins, adding a sublinear number of small perturbations. These don't change the pressure, but they make the system satisfy certain "identities" (the perturbed statistics concentrate).
2. **Prescribe branching patterns:** develop a "calculus" on finite probability trees that expands the effect of inserting a new spin into a signed sum over replica paths, using independent labels to pick out any specified replica branching pattern.
3. **Concentrate all multioverlaps** (the core trick): **shift every internal branching depth of the target tree down one level, all at once.** That way the new branchings always land on depths the old tree hasn't used; each new branching pays a small factor, which exactly cancels the small factor in the normalizing coefficient, so the error stays controllable after

division. Then **average over the branching depths** (it doesn't need to hold uniformly for every depth), and inductively make the conditional covariances decay.

4. **Recover independent cavity messages:** once the multioverlaps are concentrated, the messages at different positions can be replaced by independent copies, giving a valid trial value that is close to the true pressure.

Order of limits: for each fixed depth, first let $N \rightarrow \infty$, then deepen the hierarchy. Finally a truncation removes the "bounded interactions" assumption, leaving only first moments.

The reasoning summary ([reasoning_traces/mezard-parisi-formula.pdf](#), 5 pages) records this process. Two lines verbatim:

"Aha recursive structure: overfitting shared variable can simulate original model at smaller scale."

"Great! Need carefully prove average error little with shifted depths not rare"

The summary also candidly records failed attempts: the pinning method (fixing some spins) could not reconcile the observed distributions; a claim of "exact purity" "proves too strong"; and before "shifting the branching depths," various expansions all had errors that were too large after dividing by small exponent gaps.

8.5 Neighboring results in the same batch (OpenAI's claims; not verified for this chapter)

The same release also contains a whole group of spin glass results: family 222 (perceptron free energy, jamming exponents), 281 (full support of the zero-temperature SK Parisi measure; QAOA reaching the SK ground-state energy), 235 (computability and variance of the random k-SAT threshold; priority for the existence of the 3-SAT threshold goes to Gaia Carenini's ECCC TR26-229), 217 and others (SK fluctuations and dynamics). Section 10.3 uses 222 and 281.

9. How well has it been verified

9.1 The Lean statement

The main theorem of the challenge file [lean/ComparatorChallenges/DilutedSpin.lean](#) (154 lines):

```
theorem mezard_parisi {p : ℕ} (M : Model p) (hM : Admissible M) :  
  Tendsto (pressure M) atTop (λ N (variationalValue M))
```

Word-by-word translation:

- `{p : ℕ}`: p is a natural number, the arity of the interactions.
- `(M : Model p)`: a model, made of three things: the density `alpha`, the distribution of interactions `disorder` (together with the decomposition witnesses θ, a, b, f), and the distribution of external fields `field`.
- `(hM : Admissible M)`: the assumption that the model satisfies the conditions. The `Admissible` structure spells them out one by one: `arity : 2 ≤ p`, `even : Even p`, `density : 0 < alpha`, two first-moment integrability conditions, `factorization` ($e^\theta = a(1 + b \cdot \prod f)$, with $|b \cdot \prod f| < 1$), the f 's i.i.d. and b independent of the f 's, all moments of b integrable, and `positivity : 0 ≤ E[(-b)^n]`. These correspond one-to-one with (2.1)–(2.4) in the paper.
- `pressure M`: a sequence whose N -th term is F_N . The definition is written very concretely: take the expectation over $\text{Poisson}(\alpha N)$ interactions and i.i.d. external fields, **average uniformly** over all possible index choices (implementing "uniform sampling with replacement"), take log-sum-exp over the 2^N configurations, and divide by N .
- `variationalValue M : π r, phi M r`, i.e. the infimum over all numbers of levels r ; `phi M r` is the infimum of `functional` over all $\zeta \in \text{Hierarchy } (r+1)$ and all m satisfying `Exponents` ($0 < m_1 < \dots < m_r < 1$), and `functional` corresponds term by term to the paper's (2.11) $B_r = \ln 2 + E \ln T_0 V_s - \alpha(p-1) \cdot E \ln T_0 V_e$.
- `Tendsto ... atTop (λ ...)`: as $N \rightarrow \infty$, F_N converges to this value. This also includes "the limit exists."

The difference from Explainer 01: the quasi-Riemann statement used only Mathlib's standard definition `riemannZeta`; **here almost all the definitions (`pressure`, `Hierarchy`, `functional`, ...) are written by the challenge file itself**, so "is the statement faithful" depends on a human reading these 154 lines. I read them item by item against Section 2 of the paper and found no inconsistencies, but I am not an expert in this field $\triangle$.

9.2 The proof itself

- Solution module `0AI/Probability/DilutedSpin/`: **405 .lean files, 56,285 lines** (counted by script). grep for `sorry` / `admit` / `native_decide` / `axiom` declarations: **0**.
- The definitions in the solution module's `Model.lean` are **word-for-word identical** to those in the challenge file (I diffed the section from `Spin` to `variationalValue`).
- Comparator config `DilutedSpin.json`: only Lean's three standard axioms are allowed (`propext`, `Quot.sound`, `Classical.choice`); `enable_nanoda: false`, meaning no second, independent kernel is configured for a re-check.
- **We did not actually compile the proof or run Comparator** (the full library is very large). `formalization.yaml` is marked `review: status: unchecked` overall.

- Unlike quasi-Riemann, I found no report of any third party (someone like Goldblatt) independently running this proof successfully $\triangle$.

9.3 Human experts

As of 2026-10-08, I found no named comment or verification report on this paper from any spin glass / probability expert $\triangle$ (searched; I only found aggregator blogs restating family 221). Panchenko, Talagrand, Coja-Oghlan, and others are the natural referees.

9.4 Remaining risks

1. **Faithfulness of the statement:** Lean proves the statement written in the challenge file. A few places deserve a close look from an expert: each level of `Hierarchy` uses the Borel σ -algebra of the weak topology (the file's comments say this follows the paper); the indexing convention for m in `logMean` (`m 0`) corresponds to the paper's m_1 ; and `pressure` uses "average over all indices" in place of "random sampling." I believe these are all equivalent, but I haven't checked formally $\triangle$.
2. **Title vs. scope:** readers can easily take "the Mézard–Parisi formula" to mean the Bethe lattice (fixed-degree) case of Mézard–Parisi 2001, or to include 3-SAT. Neither is included (Section 8.3). This is the "spec vs. intent" gap that keeps recurring in this series.
3. **Readability of the proof:** a 36-page paper against 56,000 lines of Lean; the paper is written in summary style. It may take humans quite some time to understand why the "shifting the branching depths" trick works.

10. What it would mean if true (and what it would not mean)

10.1 For mathematics

1. **A roughly 20-year-old open conjecture (Panchenko–Talagrand 2004) is resolved in its original model class**, and it needs only first moments, a weaker assumption than the original problem set.
2. **The new technique may matter more than the conclusion:** "shift the branching depths + average over depths" to control all multioverlaps. If it holds up, it may generalize to other sparse models (fixed degree, odd arity). The paper itself says the lower-bound part **holds for any bounded interaction that is invariant under permutations of its inputs**, without needing the decomposition or the positivity condition (end of §2.2 of the paper); what is stuck is the upper bound. In other words, if someone later extends the interpolation upper bound to odd arity, the lower-bound half is already done.
3. **What it does not mean:** the formula for the random 3-SAT threshold, a complete solution of the Bethe lattice spin glass, and "ultrametricity" on sparse models are all still unsolved.

10.2 For physics

Physicists already "knew" in 2001 that this formula was right, and used it to compute a large number of numerical predictions. The theorem doesn't change any physical prediction. Its role is to **upgrade the cavity method from "a trustworthy heuristic" to a theorem** for one class of sparse systems. This is the same kind of significance as Talagrand 2006 for SK, but with a narrower scope.

10.3 For ML and neuroscience (the honest version)

Connections that really exist:

- **A Hopfield network (1982) is an Ising system.** Energy $E = -\frac{1}{2} \sum_{i,j} W_{ij} \sigma_i \sigma_j$, and the Hebb rule $W_{ij} = (1/N) \sum_{\mu} \xi_i^{\mu} \xi_j^{\mu}$ stores P patterns. When too many are stored, the "crosstalk" from the other patterns acts like random couplings and creates frustration, and the system falls into "spin glass states" that don't correspond to any memory. Amit–Gutfreund–Sompolinsky 1985 used the replica method to compute the capacity **$\alpha_c = P/N \approx 0.138$** . Script simulation ($N = 1000$, zero-temperature asynchronous updates, starting from a stored pattern, 30 runs per α):

$\alpha = P/N$	0.05	0.10	0.12	0.13	0.14	0.15	0.17	0.20
Mean final overlap m	1.000	0.998	0.992	0.974	0.941	0.917	0.583	0.363

Retrieval quality clearly collapses between 0.13 and 0.17, consistent with the 0.138 prediction. At $N = 1000$ the transition is smeared out by finite-size effects, and past 0.138 the overlap doesn't drop to zero right away, so this table is only a qualitative check. Hopfield and Hinton won the 2024 Nobel Prize in Physics.

- **Perceptron capacity:** Gardner 1988 used the replica method to compute the capacity of the spherical perceptron, $\alpha = 2$; for the Ising perceptron (weights ± 1), Krauth–Mézard 1989 predicted about 0.833, and rigorous proofs have only recently made partial progress (Ding–Sun 2019 proved the lower bound; Huang 2024 gave a matching upper bound under a numerical assumption $\triangle$ not rechecked). Family 222 in the same batch deals with the perceptron free energy at **positive temperature**, not the zero-temperature capacity itself.
- **Random optimization landscapes and algorithms:** Montanari 2019 (arXiv:1812.10897) gave a polynomial-time algorithm that approximately finds the SK ground state, **provided** the zero-temperature Parisi measure has "no overlap gap." Family 281 in the same batch claims to prove this full-support property (OpenAI's claim, not verified $\triangle$). If it holds, "the SK ground state can be approximated efficiently" becomes unconditional. This is the most direct answer spin glass theory gives to "can random landscapes be optimized efficiently," and it is closer to ML's concerns than 221 itself.

- **Inference on sparse graphs:** LDPC decoding, community detection, and the information-theoretic thresholds of sparse constraint satisfaction problems all rely on cavity / BP-type variational formulas. 221 adds one more rigorous cornerstone to arguments of the form "physics prediction = true threshold" (speculative extrapolation).

Where to discount:

- **221 does not cover Hopfield.** Hopfield is fully connected (dense), and its couplings are determined by the patterns, so different edges are not independent; 221 requires sparse models with independent interactions. The "sparse Hopfield" models in neuroscience (for example the strongly diluted model of Derrida–Gardner–Zippelius 1987) also don't fit the Panchenko–Talagrand decomposition.
- Analogies like **deep network loss landscape $\approx$ spherical spin glass** (Choromanska et al. 2015) rely on strong simplifying assumptions and are **heuristic**. This theorem won't tell you where SGD converges or how well a model generalizes.
- The title of deck §13, "from magnets to the energy landscapes of neural networks," is about an **intellectual lineage** (energy functions, the replica method, rugged landscapes), not a claim that this theorem has direct consequences for neural networks.

10.4 For AI

This is a textbook case of "AI fills in a missing step in a classic research program": the problem has a clear origin (the conjecture after Theorem 4 of PT 2004), a Lean statement, and a candid reasoning summary. It is also a teaching case for "spec vs. intent": the scope Lean covers (Poisson, even arity, positivity) is narrower than the title suggests, and judging that gap takes domain knowledge; Lean can't do it for you.

11. Self-test

Try to answer these after reading. If you can't, go back and reread the corresponding section.

1. Why do physicists compute $E[\ln Z]$ rather than the easier $\ln E[Z]$? When do the two differ? (Section 3.3)
2. Which step of the replica trick has no mathematical justification? (Section 4.1)
3. What is wrong with the replica-symmetric SK solution? Give a clearly unphysical number. (Section 4.2)
4. Why is cavity / BP exact on trees but inexact on graphs with short cycles? Why is there still hope for sparse random graphs? (Sections 5.2–5.3)
5. What does "hierarchical" refer to in the hierarchical cavity formula? How is **Hierarchy** defined in Lean? (Sections 5.4, 9.1)

6. What is the relationship between the $\pi^2/6$ of the random assignment problem and the formula OpenAI proved? (Sections 0, 6)
7. In the Panchenko–Talagrand conjecture, which half was proved long ago? Why is the other half harder in sparse models than in SK? (Sections 7.2–7.3)
8. List three classes of models that OpenAI's theorem does **not** cover. (Section 8.3)
9. Why can't you say "this theorem proves such-and-such a property of Hopfield networks"? (Section 10.3)

Appendix: answers to the checks

- **Check 1:** $e^1 \approx 2.7183$, $e^{-1} \approx 0.3679$, $2e + 2e^{-1} \approx 6.1724$, so $P(+,+) \approx 2.7183 / 6.1724 \approx 0.4404$ (script: 0.44040).
- **Check 2:** $H = \sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3$. For $(+,+,+)$ and $(-,-,-)$, all three terms are $+1$, so $H = +3$. The other 6 configurations all have "two of one sign, one of the other": the same-sign pair contributes $+1$, and the other two pairs contribute -1 each, so $H = -1$. So there are 2 with $+3$ and 6 with -1 .
- **Check 3** (the example from Section 5.3): only spin 1 has an external field. $Z = \sum e^{(\sigma_1\sigma_2 + 0.5\sigma_1)}$. The total weight for $\sigma_2 = +1$ is $e^{(1.5)} + e^{(-1.5)}$, and for $\sigma_2 = -1$ it is $e^{(-0.5)} + e^{(0.5)}$. $m_2 = [e^{1.5} + e^{-1.5} - e^{0.5} - e^{-0.5}] / [e^{1.5} + e^{-1.5} + e^{0.5} + e^{-0.5}] = (4.7048 - 2.2553)/(4.7048 + 2.2553) \approx 0.3519 = \tanh(1) \cdot \tanh(0.5)$.
- **Check 4:** $a + d$ and $b + c$ are two independent $\text{Gamma}(2, 1)$ variables, with $P(X > t) = (1+t)e^{-t}$. So $P(\min > t) = (1+t)^2 e^{-2t}$, and $E[\min] = \int_0^\infty (1 + 2t + t^2)e^{-2t} dt = 1/2 + 2 \cdot (1/4) + 2/8 = 5/4$.
- **Check 5:** when $s_1 s_2 = +1$, the right-hand side $= \cosh(\beta J) + \sinh(\beta J) = e^{(\beta J)}$; when $s_1 s_2 = -1$, the right-hand side $= \cosh(\beta J) - \sinh(\beta J) = e^{(-\beta J)}$. In both cases it equals the left-hand side.

Appendix: Glossary

Term	One-line explanation
Spin	A variable taking only ± 1 , representing a tiny compass needle (or a binary neuron)
Ising model	A spin system with energy $-\sum J_{ij} \sigma_i \sigma_j$
Boltzmann distribution	$P(\sigma) \propto e^{(-\beta H(\sigma))}$, i.e. a softmax with $-\beta \cdot \text{energy}$ as the logit
Partition function Z	The sum of the weights of all configurations; the normalizing constant
Free energy / pressure	$F = -(1/\beta) \ln Z$; pressure $= (1/N) \ln Z$; they differ by a factor
Frustration	A situation where it is impossible to satisfy all interactions at once
Spin glass	A spin system with randomly signed couplings and frustration everywhere
Quenched / annealed average	$E[\ln Z] / \ln E[Z]$; physics needs the former
Replica method	A non-rigorous trick that computes $E[\ln Z]$ via $\ln Z = \lim (Z^n - 1)/n$
Overlap Q_{ab}	The similarity of two replica configurations, $(1/N) \sum \sigma_i^a \sigma_i^b$
Multioverlap	The spatial average of a product of spins across several replicas; sparse models require controlling all of them
RSB (replica symmetry breaking)	Replicas form a hierarchy (a tree), corresponding to many pure states
Ultrametric	A distance structure where, among any three points, the two largest distances are equal, like a taxonomy tree
SK model	A dense spin glass in which every pair of spins has a random coupling
Viana–Bray model	A Poisson-diluted two-body spin glass
Bethe lattice	A sparse (random regular) graph in which every vertex has the same fixed degree
Cavity method / BP	Remove a spin and compute its state from messages sent by its neighbors; exact on trees
Hierarchical cavity formula	A variational formula expressing the free energy in terms of "distributions of distributions of messages..."
Panchenko–Talagrand class	Even-arity interactions with $e^\theta = a(1 + b \cdot \Pi f)$ and $E[(-b)^n] \geq 0$
Interpolation method	Guerra's trick: interpolate continuously between the real model and a trial model to obtain an upper bound

Appendix: Further reading

- **Mézard & Montanari**, *Information, Physics, and Computation* (Oxford University Press, 2009): covers spin glasses, the cavity method, BP, random k -SAT, and coding in one book, written for CS readers; Chapters 14–19 cover exactly Sections 5–7 of this chapter. Most recommended.

- **Panchenko**, *Introduction to the SK model* (survey, arXiv:1412.0170):
<https://arxiv.org/abs/1412.0170> ; the full version is his book *The Sherrington–Kirkpatrick Model* (Springer, 2013).
- **Nobel 2021 popular information** (the Parisi part):
<https://www.nobelprize.org/prizes/physics/2021/popular-information/>
- **Mézard & Parisi 2001**, *The Bethe lattice spin glass revisited*, EPJ B 20:217 (arXiv:cond-mat/0009418 $\triangle$ ID from memory, please verify).
- **Panchenko & Talagrand 2004**, *Bounds for diluted mean-fields spin glass models*, PTRF 130:319, <https://doi.org/10.1007/s00440-004-0342-2> .
- **Panchenko 2016**, *Structure of finite-RSB asymptotic Gibbs measures in the diluted spin glass models*: <https://arxiv.org/abs/1406.4702>
- **Aldous 2001**, *The $\zeta(2)$ limit in the random assignment problem*, Random Structures & Algorithms 18:381–418.
- **Montanari 2019**, *Optimization of the Sherrington–Kirkpatrick Hamiltonian*:
<https://arxiv.org/abs/1812.10897>
- **Concetti 2025** (random regular graphs, the open part): <https://arxiv.org/abs/2508.17049>
- The original paper: <https://github.com/openai/math/blob/main/preprints/The-Mezard-Parisi-formula-for-diluted-spin-glasses-September-23-2026/paper.pdf>
- The Lean statement:
<https://github.com/openai/math/blob/main/lean/ComparatorChallenges/DilutedSpin.lean> ;
scope notes: <https://github.com/openai/math/blob/main/lean/docs/221.md>
- The reasoning summary:
https://github.com/openai/math/blob/main/reasoning_traces/mezard-parisi-formula.pdf

33. What people said

WHO	QUOTE	SOURCE
Levent Alpöge (Anthropic)	"It's obviously the most significant moment in mathematical history." (later adding there would be "sad stories" of people being scooped)	X [R]
Terence Tao	"Problems are being solved autonomously by AI prompters who have no interest in the broader field itself"; proposes Math 1.0 → 2.0	Fortune [R]
Dan Litt (Toronto)	"My view is that this is great for mathematics," while worrying that an "AI has solved math" perception could discourage young people	Fortune [R]
Tristan Buckmaster (NYU)	"There's likely to be a bunch of results where they take someone's work and then take it to completion"	Fortune [R]
Scott Aaronson	Post "The Mathocalypse"; on $L = BPL$: "its truth was never in serious doubt"	blog [R]
Lance Fortnow	"if they hold up, we've seen more progress in TCS in the last 24 hours than in the previous three decades combined"	Computational Complexity blog [R]
Mark Braverman	"Math by press release is not that healthy for math."	Quanta [R]
François Chollet	Asks whether gains in RLVR-friendly domains generalize, or whether non-verifiable domains stay bottlenecked on human data	X (paraphrase) [R]
AHM	"Releasing over 700 files at once is not a demonstration of scholarship, but a demonstration of power."; urges mathematicians to stop working with OpenAI	reposted on Tao's blog (Tao notes he does not endorse it) [P]
Álvaro Lozano-Robledo	"Keep calm and carry on studying math."; LLMs "may work within the confines of the convex hull of ideas that are currently available in the literature"	guest post on Tao's blog [P]
Andrew Sutherland (MIT)	"We should ask for receipts."	via Scientific American [R]

Note: "The Future of Mathematics" on Tao's blog (10-05) is a guest post by Jeremy Avigad, not by Tao; some coverage conflated the two.

34. Open questions and how to keep learning

34.1 What to watch

1. Will OpenAI publish the list of ~4,000 problems and the failures? (This decides whether "what is the real capability?" can be answered.)
2. How many errors like the Weil-class paper remain in the ~58% that is unformalized? How will the withdrawal rate evolve?
3. Who reviews the faithfulness of the Lean statements? When does `review: unchecked` change? Who audits the `lana-agents` dependencies and the large patches?
4. When will the first human domain expert give an independent "read and confirmed" verdict on a headline result (quasi-RH, UGC, Thompson F)?
5. What priority norms should govern "completing someone else's work" (the Buckmaster episode)?
6. Chromatic number of the plane: what does a finite unit-distance graph needing 6 colors look like?

34.2 Learning path (starting from linear algebra + calculus)

STAGE	MATERIAL	WHY
1 · What math is	Gowers, <i>Mathematics: A Very Short Introduction</i>	150 pages by a Fields medalist: abstraction, proof, what "hard" means
1 · Roots of this debate	Thurston, On Proof and Progress in Mathematics; Gowers, The Two Cultures of Mathematics; Tao, What is good mathematics?	after these, "Math 1.0 → 2.0" makes complete sense
2 · Touch a proof	Natural Number Game (Kevin Buzzard, free online)	prove commutativity of addition from scratch in Lean — then §3 makes full sense
2 · Learn proofs systematically	MIT 6.042 Mathematics for Computer Science (OCW + free textbook)	designed for CS backgrounds: proof methods, discrete math, probability
3 · Intuition	3Blue1Brown: <i>Essence of linear algebra</i> ; visualizing the Riemann zeta function	the figure in §4 is a static version of that episode
3 · Following the news	Quanta Magazine (math); <i>The Princeton Companion to Mathematics</i> (as a reference)	every piece explains why a problem is hard and where people were stuck
4 · ML-adjacent	Bubeck, <i>Convex Optimization: Algorithms and Complexity</i> ; Prince, <i>Understanding Deep Learning</i> (free)	convergence proofs for optimization; intro to deep-learning theory

35. Glossary and sources

35.1 Glossary

TERM	IN ONE LINE
Lean / Mathlib	a proof assistant / its community math library
sorry / axiom	in Lean: "leave this step empty" / "assume this is true" — both weaken a proof
Comparator	tool checking "the theorem proven = the challenge statement, using only allowed axioms"
zero-free region	a region where ζ or an L-function has no zeros
irrationality exponent μ	how "unusually well" a number can be approximated by fractions
ω	the optimal exponent of matrix multiplication, $2 \leq \omega \leq 3$
approximation ratio	the fraction of the optimum a polynomial-time algorithm guarantees
L / RL / BPL	log-space deterministic / one-sided randomized / two-sided randomized computation
amenable	a group admits a translation-invariant "average"
blowup	a solution of a smooth equation going to infinity in finite time
replica symmetry breaking (RSB)	Parisi's description of the hierarchical state structure of spin glasses
galactic algorithm	asymptotically faster, but only better at astronomical sizes

35.2 Main sources

Primary: openai/math (README, history.md, overview.pdf, CONTENTS.md, lean/, preprints/, reasoning_traces/; snapshot fd4aeeb, 2026-10-08) · openai/NavierStokesAndEuler · Clay NS official statement (Fefferman) · leanprover/comparator · independent re-checks davegoldblatt/openai-zeta-proof-check, erenciracioglu-dotcom/openai-math-9-4-lean-check · Epoch FrontierMath Erdős (arXiv:2609.25050).

Coverage and commentary: Fortune 10-07 · Nature (paywalled, full text not read [△](#)) · Scientific American · Quanta 09-08 · Quanta 10-07 · NPR 09-22 · Aaronson · Kalai · Tao's blog · Latent Space.

Classic references: Strassen 1969; Coppersmith–Winograd 1990; Alman et al. arXiv:2404.16349; Ambainis–Filmus–Le Gall arXiv:1411.5414; Goemans–Williamson 1995;

Khot 2002; KKMO 2007; Raghavendra 2008; Khot–Minzer–Safra 2018; Nisan 1992; Saks–Zhou 1999; Reingold 2008; Roth 1955; Zeilberger–Zudilin [arXiv:1912.06345](#); Meiburg [arXiv:2208.13356](#); Guth–Maynard [arXiv:2405.20552](#); de Grey [arXiv:1804.02385](#); Moore [arXiv:1102.0747](#); Tao 2016 [arXiv:1402.0290](#); Glassey–Strauss 1986; Parisi 1979; Talagrand 2006; Mézard–Parisi 2001; Ding–Sly–Sun 2015.